

Fermi Notes on Physics

Provisional Draft

Introductory Undergraduate
Lectures on Astrophysics
and Cosmology

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Observational Foundations and Instrumentation

1.1 Magnitudes, Colours, and the Distance Ladder

Parallax, Luminosity, and Flux at School

Parallax and magnitudes are part of school astronomy, and we recall them briefly. As the Earth moves round the Sun, a nearby star appears to move back and forth against the much more distant stars behind it. The angle p subtended at the star by the radius of the Earth's orbit, which is the distance from the Earth to the Sun, is called the parallax of the star, as Figure 1 shows.

The astronomical unit is defined as the mean distance from the Earth to the Sun, $1 \text{ AU} \approx 1.50 \times 10^{11} \text{ m}$. Since the parallax is small, less than a second of arc for every star, $\tan p \approx p$ with p in radians, and the distance to the star is $d = 1 \text{ AU}/p$. The parsec is defined as the distance at which one astronomical unit subtends an angle of one second of arc, so that

$$\frac{d}{1 \text{ pc}} = \frac{1}{p/1''}, \quad (1.1)$$

with $1 \text{ pc} \approx 3.09 \times 10^{16} \text{ m} \approx 3.26 \text{ light-years}$.

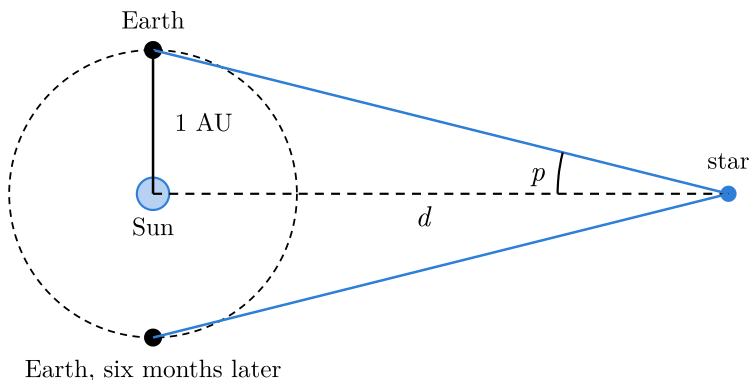


Figure 1. The parallax of a star. The angle p is the angle subtended at the star by the radius of the Earth's orbit, one astronomical unit, and it is half the angle through which the direction of the star appears to swing over six months. The angle is greatly exaggerated here, since for every star it is far too small to draw.

The luminosity L of a star is the energy it radiates per unit time, and the flux F that reaches us is the energy crossing unit area per unit time, at right angles to the line of sight. If the star radiates equally in all directions and nothing absorbs the light on the way, the energy crosses a sphere of area $4\pi d^2$, and

$$F = \frac{L}{4\pi d^2}, \quad (1.2)$$

which is the inverse-square law. The luminosity of the Sun is $L_{\odot} \approx 3.8 \times 10^{26}$ W.

The Magnitude Scale

The brightness of a star is usually quoted as a magnitude, on a scale that goes back to the Greek astronomers, who sorted the stars visible to the eye into six classes, from the first magnitude for the brightest to the sixth for the faintest. The response of the eye

to brightness is roughly logarithmic, and in 1856 Pogson made the scale exact by defining a difference of five magnitudes to be a ratio of one hundred in flux. The apparent magnitudes m_1 and m_2 of two stars whose fluxes are F_1 and F_2 are therefore related by

$$m_1 - m_2 = -2.5 \log_{10} \frac{F_1}{F_2}, \quad (1.3)$$

so that one magnitude is a ratio of $100^{1/5} \approx 2.512$ in flux, and a brighter star has the smaller magnitude. The relation fixes only differences, and the zero of the scale is set by a reference: $m = -2.5 \log_{10}(F/F_0)$, where F_0 is the flux of a star of magnitude zero. In the system most often used for visible light, F_0 is chosen so that the bright star Vega has a magnitude close to zero.

The absolute magnitude $\mathcal{M}$ of a star is defined as the apparent magnitude it would have at a distance of 10 pc. Since the flux goes as $1/d^2$, the flux at the true distance d is $(10 \text{ pc}/d)^2$ times the flux at 10 pc, and the definition of the magnitude scale gives

$$m - \mathcal{M} = -2.5 \log_{10} \left(\frac{10 \text{ pc}}{d} \right)^2 = 5 \log_{10} \frac{d}{10 \text{ pc}}. \quad (1.4)$$

The difference $m - \mathcal{M}$ is called the distance modulus, and measuring m for a star whose $\mathcal{M}$ is known gives its distance. The relation assumes that no light is absorbed on the way, which is a valid approximation for the nearest stars.

Photometric Bands and Bolometric Magnitudes

A detector does not respond equally to all wavelengths, and astronomers usually measure a flux through a filter, a piece of coloured glass or a glass plate with thin coatings that transmits light only in a chosen range of wavelengths. Such a range is called a photometric band. In the system of Johnson and Morgan the B band is centred near $\lambda_B \approx 440 \text{ nm}$ in the blue and the V band, for "visual", near $\lambda_V \approx 550 \text{ nm}$ in the yellow-green, with widths of about 90 nm, and there are further bands in the ultraviolet, the red, and the infrared. A magnitude measured in a band carries its

name, as in m_B , m_V , and $\mathcal{M}_V$, and each band has a separate zero point F_0 .

The magnitude that corresponds to the flux integrated over all wavelengths is called the bolometric magnitude, m_{bol} , and the absolute bolometric magnitude $\mathcal{M}_{\text{bol}}$ is a measure of the luminosity. For two stars at the same distance the ratio of the fluxes is the ratio of the luminosities, so comparing a star with the Sun gives

$$\mathcal{M}_{\text{bol}} - \mathcal{M}_{\text{bol},\odot} = -2.5 \log_{10} \frac{L}{L_{\odot}}, \quad (1.5)$$

where $\mathcal{M}_{\text{bol},\odot} \approx 4.74$ is the absolute bolometric magnitude of the Sun. A bolometric flux is hard to measure from the ground, since the atmosphere absorbs much of the ultraviolet and the infrared, and it is often obtained from the V magnitude by adding a correction. The quantity

$$\text{BC} = m_{\text{bol}} - m_V = \mathcal{M}_{\text{bol}} - \mathcal{M}_V \quad (1.6)$$

is called the bolometric correction. The difference of apparent magnitudes and the difference of absolute magnitudes in this display are equal because the distance modulus is the same for every band, so the bolometric correction depends on the spectrum of the star alone. Its zero point is fixed by convention, which makes it close to zero for stars somewhat hotter than the Sun and negative for stars hotter or cooler than these, since hot stars put much of their radiation into the ultraviolet and cool stars into the infrared, outside the V band. In Problem 3 you will find the temperature at which a black body puts the largest fraction of its radiation into a narrow band.

Colour Index and Colour Temperature

The difference $m_B - m_V$ between the magnitudes of a star in two bands is called a colour index, in this case the B – V colour index. By the definition of the magnitude scale, $m_B - m_V = -2.5 \log_{10}(F_B/F_V) + C_{\text{BV}}$, where F_B and F_V are the fluxes through the two filters and the constant C_{BV} comes from the two zero points.

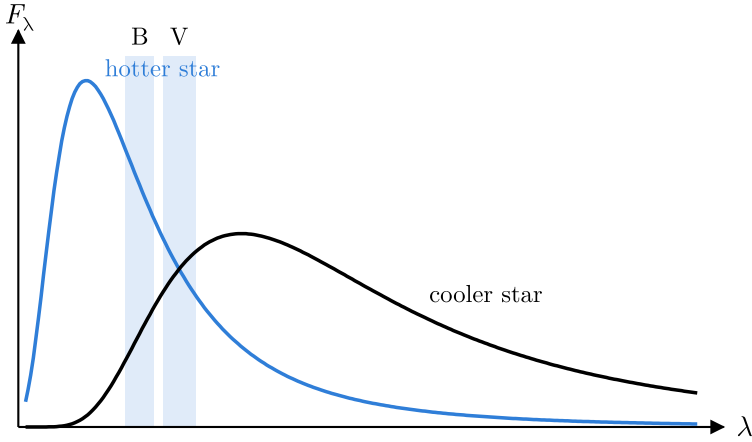


Figure 2. The flux per unit wavelength F_λ emitted by black bodies at two temperatures, each curve scaled to the same value in the V band. The shaded strips mark the B and V bands. The hotter body (accent) sends more flux through the B band than through the V band, and the cooler body (solid) less, so the cooler body has the larger colour index $m_B - m_V$.

A hot star is bluer than a cool one, and has the smaller colour index, as Figure 2 shows.

To see how the colour index depends on temperature, let us take the star to be a black body of radius R and temperature T ; blackbody radiation is treated in detail in Section 5.6 of Vol. 2 of the Fermi Notes on Physics. Let F_λ be the flux per unit wavelength received from the star, so that $F_\lambda d\lambda$ is the flux carried by wavelengths between λ and $\lambda + d\lambda$. In Problem 2 of Section 1.2 of Vol. 3 of the Fermi Notes on Physics the energy density of blackbody radiation per unit wavelength was found to be $u_\lambda = (8\pi hc/\lambda^5)/(e^{hc/\lambda k_B T} - 1)$, and by the argument of Section 5.6 of Vol. 2, applied to each range of wavelength separately, the flux leaving a surface that emits as a black body is $c/4$ times the energy density, so that

$$F_{s,\lambda}(T) = \frac{c}{4} u_\lambda = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1}. \quad (1.7)$$

Integrating over all wavelengths gives $\sigma_{\text{SB}} T^4$, where σ_{SB} is the

Stefan–Boltzmann constant, which Vol. 2 writes as σ . The luminosity of the black body is the flux $\sigma_{\text{SB}}T^4$ leaving each unit area of its surface multiplied by its area, $L = 4\pi R^2\sigma_{\text{SB}}T^4$, so the flux at distance d is $L/4\pi d^2 = (R/d)^2\sigma_{\text{SB}}T^4$, and every range of wavelength is diluted by the same factor, so $F_\lambda = (R/d)^2F_{\text{s},\lambda}$. The peak of $F_{\text{s},\lambda}$ lies at $\lambda_{\text{max}} = hc/(\zeta_{\text{p}}k_{\text{B}}T)$, where ζ_{p} is the pure number close to five found in the same problem of Vol. 3, so $\lambda_{\text{max}}T \approx 2.9 \times 10^{-3}$ m K, which is Wien’s displacement law in its school form.

For bands narrow enough that F_λ changes little across each of them, $F_{\text{B}} \approx F_\lambda(\lambda_{\text{B}}) \Delta\lambda_{\text{B}}$ and $F_{\text{V}} \approx F_\lambda(\lambda_{\text{V}}) \Delta\lambda_{\text{V}}$, where $\Delta\lambda_{\text{B}}$ and $\Delta\lambda_{\text{V}}$ are the widths of the bands. In the ratio of the two fluxes the factor $(R/d)^2$ cancels, which is why the colour index of a star does not depend on its size or its distance:

$$\frac{F_{\text{B}}}{F_{\text{V}}} = \frac{\Delta\lambda_{\text{B}}}{\Delta\lambda_{\text{V}}} \left(\frac{\lambda_{\text{V}}}{\lambda_{\text{B}}} \right)^5 \frac{e^{hc/\lambda_{\text{V}}k_{\text{B}}T} - 1}{e^{hc/\lambda_{\text{B}}k_{\text{B}}T} - 1}. \quad (1.8)$$

When $hc/\lambda k_{\text{B}}T \gg 1$ in both bands, the 1 in each denominator can be neglected, which is called the Wien approximation. Since $\log_{10} e^x = x/\ln 10$, the colour index then becomes

$$m_{\text{B}} - m_{\text{V}} \approx C' + \frac{T_{\text{BV}}}{T}, \quad T_{\text{BV}} = \frac{2.5}{\ln 10} \frac{hc}{k_{\text{B}}} \left(\frac{1}{\lambda_{\text{B}}} - \frac{1}{\lambda_{\text{V}}} \right), \quad (1.9)$$

where the constant C' collects C_{BV} and the logarithms of the ratios of the wavelengths and of the widths. Since $\lambda_{\text{B}} < \lambda_{\text{V}}$, the constant T_{BV} is positive, and the colour index falls as the temperature rises. With the central wavelengths of the B and V bands put in, T_{BV} is about 7×10^3 K. The zero points are chosen so that the colour index of Vega is zero, and if Vega is taken to be a black body at a temperature T_0 , then $C' = -T_{\text{BV}}/T_0$ and

$$m_{\text{B}} - m_{\text{V}} \approx T_{\text{BV}} \left(\frac{1}{T} - \frac{1}{T_0} \right). \quad (1.10)$$

The temperature of the black body that has the same colour index as a star is called the colour temperature of the star. Solving the

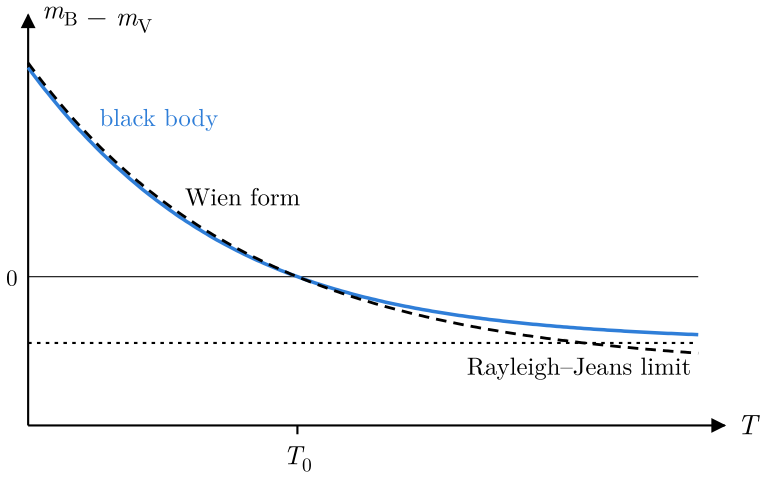


Figure 3. The B - V colour index of a black body against its temperature, on a logarithmic scale of temperature, with the zero point chosen so that the colour index vanishes at the temperature T_0 . The solid curve is computed from the Planck spectrum, the dashed curve is the Wien form $T_{BV}(1/T - 1/T_0)$, and the dotted line is the constant that the colour index approaches in the Rayleigh-Jeans limit.

last relation for T gives the colour temperature, as long as the Wien approximation holds, which requires $T \ll hc/\lambda_V k_B \approx 2.6 \times 10^4$ K, and Figure 3 compares the Wien form with the colour index computed from the full Planck spectrum. For the hottest stars both bands lie on the long-wavelength side of the peak, where the Rayleigh–Jeans form of the spectrum holds, and you will show in Problem 2 that the colour index then tends to a constant that does not depend on the temperature. A small error in the measured colour of a very hot star therefore gives a large error in its colour temperature, and the B – V colour index is a good thermometer only for stars of moderate temperature.

Effective Temperature

A star is not a black body. Its atmosphere absorbs strongly at some wavelengths, as the dark lines in its spectrum show, so its spectrum differs from the Planck spectrum, and a colour temperature depends on the pair of bands used. The effective temperature T_{eff} of a star, which depends on no choice of band, is defined by

$$L = 4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4, \quad (1.11)$$

so that it is the temperature of a black body of the same radius and luminosity, and the flux leaving each unit area of the surface of the star is $\sigma_{\text{SB}} T_{\text{eff}}^4$. With the measured values of $L_{\odot}$ and $R_{\odot} \approx 7.0 \times 10^8$ m put in, the effective temperature of the Sun comes to about 5.8×10^3 K. Solving the definition for the radius gives

$$R = \frac{1}{T_{\text{eff}}^2} \left(\frac{L}{4\pi\sigma_{\text{SB}}} \right)^{1/2}, \quad (1.12)$$

and this is how the radii of most stars are found, since the disc of a star is too small to be resolved by a telescope in all but a few cases. Comparing a star with the Sun, and using the relation between bolometric magnitude and luminosity, gives

$$\mathcal{M}_{\text{bol}} - \mathcal{M}_{\text{bol},\odot} = -5 \log_{10} \frac{R}{R_{\odot}} - 10 \log_{10} \frac{T_{\text{eff}}}{T_{\text{eff},\odot}}, \quad (1.13)$$

so that of two stars with the same effective temperature, the brighter is the larger.

The Distance Ladder

The distances to the stars and galaxies are found by a chain of methods, each of which reaches further than the one before and is calibrated by it. The chain is called the distance ladder, and each method is called a rung. The rungs described in this section are the following:

1. Trigonometric parallax, for nearby stars.
2. Main-sequence fitting, for star clusters.
3. The period–luminosity relation of Cepheid variable stars, for the nearer galaxies.
4. Supernovae of Type Ia, for galaxies far beyond those.

The list is not complete, and other methods, such as those based on the variable stars called RR Lyrae stars, lie between these rungs.

The first rung rests on the parallax, and its reach is set by the smallest parallax that can be measured. Differentiating $d = 1/p$, with d in parsecs and p in seconds of arc, gives $\delta d = -\delta p/p^2$, so an error δp in the parallax makes an error of size $\delta p/p^2$ in the distance, and the fractional uncertainty in the distance is

$$\frac{\delta d}{d} = \frac{\delta p}{p}, \quad (1.14)$$

and if the angle p is measured with a fixed uncertainty δp , the fractional uncertainty grows in proportion to the distance. From the ground, where the atmosphere blurs the images of stars, parallaxes below about a hundredth of a second of arc are hard to measure. The Gaia satellite, above the atmosphere, has measured the parallaxes of over a billion stars, with uncertainties of order 10^{-5} seconds of arc for the brighter ones, which gives distances good to ten per cent out to a few kiloparsecs.

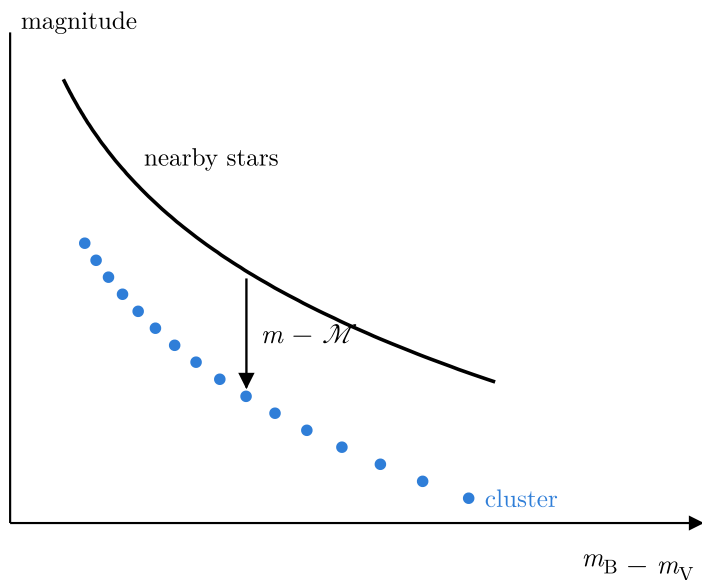


Figure 4. Main-sequence fitting. The solid curve is the main sequence of nearby stars whose distances are known from their parallaxes, drawn in absolute magnitude, and the dots are the stars of a cluster, drawn in apparent magnitude, with brighter stars higher in the figure. The vertical shift that brings the curve and the dots into coincidence is the distance modulus $m - M$ of the cluster.

Main-Sequence Fitting

A star cluster is a group of stars that formed together, and its stars all lie at nearly the same distance from us, since the size of a cluster is small compared with its distance. A plot of the apparent magnitude m_V of each star against its colour index is called a colour–magnitude diagram, and it is the observational form of the Hertzsprung–Russell diagram that you have met at school, with the colour index standing in for the temperature and the magnitude for the luminosity. Most of the stars of a cluster lie on the main sequence, and for nearby stars, whose parallaxes are known, the main sequence can be drawn in absolute magnitude. Since every star of the cluster has the same distance modulus, the main sequence of the cluster is the same curve shifted vertically by $m - \mathcal{M}$, as shown in Figure 4, and sliding one curve over the other until they coincide gives the distance modulus and hence the distance. In Figure 4 the main sequence is drawn schematically for black bodies whose luminosities and radii go as $M^{3.5}$ and $M^{0.8}$ with mass, which is found by observation to describe main-sequence stars roughly. The method assumes that the stars of the cluster have the same composition as the nearby stars, since the position of the main sequence depends on the abundance of the heavier elements, and that nothing dims or reddens the light of the cluster on its way to us.

Cepheid Variables

A Cepheid variable is a giant star whose brightness rises and falls regularly, with a period P_C that lies between about a day and a hundred days, because the star pulsates in and out. The mechanism of the pulsation is not treated in this course. In 1912 Henrietta Leavitt, studying Cepheids in the Small Magellanic Cloud, a small galaxy near the Milky Way, found that the brighter Cepheids have the longer periods. Since the stars of the Small Magellanic Cloud are all at nearly the same distance from us, Leavitt’s discovery is a relation between luminosity and period. It is found by observation that the mean absolute magnitude of a Cepheid is related to its period by

$$\mathcal{M}_V = \alpha \log_{10} \frac{P_C}{1 \text{ day}} + \beta, \quad (1.15)$$

with α negative, and this is called the period–luminosity relation. The constants α and β are found from Cepheids whose distances are known from their parallaxes, or from the clusters in which some of them lie, and this is how the third rung is calibrated by the first two. Cepheids are thousands of times as luminous as the Sun, and with telescopes in space they can be seen in galaxies tens of megaparsecs away.

The Distance to a Cepheid

As an example, consider a Cepheid in a distant galaxy whose brightness is observed to vary with the period P_C about a mean apparent magnitude m_V . The period–luminosity relation gives the absolute magnitude, and the distance modulus gives

$$d = 10 \text{ pc} \times 10^{[m_V - \alpha \log_{10}(P_C/1 \text{ day}) - \beta]/5}. \quad (1.16)$$

An error $\delta\beta$ in the calibration makes the same error in $\mathcal{M}_V$ for every Cepheid. The distance goes as $10^{-\mathcal{M}_V/5} = e^{-(\ln 10)\mathcal{M}_V/5}$, so its logarithmic derivative gives

$$\frac{\delta d}{d} = -\frac{\ln 10}{5} \delta\beta \approx -0.46 \delta\beta. \quad (1.17)$$

An error in the zero point of one rung therefore changes every distance measured with it by the same fraction, and the error is carried into every rung calibrated from it.

Supernovae of Type Ia

A supernova is a star that explodes and for a few weeks shines about as brightly as a whole galaxy, and those of Type Ia are the supernovae whose spectra show no lines of hydrogen and a strong line of silicon. It is found by observation that supernovae of Type Ia reach nearly the same luminosity at their peak, and that the small differences between them are related to the rate at which they fade, the brighter ones fading more slowly. A source of known luminosity, whose distance follows from its apparent magnitude, is called a standard candle, and once its peak luminosity is corrected

for the rate of fading, a supernova of Type Ia is a standard candle. The peak luminosity is calibrated with supernovae in galaxies whose distances are known from Cepheids, and supernovae of Type Ia can be observed out to thousands of megaparsecs.

1.2 Telescopes, Detectors, and Observing across the Spectrum

Telescopes at School

At school you met the two kinds of optical telescope. A refracting telescope forms an image with a lens, called the objective, and a reflecting telescope with a concave mirror. The power of either to collect light goes as the square of the diameter D of the objective, and its ability to separate two stars close together on the sky is limited by diffraction, as the Rayleigh criterion expresses. All the largest telescopes are reflectors, since a mirror can be supported across its whole back, while a lens is held at its edge and sags under its weight, and since a mirror reflects all colours to the same focus. This section first works out how many photons a telescope collects, then describes the limits that diffraction and the atmosphere place on the sharpness of its images, then the detectors that count the photons and the precision of the count, and finally the telescopes that observe at wavelengths the atmosphere does not transmit, among them the radio interferometers.

Collecting Area and the Rate of Arrival of Photons

Let F_λ be the flux per unit wavelength from a star, measured above the atmosphere. A telescope of diameter D has a collecting area $A = \pi D^2/4$, less the small area blocked by any mirror or support in the path of the light, and in a band of width $\Delta\lambda$ centred on λ it collects the energy $F_\lambda \Delta\lambda A$ per unit time. Each photon carries the energy $h\nu = hc/\lambda$, so the photons arrive at the rate $F_\lambda \Delta\lambda A\lambda/hc$. Some are absorbed in the atmosphere, at the mirrors, or in the filter, and some pass through the detector unregistered, so let η be the fraction of the photons arriving at the top of the atmosphere that is recorded. The rate at which photons from the star are recorded is then

$$\dot{N}_* = \eta \frac{\pi D^2}{4} \frac{\lambda}{hc} F_\lambda \Delta\lambda. \quad (1.18)$$

Photons from a Star of Given Magnitude

As an example, consider a star of apparent magnitude m_V . Let $F_{\lambda,0}$ be the flux per unit wavelength of a star of magnitude zero at the centre of the V band. The definition of the magnitude scale gives $F_\lambda = F_{\lambda,0} 10^{-0.4m_V}$, and therefore

$$\dot{N}_* = \eta \frac{\pi D^2}{4} \frac{\lambda_V \Delta\lambda_V}{hc} F_{\lambda,0} 10^{-0.4m_V}. \quad (1.19)$$

With the measured zero point put in, a star of magnitude zero delivers about 10^{10} photons per second through each square metre of collecting area in the V band.

The Diffraction Limit

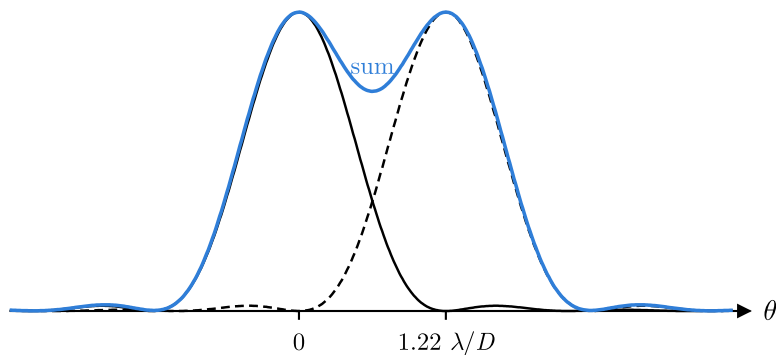


Figure 5. The Rayleigh criterion. The solid and dashed curves are the diffraction patterns of two point sources of equal brightness seen through a circular aperture of diameter D , plotted against the angle θ on the sky, and separated by the angle at which the centre of each pattern falls on the first dark ring of the other. The accent curve is their sum, which has a shallow dip between the two peaks.

Light from a star arrives at the telescope as a plane wave, and the objective, of finite diameter, diffracts it. The image of a point source is therefore a small bright disc surrounded by faint rings, which is called the Airy pattern, and its central disc ends at the first dark ring, at an angle θ from the centre given by

$$\sin \theta \approx 1.22 \frac{\lambda}{D}. \quad (1.20)$$

For a slit of width D the first minimum lies at $\sin \theta = \lambda/D$, as you have seen at school. The factor 1.22 comes from the Bessel function that describes the diffraction pattern of a circular aperture, whose properties are beyond this course. The Rayleigh criterion states that two point sources of equal brightness are just resolved when the centre of the pattern of one falls on the first dark ring of the other. At that separation the sum of the two patterns has a dip between its two peaks, as Figure 5 shows. The smallest angle that a telescope can resolve is therefore about $1.22\lambda/D$, which is called its diffraction limit. For a telescope with a mirror eight metres across, observing in visible light, the diffraction limit is of order a hundredth of a second of arc.

The Blurring of Images by the Atmosphere

Turbulence in the air above a telescope mixes pockets of air at slightly different temperatures, and hence with slightly different refractive indices, so the plane wave from a star that enters the top of the atmosphere reaches the ground with its wavefront wrinkled. Over a patch of the wavefront smaller than a certain length the wavefront is nearly flat, and this length is called the Fried parameter, ℓ_F . At good observing sites ℓ_F is some ten to twenty centimetres for visible light. A telescope larger than ℓ_F receives many such patches, each of which forms a separate diffraction pattern, of angular size about λ/ℓ_F , at a slightly different position, and the positions change many times a second as the turbulence moves. In an exposure longer than a fraction of a second the image of a star is therefore a blurred disc of angular diameter about λ/ℓ_F , which is called the seeing disc. At good sites it is between about half a second and one second of arc, and a large telescope on the ground

resolves no finer detail than one whose diameter is ℓ_F .

There are two ways around this limit that are in common use. The first is to place the telescope above the atmosphere, as was done with the Hubble Space Telescope, whose images are limited by diffraction. The second is adaptive optics, in which the light is reflected from a thin mirror whose shape is changed hundreds of times a second by small pistons behind it, so as to cancel the wrinkles in the wavefront, which are measured on a bright star close to the object being observed, or on an artificial star made by shining a laser into the upper atmosphere.

Charge-Coupled Devices and the Counting of Photons

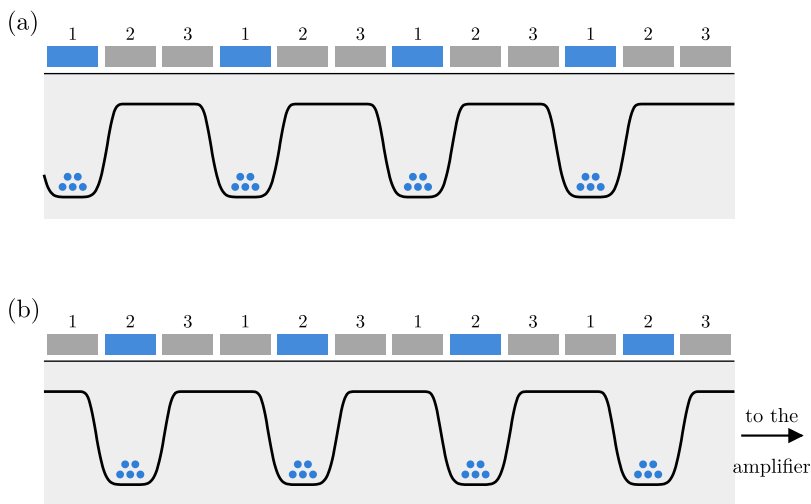


Figure 6. The transfer of charge along a row of pixels of a charge-coupled device, seen in cross-section, with the silicon shaded. The electrodes are connected in three sets, labelled 1, 2, and 3, and the set held at the higher voltage (accent) makes a well in the potential energy of an electron (curve) under each of its electrodes, in which the freed electrons (dots) collect. (a) During the exposure the wells lie under the electrodes of set 1. (b) Raising the voltage of set 2 and then lowering that of set 1 moves every well, with the charge in it, one electrode towards the amplifier at the end of the row.

Most images at visible wavelengths are now recorded with a charge-coupled device, which you met at school: a thin chip of silicon whose surface is divided into a grid of picture elements, called pixels. A photon absorbed in the silicon frees an electron from its bond in the crystal, provided the energy of the photon is larger than about 1.1 eV, which is true for visible light and the near infrared. Electrodes on the surface of the chip, separated from the silicon by a thin insulating layer, are held at voltages that make a well in the potential energy of an electron under every pixel, and the freed electrons collect in these wells during the exposure. To read the image out, the voltages on the electrodes are changed in turn so that the wells move along the rows, carrying the packets of charge with them, as Figure 6 shows, until each packet reaches an amplifier at the edge of the chip, which measures its charge.

A charge-coupled device records up to about ninety per cent of the photons that fall on it, where a photographic plate recorded only a few per cent, and the charge in a pixel is proportional to the number of photons over a wide range. Some electrons do not come from the light, however. Electrons freed by the thermal motion of the atoms of the silicon collect in the wells at a steady rate, which is called the dark current, and it is reduced by cooling the chip, often with liquid nitrogen. The amplifier adds a small random error to each measured charge, which is called the read noise.

The Signal-to-Noise Ratio

Photons from a steady source arrive independently of one another at a constant mean rate, and the number recorded in a fixed time therefore follows the Poisson distribution of Section 8.1 of Vol. 4 of the Fermi Notes on Physics. For that distribution the variance is equal to the mean, which Vol. 4 writes as μ and we write as $\bar{N}$. The number of photons recorded from a star in an exposure of duration t has the mean $\bar{N}_* = \dot{N}_* t$ and the standard deviation $\sqrt{\bar{N}_*}$, and the ratio of the mean to the standard deviation, which is called the signal-to-noise ratio and written S/N, is

$$S/N = \frac{\bar{N}_\star}{\sqrt{\bar{N}_\star}} = \sqrt{\dot{N}_\star t}. \quad (1.21)$$

The night sky glows faintly, and the pixels in which the image of the star falls also collect photons from the sky, at a rate $\dot{N}_{\text{sky}}$, and electrons from the dark current, at a rate $\dot{N}_{\text{dk}}$. The mean of these two contributions is measured from nearby pixels that contain no star, over an area so much larger than the image that the uncertainty in the mean is small, and it is subtracted. The total count in the image is the sum of three independent Poisson counts, and it is itself a Poisson count whose mean is the sum of their means, as you showed in Problem 5 of Section 8.1 of Vol. 4. Its standard deviation is therefore $\sqrt{(\dot{N}_\star + \dot{N}_{\text{sky}} + \dot{N}_{\text{dk}})t}$, and

$$S/N = \frac{\dot{N}_\star \sqrt{t}}{\sqrt{\dot{N}_\star + \dot{N}_{\text{sky}} + \dot{N}_{\text{dk}}}}. \quad (1.22)$$

The read noise adds its variance to the denominator in the same way, since the variances of independent quantities add, as shown in Section 8.1 of Vol. 4. In every case the signal-to-noise ratio grows as the square root of the exposure time. For a bright star, whose photons outnumber those of the background, it is $\sqrt{\dot{N}_\star t}$, the value for the star alone. For a faint star, whose photons are few compared with those of the sky, it is $\dot{N}_\star \sqrt{t/\dot{N}_{\text{sky}}}$, and the time needed to reach a given signal-to-noise ratio goes as $\dot{N}_{\text{sky}}/\dot{N}_\star^2$. You will show in Problem 3 that, for such a star, halving the diameter of the seeing disc shortens the exposure as much as doubling the diameter of the telescope, which is one reason for building telescopes at sites with good seeing and for placing them in space.

Observing through the Atmosphere

The atmosphere transmits only some wavelengths, as Figure 7 shows. Gamma rays and X-rays are absorbed high in the atmosphere by the atoms and molecules of the air, which they ionise, and ultraviolet light of wavelengths shorter than about 300 nm is

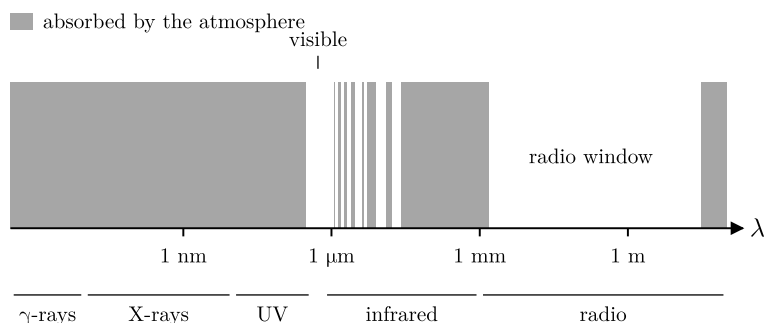


Figure 7. The transparency of the Earth's atmosphere to electromagnetic radiation, drawn schematically against the wavelength on a logarithmic scale. The shaded regions mark the wavelengths that are absorbed before they reach the ground, or that are reflected back into space at the longest wavelengths. The gaps are the windows through which telescopes on the ground can observe, namely the visible window, the narrow windows in the near and mid infrared, and the radio window.

absorbed by the ozone in the stratosphere. The visible window runs from there to about one micrometre. In the infrared, water vapour and carbon dioxide absorb strongly in a series of bands, leaving narrow windows between them, which are best used from high, dry sites. From a few tens of micrometres to about a millimetre the atmosphere is almost opaque. The radio window runs from about a millimetre to some tens of metres. Beyond that, the ionosphere, the layer of the upper atmosphere in which ultraviolet light from the Sun has ionised the gas, reflects radio waves back into space; the theory of waves in an ionised gas, which accounts for this reflection, is not treated in this course.

Telescopes in Space

A telescope above the atmosphere can observe at every wavelength, free of the seeing disc, and ultraviolet astronomy at wavelengths shorter than the ozone cut-off is done entirely from space. In the infrared there is a further reason to leave the ground, which is that the atmosphere and the telescope glow. The atmosphere and the telescope are both at a temperature of a few hundred kelvin, and by Wien's displacement law of Section 1.1 their radiation peaks

at about ten micrometres, so an infrared telescope on the ground looks at faint sources through a bright background produced by the telescope itself and by the air. Infrared telescopes in space are therefore cooled, either with liquid helium or by shading them from the Sun and the Earth, so that they radiate little at the wavelengths they observe.

X-rays are reflected only when they strike a smooth metal surface at a grazing angle, within a degree or so of the surface, and at larger angles they pass into the mirror and are absorbed. An X-ray telescope therefore focuses them with nested mirrors shaped like shallow cones, which the X-rays strike almost edge on, and its detector records each X-ray photon separately, with its energy.

Radio Telescopes and Interferometry

A radio telescope is a large metal dish, shaped as a paraboloid, that reflects the radio waves from a source to a receiver at its focus, which converts them into an alternating voltage. Since radio wavelengths are long, the diffraction limit $1.22\lambda/D$ is large: for a dish one hundred metres across, observing at a wavelength of 21 cm, it is about ten minutes of arc, a third of the angular diameter of the Moon. Much finer resolution is obtained by combining the signals from several dishes, and an arrangement of this kind is called an interferometer.

Let two identical dishes be separated by a distance ℓ , which is called the baseline, and let a distant source lie at an angle θ to the vertical, in the plane that contains the baseline, as in Figure 8. The wave from the source reaches one dish after travelling an extra path $\ell \sin \theta$, and hence a time $\Delta t = \ell \sin \theta / c$ later than it reaches the other. For radiation of a single frequency ν , the voltages from the two receivers are

$$V_1 = V_a \cos(2\pi\nu t), \quad V_2 = V_a \cos[2\pi\nu(t - \Delta t)], \quad (1.23)$$

where V_a is their common amplitude, and they are brought along cables of equal length to a central receiver that adds them and measures the mean power, which is proportional to the mean of $(V_1 + V_2)^2$ over time. The mean of $\cos^2$ over a period is one half,

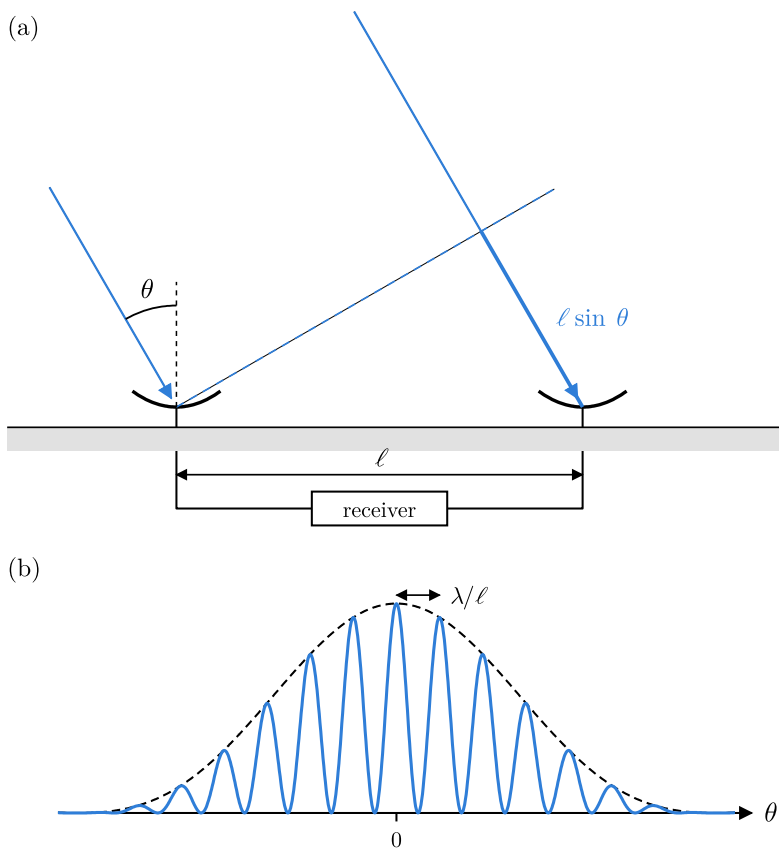


Figure 8. A radio interferometer of two dishes. (a) The two dishes are separated by the baseline ℓ , and a plane wave arrives from a direction at angle θ to the vertical. The dashed line is a wavefront, which reaches the right-hand dish after travelling the extra path $\ell \sin \theta$. (b) The mean power from the central receiver, which adds the two signals, against θ (accent). The fringes are separated by λ/ℓ , and they lie within the response of a single dish (dashed), whose width is of order λ/D .

so V_1^2 and V_2^2 each have mean $V_a^2/2$. By the identity $\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$, the cross term $2V_1V_2$ is $V_a^2[\cos(2\pi\nu\Delta t) + \cos(4\pi\nu t - 2\pi\nu\Delta t)]$, and the second cosine averages to zero. With $\nu\Delta t = \ell \sin\theta/\lambda$, the mean of the square of the sum is

$$\langle (V_1 + V_2)^2 \rangle = V_a^2 \left[1 + \cos \left(\frac{2\pi\ell \sin\theta}{\lambda} \right) \right]. \quad (1.24)$$

The response has maxima wherever $\ell \sin\theta$ is a whole number of wavelengths, and these maxima are called fringes, since they are the interference fringes of Young's two-slit experiment with the dishes in place of the slits. Near the vertical, where $\sin\theta \approx \theta$, neighbouring fringes are separated by the angle λ/ℓ , which is much smaller than the width λ/D of the response of a single dish when the baseline is much longer than the diameter of a dish, as in Figure 8. As the Earth rotates, the source moves through the fringes, and the times at which it crosses them fix its position to a small fraction of λ/ℓ . The resolution of the interferometer is therefore set by its baseline, which can be made far longer than any dish is wide.

An array of many dishes has many baselines, of different lengths and directions, and as the Earth rotates each baseline turns relative to the source. Combining the signals from all the pairs of dishes over many hours builds up an image as sharp as that of a single dish as large as the array, a method called aperture synthesis, for which Ryle received the Nobel Prize in 1974. In very-long-baseline interferometry the dishes are on different continents, and each records its signal together with the time from an atomic clock, a clock whose ticks are set by the frequency of a transition between two levels of an atom. The recordings are combined afterwards, and with baselines as long as the diameter of the Earth and wavelengths of a few centimetres or less, the resolution is better than a thousandth of a second of arc.

Problems

§1.1 Magnitudes, Colours, and the Distance Ladder

1. Consider two stars that have the same absolute magnitude and parallaxes p_1 and p_2 . Find the difference $m_1 - m_2$ of their appar-

ent magnitudes, and the ratio of the fluxes received from them.

2. Show that when $hc/\lambda k_B T \ll 1$ in both bands, the ratio F_B/F_V for a black body tends to $(\Delta\lambda_B/\Delta\lambda_V)(\lambda_V/\lambda_B)^4$, and hence that the $B - V$ colour index of a very hot black body tends to a constant. Explain with Figure 3 why the colour index is a poor thermometer for the hottest stars.
3. A black body of temperature T emits a fraction of its total flux into a narrow band of width $\Delta\lambda$ centred on λ_V . Let $x = hc/\lambda_V k_B T$. Show that the fraction is proportional to $x^4/(e^x - 1)$, and that it is largest when $x = 4(1 - e^{-x})$. Explain what this implies for the temperature at which the bolometric correction of a black body is closest to zero.
4. A star has the same effective temperature as the Sun, and its absolute bolometric magnitude is smaller than that of the Sun by Δ . Find its radius in terms of $R_\odot$.
5. A Cepheid in the Small Magellanic Cloud and a Cepheid in the Milky Way whose parallax is p have the same period, and their mean apparent magnitudes in the V band are m_1 and m_2 respectively. Find the distance to the Small Magellanic Cloud, assuming that no light is absorbed on the way.

§1.2 Telescopes, Detectors, and Observing across the Spectrum

1. Find the time needed to reach a signal-to-noise ratio $\mathcal{S}$ when observing a bright star of apparent magnitude m_V with a telescope of diameter D , using the rate of arrival of photons found in the text, and show that it goes as $10^{0.4m_V}/D^2$.
2. Show that the uncertainty in a magnitude found from a measured flux is $\delta m = (2.5/\ln 10) \delta F/F$, and hence that $\delta m \approx 1.09/(S/N)$.
3. A faint star is observed against the sky, whose brightness per unit solid angle is fixed. The light of the star is spread over a seeing disc of angular diameter θ_s , and the sky photons that compete with it are those that fall within the same disc. Show that the exposure time needed to reach a given signal-to-noise ratio goes as θ_s^2/D^2 for a telescope of diameter D .

4. For the interferometer of Figure 8, show that near a direction θ_1 that is not close to the vertical, the fringes are separated by $\lambda/(\ell \cos \theta_1)$, and interpret $\ell \cos \theta_1$ as the length of the baseline seen from the source.
5. A telescope at temperature T_t radiates approximately as a black body. Using Wien's displacement law of Section 1.1, find the wavelength at which its radiation per unit wavelength peaks. The telescope observes at a wavelength λ_o shorter than this peak. Explain why cooling the telescope reduces its glow at λ_o by a far larger factor than it reduces the total flux that it radiates.

Celestial Mechanics and Planetary Systems

2.1 Kepler's Laws and Orbital Elements

In this part we apply the mechanics of orbits to stars and planets. This section first reviews Kepler's laws and the solution of the two-body problem, then describes the quantities that fix an orbit in space, and finally shows how the position of a body on its orbit is found at any given time. Section 2.2 uses these results to measure the masses and radii of stars in binary systems, and Section 2.3 applies them to planets around other stars and describes how planetary systems form.

Kepler's Laws at School

Kepler found his laws of planetary motion by analysing Tycho Brahe's measurements of the positions of Mars, and they are part of school physics. They state that:

1. The orbit of each planet is an ellipse with the Sun at one focus.
2. The line joining the Sun and a planet sweeps out equal areas in equal times.
3. The square of the orbital period of a planet is proportional to

the cube of the semi-major axis of its orbit.

Each of these laws was derived from Newton's law of gravitation in Vol. 1 of the Fermi Notes on Physics, and we recall the results that this part will use. Section 5.3 of Vol. 1 showed that two bodies of masses M_1 and M_2 , attracting each other, can be replaced by a single body of reduced mass

$$\mu_r = \frac{M_1 M_2}{M_1 + M_2} \quad (2.1)$$

moving about a fixed centre, where the position $\mathbf{r}$ of the single body is the position of body 1 relative to body 2. Vol. 1 writes μ for the reduced mass; in these notes μ is the mean molecular weight, so we write μ_r . Section 5.4 of Vol. 1 showed that a bound relative orbit is an ellipse,

$$r = \frac{a(1 - e^2)}{1 + e \cos \theta}, \quad (2.2)$$

where a is the semi-major axis, e the eccentricity, and θ the angle at the focus measured from the point of closest approach. The point of closest approach, at $r = a(1 - e)$, is called the pericentre, and the furthest point, at $r = a(1 + e)$, is called the apocentre; for an orbit about the Sun they are the perihelion and the aphelion, and for a binary star the periastron and the apastron. The angle θ is called the true anomaly, an old word for an angle that measures the position of a body on its orbit.

The same section of Vol. 1 derived Kepler's third law with both masses,

$$P_{\text{orb}}^2 = \frac{4\pi^2}{G(M_1 + M_2)} a^3, \quad (2.3)$$

where P_{orb} is the orbital period, and Section 5.5 of Vol. 1 showed that the energy of the orbit is $E = -GM_1 M_2 / 2a$. The same section combines this with conservation of energy to give the relative speed at any point of the orbit,

$$v^2 = G(M_1 + M_2) \left(\frac{2}{r} - \frac{1}{a} \right), \quad (2.4)$$

where v is the speed of body 1 relative to body 2, so the relative speed is greatest at pericentre and least at apocentre. Each body moves about the centre of mass on an ellipse of the same shape as the relative orbit, with semi-major axes

$$a_1 = \frac{M_2}{M_1 + M_2} a, \quad a_2 = \frac{M_1}{M_1 + M_2} a, \quad (2.5)$$

since, as Section 5.3 of Vol. 1 showed, the positions of the two bodies measured from the centre of mass are fixed multiples of $\mathbf{r}$, and they lie on opposite sides of it.

In the solar system it is convenient to measure distances in astronomical units, times in years, and masses in solar masses. The astronomical unit of Section 1.1, 1 AU, is very close to the semi-major axis of the orbit of the Earth. Applying the third law to the Earth, whose mass is about $3 \times 10^{-6} M_\odot$ and can be neglected beside the mass of the Sun, gives $(1 \text{ yr})^2 = 4\pi^2(1 \text{ AU})^3/GM_\odot$. Dividing the third law by this relation gives

$$\frac{M_1 + M_2}{M_\odot} = \frac{(a/1 \text{ AU})^3}{(P_{\text{orb}}/1 \text{ yr})^2}, \quad (2.6)$$

a form that is convenient for binary stars, whose masses are measured in solar masses and whose periods in years.

The Orbital Elements

An ellipse lying in a known plane is fixed by its size, its shape, and the direction of its major axis. To describe an orbit in space we also need to know how its plane is tilted, and where the body is on the orbit at some instant. The quantities that fix all of these are called the orbital elements.

Positions are measured from a reference plane through the focus. For the planets it is the plane of the orbit of the Earth, and for a

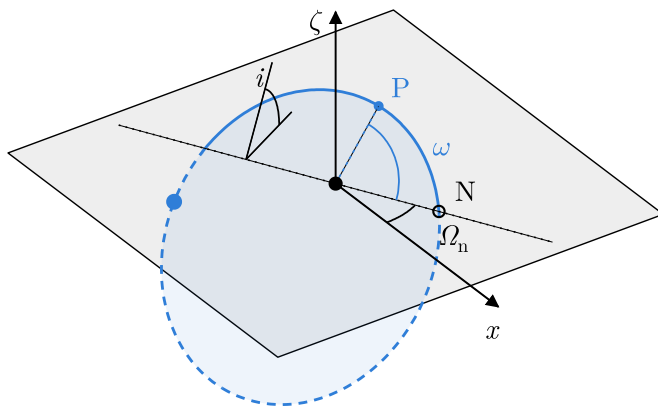


Figure 9. The angles that fix the orientation of an orbit in space. The reference plane is shaded grey and the plane of the orbit is shaded in the accent colour, and the part of the orbit below the reference plane is dashed. The orbit crosses the reference plane on the line of nodes (dotted), at the ascending node N where ζ increases. The longitude of the ascending node Ω_n is measured in the reference plane from the reference direction x to N , the argument of pericentre ω in the plane of the orbit from N to the pericentre P , and the inclination i is the angle between the two planes, drawn between two lines perpendicular to the line of nodes. The central body sits at the focus, and the orbiting body is the accent dot.

binary star it is the plane of the sky, which is the plane through the primary star perpendicular to the line of sight. A reference direction is chosen in that plane; for a binary star it is the direction of north on the sky. Let ζ be the coordinate perpendicular to the reference plane; for a binary star we take ζ to increase away from the observer, so that a body with increasing ζ is receding. The plane of the orbit meets the reference plane along a line through the focus, called the line of nodes, and the orbit crosses the reference plane at the two ends of this line. The node at which the body crosses with ζ increasing is called the ascending node. With these definitions, the orbital elements are as follows, and Figure 9 shows the angles among them:

1. The semi-major axis a .
2. The eccentricity e .
3. The inclination i , the angle between the plane of the orbit and the reference plane, which runs from zero to π . An orbit with $i = 0$ lies in the reference plane, and for a binary star an orbit with $i = \pi/2$ is seen edge on.
4. The longitude of the ascending node Ω_n , the angle in the reference plane from the reference direction to the ascending node. It is usually written Ω , which in these notes is the angular speed of rotation of a star.
5. The argument of pericentre ω , the angle in the plane of the orbit from the ascending node to the pericentre, measured in the direction of motion.
6. The time of pericentre passage t_p , an instant at which the body is at pericentre.

The period follows from a and the masses by the third law. For a binary star the masses are what we wish to find, so in practice the period is often listed among the elements, and the sum of the masses is found from it.

The height of the body above the reference plane follows from these angles. Let $\hat{\mathbf{n}}$ be the unit vector along the line of nodes pointing towards the ascending node, and let $\hat{\mathbf{m}}$ be the unit vector in the plane of the orbit perpendicular to $\hat{\mathbf{n}}$, pointing in the direction in

which the body moves as it passes the ascending node. The body is at the angle $\omega + \theta$ from the ascending node, measured in the plane of the orbit, so its position relative to the focus is

$$\mathbf{r} = r \cos(\omega + \theta) \hat{\mathbf{n}} + r \sin(\omega + \theta) \hat{\mathbf{m}}. \quad (2.7)$$

The vector $\hat{\mathbf{m}}$ is perpendicular to the line of nodes and makes the angle i with the reference plane, so its component perpendicular to that plane is $\sin i$, while $\hat{\mathbf{n}}$ lies in the reference plane. The height of the body is therefore

$$\zeta = r \sin(\omega + \theta) \sin i. \quad (2.8)$$

The components of $\mathbf{r}$ in the reference plane follow in the same way and involve Ω_n as well. For a binary star they give the position of the companion on the sky, while ζ is its distance behind the plane of the sky, and the rate of change $d\zeta/dt$ is its radial velocity relative to the primary.

The Eccentric Anomaly

It remains to find where the body is at a given time, which means finding θ and r as functions of t . Kepler's second law says that the area swept out by the line from the focus is proportional to the time. The whole ellipse, whose semi-minor axis is $a\sqrt{1-e^2}$ and whose area is therefore $\pi a^2\sqrt{1-e^2}$, is swept out in one period, so the area swept out since pericentre is $\pi a^2\sqrt{1-e^2}(t-t_p)/P_{\text{orb}}$. Let

$$\eta = \frac{2\pi(t-t_p)}{P_{\text{orb}}}, \quad (2.9)$$

which is called the mean anomaly. It is the angle that would have been swept out since t_p by a body moving at constant angular speed on a circle with the same period, and it grows uniformly with time.

To turn the area into an angle we use the construction of Figure 10. Let C be the centre of the ellipse, F the focus at which the other body sits, P the pericentre, and B the body. The circle of radius a

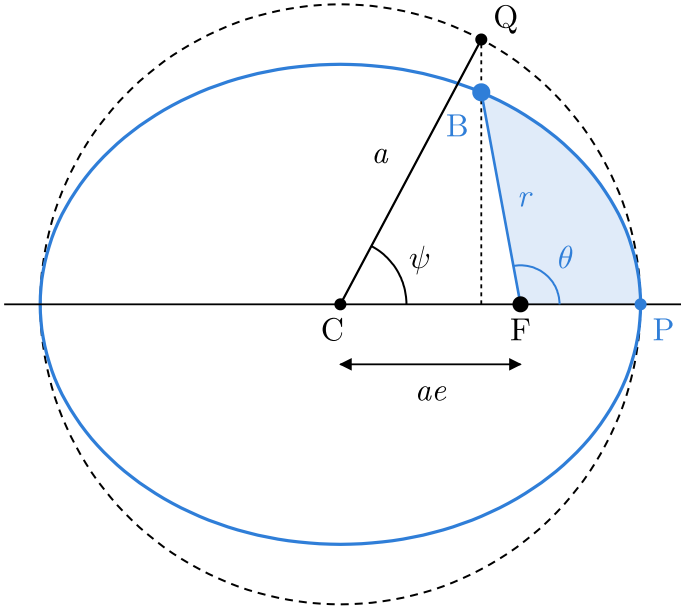


Figure 10. The eccentric anomaly. The auxiliary circle (dashed) has radius a and is centred on the centre C of the elliptical orbit, whose focus is F and whose pericentre is P . The line through the body B perpendicular to the major axis meets the circle at Q . The eccentric anomaly ψ is the angle PCQ , and the true anomaly θ is the angle PFB . The shaded region FPB is the area swept out since pericentre.

centred on C is called the auxiliary circle, and it touches the ellipse at the two ends of the major axis. The line through B perpendicular to the major axis meets the auxiliary circle at Q, and the angle ψ between CP and CQ is called the eccentric anomaly. In books on astronomy the mean and eccentric anomalies are usually written M and E , symbols that these notes keep for mass and energy.

With the origin at C and the x -axis along CP, the ellipse is $x^2/a^2 + y^2/a^2(1 - e^2) = 1$, which is the auxiliary circle compressed in the y -direction by the factor $\sqrt{1 - e^2}$. The point Q is $(a \cos \psi, a \sin \psi)$, and B, which has the same x -coordinate, is $(a \cos \psi, a\sqrt{1 - e^2} \sin \psi)$. The focus F lies at $(ae, 0)$, since, as Section 5.4 of Vol. 1 showed, the focus is a distance ae from the centre. Measured from F, the coordinates of B are

$$r \cos \theta = a(\cos \psi - e), \quad r \sin \theta = a\sqrt{1 - e^2} \sin \psi. \quad (2.10)$$

Squaring and adding the two components gives $r^2 = a^2(1 - 2e \cos \psi + e^2 \cos^2 \psi)$, which is a perfect square, and therefore

$$r = a(1 - e \cos \psi). \quad (2.11)$$

We now consider the area FPB, which is shaded in Figure 10, bounded by the segment FP, the arc of the ellipse from P to B, and the segment BF. Compressing the plane in the y -direction by the factor $\sqrt{1 - e^2}$ turns the auxiliary circle into the ellipse and the point Q into B, and it leaves F and P in place, since they lie on the x -axis. It therefore turns the region FPQ, bounded by FP, the arc of the circle from P to Q, and QF, into the region FPB, and it multiplies every area by $\sqrt{1 - e^2}$. The region FPQ is the sector CPQ of the circle, of area $\frac{1}{2}a^2\psi$, less the triangle CFQ, whose base CF has length ae and whose height is $a \sin \psi$. Hence

$$\begin{aligned} \text{area FPB} &= \sqrt{1 - e^2} \left(\frac{a^2\psi}{2} - \frac{a^2e \sin \psi}{2} \right) \\ &= \frac{a^2\sqrt{1 - e^2}}{2} (\psi - e \sin \psi). \end{aligned} \quad (2.12)$$

For ψ between π and 2π the point Q lies below the axis, the triangle lies outside the sector and is added to it, and since $\sin \psi$ is then negative the same expression results. Setting the area equal to $\pi a^2 \sqrt{1-e^2} (t - t_p) / P_{\text{orb}} = \frac{1}{2} a^2 \sqrt{1-e^2} \eta$ gives

$$\psi - e \sin \psi = \eta, \quad (2.13)$$

which is called Kepler's equation.

Solving Kepler's Equation

Kepler's equation gives the time at which the body reaches a given eccentric anomaly directly, but the question usually asked runs the other way: given the time, and hence η , we want ψ . The equation cannot be solved for ψ in terms of elementary functions, and it is usually solved numerically. The function $f(\psi) = \psi - e \sin \psi - \eta$ has the derivative $1 - e \cos \psi$, which is positive for $e < 1$, so f increases steadily and has exactly one root. The Newton-Raphson method of Section 9.2 of Vol. 4 of the Fermi Notes on Physics finds it by the iteration

$$\psi_{k+1} = \psi_k - \frac{\psi_k - e \sin \psi_k - \eta}{1 - e \cos \psi_k}, \quad (2.14)$$

starting from $\psi_0 = \eta$. The denominator never vanishes for $e < 1$, so every step of the iteration is defined, and it converges quickly for most orbits. For e close to one, a starting value of $\psi_0 = \pi$ is often safer. Once ψ is known, dividing the first component of the position of B by $r = a(1 - e \cos \psi)$ gives

$$\cos \theta = \frac{\cos \psi - e}{1 - e \cos \psi}, \quad (2.15)$$

with θ and ψ in the same half of the orbit, since $\sin \theta$ and $\sin \psi$ have the same sign.

The position of a body at the time t is therefore found in four steps: first η from t and t_p , then ψ from Kepler's equation, then r and θ from ψ , and finally the position in space from a , e , i , Ω_n , and ω . A

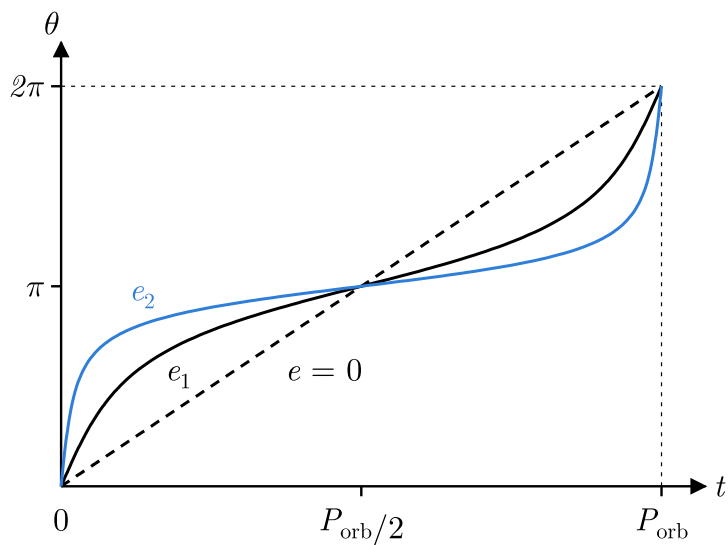


Figure 11. The true anomaly θ over one period, found by solving Kepler's equation, for a circular orbit (dashed) and for orbits of eccentricities e_1 and e_2 , with $0 < e_1 < e_2$. Time is measured from pericentre. The more eccentric the orbit, the faster θ changes near pericentre and the more slowly near apocentre.

table of positions of a body computed in this way at a series of times is called an ephemeris. Figure 11 shows the true anomaly found in this way over one period. For a circular orbit θ grows uniformly. For an eccentric orbit it grows quickly near pericentre, where the body is close to the focus and, by the second law, has to sweep out area quickly with a short radius, and slowly near apocentre.

The Time Spent Close to the Focus

As an example, let us find the fraction of its period that a body spends on the half of the orbit nearer the focus, which is the part with $-\pi/2 < \theta < \pi/2$, cut off by the line through the focus perpendicular to the major axis. At $\theta = \pi/2$ we have $\cos \theta = 0$, so the expression for $\cos \theta$ gives $\cos \psi = e$ and $\sin \psi = \sqrt{1 - e^2}$. Kepler's equation then gives the mean anomaly at that point,

$$\eta = \arccos e - e\sqrt{1 - e^2}. \quad (2.16)$$

The orbit is symmetric about the major axis, so the body takes as long to go from $\theta = -\pi/2$ to pericentre as from pericentre to $\theta = \pi/2$, and the fraction of the period spent on the near half is $2\eta/2\pi$, or

$$f_{\text{near}} = \frac{1}{\pi} \left(\arccos e - e\sqrt{1 - e^2} \right). \quad (2.17)$$

For a circular orbit $\arccos 0 = \pi/2$ and $f_{\text{near}} = \frac{1}{2}$, as it must be. For a small eccentricity, the Taylor expansion $\arccos e \approx \pi/2 - e$ together with $e\sqrt{1 - e^2} \approx e$ gives $f_{\text{near}} \approx \frac{1}{2} - 2e/\pi$, and as e approaches one, both terms in the bracket approach zero and so does f_{near} . A body on a very eccentric orbit therefore spends only a small part of its period near the focus. For Halley's comet, whose orbit has $e \approx 0.97$ and a period of about 76 years, the result is a few months.

2.2 Binary Stars and the Masses and Radii of Stars

The mass of an isolated star cannot be measured, since a star shows its mass only through its gravitational effect on other bodies. For-

unately many stars are not isolated; roughly half of the stars like the Sun have at least one stellar companion, and a pair of stars in orbit about their common centre of mass is called a binary star. The orbits of binary stars are the source of almost all the directly measured masses of stars. This section treats binaries according to the way in which they are detected: visual binaries, whose two stars are seen separately; spectroscopic binaries, whose orbital motion is found from the Doppler shifts of their spectral lines; and eclipsing binaries, whose stars pass in front of one another. A single system can belong to more than one of these classes, and there are other kinds; in an astrometric binary, for example, only one star is seen, and its companion is inferred from the wavy path of the visible star across the sky.

Visual Binaries

In a visual binary the two stars are far enough apart to be resolved by a telescope, and the position of the fainter star relative to the brighter one can be measured over many years. The fainter star traces out an ellipse on the sky, as in Figure 12, which is the projection of the true relative orbit onto the plane of the sky. A projection of this kind is a linear map from the plane of the orbit to the plane of the sky, so it turns an ellipse into an ellipse and the centre of one into the centre of the other. It also multiplies every area in the plane of the orbit by the same factor, $|\cos i|$, so Kepler's second law holds on the sky as well, and the line from the brighter star to the fainter sweeps out equal areas in equal times. Projection does not, however, carry the foci of an ellipse to the foci of its image. Unless the orbit happens to lie in the plane of the sky, the brighter star is displaced from the focus of the apparent ellipse, and Figure 12 shows this displacement. The eccentricity follows from the displacement directly, since a projection preserves the ratio of lengths along a line, and the brighter star lies on the projection of the true major axis at a fraction e of the distance from the centre to the projected pericentre. The elements i , ω , and Ω_n are found by seeking the ellipse in space, with the brighter star at a focus, whose projection matches the apparent orbit. Positions alone cannot tell which of the two nodes is the ascending node, since an orbit and its mirror image in the plane of the sky have the same projection, and radial

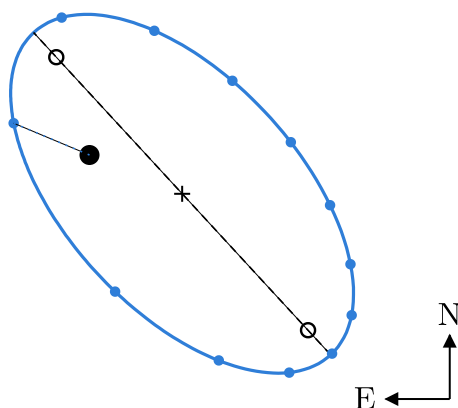


Figure 12. The apparent orbit of a visual binary, which is the projection of the relative orbit onto the plane of the sky, with north up and east to the left. The dots mark positions of the fainter star at equal intervals of time. The brighter star (filled circle) lies at the projection of the focus of the true orbit, away from the two foci of the apparent ellipse (open circles). The cross marks the centre of the apparent ellipse and the dashed line its major axis, and the dotted line joins the brighter star to the projected periastron.

velocities are needed to decide.

The fit gives the angular semi-major axis α of the true relative orbit, and the distance d to the system is found from its parallax p , as in Section 1.1. When α is small and measured in radians, the semi-major axis is $a = \alpha d$. As Section 1.1 showed, a star at a parallax of p seconds of arc lies $1/p$ parsecs away, and since the parsec is the distance at which one astronomical unit subtends one second of arc, a length that subtends α seconds of arc at that distance is α/p astronomical units. The form of Kepler's third law found in Section 2.1 then gives

$$\frac{M_1 + M_2}{M_\odot} = \frac{(\alpha/p)^3}{(P_{\text{orb}}/1 \text{ yr})^2}, \quad (2.18)$$

with α and p in the same units. Since the parallax enters as its cube, a fractional error in p produces three times that fractional error in the mass.

To find the two masses separately we need the orbits of the two stars about their centre of mass, which are measured against the background of more distant stars. By Section 5.3 of Vol. 1, the centre of mass moves in a straight line at constant speed, and each star moves about it on an ellipse, with semi-major axes in the ratio $a_1/a_2 = M_2/M_1$. The two orbits lie in the same plane and are projected by the same factor, so the ratio of their angular semi-major axes on the sky is also M_2/M_1 , for any inclination. Together with the sum, this ratio gives both masses. As an example, consider Sirius, whose brighter star Bessel found in 1844 to be moving on a wavy path, so that Sirius was an astrometric binary until its faint companion was first seen in 1862. The orbit, with a period of about 50 years, gives masses of about $2.1 M_\odot$ and $1.0 M_\odot$.

Spectroscopic Binaries

Most binaries are too distant, or their stars too close together, to be resolved, but their orbital motion can still be detected in their spectra. A spectrograph is an instrument that spreads the light of a star into a spectrum, usually with a diffraction grating, and records it on a charge-coupled device, so that the wavelength of each

absorption line can be measured. The component of the velocity of a star along the line of sight is called its radial velocity v_r , and it is taken to be positive for a receding star. By the Doppler shift for speeds much less than c , which is familiar from school, a line emitted at the wavelength λ_0 by a star of radial velocity v_r is observed at $\lambda_0 + \Delta\lambda$, with

$$v_r = c \frac{\Delta\lambda}{\lambda_0}. \quad (2.19)$$

A binary whose motion is found in this way is called a spectroscopic binary. If the lines of both stars are seen, it is called double-lined; if one star is much fainter than the other and only the lines of the brighter star are seen, it is called single-lined.

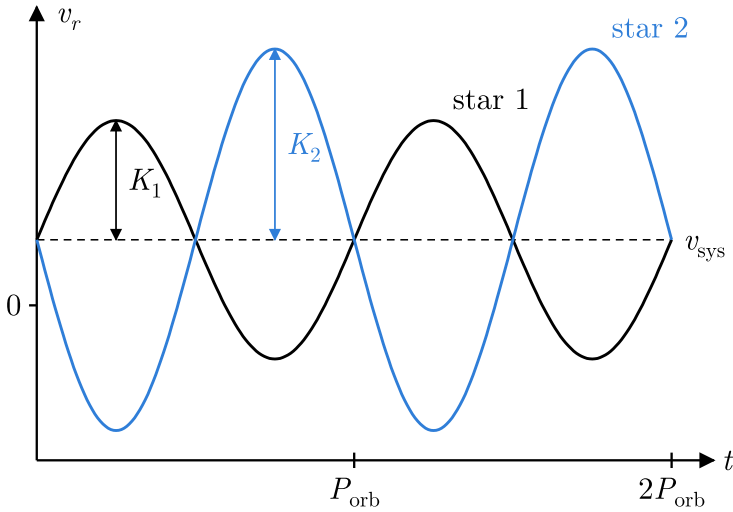


Figure 13. The radial velocities of the two stars of a double-lined spectroscopic binary on circular orbits, over two periods. The curves are sinusoids in opposite phase about the radial velocity v_{sys} of the centre of mass, with semi-amplitudes K_1 and K_2 . Star 2, which has the larger semi-amplitude, is the less massive.

Let us take first a pair of stars on circular orbits. Star 1 moves on a circle of radius a_1 about the centre of mass, at the speed $2\pi a_1/P_{\text{orb}}$.

As you showed in Problem 5 of Section 2.1, its distance behind the plane of the sky changes sinusoidally, with amplitude $a_1 \sin i$, and its radial velocity is therefore

$$v_{r1} = v_{\text{sys}} + K_1 \cos \frac{2\pi(t - t_0)}{P_{\text{orb}}}, \quad K_1 = \frac{2\pi a_1 \sin i}{P_{\text{orb}}}, \quad (2.20)$$

where t_0 is an instant at which star 1 passes its ascending node, and v_{sys} is the radial velocity of the centre of mass, which the Doppler shift includes. The constant K_1 is called the semi-amplitude of the radial-velocity curve. Star 2 is always on the opposite side of the centre of mass, so its radial velocity is $v_{r2} = v_{\text{sys}} - K_2 \cos[2\pi(t - t_0)/P_{\text{orb}}]$, with $K_2 = 2\pi a_2 \sin i / P_{\text{orb}}$. Figure 13 shows the two curves, which are sinusoids in opposite phase about v_{sys} , crossing whenever the two stars are moving across the line of sight.

The ratio of the semi-amplitudes gives the ratio of the masses,

$$\frac{K_1}{K_2} = \frac{a_1}{a_2} = \frac{M_2}{M_1}, \quad (2.21)$$

so the star with the larger swing in velocity is the less massive. Adding the two semi-amplitudes gives $K_1 + K_2 = 2\pi a \sin i / P_{\text{orb}}$, where $a = a_1 + a_2$ is the radius of the relative orbit. Multiplying Kepler's third law, $M_1 + M_2 = 4\pi^2 a^3 / GP_{\text{orb}}^2$, by $\sin^3 i$ and substituting $a \sin i = P_{\text{orb}}(K_1 + K_2)/2\pi$ gives

$$(M_1 + M_2) \sin^3 i = \frac{P_{\text{orb}}}{2\pi G} (K_1 + K_2)^3. \quad (2.22)$$

Since M_1 is the fraction $K_2/(K_1 + K_2)$ of the total, the masses of a double-lined binary are

$$\begin{aligned} M_1 \sin^3 i &= \frac{P_{\text{orb}}}{2\pi G} K_2 (K_1 + K_2)^2, \\ M_2 \sin^3 i &= \frac{P_{\text{orb}}}{2\pi G} K_1 (K_1 + K_2)^2. \end{aligned} \quad (2.23)$$

The spectrum alone does not give the inclination, so it gives only the combinations $M_1 \sin^3 i$ and $M_2 \sin^3 i$, which are lower limits on the masses.

The Mass Function

For a single-lined binary only K_1 and P_{orb} are measured. The semi-major axis of the orbit of star 1 is $a_1 = aM_2/(M_1 + M_2)$, and $a_1 \sin i = P_{\text{orb}}K_1/2\pi$. Cubing the second relation, substituting the first, and using the third law in the form $a^3 = G(M_1 + M_2)P_{\text{orb}}^2/4\pi^2$ gives

$$\frac{M_2^3}{(M_1 + M_2)^3} \frac{G(M_1 + M_2)P_{\text{orb}}^2}{4\pi^2} \sin^3 i = \frac{P_{\text{orb}}^3 K_1^3}{8\pi^3}, \quad (2.24)$$

which can be rewritten as

$$f_{\text{m}} = \frac{M_2^3 \sin^3 i}{(M_1 + M_2)^2} = \frac{P_{\text{orb}} K_1^3}{2\pi G}. \quad (2.25)$$

The quantity f_{m} is called the mass function, and it is found from the measured period and semi-amplitude alone. Since $\sin i \leq 1$ and $M_2/(M_1 + M_2) < 1$, the mass function is less than M_2 , so it sets a lower limit on the mass of the unseen star. If M_1 is known from the spectrum of the visible star, the mass function gives M_2 for any assumed inclination, and when $M_2 \ll M_1$ it gives $M_2 \sin i$ directly.

When the inclinations are unknown, the masses of a large sample of binaries can still be studied statistically, since the orbits of binaries are oriented at random; you will show in Problem 1 that the mean value of $\sin^3 i$ over random orientations is $3\pi/16$.

Many binaries have eccentric orbits, and their radial-velocity curves are no longer sinusoids. You will show in Problem 2 that for an orbit of eccentricity e and argument of periastron ω the radial velocity of star 1 is

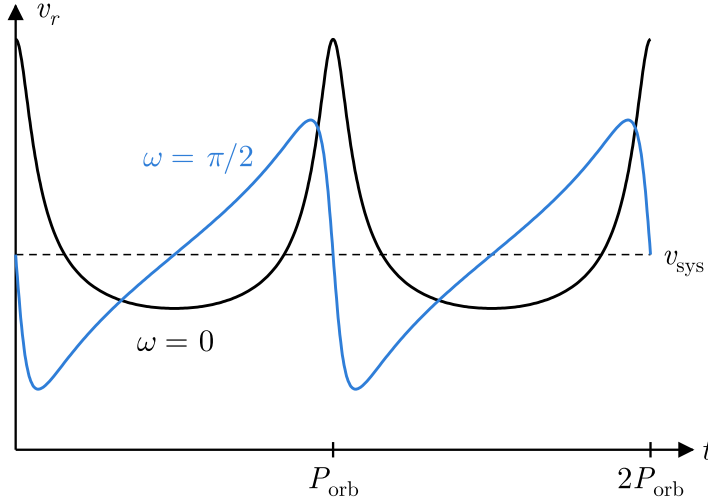


Figure 14. The radial velocity of one star of a binary with an eccentric orbit, over two periods, for two values of the argument of periastron ω and the same eccentricity. Time is measured from periastron. For $\omega = 0$ the curve has a sharp maximum at periastron and a broad, shallow minimum.

$$v_{r1} = v_{\text{sys}} + K_1 [\cos(\omega + \theta) + e \cos \omega], \quad K_1 = \frac{2\pi a_1 \sin i}{P_{\text{orb}} \sqrt{1 - e^2}}, \quad (2.26)$$

where θ is the true anomaly found from the time as in Section 2.1. Figure 14 shows two such curves with the same eccentricity. With $\omega = 0$ the periastron lies at the ascending node, where the star is receding fastest, and the curve has a sharp maximum and a broad, shallow minimum; with $\omega = \pi/2$ the star passes periastron where it is furthest behind the plane of the sky and moving across the line of sight, so the curve falls quickly from its maximum to its minimum through periastron and rises slowly back through apastron. The shape of the curve therefore gives e and ω , and the mass function becomes $f_{\text{m}} = (1 - e^2)^{3/2} P_{\text{orb}} K_1^3 / 2\pi G$.

Eclipsing Binaries

If the line of sight lies close to the plane of the orbit, the stars pass in front of each other once in each half period, and the total brightness of the system dips during each eclipse. A binary of this kind is called an eclipsing binary, and the record of its brightness against time is called its light curve. For a circular orbit of radius a , the smallest separation of the two stars on the sky is $a \cos i$, and you will show in Problem 3 that eclipses occur only if $a \cos i < R_1 + R_2$, where R_1 and R_2 are the radii of the stars. The radii are usually much smaller than a , so an eclipsing binary has i close to $\pi/2$ and $\sin i$ close to one. If the binary is also a double-lined spectroscopic binary, the uncertainty of the inclination then has little effect, and the expressions for $M_1 \sin^3 i$ and $M_2 \sin^3 i$ give the two masses.

The simplest case is a circular orbit seen exactly edge on, with each stellar disc taken to be uniformly bright. Let R_s and R_l be the radii of the smaller and the larger star. Near an eclipse the stars move almost at right angles to the line of sight, and since the radii are much smaller than a , the smaller star moves across the disc of the larger in a straight line at the relative speed $v = 2\pi a/P_{\text{orb}}$. With $\sin i = 1$ this speed is $K_1 + K_2$. The eclipse has four instants of contact, marked in Figure 15: at t_a the disc of the smaller star first touches the limb of the larger, which is the edge of its disc; at t_b it lies entirely inside the disc of the larger; at t_c its leading edge reaches the opposite limb; and at t_d it leaves the disc. Between t_a and t_b the smaller star moves through a distance equal to its diameter, and between t_a and t_c its leading edge moves across the diameter of the larger star, so

$$R_s = \frac{v}{2} (t_b - t_a) , \quad R_l = \frac{v}{2} (t_c - t_a) . \quad (2.27)$$

The depths of the two eclipses give the ratio of the effective temperatures. The flux received from a star of radius R and effective temperature T_{eff} at the distance d is its luminosity $4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4$, defined in Section 1.1, spread over a sphere of area $4\pi d^2$, which is $R^2 \sigma_{\text{SB}} T_{\text{eff}}^4 / d^2$. The disc of the star has area πR^2 , so this flux is $\sigma_{\text{SB}} T_{\text{eff}}^4 / \pi d^2$ for each unit of area of the disc, and for a uniformly bright disc a part of area A contributes $A \sigma_{\text{SB}} T_{\text{eff}}^4 / \pi d^2$. Let T_s and

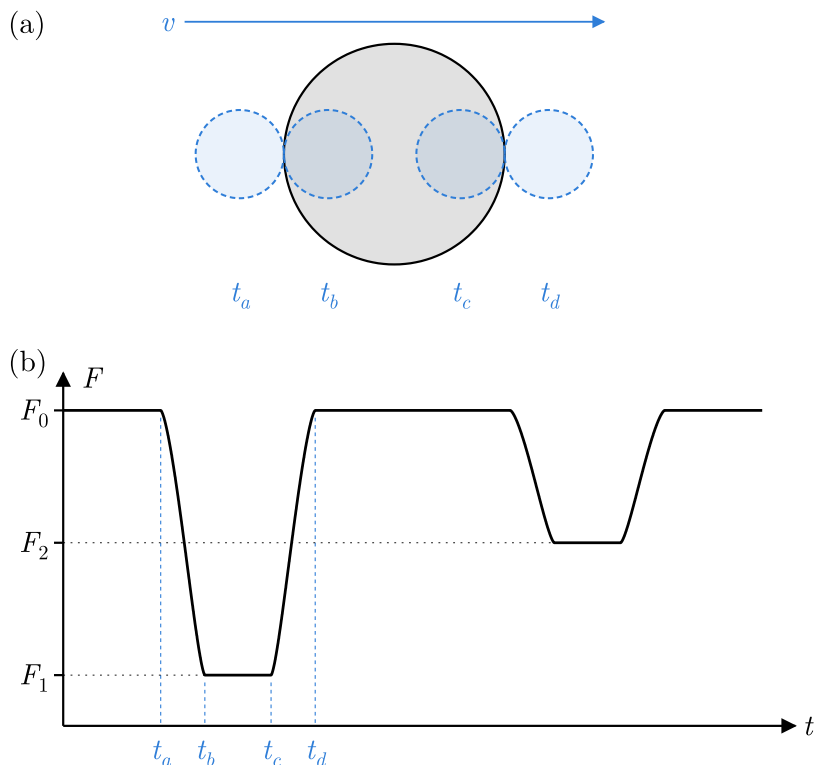


Figure 15. (a) The smaller star of an eclipsing binary crossing the disc of the larger star (shaded), drawn at the four instants of contact t_a , t_b , t_c , and t_d for an orbit seen edge on. The arrow shows the direction of the relative velocity v . (b) The light curve over one period, for uniformly bright discs with the smaller star the cooler. The primary minimum F_1 occurs when the smaller star passes in front of the larger, as in (a), and the secondary minimum F_2 when it passes behind. Both minima are flat between the second and third contacts, when the hidden area is constant. The eclipses are drawn much wider, as a fraction of the period, than in a real binary, in which the radii are much smaller than the separation.

T_1 be the effective temperatures of the two stars and F_0 the flux received outside eclipse. When the smaller star passes in front of the larger, it hides an area πR_s^2 of the larger star, and when it passes behind, the whole of its disc, also of area πR_s^2 , is hidden. The fluxes lost in the two eclipses are therefore

$$\begin{aligned}\Delta F_{\text{front}} &= \frac{R_s^2}{d^2} \sigma_{\text{SB}} T_1^4 \quad (\text{smaller star in front}), \\ \Delta F_{\text{behind}} &= \frac{R_s^2}{d^2} \sigma_{\text{SB}} T_s^4 \quad (\text{smaller star behind}),\end{aligned}\tag{2.28}$$

and their ratio is

$$\frac{\Delta F_{\text{behind}}}{\Delta F_{\text{front}}} = \left(\frac{T_s}{T_1} \right)^4.\tag{2.29}$$

The same area is hidden in both eclipses, so the deeper of the two, which is called the primary minimum, occurs when the hotter star is hidden, and the shallower is called the secondary minimum. In Figure 15 the smaller star is the cooler, the primary minimum F_1 occurs when the smaller star is in front, and the secondary minimum F_2 when it is behind, so $(F_0 - F_2)/(F_0 - F_1) = (T_s/T_1)^4$. When the brightness is measured in a single photometric band rather than over all wavelengths, the ratio of the depths is the ratio of the fluxes that unit areas of the two surfaces emit in that band, which still increases with the ratio of the temperatures.

Real stellar discs are not uniformly bright, and if the orbit is not exactly edge on the smaller star may cover the larger only partly, so that the bottom of the eclipse is rounded. The fitting of models of the two stars to such light curves is not treated in this course.

A Double-Lined Eclipsing Binary

As an example of what a single system can give, consider a binary that is both double-lined and eclipsing, with circular orbits and $i = \pi/2$, for which the period P_{orb} , the semi-amplitudes K_1 and K_2 , the contact times, and the depths of the two eclipses have been

measured. The masses follow from the expressions for $M_1 \sin^3 i$ and $M_2 \sin^3 i$ with $\sin i = 1$,

$$M_1 = \frac{P_{\text{orb}}}{2\pi G} K_2 (K_1 + K_2)^2, \quad M_2 = \frac{P_{\text{orb}}}{2\pi G} K_1 (K_1 + K_2)^2. \quad (2.30)$$

With $v = K_1 + K_2$, the radii are $R_s = \frac{1}{2}(K_1 + K_2)(t_b - t_a)$ and $R_l = \frac{1}{2}(K_1 + K_2)(t_c - t_a)$. Spectra taken during the eclipses show which star is hidden at each minimum, and so which radius belongs to which mass. The depths give T_s/T_l , and if the effective temperature of one star is found from its colour, as in Section 1.1, the other follows. The luminosity of each star is then $L = 4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4$, and the distance is not needed for any of these results. A single system of this kind therefore gives the mass, the radius, and the luminosity of both its stars, and for the best-measured systems the masses and radii are known to within a few per cent or better.

The Mass–Luminosity Relation

When the masses and luminosities found from binaries are plotted against each other for stars on the main sequence, the points are found to lie close to a single curve, which Figure 16 approximates by power laws joined end to end. This relation, found by observation, is called the mass–luminosity relation. Over a wide range of masses around that of the Sun it can be written approximately as

$$\frac{L}{L_{\odot}} \approx \left(\frac{M}{M_{\odot}} \right)^{3.5}, \quad (2.31)$$

with the exponent closer to four for stars of one or two solar masses, smaller for stars of low mass, and approaching one for the most massive stars. The masses of main-sequence stars run from about $0.08 M_{\odot}$ to more than $100 M_{\odot}$, and over this range the luminosity changes by more than ten powers of ten. The radii of main-sequence stars are also found by observation to increase with mass, roughly as $R \propto M^{0.8}$ for stars not far from the mass of the Sun. The relation holds only for stars on the main sequence; the giants and the white dwarfs of the Hertzsprung–Russell diagram, as it is drawn at school, lie far from it.

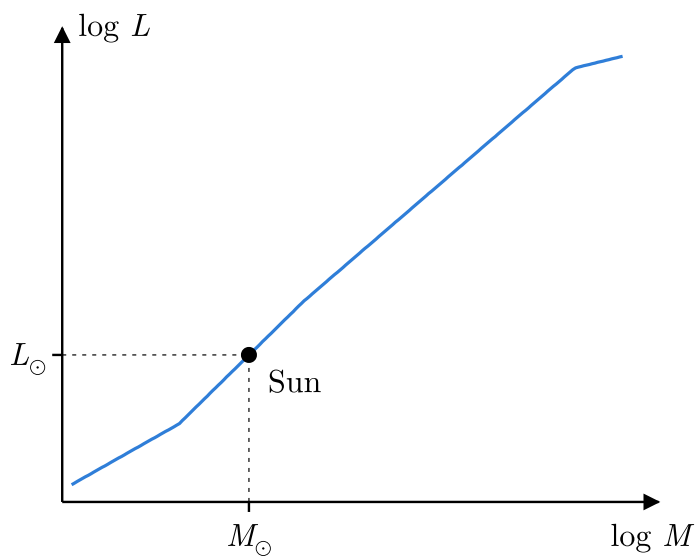


Figure 16. The mass–luminosity relation of main-sequence stars on logarithmic axes, approximated by power laws joined end to end. The Sun is marked. The slope is steepest for stars of about the mass of the Sun and is smallest at the highest masses.

2.3 Exoplanets and the Formation of Planetary Systems

A planet that orbits a star other than the Sun is called an exoplanet. The first to be found around a star like the Sun was reported by Mayor and Queloz in 1995, orbiting the star 51 Pegasi, and by the middle of the 2020s more than five thousand had been confirmed. A planet is very much fainter than its star and lies very close to it on the sky; the light that Jupiter reflects is about a thousand millionth of the light of the Sun. Most exoplanets have therefore been found from their effects on the light of the star. This section first describes the four methods that have found most of the known planets, which are the radial velocity of the star, transits, direct imaging, and gravitational microlensing, and then describes how a planetary system forms, from the collapse of a rotating cloud of gas to the growth of planets in a disc.

The Radial Velocity of the Star

A star and its planet orbit their common centre of mass, so the star moves on a small orbit about that point, and its radial velocity can be measured as for a single-lined spectroscopic binary. Let M be the mass of the star and M_{pl} the mass of the planet. The mass function derived in Section 2.2, with $M_1 = M$ and $M_2 = M_{\text{pl}}$, is

$$\frac{M_{\text{pl}}^3 \sin^3 i}{(M + M_{\text{pl}})^2} = \frac{P_{\text{orb}} K^3}{2\pi G}, \quad (2.32)$$

for a circular orbit, where K is the semi-amplitude of the radial velocity of the star. A planet has a much smaller mass than its star, so $(M + M_{\text{pl}})^2 \approx M^2$, and taking the cube root gives

$$M_{\text{pl}} \sin i \approx \left(\frac{P_{\text{orb}} M^2}{2\pi G} \right)^{1/3} K. \quad (2.33)$$

The mass of the star can be estimated from its spectrum, by comparison with stars of the same spectral class whose masses are known,

so the measured K and P_{orb} give $M_{\text{pl}} \sin i$, which is a lower limit on the mass of the planet.

The semi-amplitude, found by solving the same relation for K , is $K \approx (2\pi G/P_{\text{orb}})^{1/3} M_{\text{pl}} \sin i / M^{2/3}$, so it increases with the mass of the planet and decreases slowly with the period. Jupiter moves the Sun with a semi-amplitude of about 12.5 ms^{-1} over its period of twelve years, and the Earth moves it at about 0.09 ms^{-1} . A speed of a metre per second shifts a spectral line by about three parts in a thousand million of its wavelength. Shifts this small can be measured because they are found from thousands of lines simultaneously, and because the spectrum of the star is compared with reference lines of known wavelength that pass through the spectrograph with the starlight, such as the absorption lines of a cell of iodine vapour placed in the beam. The best spectrographs reach a precision somewhat below a metre per second. Since K is largest for massive planets with short periods, the first planets found were mostly giant planets with periods of a few days, which are called hot Jupiters. For an eccentric orbit the radial-velocity curve has the shapes of Figure 14 in Section 2.2.

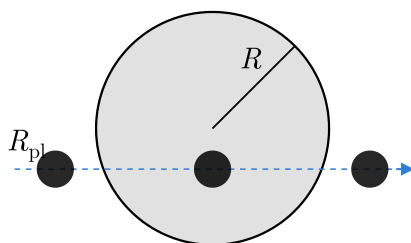
Transits

If the orbit of a planet is seen nearly edge on, the planet passes in front of its star once in every period, and the star dims slightly while the planet crosses it, as Figure 17 shows. The passage is called a transit, and it is the eclipse of Section 2.2 with a dark planet in the role of the smaller star. Taking the disc of the star to be uniformly bright, and the planet to be wholly inside the disc of the star, the planet hides an area πR_{pl}^2 of the disc of the star, whose area is πR^2 , where R_{pl} is the radius of the planet and R that of the star. The fractional drop in the flux, which is called the depth of the transit, is therefore the ratio of these two areas,

$$\delta = \left(\frac{R_{\text{pl}}}{R} \right)^2. \quad (2.34)$$

A planet the size of Jupiter crossing a star like the Sun has a depth of about one per cent, and a planet the size of the Earth a depth

(a)



(b)

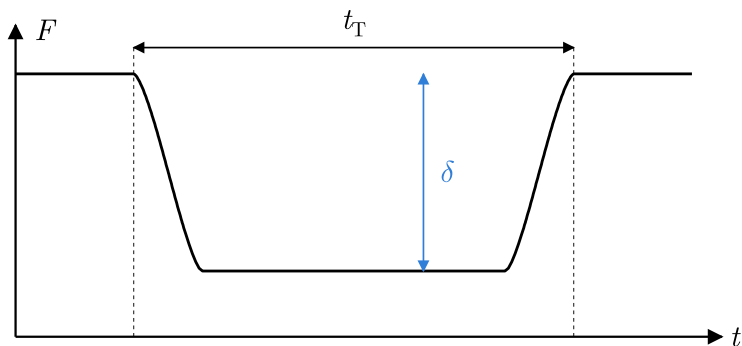


Figure 17. (a) A planet crossing the disc of its star (shaded) along a chord, drawn at three positions and at a larger size relative to the star than most transiting planets have. (b) The light curve during the transit, for a uniformly bright star. The flux falls by the depth δ and recovers after the duration t_T .

of less than one part in ten thousand. The flickering of starlight in the atmosphere of the Earth limits measurements from the ground, and most small transiting planets have been found by telescopes in space, such as the Kepler space telescope, which measured the brightness of about a hundred and fifty thousand stars every half hour for four years.

A transit is seen only if the orbit is close to edge on, and you will show in Problem 2 that the probability that a randomly oriented planet on a circular orbit of radius a transits is about R/a . For a planet at the distance of the Earth from a star like the Sun this is about half a per cent, so a transit survey has to watch a great many stars.

A planet that both transits and shows a radial-velocity curve has $\sin i$ close to one, so its mass follows from K and its radius from δ , and together they give its mean density. The density distinguishes a planet made mostly of hydrogen and helium, such as Jupiter, whose mean density is close to that of water, from a planet made mostly of rock and iron, such as the Earth, whose mean density is about five times that of water.

Direct Imaging and Gravitational Microlensing

A planet can be photographed directly if its angular distance from its star, about a/d for a planet at the distance a from a star at the distance d , exceeds the diffraction limit of the telescope found in Section 1.2. The glare of the star is suppressed by a coronagraph, an instrument that blocks the light of the star with a small opaque mask at the focus of the telescope, so that the much fainter light around it can be recorded. The most favourable targets are young giant planets far from their stars, which are still hot from their formation and radiate strongly in the infrared, where the planet is much less faint relative to its star than in visible light; an example is the system of giant planets around the young star HR 8799, the first of which were imaged in 2008.

The gravity of a star bends the light that passes close to it, an effect described by general relativity, which is not treated in this course. When a foreground star passes almost exactly in front of a more distant star, the bending brightens the background star for a period

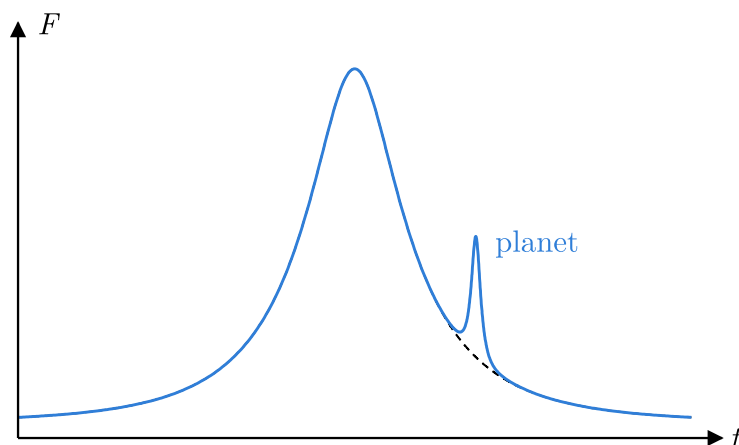


Figure 18. The brightness of a background star during a microlensing event (accent), with the brief extra brightening produced by a planet of the foreground star. The dashed curve shows the event without the planet. The effect of the planet is modelled here, approximately, as that of a separate small lens whose brightening is added to that of the star.

of weeks, and the brightness rises and falls symmetrically about the time of closest alignment. This effect is called gravitational microlensing. If the foreground star has a planet, and the planet passes close to the line of sight during the event, it adds a brief extra brightening lasting from hours to days, as Figure 18 shows. The method finds planets at a few astronomical units from faint and distant stars, which other methods cannot reach, but each event happens only once, and surveys have to watch many millions of stars in the crowded fields towards the centre of the Galaxy, where such alignments are most frequent.

The Collapse of a Rotating Cloud

The planets of the solar system all orbit the Sun in the same sense, in orbits that lie close to one plane, and the Sun rotates in the same sense, with its equator close to that plane. These facts led Kant and Laplace in the eighteenth century to propose that the Sun and the planets formed together from a rotating, flattened cloud of

gas. Discs of gas and dust are now observed around many young stars, and images made with interferometers of the kind described in Section 1.2, working at millimetre wavelengths, show some of them marked with bright rings and dark gaps. How a cloud of interstellar gas begins to collapse into a star, and the distribution of the masses of the stars formed, are not treated in this course. We ask here only why a collapsing cloud forms a disc, and how large the disc is.

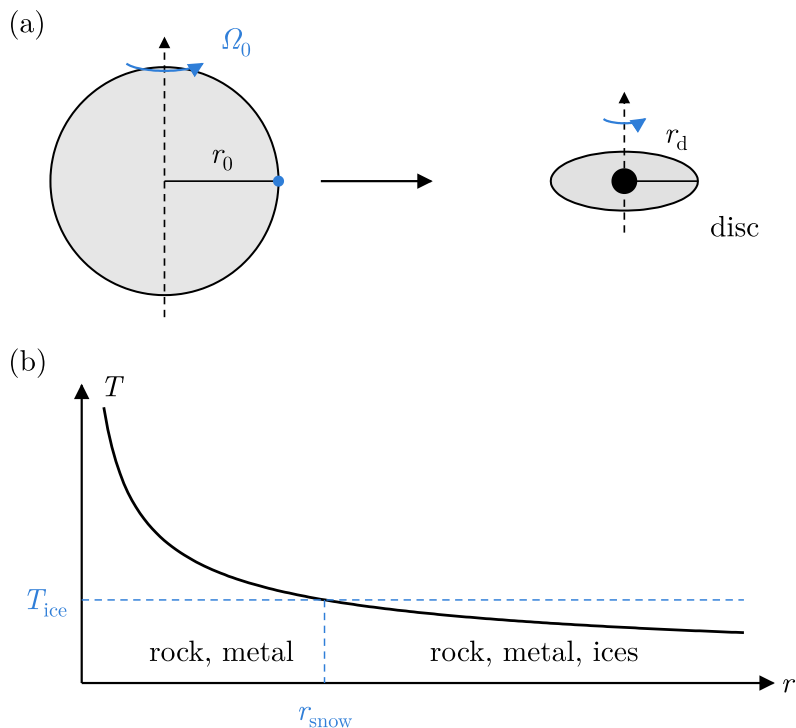


Figure 19. (a) A cloud of radius r_0 rotating with angular speed Ω_0 about the dashed axis collapses into a star surrounded by a disc of radius r_d in its equatorial plane. (b) The temperature T of dust grains in the disc at the distance r from the star, which goes as $r^{-1/2}$. Water is found as ice beyond the snow line r_{snow} , where the temperature falls below T_{ice} .

Let us consider a cloud of mass M and radius r_0 , rotating slowly as a rigid body with the angular speed Ω_0 , as in Figure 19(a), and follow a small parcel of gas that starts at the edge of the cloud in

its equatorial plane. Let J be its angular momentum per unit mass about the axis of rotation, which is initially

$$J = \Omega_0 r_0^2. \quad (2.35)$$

While the cloud is roughly spherical, the gravitational force on the parcel is that of the mass inside it acting as a point at the centre, as Section 5.2 of Vol. 1 showed. This force points towards the centre and exerts no torque about it, so, as in Section 5.3 of Vol. 1, the angular momentum of the parcel is conserved as it falls. At the radius r its speed about the axis is J/r , and moving on a circle at this speed requires a centripetal acceleration J^2/r^3 . Gravity supplies the acceleration GM/r^2 , which grows more slowly as r decreases than J^2/r^3 does, so the parcel falls inwards until the two accelerations are equal and then moves on a circle. The radius at which this happens is

$$r_d = \frac{J^2}{GM} = \frac{\Omega_0^2 r_0^4}{GM}. \quad (2.36)$$

The ratio $r_d/r_0 = \Omega_0^2 r_0 / (GM/r_0^2)$ is the ratio of the centripetal acceleration of the edge of the original cloud to the gravitational acceleration there, which is small for a slowly rotating cloud. For the slowly rotating dense cores of interstellar clouds in which stars are observed to form, r_d comes out at tens to hundreds of astronomical units, comparable with the size of the solar system.

Gas that starts away from the equatorial plane also conserves the component of its angular momentum along the axis, since gravity exerts no torque about the centre, and that component fixes the radius at which the gas settles. Its motion parallel to the axis is not opposed by rotation, so it falls towards the equatorial plane, where streams arriving from above and below collide and lose their motion along the axis, and the gas settles into a thin rotating disc. The gas with the least angular momentum, which starts close to the axis, falls to the centre and forms the star. The processes by which gas in the disc loses angular momentum and moves inwards onto the star are not treated in this course.

The Temperature of the Disc and the Snow Line

The kind of solid that can exist at a given place in the disc depends on the temperature there. To estimate it, consider a small grain of dust at the distance r from a star of luminosity L . We suppose that the grain sees the star directly, with no dust between them to absorb the starlight, that it absorbs all the starlight falling on it, and that it radiates as a blackbody at a single temperature T . The flux of starlight at the distance r is $L/4\pi r^2$, so a grain presenting a cross-sectional area A_g to the star absorbs the power $A_g L/4\pi r^2$. A spherical grain has a surface area four times its cross-sectional area, and by the Stefan–Boltzmann law of Section 5.6 of Vol. 2 of the Fermi Notes on Physics it radiates the power $4A_g \sigma_{\text{SB}} T^4$. Equating the two powers gives

$$T^4 = \frac{L}{16\pi\sigma_{\text{SB}}r^2}, \quad (2.37)$$

and writing $L = 4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4$, as in Section 1.1, gives

$$T = T_{\text{eff}} \left(\frac{R}{2r} \right)^{1/2}. \quad (2.38)$$

The temperature does not depend on the size of the grain, and it goes as $r^{-1/2}$ with distance from the star, as in Figure 19(b). For the present Sun it gives about 280 K at the distance of the Earth. The heating of the disc by the gas flowing through it towards the star, which adds to this temperature close to the star, is not treated in this course.

At the low pressures of the disc, water vapour condenses directly into ice below a temperature T_{ice} of about 150 K. The distance beyond which water is found as ice is called the snow line, and setting $T = T_{\text{ice}}$ gives

$$r_{\text{snow}} = \frac{R}{2} \left(\frac{T_{\text{eff}}}{T_{\text{ice}}} \right)^2, \quad (2.39)$$

which for the Sun comes to about three astronomical units, between the orbits of Mars and Jupiter. Inside the snow line the solids are rock and metal, such as silicates and iron, which condense at higher temperatures. Outside it, water ice condenses as well, and since oxygen is much more abundant in the gas than the elements that form rock, the mass of solid material available beyond the snow line is several times larger than inside it.

The Growth of Planets

In the disc, grains of dust collide and stick together into larger and larger aggregates. How these grow into bodies kilometres across, which are called planetesimals, is not yet fully understood. Once planetesimals have formed they grow further by collisions, in which the gravity of the growing body plays an important part.

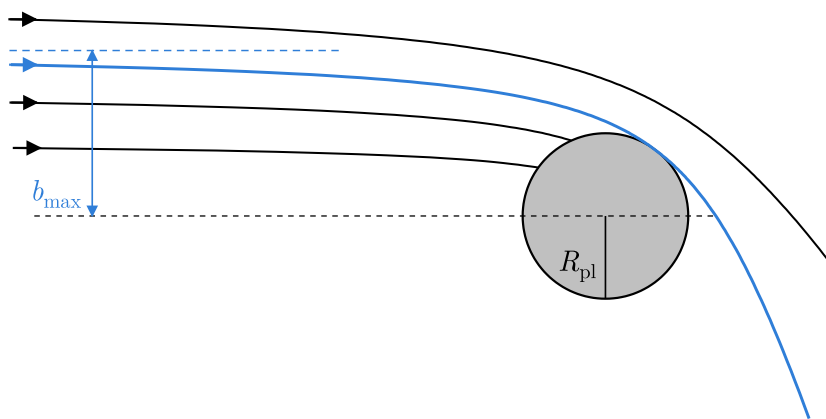


Figure 20. Gravitational focusing. Small bodies approach a body of radius R_{pl} (shaded) from the left, on hyperbolic paths bent by its gravity, computed for an escape speed from its surface larger than v_{∞} . The path whose impact parameter is b_{max} (accent), measured from its asymptote (dashed), grazes the surface, and every path with a smaller impact parameter strikes it.

Let us now take a body of mass M_{pl} and radius R_{pl} , and small bodies approaching it with the speed v_{∞} relative to it while they are still far away. Let b be the impact parameter of a small body,

the distance by which it would miss the centre of the large body if it moved in a straight line. Gravity bends the paths towards the large body, as Figure 20 shows, so a small body can strike it even if b is larger than R_{pl} . On the path that just grazes the surface, the small body passes the surface at the speed v_{max} , moving at right angles to the radius. Conservation of angular momentum per unit mass, from the approach to the grazing point, gives

$$bv_{\infty} = R_{\text{pl}}v_{\text{max}} , \quad (2.40)$$

and conservation of energy per unit mass gives $\frac{1}{2}v_{\infty}^2 = \frac{1}{2}v_{\text{max}}^2 - GM_{\text{pl}}/R_{\text{pl}}$. Let $v_{\text{esc}} = (2GM_{\text{pl}}/R_{\text{pl}})^{1/2}$ be the escape speed from the surface, found in Section 5.1 of Vol. 1. The energy equation then gives $v_{\text{max}}^2 = v_{\infty}^2 + v_{\text{esc}}^2$, and the grazing path has the impact parameter

$$b_{\text{max}} = R_{\text{pl}} \left(1 + \frac{v_{\text{esc}}^2}{v_{\infty}^2} \right)^{1/2} . \quad (2.41)$$

A body with a smaller impact parameter comes closer to the centre and strikes the surface, so the large body collects every small body with $b < b_{\text{max}}$, and it has the effective cross-section

$$\sigma = \pi R_{\text{pl}}^2 \left(1 + \frac{v_{\text{esc}}^2}{v_{\infty}^2} \right) . \quad (2.42)$$

The enlargement of the cross-section by the factor in brackets is called gravitational focusing.

The rate at which the body grows follows from this cross-section. Let ρ_{sol} be the mass per unit volume of the small bodies in the disc near the growing body. In the time dt the body sweeps up the small bodies in a volume $\sigma v_{\infty} dt$, so

$$\frac{dM_{\text{pl}}}{dt} = \rho_{\text{sol}} \pi R_{\text{pl}}^2 v_{\infty} \left(1 + \frac{v_{\text{esc}}^2}{v_{\infty}^2} \right) . \quad (2.43)$$

The rate takes a simple form in two limits. If the random speeds are large, $v_\infty \gg v_{\text{esc}}$, focusing is weak and the rate goes as R_{pl}^2 . Writing $M_{\text{pl}} = \frac{4}{3}\pi R_{\text{pl}}^3 \rho_{\text{pl}}$, where ρ_{pl} is the density of the growing body, gives $dM_{\text{pl}}/dt = 4\pi R_{\text{pl}}^2 \rho_{\text{pl}} dR_{\text{pl}}/dt$, and so $dR_{\text{pl}}/dt = \rho_{\text{sol}} v_\infty / 4\rho_{\text{pl}}$, which is the same for bodies of every size. If instead the random speeds are small, $v_\infty \ll v_{\text{esc}}$, focusing is strong, and substituting $v_{\text{esc}}^2 = 2GM_{\text{pl}}/R_{\text{pl}}$ gives

$$\frac{dM_{\text{pl}}}{dt} \approx \frac{2\pi G \rho_{\text{sol}}}{v_\infty} M_{\text{pl}} R_{\text{pl}}, \quad (2.44)$$

which goes as $M_{\text{pl}}^{4/3}$, since R_{pl} goes as $M_{\text{pl}}^{1/3}$. The rate of growth per unit mass then goes as $M_{\text{pl}}^{1/3}$, so a body that is already larger than its neighbours grows faster in proportion to its mass and pulls further ahead of them, a process called runaway growth; you will compare the two limits in Problem 5. In the inner solar system runaway growth is thought to have produced bodies with masses between those of the Moon and Mars, which then collided with one another over tens of millions of years to form the rocky planets.

Beyond the snow line there is more solid material, and the cores that grow there can reach about ten times the mass of the Earth. A core of that mass attracts the gas of the disc strongly enough to gather it faster and faster, and acquires a massive envelope of hydrogen and helium. This account of the formation of giant planets is called the core-accretion model. It requires the core to grow before the gas of the disc disperses, which observations of young stars in clusters of known age suggest takes a few million years. Jupiter and Saturn are thought to have formed in this way, while Uranus and Neptune, whose cores grew more slowly in the sparser outer disc, gathered much less gas. The hot Jupiters orbit where it is far too hot for ice and the solid material is far too sparse for a giant core to grow, which suggests that planets can move inwards after they form; the migration of planets is not treated in this course.

Problems

§2.1 Kepler's Laws and Orbital Elements

1. Show from the coordinates of B in Figure 10 that $\tan(\theta/2) = \sqrt{(1+e)/(1-e)} \tan(\psi/2)$.
2. Show that to first order in e the solution of Kepler's equation is $\psi \approx \eta + e \sin \eta$, and use the result of Problem 1 to show that $\theta \approx \eta + 2e \sin \eta$.
3. Show that the time average of the distance r over one period is $a(1 + \frac{1}{2}e^2)$.
4. Find the fraction of the period that elapses between pericentre and the first instant at which $r = a$.
5. A body moves on a circular orbit of radius a with inclination i . Show that its height above the reference plane varies sinusoidally with time, with amplitude $a \sin i$, and find the largest value of $d\zeta/dt$.

§2.2 Binary Stars and the Masses and Radii of Stars

1. The normal to the plane of a randomly oriented orbit is equally likely to point in any direction. Show that the probability that the inclination lies between i and $i + di$, for $0 \leq i \leq \pi/2$, is $\sin i di$, and that the mean value of $\sin^3 i$ is $3\pi/16$.
2. Differentiate $\zeta = r \sin(\omega + \theta) \sin i$ with respect to time, using $r = a(1 - e^2)/(1 + e \cos \theta)$ and the constancy of $r^2 d\theta/dt$, to show that the radial velocity of a body on an elliptical orbit is $K[\cos(\omega + \theta) + e \cos \omega]$ with $K = 2\pi a \sin i / P_{\text{orb}} \sqrt{1 - e^2}$. Hence show that the mass function of an eccentric single-lined binary is $(1 - e^2)^{3/2} P_{\text{orb}} K_1^3 / 2\pi G$.
3. Stars of radii R_1 and R_2 move on circular orbits with relative separation a and inclination i . Show that their separation on the sky is smallest when they are at the ends of the line through the centre of mass perpendicular to the line of nodes, and that eclipses occur only if $a \cos i < R_1 + R_2$. Deduce that, for random orientations, the probability that a binary eclipses is $(R_1 + R_2)/a$.

4. In an astrometric binary the visible star, of known mass M_1 , moves on an ellipse of angular semi-major axis α_1 seconds of arc about the centre of mass, with period P_{orb} in years, and the parallax is p seconds of arc. Find an equation for the mass M_2 of the unseen companion in solar masses, and compare it with the mass function.
5. Show that the light curve of an eclipsing binary of the kind shown in Figure 15 gives the ratio R_s/R_l without any measurement of velocity, and that together with the period it gives $(R_s + R_l)/a$.

§2.3 Exoplanets and the Formation of Planetary Systems

1. A planet of mass M_{pl} moves on a circular orbit of period P_{orb} about a star of mass M . Show that the semi-amplitude of the radial velocity of the star is $K = (2\pi G/P_{\text{orb}})^{1/3} M_{\text{pl}} \sin i / (M + M_{\text{pl}})^{2/3}$, and find the ratio of the semi-amplitudes produced by two planets of equal mass whose periods are P_1 and P_2 .
2. Show that the probability that a planet on a randomly oriented circular orbit of radius a transits its star is $(R + R_{\text{pl}})/a$, and hence about R/a .
3. A planet on a circular orbit of radius a crosses the centre of the disc of its star, with $R_{\text{pl}} \ll R \ll a$. Show that the transit lasts $t_{\text{T}} \approx P_{\text{orb}} R/\pi a$, and use Kepler's third law with $M_{\text{pl}} \ll M$ to show that the mean density of the star is $3P_{\text{orb}}/\pi^2 G t_{\text{T}}^3$.
4. A planet on a circular orbit of radius a reflects the fraction A_{b} of the starlight falling on it, absorbs the rest, and, because it rotates rapidly, radiates as a blackbody from its whole surface at a single temperature. Show that its temperature is $T_{\text{eff}}(1 - A_{\text{b}})^{1/4}(R/2a)^{1/2}$. How is the result changed for a planet that always turns the same face to its star and radiates only from that face?
5. Bodies of masses $M_{\text{A}} > M_{\text{B}}$ grow in the same region of a disc. Show that if focusing is weak the difference of their radii stays constant, so that the ratio $M_{\text{A}}/M_{\text{B}}$ decreases towards one, while if focusing is strong the ratio $M_{\text{A}}/M_{\text{B}}$ increases.

Radiation, Radiative Transfer, and Stellar Atmospheres

3.1 Stellar Spectra and Their Classification

In this part we turn from the orbits and masses of stars to the light they emit, and to what the spectrum of that light tells us about the gas from which it escapes. Section 3.1 first reviews the spectra met at school and describes how stars are classified by their spectra. Section 3.2 then accounts for the sequence of spectral classes with the Boltzmann and Saha equations, Section 3.3 sets up the equation of radiative transfer, and finally Section 3.4 applies that equation to the atmosphere of a star and to the formation of its absorption lines.

Spectra at School

The spectra of hot bodies and of glowing gases were introduced at school, and we recall the main facts briefly. A hot solid or a hot, dense gas emits light at every wavelength, with a distribution close to the spectrum of blackbody radiation, which is treated in detail in Section 5.6 of Vol. 2 of the Fermi Notes on Physics. A hot gas of low density emits light only at particular wavelengths characteristic of its atoms, and the bright lines at those wavelengths are called emission lines. As in Section 1.3 of Vol. 3 of the Fermi

Notes on Physics, a photon emitted when an atom passes from a level of energy E_u to a lower level of energy E_l has the frequency $\nu = (E_u - E_l)/h$ and the wavelength $\lambda = c/\nu$.

In 1814 Fraunhofer mapped several hundred dark lines in the spectrum of the Sun, and in 1859 Kirchhoff and Bunsen showed that some of them lie at the wavelengths of the bright lines that particular elements emit in the laboratory, such as the yellow lines of sodium. Kirchhoff summarised the results in three statements, which were found by experiment and are known as Kirchhoff's laws of spectral analysis:

1. a hot solid, liquid, or dense gas emits a continuous spectrum;
2. a hot gas of low density emits a spectrum of bright lines, called an emission spectrum;
3. a cooler gas of low density lying in front of a source of a continuous spectrum removes light from that spectrum at the wavelengths of the emission lines of the gas, and so produces a continuous spectrum crossed by dark lines, called an absorption spectrum.

The three arrangements are drawn in Figure 21. The spectrum of almost every star is of the third kind, and the dark lines of the solar spectrum are called Fraunhofer lines. By the third of Kirchhoff's laws, the light of a star therefore passes, on its way out, through gas that is cooler than the layers beneath it.

The spectrum of a star is recorded with a spectrograph, which is an instrument that passes the light collected by a telescope through a narrow slit and then through a prism or a diffraction grating. The prism or the grating sends light of different wavelengths in different directions, so that the images of the slit at different wavelengths lie side by side across a detector such as the charge-coupled device of Section 1.2, and a spectral line is the image of the slit at the wavelength of the line.

The Absorption Lines of a Star

An atom in a level of energy E_l can absorb a photon of frequency $\nu = (E_u - E_l)/h$ and pass to the level of energy E_u . The excited

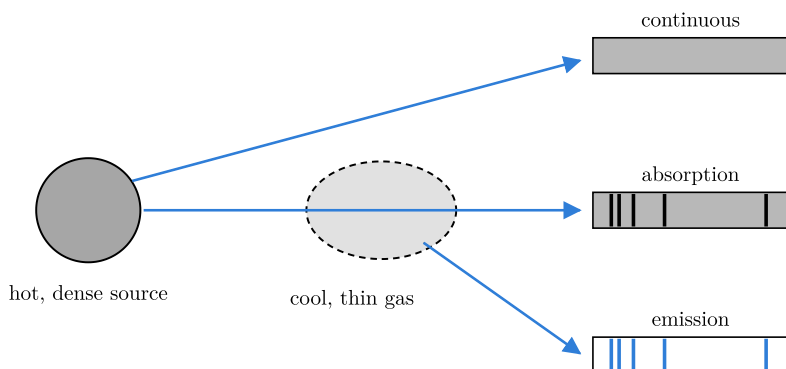


Figure 21. The three kinds of spectrum described by Kirchhoff's laws. A hot, dense source (heavily shaded) seen directly gives a continuous spectrum, and seen through a cooler cloud of gas of low density (lightly shaded) it gives an absorption spectrum. The cloud seen against an empty background gives an emission spectrum, whose bright lines lie at the wavelengths of the dark lines.

atom soon gives up the energy again, either by emitting a photon, which may go in any direction, or by passing the energy to a neighbouring particle in a collision. Of the photons of that frequency that were travelling towards the observer, many are removed and few are replaced, so the light that reaches the observer is weaker at the frequency of the line than at neighbouring frequencies, and an absorption line forms.

The strength of a line depends on how many atoms along the line of sight are in the lower level of the transition. That number depends on the abundance of the element, and also on the temperature and the density of the gas, which fix how the atoms of the element are shared among their energy levels and among their stages of ionisation. The stage of ionisation is written as the symbol of the element followed by a roman numeral, I for the neutral atom, II for the atom with one electron removed, and so on, so that H I is neutral hydrogen, H II is a hydrogen atom stripped of its electron, which is a proton, He II is singly ionised helium, and Ca II is singly ionised calcium.

The Balmer Series

The energy levels of hydrogen, the most abundant element in stars, were found from the Bohr model in Section 1.3 of Vol. 3 and from the Schrödinger equation in Section 5.2 of Vol. 3. In these notes n denotes a number density, so we write k for the principal quantum number, which Vol. 3 writes n . We also write χ_{H} for the Rydberg energy of Vol. 3, which is the ionisation energy of hydrogen in its ground state, in keeping with the symbol χ used in these notes for ionisation energies. The levels are

$$E_k = -\frac{\chi_{\text{H}}}{k^2}, \quad k = 1, 2, 3, \dots, \quad (3.1)$$

with $\chi_{\text{H}} \approx 13.6$ eV. The Balmer series is the set of transitions whose lower level is $k = 2$, and its wavelengths are given by the Rydberg formula of Section 1.3 of Vol. 3,

$$\frac{1}{\lambda} = \mathcal{R}_{\text{H}} \left(\frac{1}{4} - \frac{1}{k^2} \right), \quad k = 3, 4, 5, \dots, \quad (3.2)$$

where $\mathcal{R}_{\text{H}} = \chi_{\text{H}}/hc$ is the Rydberg constant. The lines with $k = 3, 4, 5, 6, \dots$ are called $\text{H}\alpha$, $\text{H}\beta$, $\text{H}\gamma$, $\text{H}\delta$, and so on. $\text{H}\alpha$ lies in the red, $\text{H}\beta$ in the blue-green, and the later lines crowd together as k grows, approaching a limiting wavelength in the near ultraviolet, $4/\mathcal{R}_{\text{H}} \approx 365$ nm, which is called the Balmer limit. In a stellar spectrum the Balmer lines appear in absorption, and an atom can absorb a photon of a Balmer line only if it is already in the level $k = 2$.

A photon with a wavelength shorter than the Balmer limit has an energy greater than $\chi_{\text{H}}/4$, enough to remove the electron from an atom in the level $k = 2$, so it can be absorbed at any such wavelength, the energy left over going into the motion of the freed electron. The spectrum of a star with many hydrogen atoms in the level $k = 2$ therefore drops at the Balmer limit, as Figure 22 shows, and the drop is called the Balmer jump.

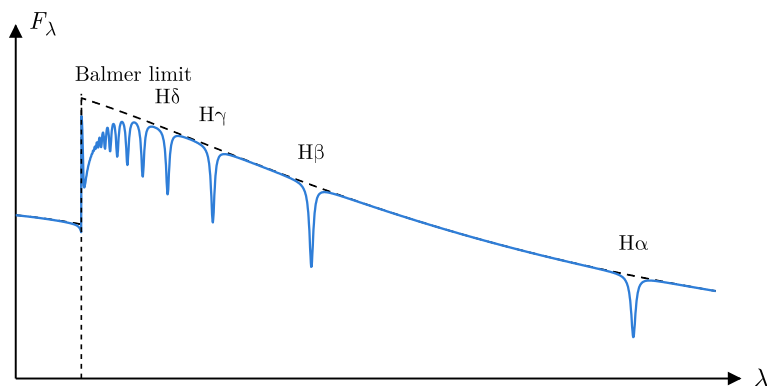


Figure 22. The spectrum of an A star, drawn schematically as a Planck curve (dashed) crossed by the absorption lines of the Balmer series (accent), whose wavelengths are computed from the Rydberg formula. The lines crowd towards the Balmer limit, and at wavelengths shorter than the limit the continuum is lower, because photons of those wavelengths can ionise hydrogen atoms in the level $k = 2$.

The Spectral Sequence

Stellar spectra were first classified on a large scale at the Harvard College Observatory at the end of the nineteenth century. Williamina Fleming sorted them into classes labelled A, B, C, and so on, roughly in order of decreasing strength of the Balmer lines. Annie Jump Cannon found that the spectra form a smoother sequence when some of these classes are dropped and the rest are reordered, and she arrived at the order O, B, A, F, G, K, M, with each class divided into subclasses numbered from 0 to 9, so that the Sun is of class G2. With this scheme she classified about a quarter of a million stars for the Henry Draper Catalogue, published between 1918 and 1924.

The features that distinguish the classes are the following:

1. O: lines of ionised helium, He II, and weak Balmer lines;
2. B: lines of neutral helium, He I, and stronger Balmer lines;
3. A: the strongest Balmer lines, with lines of singly ionised metals

such as Ca II beginning to appear;

4. F: weaker Balmer lines, with the lines of Ca II growing stronger and lines of neutral metals appearing;
5. G: strong lines of Ca II and many lines of neutral metals such as iron, with the Balmer lines weaker still;
6. K: strong lines of neutral metals, weak Balmer lines, and the first bands of molecules;
7. M: strong bands of molecules, among them titanium oxide, together with lines of neutral metals.

Astronomers call every element heavier than helium a metal, which is why calcium and iron are listed as metals here. The sequence has since been extended at its cool end by the classes L, T, and Y, which contain stars and brown dwarfs cooler than the M stars, and there are further classes for stars whose spectra do not fit into it, such as the carbon stars.

The order of the classes is also the order of their colours. The $B - V$ colour index of Section 1.1 increases steadily from the O stars, which are blue, to the M stars, which are red, and by Wien's law a redder colour means a lower temperature. The effective temperatures run from above about 3×10^4 K for the O stars to below about 3.5×10^3 K for the M stars, with the Sun at about 5.8×10^3 K. The lines of ionised atoms are strongest in the hottest stars, where collisions are energetic enough to remove electrons from atoms, and the bands of molecules are found only in the coolest, since in a hotter gas collisions break molecules apart. In 1925 Cecilia Payne showed that the differences between the classes come mainly from differences of temperature, and that the compositions of the stars are broadly similar and dominated by hydrogen. The Balmer lines, however, are strongest in the A stars and weaker in both hotter and cooler stars.

Luminosity Classes

Stars of the same spectral class can differ in luminosity by a large factor. Their effective temperatures are similar, so the relation $L = 4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4$ of Section 1.1 shows that the more luminous stars

are the larger, with R proportional to $L^{1/2}$ at a fixed effective temperature. At a given spectral class the lines of a more luminous star are observed to be narrower, and in 1943 Morgan, Keenan, and Kellman used this difference in width, and other differences between the spectra like it, to define a second classification, by luminosity class. The classes of what is now called the Morgan–Keenan system are these:

1. Ia and Ib, the more luminous and the less luminous supergiants;
2. II, the bright giants;
3. III, the giants;
4. IV, the subgiants;
5. V, the main-sequence stars, which are also called dwarfs.

The scheme is sometimes extended by class VI for the subdwarfs, which lie a little below the main sequence, and white dwarfs are given a separate class, D. A full classification gives both the spectral class and the luminosity class, so that the Sun is G2 V, Aldebaran is K5 III, and Rigel is B8 Ia.

The widths of the lines depend on the luminosity class through the pressure of the gas in which they form. By the balance of forces found for the isothermal atmosphere in Section 1.2 of Vol. 2, $dP/dy = -\rho g$, where y is the height and Vol. 2 writes p for the pressure and z for the height, the pressure at any level supports the weight of the gas above it, and so equals g times the mass per unit area of that gas. The light we receive escapes from roughly the level above which there is too little gas to absorb it, and in stars of similar temperature and composition, whose gases absorb light in similar ways, the mass per unit area above that level is similar. The pressure there is therefore roughly proportional to the surface gravity $g = GM/R^2$. The masses of stars span a much smaller range than their radii, so a giant, whose radius exceeds that of a main-sequence star of the same class by a factor of ten or more, has a surface gravity smaller by a factor of a hundred or more, and a much lower pressure in its atmosphere.

An atom that absorbs a photon while another particle passes close to it has its energy levels shifted slightly by the electric field of that

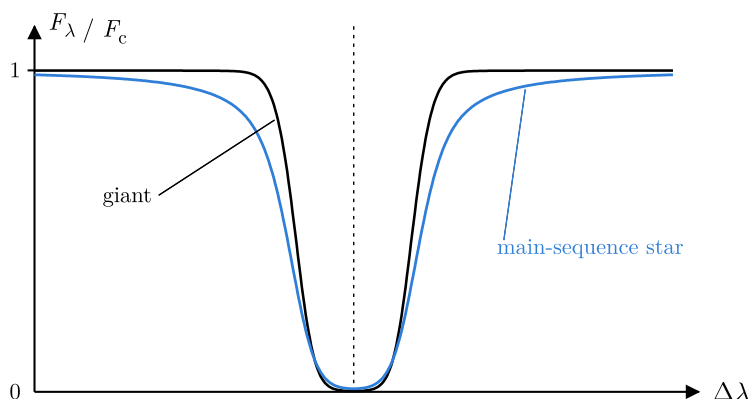


Figure 23. The same absorption line in a main-sequence star (accent) and in a giant (solid) of the same spectral class, computed for the same number of absorbing atoms and the same thermal motion. The higher pressure in the atmosphere of the main-sequence star broadens the wings of its line.

particle, so the photons it can absorb are spread over a small range of frequency. The denser the gas, the more frequent and the closer these encounters, and the broader the line, an effect called pressure broadening. Every line also has a width from the thermal motion of the atoms, which shifts the frequency at which each atom absorbs by the Doppler effect, but you will find in Problem 3 that this width depends on the temperature and on the mass of the atom, and not on the pressure. At the same spectral class the lines of a main-sequence star are therefore broader than those of a giant, as Figure 23 shows.

The Hertzsprung–Russell Diagram from Spectra

In 1911 Hertzsprung plotted the magnitudes of the stars of the Pleiades and the Hyades against their colours, and in 1913 Russell, independently, plotted the absolute magnitudes of stars of known distance against their spectral classes; both found that the stars do not fill the diagram. A plot of this kind is called a Hertzsprung–Russell diagram, and one is drawn in Figure 24, with the absolute bolometric magnitude $\mathcal{M}_{\text{bol}}$ of Section 1.1 on the vertical axis, the

brightest stars at the top, and the spectral class on the horizontal axis. Since the spectral class is a sequence of temperature, the diagram is in effect a plot of luminosity against temperature, with the temperature decreasing to the right.

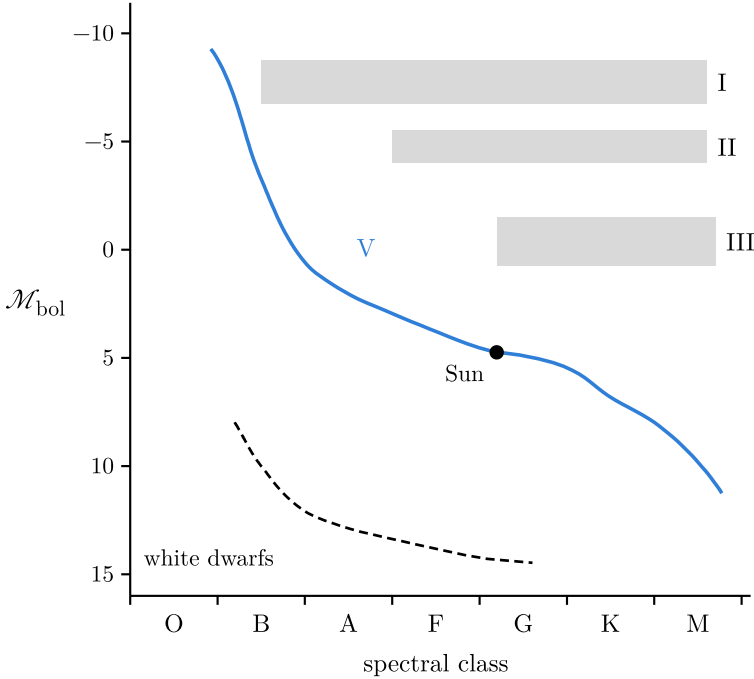


Figure 24. A Hertzsprung–Russell diagram drawn from spectral class and absolute bolometric magnitude. The main sequence, luminosity class V (accent), is drawn from $L \propto M^{3.5}$ and $R \propto M^{0.8}$, with the Sun marked. The giants (III), the bright giants (II), and the supergiants (I) are drawn schematically as shaded bands of constant luminosity, and the white dwarfs lie along a line of constant radius (dashed), found from $L = 4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4$.

About nine in ten of the stars near the Sun lie along a band that runs from the luminous O stars at the upper left to the faint M stars at the lower right, and this band is called the main sequence; its stars are of luminosity class V. The giants and supergiants lie above the main sequence, most of them towards the cool end, and the white dwarfs lie below it, since they are hot but so small that their luminosities are low. A spectrum classified by both spectral

class and luminosity class therefore places the star on the diagram.

This placing gives the distance to a star that is too far away for its trigonometric parallax to be measured. The spectral class and the luminosity class found from the spectrum, together with a diagram calibrated with stars of known distance, give the absolute magnitude $\mathcal{M}$ of the star. Measuring its apparent magnitude m in the same band and using the distance modulus of Section 1.1,

$$m - \mathcal{M} = 5 \log_{10} \frac{d}{10 \text{ pc}}, \quad (3.3)$$

gives the distance d , provided that the absorption of light by interstellar dust along the line of sight is negligible or has been corrected for. The method is called spectroscopic parallax, although no parallax is measured, and its accuracy is limited by the spread of absolute magnitudes within each band of the diagram; you will relate that spread to the uncertainty in the distance in Problem 4.

3.2 The Boltzmann and Saha Equations

Section 3.1 described the spectral sequence, in which the lines of ionised atoms are strongest in the hottest stars, the bands of molecules appear only in the coolest, and the Balmer lines are strongest in the A stars. The strength of an absorption line depends on the number of atoms in the lower level of its transition, and that number is set by two equilibria. The first equilibrium shares the atoms of a given stage of ionisation among their energy levels, and the second shares the atoms of an element among its stages of ionisation. In this section we first find the populations of the levels from the Boltzmann factor, then derive the Saha equation for the ionisation of hydrogen, and finally combine the two equations to find how the strength of the Balmer lines depends on temperature.

The Boltzmann and Saha equations both describe a gas in thermal equilibrium. The temperature of a star changes from place to place, but in the dense layers where most lines form the particles collide often enough for the gas in each small region to be in equilibrium

at the temperature of that region. This condition is called local thermodynamic equilibrium, and we assume it throughout this part.

The Populations of the Levels

In Section 4.4 of Vol. 2 the probability that a small system in contact with a reservoir at temperature T is in a particular microstate of energy E was shown to be proportional to the Boltzmann factor $e^{-E/k_B T}$. As in the derivation of the distribution of molecular speeds in that section, we take a single atom as the small system and the rest of the gas as the reservoir. The energy of the atom is the sum of the kinetic energy of its centre of mass and the energy of its electrons, and the exponential of a sum is a product, so the Boltzmann factor is the product of a factor that depends on the velocity and a factor that depends on the level. The velocity factor is the same whichever level the atom is in, so the ratio of the numbers of atoms in two levels depends only on the level factors.

A level may contain several states of the same energy. Let g_j be the number of states in the level j , which is called its statistical weight, and let E_j be the energy of the level. Each state of the level carries the same Boltzmann factor, so the number density n_j of atoms in the level is proportional to $g_j e^{-E_j/k_B T}$, and for two levels j and j' of the same atom,

$$\frac{n_{j'}}{n_j} = \frac{g_{j'}}{g_j} e^{-(E_{j'} - E_j)/k_B T}, \quad (3.4)$$

which is called the Boltzmann equation. In hydrogen the level k contains k^2 states of the orbital motion of the electron, which is the degeneracy found in Section 5.2 of Vol. 3, and each of them combines with the two states of the spin of the electron found in Section 4.2 of Vol. 3, so that $g_k = 2k^2$, as you showed in Problem 6 of Section 5.2 of Vol. 3. For the first excited level and the ground state, $E_2 - E_1 = \frac{3}{4}\chi_H$, and the Boltzmann equation gives

$$\frac{n_2}{n_1} = 4 e^{-3\chi_H/4k_B T}. \quad (3.5)$$

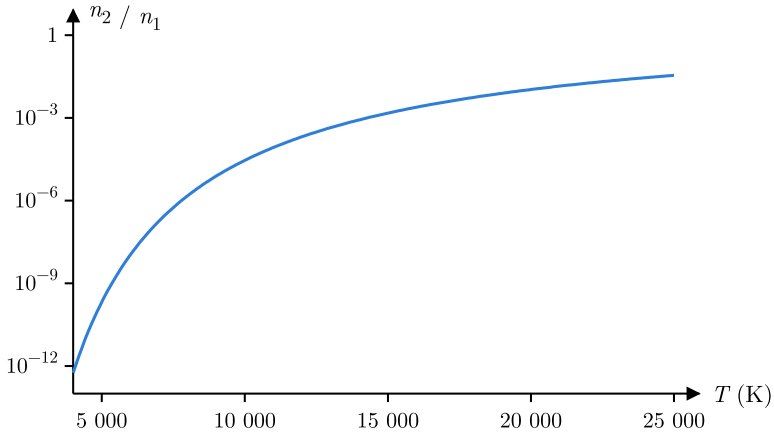


Figure 25. The ratio n_2/n_1 of the number densities of hydrogen atoms in the first excited level and in the ground state, from the Boltzmann equation, drawn on a logarithmic scale against the temperature.

The ratio is drawn in Figure 25. It is tiny at the temperatures of stellar atmospheres, because $\frac{3}{4}\chi_{\text{H}}$ is many times $k_B T$ there. At the effective temperature of the Sun only a few atoms in 10^9 are in the level $k = 2$, and even at 10^4 K the fraction is only a few parts in 10^5 . The levels $k = 1$ and $k = 2$ would be equally populated only at a temperature close to 10^5 K, as you will show in Problem 1. If the Boltzmann equation alone set the strength of the Balmer lines, they would therefore grow stronger with temperature through the whole of the spectral sequence. This is not what is observed, since they are weaker in the B and O stars than in the A stars, and the reason is that at high temperatures the hydrogen is ionised, and a hydrogen atom that has lost its electron has no level $k = 2$.

The fraction of all the atoms of a given stage of ionisation that are in the level j is found by dividing $g_j e^{-E_j/k_B T}$ by the sum of the same quantity over all the levels. For the neutral atom, with the energies measured from the ground state $j = 1$,

$$\frac{n_j}{n_1} = \frac{g_j e^{-(E_j - E_1)/k_B T}}{\mathcal{Z}_1}, \quad \mathcal{Z}_1 = \sum_j g_j e^{-(E_j - E_1)/k_B T}, \quad (3.6)$$

where n_{I} is the number density of neutral atoms in all levels. The sum $\mathcal{Z}_{\text{I}}$ is the partition function of Section 4.4 of Vol. 2, with the states grouped by level. Vol. 2 writes the partition function as Z , a letter that these notes keep for the mass fraction of the elements heavier than helium, and we write $\mathcal{Z}_{\text{I}}$ for the partition function of the neutral atom and $\mathcal{Z}_{\text{II}}$ for that of the singly ionised atom.

For hydrogen the partition function raises the following question: since $g_k = 2k^2$ increases with k while $e^{-(E_k - E_1)/k_B T}$ tends to $e^{-\chi_{\text{H}}/k_B T}$, the terms of the sum grow without limit, so does the sum diverge? It would for a single atom alone in space. In a gas, however, the radius of the Bohr orbit of the level k is $k^2 a_0$, by Section 1.3 of Vol. 3, where a_0 is the Bohr radius, and once this radius is comparable with the distance between neighbouring particles the electron is no longer bound to one nucleus. The sum therefore stops at a finite value of k , and at the temperatures of stellar atmospheres the terms after the first are small enough that $\mathcal{Z}_{\text{I}} \approx g_1 = 2$, as you will check in Problem 5.

The Saha Equation

The ionisation of hydrogen is the reaction



in which a neutral atom separates into a proton and a free electron, while the reverse process, called recombination, joins them again. Let n_{I} , n_{II} , and n_{e} be the number densities of atoms, protons, and free electrons. The reaction is in equilibrium when the condition of Section 3.4 of Vol. 2 holds, which says that the sum of the chemical potentials of the species taking part, each multiplied by its stoichiometric coefficient, is zero. Vol. 2 writes the chemical potential as μ , which in these notes is the mean molecular weight, so we write it as μ_{ch} , and for this reaction the condition reads

$$\mu_{\text{ch,I}} = \mu_{\text{ch,II}} + \mu_{\text{ch,e}}. \quad (3.8)$$

To turn this condition into a relation between the number densities, we need the chemical potential of each species.

The chemical potential of an ideal gas of particles with no internal structure was found in Section 5.1 of Vol. 2. The single-particle partition function there was V/λ_{th}^3 , where V is the volume and $\lambda_{\text{th}} = h/\sqrt{2\pi m_x k_B T}$ is the thermal wavelength of a particle of mass m_x , and the chemical potential came out as $k_B T \ln(N\lambda_{\text{th}}^3/V)$, which is $-k_B T$ times the logarithm of the single-particle partition function divided by the number of particles N . A particle with internal states has an energy that is the sum of its kinetic energy and its internal energy, and by Section 5.2 of Vol. 2 its single-particle partition function is then the product of V/λ_{th}^3 and a partition function for the internal states. Let the lowest internal state have the energy E_0 , measured from a zero common to all the species, and let $\mathcal{Z}$ be the internal partition function with the energies measured from E_0 . The single-particle partition function is then $(V/\lambda_{\text{th}}^3) \mathcal{Z} e^{-E_0/k_B T}$. Repeating the differentiation of the free energy of Section 5.1 of Vol. 2 with this partition function in place of V/λ_{th}^3 , with Stirling's approximation as there, gives $-k_B T$ times the logarithm of this partition function divided by N , which can be rewritten as

$$\mu_{\text{ch}} = E_0 + k_B T \ln \frac{n\lambda_{\text{th}}^3}{\mathcal{Z}}, \quad (3.9)$$

with $n = N/V$. For $E_0 = 0$ and $\mathcal{Z} = 1$ this is the result of Vol. 2.

We take the common zero of energy to be a proton and an electron at rest far apart, as in Figure 26. The proton then has $E_0 = 0$ and no internal states, so that $\mathcal{Z}_{\text{H}} = 1$. The proton does have two spin states, but the atom contains the proton and has them too, so they multiply both $\mathcal{Z}_{\text{I}}$ and $\mathcal{Z}_{\text{H}}$ by two and cancel from the result, and we leave them out of both. The free electron has $E_0 = 0$ and two spin states of equal energy, so its internal partition function is 2, and its thermal wavelength is $\lambda_e = h/\sqrt{2\pi m_e k_B T}$. The atom in its ground state has $E_0 = -\chi_{\text{H}}$ and the internal partition function $\mathcal{Z}_{\text{I}}$. The masses of the atom and the proton differ only by m_e , which is about one part in two thousand, so their thermal wavelengths are equal to that accuracy and cancel from the condition for equilibrium, which becomes

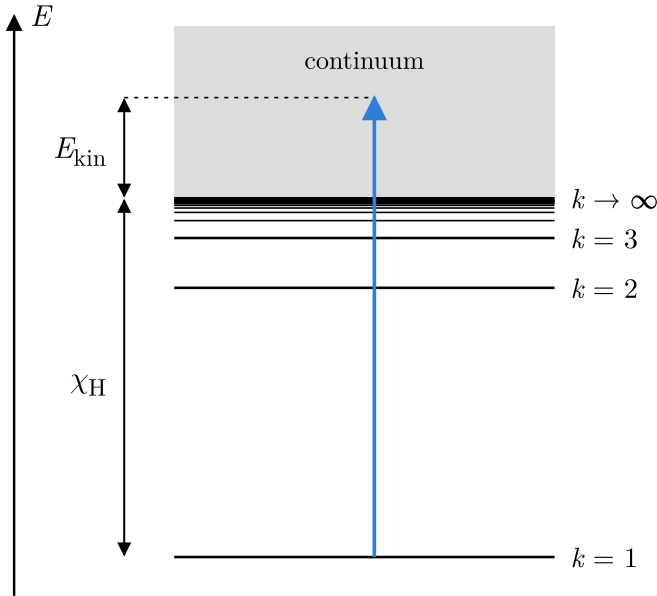


Figure 26. The energy levels of hydrogen, drawn to scale, with the level k at the energy $-\chi_H/k^2$. Above zero lies the continuum (shaded) of states in which the electron is free. Ionising an atom from its ground state into a state in which the free electron has kinetic energy E_{kin} takes the energy $\chi_H + E_{\text{kin}}$ (accent arrow).

$$-\chi_{\text{H}} + k_{\text{B}}T \ln \frac{n_{\text{I}}}{\mathcal{Z}_{\text{I}}} = k_{\text{B}}T \ln \frac{n_{\text{II}}}{\mathcal{Z}_{\text{II}}} + k_{\text{B}}T \ln \frac{n_{\text{e}}\lambda_{\text{e}}^3}{2}. \quad (3.10)$$

Collecting the logarithms on one side, using the fact that a sum of logarithms is the logarithm of a product, and taking the exponential of both sides gives

$$\frac{n_{\text{II}} n_{\text{e}}}{n_{\text{I}}} = \frac{2\mathcal{Z}_{\text{II}}}{\mathcal{Z}_{\text{I}}} \left(\frac{2\pi m_{\text{e}} k_{\text{B}}T}{h^2} \right)^{3/2} e^{-\chi_{\text{H}}/k_{\text{B}}T}, \quad (3.11)$$

which is called the Saha equation, after Meghnad Saha, who derived it in 1920. It is the law of mass action of Section 3.4 of Vol. 2 applied to ionisation. The same argument applies to the removal of an electron from any atom or ion, with χ_{H} replaced by the ionisation energy χ of the stage concerned and $\mathcal{Z}_{\text{I}}$ and $\mathcal{Z}_{\text{II}}$ by the partition functions of the two stages. In a stellar atmosphere it is often convenient to write the density of free electrons in terms of their pressure, $P_{\text{e}} = n_{\text{e}}k_{\text{B}}T$, which gives

$$\frac{n_{\text{II}}}{n_{\text{I}}} = \frac{2k_{\text{B}}T}{P_{\text{e}}} \frac{\mathcal{Z}_{\text{II}}}{\mathcal{Z}_{\text{I}}} \left(\frac{2\pi m_{\text{e}} k_{\text{B}}T}{h^2} \right)^{3/2} e^{-\chi/k_{\text{B}}T}. \quad (3.12)$$

The exponential in the Saha equation favours the atom, since ionisation takes the energy χ , while the factor $(2\pi m_{\text{e}} k_{\text{B}}T/h^2)^{3/2}/n_{\text{e}} = 1/n_{\text{e}}\lambda_{\text{e}}^3$ favours ionisation. By the interpretation given in Section 5.1 of Vol. 2, this factor is the number of cubes of side λ_{e} in the volume available to each free electron, a measure of the number of states open to an electron once it is freed, and in a stellar atmosphere it is very large. Hydrogen therefore becomes substantially ionised at temperatures at which $k_{\text{B}}T$ is still much smaller than χ_{H} .

The ionised fraction is drawn in Figure 27. With an electron pressure of about 20 N m^{-2} , which is typical of the photospheres of main-sequence stars, hydrogen is half ionised at about 10^4 K , where $k_{\text{B}}T$ is only about one sixteenth of χ_{H} , and the ionisation goes from nearly none to nearly complete over a range of temperature of a few thousand kelvin. The range is narrow because the exponent

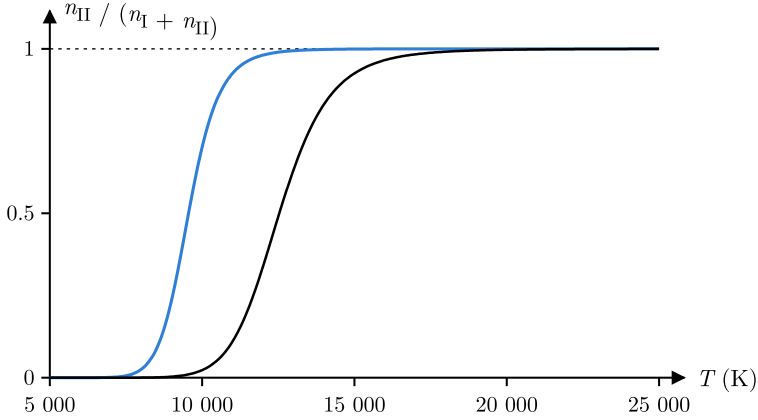


Figure 27. The fraction $n_{\text{II}}/(n_{\text{I}}+n_{\text{II}})$ of hydrogen that is ionised, from the Saha equation, against the temperature, for an electron pressure typical of the photosphere of a main-sequence star (accent) and for a larger electron pressure (solid). The ionisation goes from nearly none to nearly complete over a narrow range of temperature, and the larger electron pressure moves that range to higher temperatures.

$\chi_{\text{H}}/k_{\text{B}}T$ is large, so a small fractional change in the temperature changes the exponential by a large factor, as you will show in Problem 3. A larger electron pressure moves the range to higher temperatures, because a proton meets free electrons, and recombines, more often in a denser gas. In the tenuous atmospheres of giants and supergiants, where the pressure is low, an element is therefore more highly ionised than in a main-sequence star of the same temperature, which is one of the differences between the spectra of the luminosity classes of Section 3.1.

The Strength of the Balmer Lines

The strength of the Balmer lines depends on the number density of hydrogen atoms in the level $k = 2$, as a fraction of all the hydrogen, neutral and ionised. That fraction is the product of the fraction of neutral atoms that are in the level $k = 2$ and the fraction of all the hydrogen that is neutral,

$$\frac{n_2}{n_{\text{I}} + n_{\text{II}}} = \frac{n_2}{n_{\text{I}}} \frac{n_{\text{I}}}{n_{\text{I}} + n_{\text{II}}} = \frac{n_2}{n_{\text{I}}} \frac{1}{1 + n_{\text{II}}/n_{\text{I}}}. \quad (3.13)$$

The first factor comes from the Boltzmann equation. The atoms in every excited level, $k = 2$ included, are far fewer than those in the ground state at these temperatures, so $n_{\text{I}} \approx n_{\text{I}} + n_2$ and $n_2/n_{\text{I}} \approx (n_2/n_{\text{I}})/(1 + n_2/n_{\text{I}})$. The second factor comes from the Saha equation with $\mathcal{Z}_{\text{I}} = 2$ and $\mathcal{Z}_{\text{II}} = 1$.

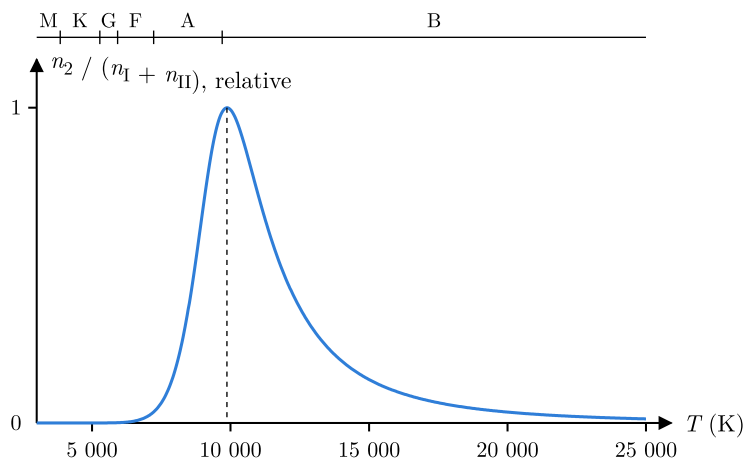


Figure 28. The fraction of all hydrogen, neutral and ionised, that consists of atoms in the level $k = 2$, from the Boltzmann and Saha equations, drawn relative to its greatest value against the temperature, for an electron pressure typical of the photosphere of a main-sequence star. The spectral classes are marked along the top. The fraction rises with the Boltzmann factor and falls as the hydrogen is ionised, and it is greatest near the temperatures of the early A stars (dashed).

The product is drawn in Figure 28. As the temperature rises from that of the Sun, the first factor rises steeply while the hydrogen remains almost entirely neutral, and the fraction in the level $k = 2$ grows. Above about 10^4 K nearly all the hydrogen is ionised, and the second factor then falls faster than the first rises, as you will show in Problem 4. The fraction is therefore greatest near 10^4 K, which is the temperature of the early A stars, and this is why the Balmer lines are strongest in the A stars. Even at the

peak only about one hydrogen atom in 10^5 is in the level $k = 2$, but hydrogen is so abundant that this is enough to produce strong lines. The curve is drawn for a single electron pressure, and the electron pressure differs from one star to another, but the position of the peak changes only slowly with it, since the electron pressure enters the Saha equation as a power while the temperature enters through an exponential.

The same reasoning applied to other elements accounts for the rest of the spectral sequence. Calcium, for example, has an ionisation energy of about 6.1 eV, so it is mostly singly ionised in the G and K stars, and the lines of Ca II, which absorb from the ground state of the ion, are strongest there. Payne used the Boltzmann and Saha equations in this way in 1925 to convert the observed strengths of the lines of each element into the numbers of its atoms.

3.3 Radiative Transfer

The light that leaves a star has been absorbed and emitted many times on its way out, and the spectrum we receive depends on how the intensity of the light changes along its path through the gas. To follow those changes we need a quantity that describes how much energy travels in each direction, and the specific intensity defined below is such a quantity. Blackbody radiation itself is treated in detail in Section 5.6 of Vol. 2, and here we need only its spectrum. In this section we first define the specific intensity and relate it to the flux, the energy density, and the pressure of radiation, then describe absorption and emission by matter, and finally set up and solve the equation of radiative transfer, which leads to Kirchhoff's law.

Specific Intensity

Let a small patch of area dA be placed in a field of radiation, and consider the radiation that crosses it in a time dt , with frequencies between ν and $\nu + d\nu$, travelling in directions within a small solid angle $d\Omega$ about a direction that makes an angle θ with the normal to the patch, as in Figure 29. The solid angle was defined in Section 1.3 of Vol. 5 of the Fermi Notes on Physics as the area that a cone cuts from a sphere centred on its vertex, divided by the square of

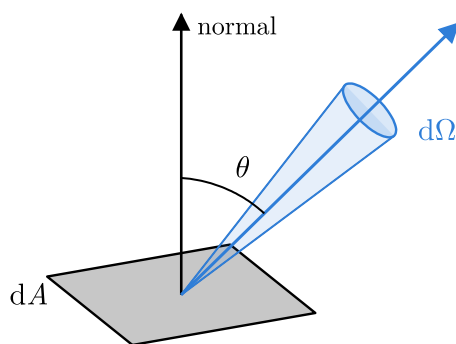


Figure 29. The definition of the specific intensity. The energy crossing a small patch of area dA (shaded) in directions within a small solid angle $d\Omega$ about a direction at an angle θ to the normal of the patch is proportional to the projected area $\cos \theta dA$ and to $d\Omega$.

the radius of the sphere, and in spherical coordinates its element is $d\Omega = \sin \theta d\theta d\phi$, as in Section 4.1 of Vol. 3. The energy crossing the patch is proportional to each of dt , $d\nu$, and $d\Omega$, and to the area of the patch as seen from the direction of travel, which is $\cos \theta dA$. We therefore write

$$dE = I_\nu \cos \theta dA dt d\Omega d\nu, \quad (3.14)$$

which defines the specific intensity I_ν , a function of position, of direction, and of frequency. Its unit is $\text{W m}^{-2} \text{Hz}^{-1} \text{sr}^{-1}$, where the steradian, sr, is the unit of solid angle. The integral of I_ν over all frequencies is written I and is called the intensity. The specific intensity describes an extended source, such as the disc of the Sun, or radiation arriving from all directions, as inside a star; for an unresolved star only the flux, defined below, can be measured.

Constancy along a Ray

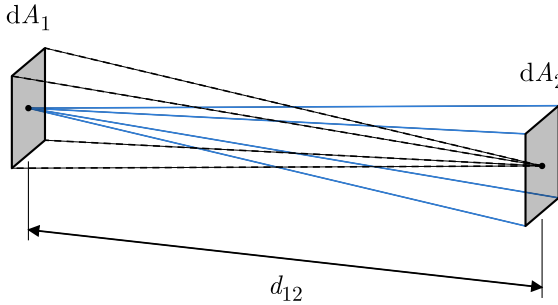


Figure 30. Small patches of areas dA_1 and dA_2 (shaded), each perpendicular to the line joining them, a distance d_{12} apart. The energy that leaves the first patch within the solid angle that the second subtends (accent) is the energy that arrives at the second patch within the solid angle that the first subtends (dashed).

In empty space the specific intensity is constant along a ray. To see why, consider two small patches of areas dA_1 and dA_2 , a distance

d_{12} apart and each perpendicular to the line joining them, as in Figure 30. Let I_1 be the specific intensity leaving the first patch in the direction of the second, and I_2 the specific intensity arriving at the second patch from the direction of the first. The second patch subtends at the first the solid angle dA_2/d_{12}^2 , so the energy that leaves the first patch in a time dt and frequency range $d\nu$ and reaches the second is

$$dE = I_1 dA_1 \frac{dA_2}{d_{12}^2} dt d\nu. \quad (3.15)$$

The first patch subtends at the second the solid angle dA_1/d_{12}^2 , so the same energy, counted as it arrives at the second patch, is $I_2 dA_2 (dA_1/d_{12}^2) dt d\nu$. In empty space no energy is lost between the patches, so the two expressions for the energy are equal, and $I_1 = I_2$. Tilting either patch multiplies both expressions by the same factor, the cosine of the angle of tilt, so the result holds for patches of any orientation.

It follows that the specific intensity of the disc of the Sun is the same at every distance from the Sun, since the flux of its light falls as the inverse square of the distance and the solid angle of its disc falls in the same way. A telescope that resolves an extended source therefore measures the specific intensity with which the light left its surface, provided that nothing absorbs or emits light along the way.

Flux, Energy Density, and Pressure

The flux F_ν is the net energy that crosses unit area of a surface in unit time and unit frequency range. By the definition of the specific intensity, the radiation crossing in directions within $d\Omega$ contributes $I_\nu \cos \theta d\Omega$, where $\cos \theta$ is negative for radiation crossing in the opposite direction, so that

$$F_\nu = \int I_\nu \cos \theta d\Omega = \int_0^{2\pi} \int_0^\pi I_\nu \cos \theta \sin \theta d\theta d\phi. \quad (3.16)$$

If the radiation is isotropic, with I_ν the same in every direction,

the flux is zero, since the integral of $\cos \theta \sin \theta$ from 0 to π vanishes, and as much energy crosses the surface in one direction as in the other. If instead the radiation leaves a surface with the same specific intensity in every outward direction and none arrives, as at the surface of a blackbody radiating into empty space, the integral over θ runs from 0 to $\pi/2$. Since $\cos \theta \sin \theta$ is the derivative of $\frac{1}{2} \sin^2 \theta$, the outward flux is then

$$F_\nu = 2\pi I_\nu \int_0^{\pi/2} \cos \theta \sin \theta \, d\theta = \pi I_\nu. \quad (3.17)$$

The factor π is half the solid angle 2π of a hemisphere, because the factor $\cos \theta$ gives radiation leaving at a slant less weight than radiation leaving along the normal.

The energy density follows from a similar argument. Radiation travelling in directions within $d\Omega$ crosses an area dA perpendicular to its direction in a time dt , and the energy that crosses in that time fills a cylinder of cross-section dA and length $c \, dt$. Dividing the energy $I_\nu \, dA \, dt \, d\Omega \, d\nu$ by the volume $c \, dt \, dA$ of the cylinder and adding the contributions of all directions, the energy per unit volume per unit frequency is

$$u_\nu = \frac{1}{c} \int I_\nu \, d\Omega, \quad (3.18)$$

which is $4\pi I_\nu/c$ for isotropic radiation.

A photon of energy E carries momentum E/c along its direction of travel, as shown in Section 1.2 of Vol. 3. Radiation crossing unit area of a surface in unit time and unit frequency range, in directions within $d\Omega$, carries the energy $I_\nu \cos \theta \, d\Omega$, and hence momentum whose component along the normal is $(I_\nu \cos \theta \, d\Omega/c) \cos \theta$. The rate at which the normal component of momentum is carried across unit area is a pressure, so the pressure of the radiation per unit frequency is

$$P_{\text{rad},\nu} = \frac{1}{c} \int I_\nu \cos^2 \theta \, d\Omega. \quad (3.19)$$

Radiation crossing in the opposite direction has $\cos \theta < 0$ and carries momentum in the opposite direction, so it adds to the pressure, which is why $\cos^2 \theta$ appears. For isotropic radiation, since $\cos^2 \theta \sin \theta$ is the derivative of $-\frac{1}{3} \cos^3 \theta$, the integral over θ from 0 to π is $2/3$, and the pressure is $4\pi I_\nu / 3c = u_\nu / 3$. This is the relation between the pressure and the energy density of isotropic radiation found in Section 5.6 of Vol. 2 from the free energy, and in Section 9.2 of Vol. 5 from the pressure of plane waves on a wall.

For blackbody radiation, which is isotropic, Section 1.2 of Vol. 3 gives the energy density per unit frequency as $(8\pi\nu^2/c^3) h\nu / (e^{h\nu/k_B T} - 1)$, where Vol. 3 writes f for the frequency. The specific intensity of blackbody radiation is therefore $c/4\pi$ times this energy density,

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1}, \quad (3.20)$$

which is called the Planck function. Integrating over frequency, the intensity of blackbody radiation is $cu/4\pi$, where u is its energy density. Section 5.6 of Vol. 2 found that the flux leaving a small hole in a cavity is $uc/4 = \sigma_{\text{SB}} T^4$, where Vol. 2 writes σ for the Stefan–Boltzmann constant, so that $u = a_{\text{rad}} T^4$ with $a_{\text{rad}} = 4\sigma_{\text{SB}}/c$, which is called the radiation constant. The integrated Planck function is therefore

$$B(T) = \int_0^\infty B_\nu(T) d\nu = \frac{\sigma_{\text{SB}} T^4}{\pi}, \quad (3.21)$$

and the flux πB leaving the surface of a blackbody, found from the outward flux πI_ν above, is $\sigma_{\text{SB}} T^4$, which is the Stefan–Boltzmann law.

Absorption and the Mean Free Path

When a beam of light passes through matter, some of its photons are absorbed, and others are scattered out of the beam into other directions. Absorption and scattering both remove energy from the beam, and in this part the word absorption is used for both processes. Let us consider a beam crossing a thin cylinder of matter of

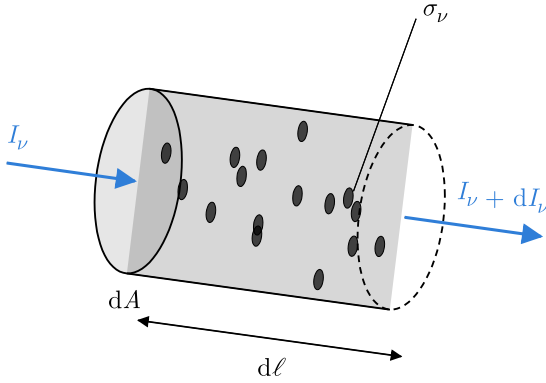


Figure 31. A beam of specific intensity I_ν crossing a thin cylinder of absorbing matter (shaded) of cross-section dA and length $d\ell$. Each absorber, drawn as a small dark disc, removes the light that falls on its cross-section σ_ν , and the beam leaves with specific intensity $I_\nu + dI_\nu$, where dI_ν is negative.

cross-section dA and length $d\ell$, as in Figure 31. Let the matter contain n absorbers per unit volume, each of which removes the light falling on an area σ_ν , which is called its cross-section at the frequency ν . The cylinder contains $n dA d\ell$ absorbers, which together block the area $n\sigma_\nu dA d\ell$, provided the cylinder is thin enough that no absorber lies in the shadow of another. The fraction of the beam removed is the fraction $n\sigma_\nu d\ell$ of the cross-section that is blocked, so

$$dI_\nu = -n\sigma_\nu I_\nu d\ell = -\kappa_\nu \rho I_\nu d\ell. \quad (3.22)$$

The product $n\sigma_\nu$ is the fraction of the beam removed per unit length, and it is called the absorption coefficient. The quantity $\kappa_\nu = n\sigma_\nu/\rho$, which is the total cross-section of the absorbers in unit mass of the matter, is called the opacity, and its unit is $\text{m}^2 \text{kg}^{-1}$. It depends on the frequency and on the temperature, the density, and the composition of the matter. As an example, consider fully ionised hydrogen. Its free electrons scatter light of every frequency, each with the same cross-section $\sigma_T \approx 6.65 \times 10^{-29} \text{ m}^2$, which is called the Thomson cross-section and comes from the classical the-

ory of the scattering of an electromagnetic wave by a free electron, which is beyond this course. There is one free electron for each mass m_{H} of the gas, so $n_{\text{e}} = \rho/m_{\text{H}}$, and the opacity is $\kappa = \sigma_{\text{T}}/m_{\text{H}}$, the same at every frequency and density.

The optical depth along a path is defined as

$$\tau_{\nu} = \int_0^{\ell} \kappa_{\nu} \rho \, d\ell' , \quad (3.23)$$

where ℓ' is the distance along the path, so that $d\tau_{\nu} = \kappa_{\nu} \rho \, d\ell$ and the equation for I_{ν} becomes $dI_{\nu}/d\tau_{\nu} = -I_{\nu}$. Integrating this first-order equation gives $I_{\nu} = I_{\nu}(0) e^{-\tau_{\nu}}$, so a beam that crosses the optical depth τ_{ν} is weakened by the factor $e^{-\tau_{\nu}}$. In a uniform medium $\tau_{\nu} = \ell/\ell_{\text{ph}}$, with

$$\ell_{\text{ph}} = \frac{1}{\kappa_{\nu} \rho} = \frac{1}{n\sigma_{\nu}} . \quad (3.24)$$

The fraction of the beam that remains after a distance ℓ is $e^{-\ell/\ell_{\text{ph}}}$, which is also the probability that a single photon travels that far without being absorbed, and the probability that it is absorbed between ℓ and $\ell + d\ell$ is $e^{-\ell/\ell_{\text{ph}}} d\ell/\ell_{\text{ph}}$. The mean distance travelled before absorption is therefore

$$\int_0^{\infty} \ell e^{-\ell/\ell_{\text{ph}}} \frac{d\ell}{\ell_{\text{ph}}} = \ell_{\text{ph}} , \quad (3.25)$$

where the integral is done by parts, differentiating ℓ and integrating the exponential, and the boundary terms vanish at both ends. The length ℓ_{ph} is called the mean free path of a photon, and the optical depth along a path is the number of mean free paths it contains. The expression $1/n\sigma_{\nu}$ has the form of the mean free path of a molecule found in Section 1.3 of Vol. 2, without the factor $\sqrt{2}$ that came there from the motion of the other molecules, since a photon moves so much faster than the atoms that they can be treated as at rest. A layer with $\tau_{\nu} \gg 1$ is called optically thick, and one with

$\tau_\nu \ll 1$ optically thin. In the photosphere of the Sun the mean free path of a photon of visible light is of order 10^2 km.

Emission and the Source Function

Matter also adds energy to a beam, by emitting photons and by scattering into the beam photons that were travelling in other directions. Let j_ν , which is called the emission coefficient, be the energy added to the beam per unit time, per unit mass, per unit frequency, and per unit solid angle about the direction of the beam. A cylinder of cross-section dA and length $d\ell$ has mass $\rho dA d\ell$ and adds the energy $j_\nu \rho dA d\ell dt d\Omega d\nu$ to the beam, and dividing by $dA dt d\Omega d\nu$ gives an increase $j_\nu \rho d\ell$ in the specific intensity. Adding the loss by absorption,

$$\frac{dI_\nu}{d\ell} = -\kappa_\nu \rho I_\nu + j_\nu \rho, \quad (3.26)$$

which is called the equation of radiative transfer. Dividing by $\kappa_\nu \rho$ and using $d\tau_\nu = \kappa_\nu \rho d\ell$,

$$\frac{dI_\nu}{d\tau_\nu} = -I_\nu + S_\nu, \quad S_\nu = \frac{j_\nu}{\kappa_\nu}, \quad (3.27)$$

where S_ν is called the source function and has the same unit as the specific intensity. The equation says that the specific intensity is drawn towards the source function, decreasing where $I_\nu > S_\nu$ and increasing where $I_\nu < S_\nu$, and that over one mean free path it changes by an amount comparable with the difference between I_ν and S_ν .

The Formal Solution

Written as $dI_\nu/d\tau_\nu + I_\nu = S_\nu$, the equation of transfer is a linear first-order equation, which can be solved with the integrating factor of Section 5.1 of Vol. 4 of the Fermi Notes on Physics. Multiplying both sides by e^{τ_ν} turns the left-hand side into $d(e^{\tau_\nu} I_\nu)/d\tau_\nu$, by the product rule, so that $d(e^{\tau_\nu} I_\nu)/d\tau_\nu = e^{\tau_\nu} S_\nu$. Integrating from the start of the path, where $\tau_\nu = 0$, to the optical depth τ_ν , and dividing by e^{τ_ν} ,

$$I_\nu(\tau_\nu) = I_\nu(0) e^{-\tau_\nu} + \int_0^{\tau_\nu} S_\nu(\tau') e^{-(\tau_\nu - \tau')} d\tau'. \quad (3.28)$$

This is called the formal solution of the equation of transfer, since it gives I_ν only once S_ν is known, and in general S_ν depends on the radiation itself, through the photons scattered into the beam. The first term is the radiation that entered the path, weakened by the whole optical depth. The second term is the sum of the contributions of every point along the path, each weakened by the optical depth $\tau_\nu - \tau'$ between that point and the end of the path.

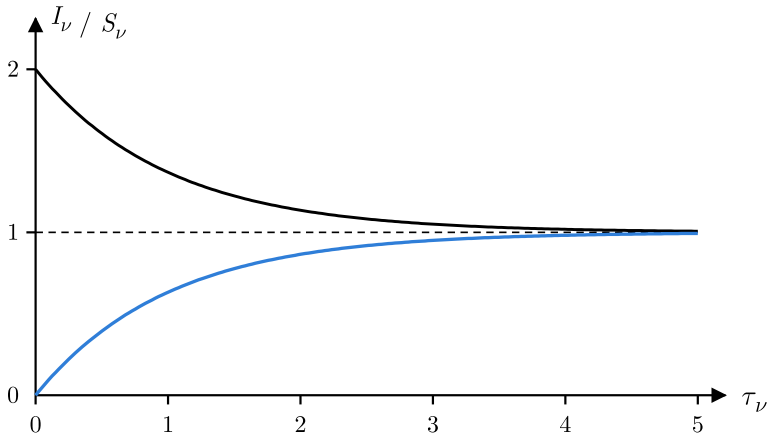


Figure 32. The specific intensity I_ν after passing through an optical depth τ_ν of matter with a uniform source function S_ν , for an incident specific intensity greater than S_ν (solid) and for none (dashed). Within a few mean free paths the specific intensity approaches S_ν from either side.

As an example, consider matter with a uniform source function. The integral is then $S_\nu(1 - e^{-\tau_\nu})$, and

$$I_\nu = I_\nu(0) e^{-\tau_\nu} + S_\nu (1 - e^{-\tau_\nu}), \quad (3.29)$$

which is drawn in Figure 32. If the matter is optically thick, $I_\nu \rightarrow S_\nu$, and the radiation that entered it is lost. If it is optically thin, expanding $e^{-\tau_\nu} \approx 1 - \tau_\nu$ gives

$$I_\nu \approx I_\nu(0) + [S_\nu - I_\nu(0)] \tau_\nu. \quad (3.30)$$

These results account for the three laws of Kirchhoff quoted in Section 3.1, since the optical depth of a gas is largest at the frequencies of its lines, where κ_ν is largest. A thin gas with nothing behind it emits $I_\nu \approx S_\nu \tau_\nu$, which is largest at the frequencies of its lines, and it shows an emission spectrum. A thin gas in front of a source with $I_\nu(0) > S_\nu$ changes the intensity by $[S_\nu - I_\nu(0)] \tau_\nu$, which is negative and largest in size at those frequencies, and it produces an absorption spectrum. A body that is optically thick at every frequency emits $I_\nu = S_\nu$, a continuous spectrum when S_ν is a smooth function of frequency.

Kirchhoff's Law

Let us consider a closed cavity whose walls are held at the temperature T . The radiation inside comes to equilibrium with the walls, and Section 5.6 of Vol. 2 showed that it is then blackbody radiation, whose spectrum depends on the temperature alone and not on the material of the walls. The specific intensity is therefore $B_\nu(T)$ at every point and in every direction. Suppose now that some matter at the same temperature is placed in the cavity. The radiation remains blackbody radiation, since the radiation in equilibrium at temperature T does not depend on the material it is in contact with. The specific intensity is then the same at every point of every ray, so $dI_\nu/d\tau_\nu = 0$ inside the matter, and the equation of transfer gives

$$S_\nu = \frac{j_\nu}{\kappa_\nu} = B_\nu(T), \quad (3.31)$$

which is called Kirchhoff's law. It says that matter that absorbs strongly at some frequency also emits strongly there, the ratio of the emission coefficient to the opacity being the Planck function.

Kirchhoff's law applies more widely than the argument suggests. The emission coefficient and the opacity are properties of the matter, fixed by the numbers of atoms and ions in each of their levels, and in local thermodynamic equilibrium these numbers are given

by the Boltzmann and Saha equations of Section 3.2 at the local temperature. The emission coefficient and the opacity then have the values they would have in a cavity at that temperature, and $S_\nu = B_\nu(T)$ holds even though the radiation is neither isotropic nor in equilibrium with the matter. The statement $I_\nu = B_\nu$ describes radiation in equilibrium, and the statement $S_\nu = B_\nu$ describes matter in local thermodynamic equilibrium; only the second holds in the atmosphere of a star, from which radiation escapes. Radiative transfer in a gas out of local thermodynamic equilibrium, as in the thinnest outer layers of a star, is not treated in this course.

With $S_\nu = B_\nu(T)$, the condition found above for an absorption spectrum can be stated in terms of temperatures. A thin layer of gas at the temperature T_g in front of an optically thick source at the temperature T_s gives dark lines when $B_\nu(T_g) < B_\nu(T_s)$, and since the Planck function increases with the temperature at every frequency, as you will show in Problem 5, this happens when the gas is cooler than the source, as the third of Kirchhoff's laws of spectral analysis states.

3.4 Stellar Atmospheres and the Formation of Lines

The atmosphere of a star is the layer of gas from which its light escapes into space. In this section we apply the equation of radiative transfer of Section 3.3 to that layer. We first set up the plane-parallel atmosphere and find the intensity that leaves it, then solve for the temperature of a grey atmosphere in the Eddington approximation, and use the result to explain the photosphere and the darkening of the limb of the Sun. Following that, we name the processes that give the gas its continuous opacity, and finally we describe how absorption lines form and how their strengths grow with the number of absorbing atoms.

The Plane-Parallel Atmosphere

This layer is thin compared with the radius of the star. By Section 1.2 of Vol. 2 the density of an isothermal layer goes as $e^{-m_{\text{av}}gy/k_B T}$, where y is the height and m_{av} is the mean mass of a particle of the gas, which can be rewritten as $e^{-y/H_{\text{sc}}}$ with

$$H_{\text{sc}} = \frac{k_B T}{m_{\text{av}} g}, \quad (3.32)$$

which is called the scale height. In the photosphere of the Sun, with a temperature of about 6×10^3 K and $g = GM_{\odot}/R_{\odot}^2 \approx 2.7 \times 10^2 \text{ m s}^{-2}$, the scale height is of order 10^2 km, a few parts in 10^4 of the radius of the Sun. Over a region of the atmosphere a few scale heights deep and of comparable width, the curvature of the surface can therefore be neglected, and the atmosphere can be treated as a stack of flat layers in which every quantity depends only on the height y . An atmosphere treated in this way is called a plane-parallel atmosphere.

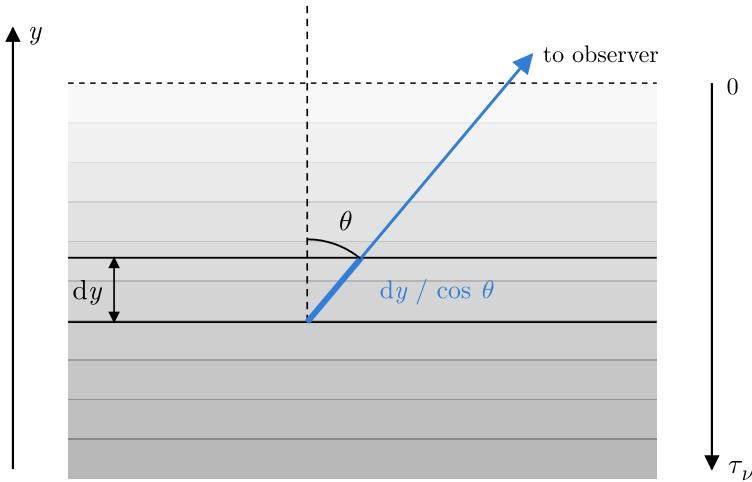


Figure 33. A plane-parallel atmosphere, with the height y measured upwards and the optical depth τ_{ν} measured downwards from the top of the atmosphere. The gas (shaded) grows denser with depth. A ray leaving at an angle θ to the vertical crosses a layer of thickness dy along a path of length $dy/\cos \theta$ (accent), and so through the optical depth $d\tau_{\nu}/\cos \theta$.

In a plane-parallel atmosphere it is convenient to measure the optical depth vertically and downwards from the top of the atmosphere, so that

$$d\tau_\nu = -\kappa_\nu \rho dy, \quad (3.33)$$

with $\tau_\nu = 0$ at the top, as in Figure 33. This is a different quantity from the optical depth along a ray used in Section 3.3. A ray travelling at an angle θ to the upward vertical covers the path $d\ell = dy/\cos\theta$ in rising through a height dy , so $d\ell = -d\tau_\nu/(\kappa_\nu \rho \cos\theta)$. Substituting this into the equation of transfer of Section 3.3, $dI_\nu/d\ell = -\kappa_\nu \rho(I_\nu - S_\nu)$, gives

$$\cos\theta \frac{dI_\nu}{d\tau_\nu} = I_\nu - S_\nu, \quad (3.34)$$

where I_ν is the specific intensity in the direction θ at the optical depth τ_ν . The equation holds for rays travelling downwards as well, which have $\cos\theta < 0$.

To find the specific intensity that leaves the top of the atmosphere in a direction with $\cos\theta > 0$, we multiply the equation by the integrating factor $e^{-\tau_\nu/\cos\theta}$, which turns it into $d(I_\nu e^{-\tau_\nu/\cos\theta})/d\tau_\nu = -(S_\nu/\cos\theta) e^{-\tau_\nu/\cos\theta}$. Integrating from $\tau_\nu = 0$ to a great depth, where $I_\nu e^{-\tau_\nu/\cos\theta}$ tends to zero because the exponential falls faster than the specific intensity grows, gives

$$I_\nu(0, \theta) = \int_0^\infty S_\nu(\tau_\nu) e^{-\tau_\nu/\cos\theta} \frac{d\tau_\nu}{\cos\theta}. \quad (3.35)$$

Each layer contributes its source function, weakened by the optical depth $\tau_\nu/\cos\theta$ along the slanting path from that layer to the top. Let us take, as an example, a source function that increases linearly with depth, $S_\nu = S_0 + S_1 \tau_\nu$. With $x = \tau_\nu/\cos\theta$ the integral becomes $\int_0^\infty (S_0 + S_1 x \cos\theta) e^{-x} dx$, and with the integrals $\int_0^\infty e^{-x} dx = 1$ and $\int_0^\infty x e^{-x} dx = 1$, which are the first two cases of the result $\int_0^\infty x^j e^{-x} dx = j!$ of Section 5.2 of Vol. 3,

$$I_\nu(0, \theta) = S_0 + S_1 \cos\theta = S_\nu(\tau_\nu = \cos\theta). \quad (3.36)$$

This result is called the Eddington–Barbier relation. The light that leaves in the direction θ carries the source function of the layer at the vertical optical depth $\cos\theta$, which is the layer from which the path to the top has unit optical depth. For a source function that is not linear the relation still holds approximately, provided the source function is close to linear over the first few units of optical depth.

The Grey Atmosphere

To find the source function we need the temperature, and to find the temperature we need to know how energy is carried through the atmosphere. The simplest model is an atmosphere whose opacity is the same at every frequency, which is called a grey atmosphere. The optical depth is then the same at every frequency, and integrating the equation of transfer over frequency gives $\cos\theta dI/d\tau = I - S$, where I and S are the specific intensity and the source function integrated over frequency. We assume that no energy is released in the atmosphere and that all of the energy is carried by radiation, a condition called radiative equilibrium, so the flux F is the same at every depth. It is equal to the flux leaving the surface, which by the definition of the effective temperature in Section 1.1 is $F = \sigma_{\text{SB}} T_{\text{eff}}^4$.

We take two averages of the equation of transfer over directions. Integrating both sides over all solid angles, the left-hand side gives $dF/d\tau$, since $F = \int I \cos\theta d\Omega$, and the right-hand side gives $\int I d\Omega - 4\pi S$, since matter at rest emits equally in all directions and S does not depend on θ . Let $\langle I \rangle = (1/4\pi) \int I d\Omega$ be the mean of the specific intensity over all directions, which is called the mean intensity. Then

$$\frac{dF}{d\tau} = 4\pi (\langle I \rangle - S) , \quad (3.37)$$

and since the flux is constant, $S = \langle I \rangle$ at every depth, which says that each layer emits as much energy as it absorbs. Multiplying the equation of transfer by $\cos\theta$ before integrating, the left-hand side gives $d(\int I \cos^2\theta d\Omega)/d\tau$, which is $c dP_{\text{rad}}/d\tau$ by the expression for the pressure of radiation in Section 3.3. The right-hand side gives

$F - S \int \cos \theta \, d\Omega = F$, since the integral of $\cos \theta$ over all directions vanishes. Hence

$$c \frac{dP_{\text{rad}}}{d\tau} = F, \quad \text{so} \quad P_{\text{rad}} = \frac{F}{c} (\tau + C), \quad (3.38)$$

where C is a constant to be determined.

To connect the pressure of radiation to the mean intensity we use the following approximation. For isotropic radiation, Section 3.3 found $P_{\text{rad}} = u/3 = 4\pi\langle I \rangle/3c$. The same relation holds exactly whenever the specific intensity is a linear function of $\cos \theta$ over all directions, $I = I_0 + I_1 \cos \theta$, since the term $I_1 \cos \theta$ contributes nothing to either $\int I \, d\Omega$ or $\int I \cos^2 \theta \, d\Omega$, being odd under the change $\theta \rightarrow \pi - \theta$. Deep in the atmosphere the radiation is nearly isotropic, with a small excess travelling outwards, which is close to this form. Eddington assumed in 1926 that $P_{\text{rad}} = 4\pi\langle I \rangle/3c$ at every depth, an assumption called the Eddington approximation, and with it the result above becomes

$$\langle I \rangle = \frac{3F}{4\pi} (\tau + C). \quad (3.39)$$

The constant C is fixed by the condition at the top of the atmosphere, where no radiation arrives from outside. If the specific intensity leaving the top is taken to be the same, I_{out} , in every outward direction, and zero in every inward direction, then the mean intensity there is $\langle I \rangle = I_{\text{out}}/2$, since the outward directions fill half of the full solid angle, and the flux is $F = \pi I_{\text{out}}$, as found in Section 3.3. The mean intensity at the top is therefore $F/2\pi$, and comparing this with the result above at $\tau = 0$ gives $3FC/4\pi = F/2\pi$, so that $C = 2/3$. This boundary condition is a further approximation, since the intensity it assumes is not linear in $\cos \theta$. The exact solution of the grey atmosphere, which is beyond this course, replaces the constant $2/3$ by a function of τ that differs from it by at most about a seventh.

In local thermodynamic equilibrium the source function is the Planck function, which for a grey atmosphere, integrated over frequency, is $S = B = \sigma_{\text{SB}} T^4/\pi$ by Section 3.3. Setting this

equal to $\langle I \rangle$, with $F = \sigma_{\text{SB}} T_{\text{eff}}^4$ and $C = 2/3$, gives $\sigma_{\text{SB}} T^4 / \pi = (3\sigma_{\text{SB}} T_{\text{eff}}^4 / 4\pi)(\tau + 2/3)$, which can be rewritten as

$$T^4 = \frac{3}{4} T_{\text{eff}}^4 \left(\tau + \frac{2}{3} \right). \quad (3.40)$$

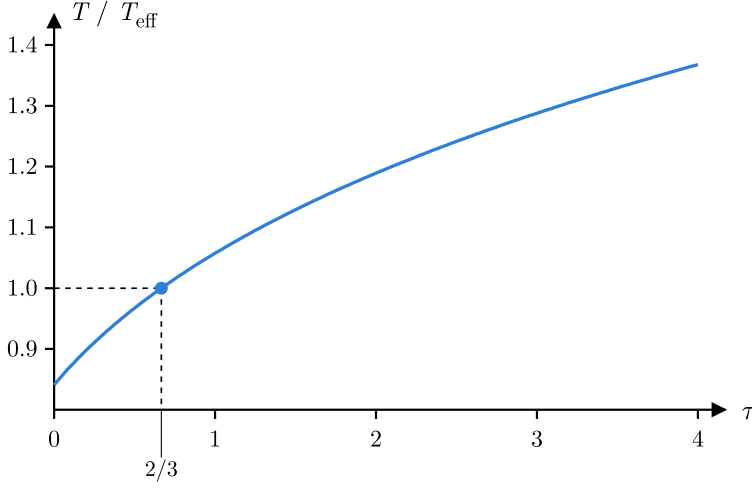


Figure 34. The temperature of a grey atmosphere in the Eddington approximation, $T^4 = \frac{3}{4} T_{\text{eff}}^4 (\tau + \frac{2}{3})$, against the optical depth. The temperature equals the effective temperature at $\tau = 2/3$ (marked), falls to $2^{-1/4} T_{\text{eff}}$ at the top, and goes as $\tau^{1/4}$ at great depth.

The temperature is drawn in Figure 34. It is equal to the effective temperature at $\tau = 2/3$, it falls to $2^{-1/4} T_{\text{eff}} \approx 0.84 T_{\text{eff}}$ at the top of the atmosphere, and it rises with depth, going as $\tau^{1/4}$ when $\tau \gg 1$.

The layer at $\tau = 2/3$ has a further property. The source function of the grey atmosphere, $S = (3\sigma_{\text{SB}} T_{\text{eff}}^4 / 4\pi)(\tau + 2/3)$, is linear in τ , so by the Eddington–Barbier relation the specific intensity leaving the top is $S_0 + S_1 \cos \theta$, and the flux leaving the top is

$$\begin{aligned} F &= 2\pi \int_0^{\pi/2} (S_0 + S_1 \cos \theta) \cos \theta \sin \theta \, d\theta \\ &= \pi \left(S_0 + \frac{2}{3} S_1 \right) \\ &= \pi S \left(\tau = \frac{2}{3} \right), \end{aligned} \quad (3.41)$$

where the integrals of $\cos \theta \sin \theta$ and $\cos^2 \theta \sin \theta$ are the ones found in Section 3.3. The flux that leaves the star is therefore the flux of a blackbody at the temperature of the layer at $\tau = 2/3$. For this reason the layer near $\tau = 2/3$ is called the photosphere, and the effective temperature of a star is the temperature of its photosphere.

Limb Darkening

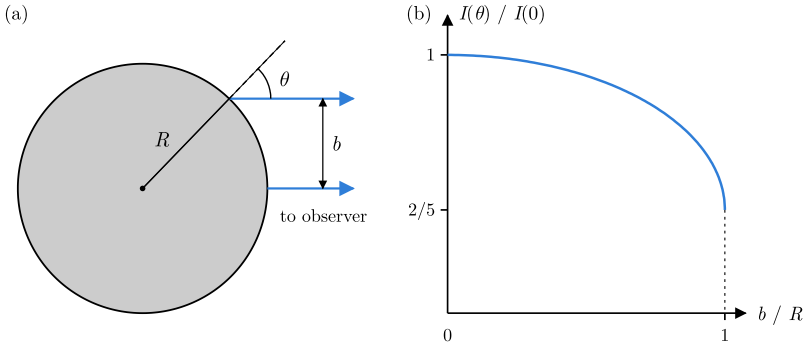


Figure 35. (a) Lines of sight to the centre of the disc of a star and to a point at the impact parameter b from the centre, where the line of sight makes the angle θ with the outward normal, so that $\sin \theta = b/R$. (b) The specific intensity across the disc of a grey atmosphere in the Eddington approximation, $I(\theta)/I(0) = (2 + 3 \cos \theta)/5$, against b/R .

Photographs of the Sun show that its disc is darker towards its edge, or limb, an effect called limb darkening. A line of sight that passes the centre of the disc at the impact parameter b meets the surface of a star of radius R at the angle θ to the outward normal, with $\sin \theta = b/R$, as Figure 35(a) shows. By the Eddington–Barbier relation the specific intensity along that line of sight is the source function at the vertical optical depth $\cos \theta$. At the centre of the disc $\cos \theta = 1$, while near the limb $\cos \theta$ is small, so the light from near the limb comes from layers higher in the atmosphere, where the temperature and the source function are lower. For the grey atmosphere in the Eddington approximation, $S_0 = \sigma_{\text{SB}} T_{\text{eff}}^4 / 2\pi$ and $S_1 = 3\sigma_{\text{SB}} T_{\text{eff}}^4 / 4\pi$, and the ratio of the specific intensity at the angle θ to that at the centre of the disc is

$$\frac{I(\theta)}{I(0)} = \frac{S_0 + S_1 \cos \theta}{S_0 + S_1} = \frac{2 + 3 \cos \theta}{5}, \quad (3.42)$$

which is drawn against b/R in Figure 35(b). At the limb the specific intensity is $2/5$ of its value at the centre. This agrees fairly well with the darkening of the Sun measured in visible light. The measured darkening is stronger at shorter wavelengths, because on the short-wavelength side of its peak the Planck function rises steeply with the temperature, so the lower temperature of the layers seen near the limb reduces the intensity more there, and the limb therefore also appears redder than the centre. Limb darkening in other stars can be measured from the shapes of the eclipses of the eclipsing binaries of Section 2.2.

The Sources of Continuous Opacity

The opacity of the gas in a real atmosphere depends on frequency, and it comes from several processes. The main ones are the following:

1. bound-bound absorption, in which an atom or ion absorbs a photon and passes from one bound level to another, which produces the absorption lines at the frequencies of the transitions;
2. bound-free absorption, also called photoionisation, in which a photon with energy greater than the ionisation energy of a level frees the electron, so that each level absorbs at every frequency above a threshold, as the level $k = 2$ of hydrogen does at wavelengths shorter than the Balmer limit of Section 3.1;
3. free-free absorption, in which a free electron passing close to an ion absorbs a photon, the ion being needed because an isolated free electron cannot absorb a photon and conserve both energy and momentum;
4. electron scattering, with the Thomson cross-section of Section 3.3, which is the same at every frequency.

The last three give the continuous opacity. The list is not complete, since molecules in the coolest stars, and the scattering of light by atoms and molecules, also contribute.

In the photosphere of the Sun, most of the continuous opacity in visible and near-infrared light comes from the negative hydrogen ion, H^- , a hydrogen atom that has bound a second electron, as Wildt pointed out in 1939. The second electron is bound with only about 0.75 eV, about a twentieth of χ_{H} , so it can be removed by the bound–free absorption of any photon with a wavelength shorter than about $1.6 \mu\text{m}$, and at longer wavelengths free–free absorption by a hydrogen atom and a passing electron takes over. The free electrons come mostly from metals such as sodium, calcium, and iron, whose ionisation energies are well below χ_{H} , so that by the Saha equation of Section 3.2 they are largely ionised in the photosphere of the Sun while hydrogen is not. Only a very small fraction of the hydrogen is in the form of H^- , but that fraction is enough to dominate the opacity. In hotter stars, broadly those of classes A and B, the absorption of hydrogen itself contributes most of the continuous opacity, and in the hottest stars electron scattering becomes a large contribution.

The Formation of Absorption Lines

At the frequency of a line the opacity is the sum of the continuous opacity κ_{c} and the opacity of the line, which is large at the centre of the line and falls away on either side. By the argument that gave the flux of the photosphere, the flux leaving the star at a given frequency is roughly π times the Planck function at the temperature of the layer at which the optical depth at that frequency is $2/3$. At a frequency inside the line the opacity is larger, so that optical depth is reached higher in the atmosphere, at a smaller continuum optical depth, where, as Figure 34 shows, the temperature is lower. Less light therefore leaves the star at the frequencies of the line than in the neighbouring continuum, and the line appears dark. The centre of the line, where the opacity is largest, forms highest in the atmosphere, and the wings form closer to the photosphere. The argument requires the temperature to fall outwards, and a layer in which the temperature rises outwards produces emission lines instead.

The shape of a line, called its profile, reflects the ways in which the frequencies at which atoms absorb are spread. The thermal motion of the atoms spreads them by the Doppler effect into a Gaussian

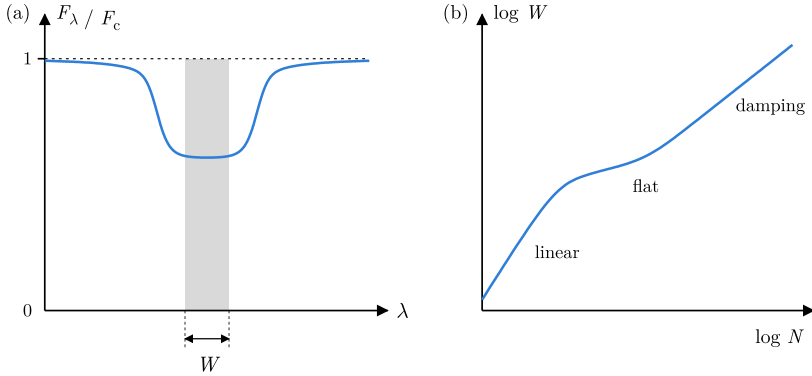


Figure 36. (a) The profile of an absorption line computed for the grey atmosphere of Figure 34 with the Eddington–Barbier relation, with the opacity of the line added to the continuous opacity. The shaded rectangle, which reaches from the continuum to zero, has the same area as the line, and its width is the equivalent width W . (b) The curve of growth, the logarithm of the equivalent width against the logarithm of the number of absorbing atoms per unit area, computed for a line formed in a layer of cool gas, with its linear, flat, and damping parts marked.

core, whose width you found in Problem 3 of Section 3.1. The pressure broadening of Section 3.1, and the natural broadening that comes from the short lifetime of the upper level, whose energy is therefore spread by the energy–time uncertainty relation of Section 3.1 of Vol. 3, give wings that fall off much more slowly than the Gaussian, as the inverse square of the distance from the centre of the line, a form whose derivation is beyond this course, and these are called damping wings. A line computed with these ingredients is drawn in Figure 36(a).

The strength of a line is measured by its equivalent width,

$$W = \int \frac{F_c - F_\lambda}{F_c} d\lambda, \quad (3.43)$$

where F_λ is the flux per unit wavelength and F_c is the flux of the continuum. The equivalent width is the width of a rectangular line that removes all the light over its width and the same total amount of light as the real line, as the shaded rectangle in Figure 36(a)

shows. A spectrograph of limited resolution blurs the profile of a line without changing the amount of light the line removes, so the equivalent width can be measured even when the profile cannot be resolved.

The way the equivalent width of a line grows with the number N of absorbing atoms per unit area along the line of sight is described by a curve called the curve of growth, which is drawn in Figure 36(b). It has three parts:

1. while the line is optically thin even at its centre, each atom absorbs independently, and W is proportional to N ; this part of the curve is called the linear part;
2. once the centre of the line becomes optically thick, the centre cannot become much darker, and further atoms deepen only the parts of the Gaussian core that are still thin, which lie ever further from the centre as N grows; because the Gaussian falls so steeply, W then grows very slowly, roughly as $\sqrt{\ln N}$; this part of the curve is called the flat part;
3. at large N the damping wings become optically thick, and W grows as $\sqrt{N}$; this part of the curve is called the damping part.

By comparing the equivalent widths of many lines of an element with the curve of growth, astronomers can find the number of atoms per unit area in the lower level of each line, and the Boltzmann and Saha equations of Section 3.2 then give the total number of atoms of the element, and hence its abundance. The position of the flat part gives the width of the Doppler core, and hence the random speeds of the atoms. Analyses of this kind, beginning with Payne's, showed that hydrogen is by far the most abundant element in the atmospheres of most stars, with helium next, and that all the heavier elements together make up only a small fraction of the mass.

Problems

§3.1 Stellar Spectra and Their Classification

1. Show that the ratio of the wavelengths of $H\alpha$ and $H\beta$ is $27/20$, and that the Balmer limit lies at $5/9$ of the wavelength of $H\alpha$.

2. A giant and a main-sequence star have the same effective temperature, and the absolute bolometric magnitude of the main-sequence star exceeds that of the giant by $\Delta\mathcal{M}$. Find the ratio of their radii and, taking their masses to be equal, the ratio of their surface gravities.
3. The atoms of mass m_{atom} in a gas at temperature T have components of velocity v_x along the line of sight distributed as in Section 1.2 of Vol. 2, and an atom moving with velocity v_x absorbs light at a wavelength shifted by $\lambda v_x/c$ from the wavelength λ of the line. Show that the light removed by a line that is weak enough for each atom to absorb independently has a Gaussian profile in wavelength, with standard deviation $(\lambda/c)\sqrt{k_B T/m_{\text{atom}}}$. Find the ratio of the widths of a line of hydrogen and a line of iron at the same wavelength and temperature, taking the mass of the iron atom to be $56 m_{\text{H}}$.
4. The absolute magnitude assigned to a star by spectroscopic parallax is uncertain by a small amount $\delta\mathcal{M}$. Show that the fractional uncertainty in its distance is $\delta d/d = (\ln 10/5) \delta\mathcal{M}$.
5. Show that a photon at the Balmer limit has the energy $\chi_{\text{H}}/4$. Find the longest wavelength of light that can ionise a hydrogen atom in the level $k = 3$, and show that it is $9/4$ of the wavelength of the Balmer limit.

§3.2 The Boltzmann and Saha Equations

1. Ignoring ionisation, find the temperature at which the number densities of hydrogen atoms in the levels $k = 1$ and $k = 2$ are equal.
2. In a gas of pure hydrogen every free electron comes from an ionised atom, so that $n_e = n_{\text{II}}$. Let x be the fraction of the hydrogen that is ionised, and $n_{\text{H}} = n_{\text{I}} + n_{\text{II}}$. Taking $\mathcal{Z}_{\text{I}} = 2$ and $\mathcal{Z}_{\text{II}} = 1$, show that $x^2/(1-x) = (1/n_{\text{H}})(2\pi m_e k_B T/h^2)^{3/2} e^{-\chi_{\text{H}}/k_B T}$.
3. Treating the partition functions as constants, show that at a fixed electron pressure $d(\ln(n_{\text{II}}/n_{\text{I}}))/d \ln T = 5/2 + \chi/k_B T$, and explain why the ionisation changes from nearly none to nearly complete over a narrow range of temperature when $\chi \gg k_B T$.

4. Show that when hydrogen is nearly all ionised and $n_2 \ll n_1$, the fraction of the hydrogen in the level $k = 2$ is approximately $(4P_e/k_B T)(h^2/2\pi m_e k_B T)^{3/2} e^{\chi_H/4k_B T}$, and hence that at a fixed electron pressure it falls as the temperature rises.
5. Suppose that the levels of hydrogen up to $k_{\max}$ are present, with $k_{\max} \gg 1$, and that $k_B T \ll \chi_H$. Show that the terms of $\mathcal{Z}_I$ with small $k \geq 2$ are each much less than the first term, and that the terms with large k add to about $\frac{2}{3} k_{\max}^3 e^{-\chi_H/k_B T}$. Taking $k_{\max}$ to be fixed by $k_{\max}^2 a_0 = n^{-1/3}$, where n is the total number density of particles, show that the terms with large k are small compared with the first term provided that $\frac{1}{3}(na_0^3)^{-1/2} e^{-\chi_H/k_B T} \ll 1$.

§3.3 Radiative Transfer

1. A sphere of radius R emits from its surface the same specific intensity I in every outward direction. Show that the flux at a distance $d \geq R$ from its centre is $\pi I(R/d)^2$, by integrating over the cone of directions in which the sphere is seen.
2. A beam of specific intensity I_ν fills a small solid angle $\Delta\Omega$ about the normal to a surface. Show that the flux, the energy density, and the pressure of the beam are approximately $I_\nu \Delta\Omega$, $I_\nu \Delta\Omega/c$, and $I_\nu \Delta\Omega/c$, so that the pressure of a parallel beam equals its energy density, three times the ratio for isotropic radiation.
3. The probability that a photon travels a distance ℓ through a uniform medium without being absorbed is $e^{-\ell/\ell_{\text{ph}}}$. Find the distance that half the photons travel without being absorbed, and the mean of the square of the distance travelled before absorption.
4. A uniform cloud has optical depth τ_ν through its thickness and source function S_ν , and there is no radiation behind it. Find the specific intensity that emerges. At the centre of a line the optical depth is τ_l , and in the neighbouring continuum it is τ_c , with the same source function. Find the ratio of the emerging intensities when both optical depths are much less than one, and when both are much greater than one.
5. A layer of gas of optical depth τ_ν at the uniform temperature T_g , in local thermodynamic equilibrium, lies in front of a source

whose specific intensity is $B_\nu(T_s)$. Show that the emerging specific intensity is $B_\nu(T_s)e^{-\tau_\nu} + B_\nu(T_g)(1 - e^{-\tau_\nu})$, and that it is less than $B_\nu(T_s)$ if and only if $T_g < T_s$. Show also that B_ν increases with T at every frequency.

§3.4 Stellar Atmospheres and the Formation of Lines

1. In the grey atmosphere of the text, find the optical depth at which $T^4 = 2T_{\text{eff}}^4$, and the ratio of the temperature at that depth to the temperature at the top of the atmosphere.
2. For a star whose disc has the limb darkening $I(\theta)/I(0) = (2 + 3\cos\theta)/5$, show that the mean specific intensity over the disc, averaged over its area, is $4/5$ of the specific intensity at the centre.
3. Show that the ratio of the scale height to the radius of a star is $k_B TR/GMm_{\text{av}}$. How does this ratio change between a main-sequence star and a giant of the same mass and photospheric temperature, and what does the change imply for the plane-parallel approximation?
4. Show that the equivalent width of an optically thin line, formed by N absorbing atoms per unit area in a layer of gas in front of a continuous source, is $W = N \int \sigma_\lambda d\lambda$, where σ_λ is the cross-section of one atom at the wavelength λ , and that it does not depend on the shape of the profile.
5. Let the source function be $S_\nu = S_0 + S_1\tau_\nu + S_2\tau_\nu^2$. Show that the specific intensity leaving the top of the atmosphere is $S_0 + S_1 \cos\theta + 2S_2 \cos^2\theta$, and find by how much this differs from the value given by the Eddington–Barbier relation.

Stellar Structure, Energy Generation, and Evolution

4.1 Stars and the Hertzsprung–Russell Diagram at School

The Hertzsprung–Russell diagram and the life cycle of a star are both part of school astronomy, and the review of them here is brief. The section then takes the diagram one step further, using the relation between luminosity, radius, and effective temperature from Section 1.1 to read the radius of a star off its position on the diagram.

The Hertzsprung–Russell Diagram

In the years around 1910, Ejnar Hertzsprung and Henry Norris Russell each plotted the brightness of stars at known distances against a measure of their surface temperature, Hertzsprung using the colours of the stars and Russell their spectral classes. A plot of this kind is called a Hertzsprung–Russell diagram. In the form usually met at school, the vertical axis is the luminosity L and the horizontal axis is the effective temperature T_{eff} , both on logarithmic scales, and the temperature increases from right to left. The reversed axis is a relic of the spectral sequence O, B, A, F, G, K, M, which was written from left to right before it was known to be a sequence of

falling temperature.

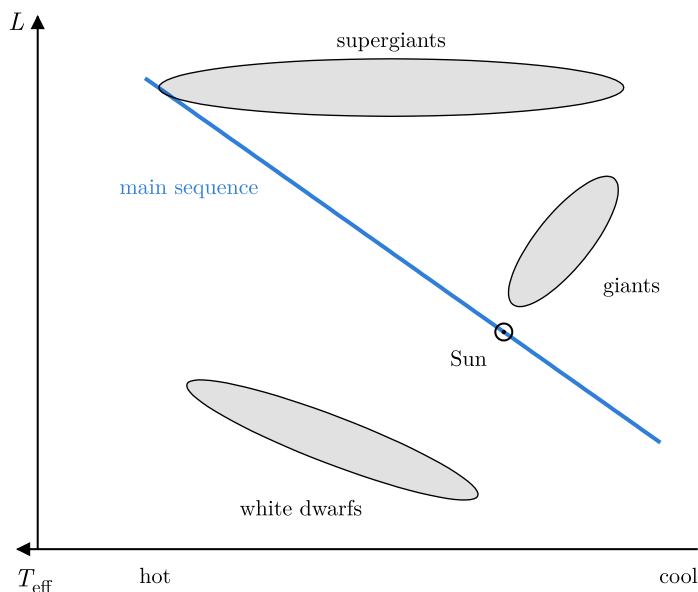


Figure 37. A schematic Hertzsprung–Russell diagram, with luminosity plotted upwards and effective temperature increasing to the left, both on logarithmic scales. The main sequence (accent) is drawn from the mass–luminosity and mass–radius relations given later in this section, with $L = 4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4$. The shaded regions mark where the giants, the supergiants, and the white dwarfs lie, and the circle marks the Sun.

Stars are not scattered at random over the diagram, as Figure 37 shows. The great majority of the stars near the Sun lie in a band running from the upper left, where the stars are hot and luminous, to the lower right, where they are cool and faint, and this band is called the main sequence. The Sun lies on it, near the middle. Above the main sequence on the right are the giants, which are cool and yet much more luminous than main-sequence stars of the same temperature, and across the top of the diagram are the supergiants, which are more luminous still. In the lower left are the white dwarfs, which are hot but faint.

The diagram is drawn in more than one way. Section 3.1 drew it with spectral class along the horizontal axis and absolute magnitude

up the vertical axis. A form that observers often use plots the absolute visual magnitude $\mathcal{M}_V$ against the $B - V$ colour index $m_B - m_V$, and it is called a colour–magnitude diagram, since both of its quantities come directly from photometry. Converting a colour–magnitude diagram into a diagram of L against T_{eff} requires the bolometric corrections and the relation between colour and effective temperature, both of which were described in Section 1.1. The diagram of luminosity against effective temperature is the one used in the rest of this part, because the equations of stellar structure predict these two quantities.

The main sequence is a sequence of mass. When the masses measured from binary stars in Section 2.2 are compared with their luminosities, the luminosity of a main-sequence star is found by observation to rise steeply with its mass, roughly as $L \propto M^{3.5}$ over much of the range, so that the most massive main-sequence stars sit at the upper left of the diagram and the least massive at the lower right. As you learned at school, a star on the main sequence is converting hydrogen into helium by nuclear fusion in its core.

The Life Cycle of a Star

The account of the life of a star given at school runs as follows. A star forms when a cloud of gas and dust, called a nebula, collapses under the gravitational attraction of its parts. As the cloud contracts, its gravitational potential energy is converted into thermal energy, and the gas at its centre grows hot. The contracting object, called a protostar, glows, although no fusion takes place in it yet. When the centre becomes hot enough for hydrogen to fuse into helium, the energy released keeps the pressure high enough to balance gravity, the contraction stops, and the star settles onto the main sequence, where it spends most of its life. A more massive star has more fuel, but it is so much more luminous that it uses up its fuel more quickly, and its life is therefore shorter.

Once the hydrogen in the core is used up, the core contracts and the outer layers of the star expand and cool, so that the star moves to the right and upwards on the diagram. A star of low mass, such as the Sun, becomes a red giant, and a star of high mass becomes a red supergiant. The red giant and the red supergiant then take the

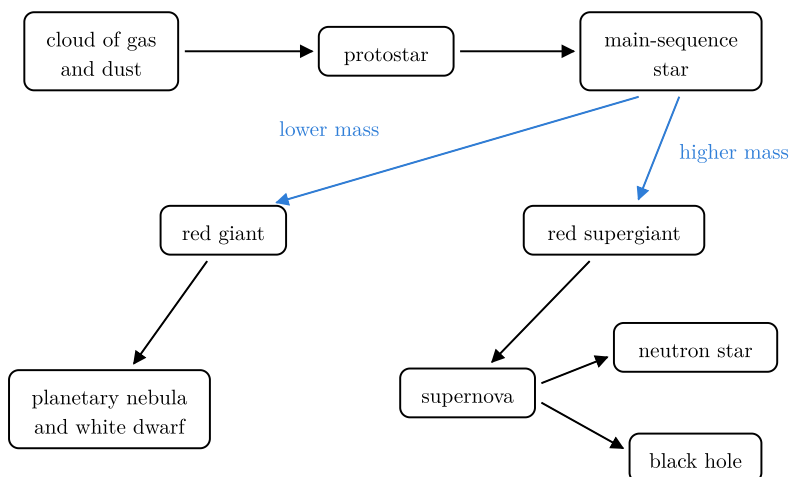


Figure 38. The life cycle of a star as it is taught at school. The path on the left is followed by stars of lower mass, such as the Sun, and the path on the right by stars of higher mass.

different paths shown in Figure 38. A star of low mass eventually loses its outer layers, which drift away as a glowing shell of gas called a planetary nebula, a name that comes from the appearance of these shells in small telescopes and has nothing to do with planets. The hot, dense core that remains is a white dwarf, which slowly cools and fades. A star of high mass fuses successively heavier elements in its core until the core is mostly iron; the core then collapses, and the star explodes as a supernova. What remains of the core is a neutron star or, if the remnant is massive enough, a black hole. The dividing line between the path of a low-mass star and that of a high-mass star is usually quoted at school as an initial mass of roughly eight times the mass of the Sun, although the boundary depends on how much mass the star loses during its life.

Each stage of this story is a place on the Hertzsprung–Russell diagram, and a star moves across the diagram as it ages. A diagram of a group of stars is therefore a picture of the stages that its members have reached at the time it is observed.

Lines of Constant Radius

In Section 1.1 the effective temperature of a star was defined by the relation

$$L = 4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4, \quad (4.1)$$

so that T_{eff} is the temperature of a blackbody of the same radius and luminosity as the star. Dividing this relation by the same relation for the Sun removes the constants, and gives

$$\frac{L}{L_{\odot}} = \left(\frac{R}{R_{\odot}} \right)^2 \left(\frac{T_{\text{eff}}}{T_{\text{eff},\odot}} \right)^4, \quad (4.2)$$

where $T_{\text{eff},\odot}$ is the effective temperature of the Sun, about 5.8×10^3 K. Since the axes of the diagram are logarithmic, we take the logarithm to base ten of both sides,

$$\log_{10} \frac{L}{L_{\odot}} = 4 \log_{10} \frac{T_{\text{eff}}}{T_{\text{eff},\odot}} + 2 \log_{10} \frac{R}{R_{\odot}}. \quad (4.3)$$

For stars of a given radius the last term is a constant, and $\log_{10} L$ is a linear function of $\log_{10} T_{\text{eff}}$ with slope four. On the Hertzsprung–Russell diagram, stars of the same radius therefore lie on a straight line, which rises by four decades in luminosity for every decade by which the temperature rises. Since the temperature increases to the left, each of these lines runs from the lower right of the diagram to the upper left. Changing the radius by a factor of ten moves the line up or down by two decades in luminosity, because the luminosity goes as R^2 at a fixed temperature.

Figure 39 draws these lines on the diagram of Figure 37, and the radius of any star can be read from its position among them. The giants, which are cool and luminous, lie between the lines for a few $R_{\odot}$ and $100 R_{\odot}$, and the most luminous supergiants reach the line for $10^3 R_{\odot}$, which is larger than the orbit of the Earth, since the Earth lies about $215 R_{\odot}$ from the Sun. The white dwarfs, which are hot and faint, lie near the line for $10^{-2} R_{\odot}$, which is close to the

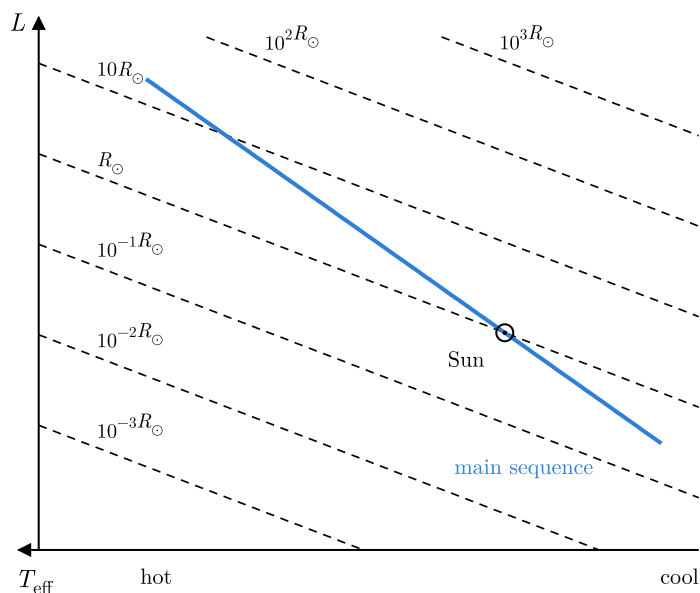


Figure 39. The Hertzsprung–Russell diagram of Figure 37 with lines of constant radius (dashed), found from $L = 4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4$. Each line is labelled by its radius in units of $R_{\odot}$. The main sequence (accent) is steeper than the lines of constant radius, so the radius of a main-sequence star increases up the sequence.

radius of the Earth. The names giant and dwarf reflect these sizes. Hertzsprung noticed early in the twentieth century that stars of the same spectral class could differ enormously in luminosity, and since stars of the same class have about the same surface temperature, the more luminous of them must be the larger. Radii measured more directly, from eclipsing binaries in Section 2.2 and, for a few nearby giants, from the angular sizes of their discs, agree with those read off the diagram.

As an example, consider two stars of the same effective temperature, a main-sequence star of luminosity L_1 and radius R_1 , and a giant of luminosity L_2 and radius R_2 . With T_{eff} the same for the main-sequence star and the giant, the relation above gives $L_2/L_1 = (R_2/R_1)^2$, and therefore

$$\frac{R_2}{R_1} = \left(\frac{L_2}{L_1} \right)^{1/2}. \quad (4.4)$$

A giant a hundred times more luminous than a main-sequence star of the same colour is ten times larger. The giant is also far less dense, since its mass is not very different from that of the main-sequence star, while its volume is a thousand times larger.

The main sequence itself can be drawn from the relations between mass, luminosity, and radius that are found by observation. Over the range of masses near that of the Sun, the luminosities and the radii of main-sequence stars measured from binaries are roughly described by

$$L \propto M^a, \quad R \propto M^b, \quad (4.5)$$

with $a \approx 3.5$ and $b \approx 0.8$. Substituting these relations into $T_{\text{eff}}^4 \propto L/R^2$ gives $T_{\text{eff}}^4 \propto M^{a-2b}$, so that $M \propto T_{\text{eff}}^{4/(a-2b)}$, and substituting this back into the mass–luminosity relation gives

$$L \propto T_{\text{eff}}^{4a/(a-2b)}. \quad (4.6)$$

With the values of a and b above, the exponent is about 7.4. On the

logarithmic diagram the main sequence is therefore a line of slope about 7.4, steeper than the lines of constant radius, whose slope is four. The main sequence crosses the lines of constant radius from smaller to larger radii as it rises, as Figure 39 shows, which is another way of saying that b is positive, and the main sequences drawn in Figures 37 and 39 were computed in this way. The relations $L \propto M^a$ and $R \propto M^b$ are only approximate descriptions of the measurements, with exponents that change along the sequence, so the straight main sequence of these figures is a schematic one.

4.2 Hydrostatic Equilibrium and the Virial Theorem

A star is a ball of gas held together by gravity and supported by the pressure of the gas and of the radiation inside it. In this section we first derive the equations that express this balance for a spherical star, then use them to estimate the pressure and the temperature at the centre of the Sun, and finally derive the virial theorem for a star, which relates its thermal energy to its gravitational energy and governs what happens to it when it contracts. Throughout, the star is taken to be spherically symmetric, so that its density, pressure, and temperature depend only on the distance r from its centre. This is a valid approximation for most stars, because the kinetic energy of their rotation and the energy of their magnetic fields are small compared with their gravitational energy.

The Equation of Mass Continuity

Let $\rho(r)$ be the density at a distance r from the centre, and let M_r be the mass inside the sphere of radius r . The shell between r and $r + dr$, shown in Figure 40, has an area $4\pi r^2$ and a thickness dr , so provided dr is small compared with r its volume is $4\pi r^2 dr$, and it adds a mass $4\pi r^2 \rho dr$ to M_r . Therefore

$$\frac{dM_r}{dr} = 4\pi r^2 \rho, \quad (4.7)$$

which is called the equation of mass continuity. The mass inside radius r is zero at the centre and equal to the mass M of the star at its surface, $r = R$.

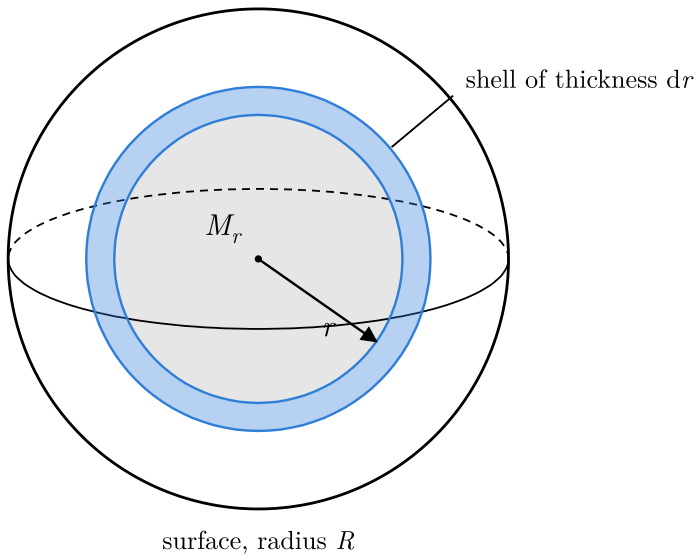


Figure 40. A star of radius R cut into spherical shells. The shaded sphere of radius r contains the mass M_r , and the shell of thickness dr just outside it (accent) has volume $4\pi r^2 dr$. The ellipse on the surface of the star is drawn to show the sphere in perspective.

Hydrostatic Equilibrium

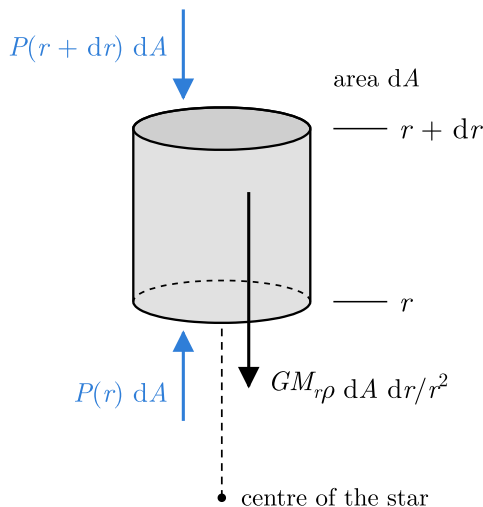


Figure 41. A small cylinder of gas inside a star, with its axis along the radius, its faces of area dA at r and $r + dr$, and the centre of the star below it. The gas beneath pushes it outwards with the force $P(r) dA$, the gas above pushes it inwards with the force $P(r + dr) dA$, and gravity pulls it towards the centre with the force $GM_r \rho dA dr / r^2$.

Let us consider a small cylinder of gas inside the star, with its axis along the radius, its lower face at r , its upper face at $r + dr$, and a cross-sectional area dA , as in Figure 41. Its mass is $\rho dA dr$. By the shell theorem of Section 5.2 of Vol. 1 of the Fermi Notes on Physics, the shells of the star outside radius r exert no net gravitational force on the cylinder, and the mass M_r inside that radius attracts it as though all of it were concentrated at the centre. The gravitational force on the cylinder is therefore $GM_r \rho dA dr / r^2$, directed towards the centre. The pressure of the gas below pushes the lower face outwards with a force $P(r) dA$, and the pressure of the gas above pushes the upper face inwards with a force $P(r + dr) dA$, while the forces on the curved side are perpendicular to the radius and cancel in opposite pairs. If the star is neither expanding nor contracting, the radial forces on the cylinder add to zero,

$$P(r) dA - P(r + dr) dA - \frac{GM_r \rho}{r^2} dA dr = 0. \quad (4.8)$$

Dividing by $dA dr$ and letting dr tend to zero gives

$$\frac{dP}{dr} = -\frac{GM_r \rho}{r^2}, \quad (4.9)$$

which is called the equation of hydrostatic equilibrium. It is the equation $dP/dy = -\rho g$ of the isothermal atmosphere in Section 1.2 of Vol. 2 of the Fermi Notes on Physics, with y the height, as in Section 3.1, with the uniform acceleration g replaced by the local acceleration due to gravity GM_r/r^2 , which changes with depth. The pressure therefore decreases outwards everywhere in the star, and at each radius its gradient is just steep enough to hold up the weight of the layers above.

The Central Pressure and Temperature of the Sun

The equation of hydrostatic equilibrium already gives a useful estimate of the pressure at the centre of a star. Dividing it by the equation of mass continuity, and using the chain rule, gives the rate at which the pressure changes with the mass enclosed,

$$\frac{dP}{dM_r} = -\frac{GM_r}{4\pi r^4}. \quad (4.10)$$

The pressure at the surface is negligible compared with that at the centre, so integrating from the centre, where $M_r = 0$, to the surface, where $M_r = M$, gives the central pressure P_c as the integral of $GM_r/4\pi r^4$ over M_r . We do not know how r depends on M_r until the structure of the star is known, but every layer lies inside the star, so $r \leq R$ and $1/r^4 \geq 1/R^4$ throughout. Therefore

$$P_c = \int_0^M \frac{GM_r}{4\pi r^4} dM_r > \int_0^M \frac{GM_r}{4\pi R^4} dM_r = \frac{GM^2}{8\pi R^4}. \quad (4.11)$$

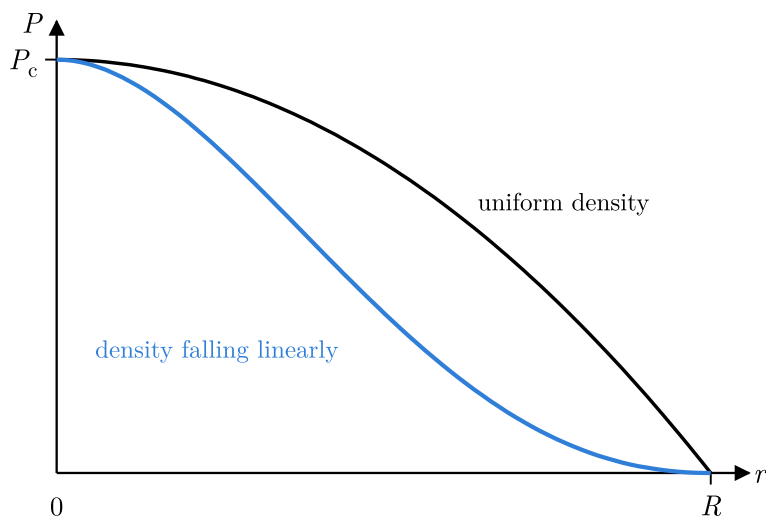


Figure 42. The pressure P inside two model stars of the same mass and radius, each divided by the central pressure P_c of that star and plotted against the distance r from the centre. The upper curve is for a star of uniform density, and the lower curve for a star whose density falls linearly from the centre to zero at the surface, which is treated in Problem 2. The pressure in the second star falls more quickly with r because more of its mass is concentrated near the centre.

For the Sun this lower limit is about 4×10^{13} Pa, several hundred million times the pressure of the Earth's atmosphere at sea level. A star of uniform density has a central pressure $3GM^2/8\pi R^4$, three times the lower limit, as you will show in Problem 1. A star whose density rises towards its centre has a larger central pressure still, since more of its mass sits at small r , where $1/r^4$ is large; in Problem 2 you will find that a star whose density falls linearly from the centre to the surface has a central pressure $10/3$ times that of a uniform star of the same mass and radius, and Figure 42 compares the pressures inside these two stars. Detailed models of the Sun give a central pressure of about 2×10^{16} Pa, because the central density of the Sun, about $1.5 \times 10^5 \text{ kg m}^{-3}$, is about a hundred times its mean density.

To turn the pressure into a temperature we need the relation between pressure, density, and temperature in the gas, which is taken up in the next subsection. For an ideal gas the pressure is $P = nk_B T$, where n is the number of particles per unit volume, as in Section 1.2 of Vol. 2. Let the mean mass of these particles be μm_H , where m_H is the mass of the hydrogen atom and μ is a number called the mean molecular weight, so that $n = \rho/\mu m_H$ and $T = \mu m_H P/\rho k_B$. For a star of uniform density, $\rho = 3M/4\pi R^3$, and with the central pressure $3GM^2/8\pi R^4$ the central temperature is

$$T_c = \frac{\mu m_H}{k_B} \frac{3GM^2}{8\pi R^4} \frac{4\pi R^3}{3M} = \frac{\mu m_H GM}{2k_B R}. \quad (4.12)$$

With μ about 0.6, the value found below for the gas inside the Sun, this gives about 7×10^6 K, which is of the same order as the central temperature of about 1.6×10^7 K found from detailed models. At such temperatures the typical kinetic energy of a particle, $k_B T$, is about a kiloelectronvolt.

The Equation of State

A relation between the pressure, the density, and the temperature of a substance is called its equation of state. The gas inside a star is fully ionised, because $k_B T$ in the interior is about a hundred times the ionisation energy of hydrogen, 13.6 eV, and larger than the energies needed to remove all the electrons from light elements

such as carbon and oxygen; only the heaviest elements keep some of their innermost electrons. The gas is therefore a mixture of bare nuclei and free electrons. It behaves as an ideal gas, even at the high densities of the interior, for two reasons. The nuclei and the electrons are far smaller than the distances between them, so they occupy a negligible fraction of the volume, and the Coulomb energy of a pair of neighbouring particles, of order $e^2/4\pi\epsilon_0 d$ for a spacing $d = n^{-1/3}$, is only about a twentieth of $k_B T$ at the centre of the Sun, so the forces between the particles have little effect on their motion. The pressure of the gas is therefore

$$P_{\text{gas}} = nk_B T = \frac{\rho k_B T}{\mu m_H}. \quad (4.13)$$

The mean molecular weight depends on the composition of the gas. Let X , Y , and Z be the fractions by mass of hydrogen, of helium, and of all heavier elements, so that $X + Y + Z = 1$, and let us count the particles in a unit mass of gas, taking the mass of a nucleus of mass number A_n to be $A_n m_H$. The hydrogen contributes X/m_H nuclei, each with one electron, and so $2X/m_H$ particles. The helium contributes $Y/4m_H$ nuclei, each with two electrons, and so $3Y/4m_H$ particles. A heavier nucleus of mass number A_n has about $A_n/2$ protons, so the heavier elements contribute about $Z/A_n m_H$ nuclei, each with about $A_n/2$ electrons, and for A_n much larger than one the number of particles is close to $Z/2m_H$. The number of particles in unit mass is $1/\mu m_H$, so

$$\frac{1}{\mu} = 2X + \frac{3}{4}Y + \frac{1}{2}Z. \quad (4.14)$$

Fully ionised hydrogen has $\mu = \frac{1}{2}$, since the proton and the electron share the mass of a hydrogen atom between them, and fully ionised helium has $\mu = \frac{4}{3}$. The Sun began its life with $X \approx 0.71$, $Y \approx 0.27$, and $Z \approx 0.02$, which gives $\mu \approx 0.61$, and as hydrogen is converted into helium in the core, μ there increases.

The radiation inside a star also exerts a pressure. Blackbody radiation is treated in detail in Section 5.6 of Vol. 2, where its pressure was shown to be one third of its energy density, $P_{\text{rad}} = u/3$, and

its energy density was found to be $u = 4\sigma_{\text{SB}}T^4/c$; Vol. 2 writes the Stefan–Boltzmann constant as σ , and these notes write σ_{SB} , because σ is kept for a cross-section. The radiation deep inside a star is very close to blackbody radiation at the local temperature, because a photon travels only a short distance before it is absorbed, as Section 3.3 showed, and in that distance the temperature hardly changes. The pressure of the radiation is therefore

$$P_{\text{rad}} = \frac{1}{3}a_{\text{rad}}T^4, \quad a_{\text{rad}} = \frac{4\sigma_{\text{SB}}}{c}, \quad (4.15)$$

where a_{rad} is called the radiation constant. The total pressure is $P = P_{\text{gas}} + P_{\text{rad}}$. At the centre of the Sun the radiation contributes less than a tenth of a per cent of the pressure. The ratio of the two pressures goes as T^3/ρ , and you will show in Problem 3 that in stars of similar structure it grows as M^2 , so the pressure of radiation becomes a large part of the total pressure in the most massive stars.

The Virial Theorem for a Star

In Section 5.3 of Vol. 1 the virial theorem was derived for a body in a bound orbit under gravity: the time average of the kinetic energy is minus one half of the time average of the potential energy. A star in hydrostatic equilibrium obeys a theorem of the same form, and we now derive it from the equation of hydrostatic equilibrium.

Let us first find the gravitational potential energy E_{grav} of a star, which is the work that would have to be done against gravity to assemble it from matter dispersed to infinity, a negative quantity, since gravity itself pulls the matter inwards. We build the star, in imagination, shell by shell from the centre outwards. When the mass inside radius r is M_r , a new shell of mass dM_r brought in from infinity to radius r is attracted only by the mass already inside, which by the shell theorem acts as a point mass M_r at the centre, so its potential energy is $-GM_r dM_r/r$. Adding the shells,

$$E_{\text{grav}} = - \int_0^M \frac{GM_r}{r} dM_r. \quad (4.16)$$

As an example, consider a star of uniform density. Then $M_r = Mr^3/R^3$ and $dM_r = 3Mr^2 dr/R^3$, so

$$E_{\text{grav}} = -\frac{3GM^2}{R^6} \int_0^R r^4 dr = -\frac{3GM^2}{5R}. \quad (4.17)$$

A star whose mass is more concentrated towards its centre has a more negative gravitational energy, but for any star E_{grav} is $-GM^2/R$ multiplied by a number of order one.

To relate E_{grav} to the pressure, we multiply the equation of hydrostatic equilibrium by $4\pi r^3$ and integrate over the star. The right-hand side becomes $-\int_0^R (GM_r/r) 4\pi r^2 \rho dr$, which is E_{grav} , since $4\pi r^2 \rho dr = dM_r$. On the left-hand side we integrate by parts, with $4\pi r^3$ as the factor that is differentiated,

$$\int_0^R 4\pi r^3 \frac{dP}{dr} dr = \left[4\pi r^3 P \right]_0^R - \int_0^R 12\pi r^2 P dr. \quad (4.18)$$

The boundary term vanishes at both limits, because r^3 is zero at the centre and the pressure is zero at the surface. Equating the two sides and writing $4\pi r^2 dr$ as the volume of a shell gives

$$3 \int_0^R P 4\pi r^2 dr = -E_{\text{grav}}. \quad (4.19)$$

The integral on the left is the pressure integrated over the volume of the star. For an ideal gas it is related to the thermal energy of the gas by a result of Section 2.2 of Vol. 2. There the difference $C_p - C_V$ between the heat capacities of an ideal gas was found to be the number of moles multiplied by the gas constant, which is Nk_B for a gas of N particles, and with $\gamma = C_p/C_V$ this gives $C_V = Nk_B/(\gamma - 1)$. The internal energy of the gas is therefore $C_V T = Nk_B T/(\gamma - 1) = P\mathcal{V}/(\gamma - 1)$, where $\mathcal{V}$ is the volume of the gas, and the thermal energy per unit volume is $P/(\gamma - 1)$. If the pressure of radiation is neglected, the thermal energy E_{th} of the whole star is therefore $1/(\gamma - 1)$ times the integral on the left, and

$$3(\gamma - 1)E_{\text{th}} + E_{\text{grav}} = 0. \quad (4.20)$$

For the fully ionised gas of a stellar interior, whose particles have only their translational energy, $\gamma = \frac{5}{3}$, and

$$2E_{\text{th}} + E_{\text{grav}} = 0. \quad (4.21)$$

This relation is called the virial theorem for a star. It has the same form as the result of Section 5.3 of Vol. 1, with E_{th} , which here is the kinetic energy of the random motion of the particles of the gas, in place of the time-averaged kinetic energy of the orbiting body. The total energy of the star is

$$E_{\text{tot}} = E_{\text{th}} + E_{\text{grav}} = \frac{1}{2}E_{\text{grav}} = -E_{\text{th}}, \quad (4.22)$$

which is negative, as it must be for a bound system.

The virial theorem gives the mean temperature of a star without any knowledge of its detailed structure. The thermal energy of a gas of $M/\mu m_{\text{H}}$ particles at a mean temperature $\bar{T}$ is $\frac{3}{2}(M/\mu m_{\text{H}})k_B\bar{T}$, and for a star of uniform density the virial theorem sets this equal to $3GM^2/10R$, so that

$$\bar{T} = \frac{\mu m_{\text{H}} GM}{5k_B R}, \quad (4.23)$$

which for the Sun is about 3×10^6 K. A star whose mass is more concentrated towards its centre has a larger $|E_{\text{grav}}|$ and so a higher mean temperature.

For a gas with a general ratio of heat capacities, the relation $3(\gamma - 1)E_{\text{th}} = -E_{\text{grav}}$ gives a total energy $E_{\text{tot}} = (3\gamma - 4)E_{\text{grav}}/3(\gamma - 1)$, which is negative only if $\gamma > \frac{4}{3}$. As γ approaches $\frac{4}{3}$, the total energy approaches zero, and the star is only loosely bound. Radiation, whose energy density u is three times its pressure, behaves in this respect as a gas with $\gamma = \frac{4}{3}$, since that is the value for which

$P/(\gamma - 1) = 3P$, so a star in which the pressure of radiation is a large fraction of the total pressure is only loosely bound.

Contraction and the Kelvin–Helmholtz Time Scale

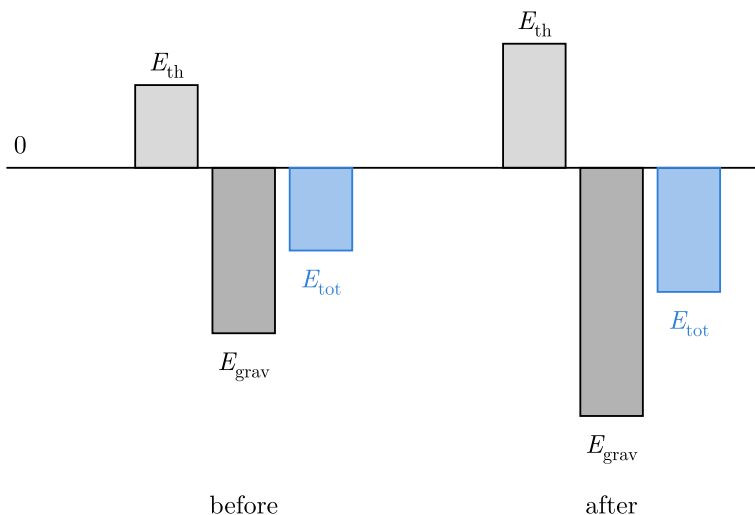


Figure 43. The thermal energy E_{th} , the gravitational energy E_{grav} , and the total energy E_{tot} (accent) of a star obeying the virial theorem, before and after it contracts. The gravitational energy becomes more negative; half of the energy released goes into the thermal energy, which rises, and half is radiated away, so that the total energy falls by the same amount as the thermal energy rises.

Suppose that a star radiates a luminosity L and has no source of energy other than its gravity, as a protostar does in the life cycle of Section 4.1. Its total energy then falls at the rate L . Since $E_{\text{tot}} = \frac{1}{2}E_{\text{grav}}$, the gravitational energy falls at twice this rate, and the star contracts; since $E_{\text{th}} = -E_{\text{tot}}$, the thermal energy rises at the rate L , and the star grows hotter. Of the gravitational energy released by the contraction, half is radiated away and half goes into heating the gas, as Figure 43 shows. A star therefore has a negative heat capacity; when it loses energy, it grows hotter. The reason is that the gravitational field inside the star grows stronger as it contracts, and its particles must then move faster to hold it

up. This is how a protostar heats its centre to the temperature at which fusion begins. The same argument, run in reverse, shows that a star whose nuclear reactions release energy faster than it is radiated expands and cools, which slows the reactions, so nuclear burning in a star in hydrostatic equilibrium tends to be stable.

The time for which gravitational contraction alone could supply the present luminosity of a star is called the Kelvin–Helmholtz time scale,

$$t_{\text{KH}} = \frac{|E_{\text{grav}}|}{2L} \approx \frac{3GM^2}{10RL}, \quad (4.24)$$

where the second form uses the gravitational energy of a uniform star. For the Sun it comes to about 10^7 years. In the nineteenth century Kelvin and Helmholtz proposed that the Sun shines by contracting, and the age of the Sun they obtained in this way was tens of millions of years at most, far shorter than the times that geologists and Darwin needed for the formation of the rocks and the evolution of life. Radioactive dating of the oldest rocks now gives the age of the solar system as about 4.6×10^9 years, several hundred times t_{KH} , so the Sun must have a source of energy other than contraction, which, as you know from school, is nuclear fusion.

A second time scale describes how quickly a star responds when its hydrostatic equilibrium is disturbed. Suppose that the pressure inside a star were suddenly removed. The surface would then fall inwards with an acceleration of about GM/R^2 , and it would reach the centre in a time of order $\sqrt{R/(GM/R^2)}$. This time is called the dynamical time scale,

$$t_{\text{dyn}} = \sqrt{\frac{R^3}{GM}}, \quad (4.25)$$

which is of order $(G\bar{\rho})^{-1/2}$ for a star of mean density $\bar{\rho}$, and you will find the exact time for the collapse of a uniform sphere in Problem 5. For the Sun t_{dyn} is about half an hour. The time taken by a sound wave to cross the star is of the same order, since the speed of sound in the gas is of order $\sqrt{P/\rho}$, which by the estimates above

is of order $\sqrt{GM/R}$. Any departure from hydrostatic equilibrium is therefore corrected within a time of order t_{dyn} . Since t_{dyn} is so much shorter than t_{KH} , a star that contracts slowly as it radiates passes through a sequence of states each of which is very close to hydrostatic equilibrium.

4.3 Energy Transport and the Equations of Stellar Structure

The equations of Section 4.2 describe the mechanical balance of a star, but they contain the temperature only through the equation of state, and they say nothing about the energy that the star radiates. In this section we first write down how the luminosity grows through the star as energy is released in it, then work out how that energy is carried outwards, by radiation and by convection, and collect the results into the equations of stellar structure. Following that, we solve a simplified form of the equations for a family of model stars called polytropes, and finally we use the equations to find how the luminosity of a main-sequence star depends on its mass.

The Luminosity Gradient

Let L_r be the energy per unit time that flows outwards through the sphere of radius r , and let ϵ be the rate at which energy is released per unit mass of the gas by nuclear reactions. In a star in thermal equilibrium, in which no layer is gaining or losing thermal energy, the energy leaving the shell between r and $r+dr$ exceeds the energy entering it by the energy released inside it, which is ϵ multiplied by the mass $4\pi r^2 \rho dr$ of the shell. Therefore

$$\frac{dL_r}{dr} = 4\pi r^2 \rho \epsilon. \quad (4.26)$$

The luminosity L_r is zero at the centre and equal to the luminosity L of the star at its surface. Outside the region in which the nuclear reactions take place, ϵ is zero and L_r is constant. In a star that is contracting or expanding, as a protostar does, the gravitational energy released in each layer adds to ϵ a further term, and that term is left out here, since it is small for a star on the main sequence.

Radiative Diffusion

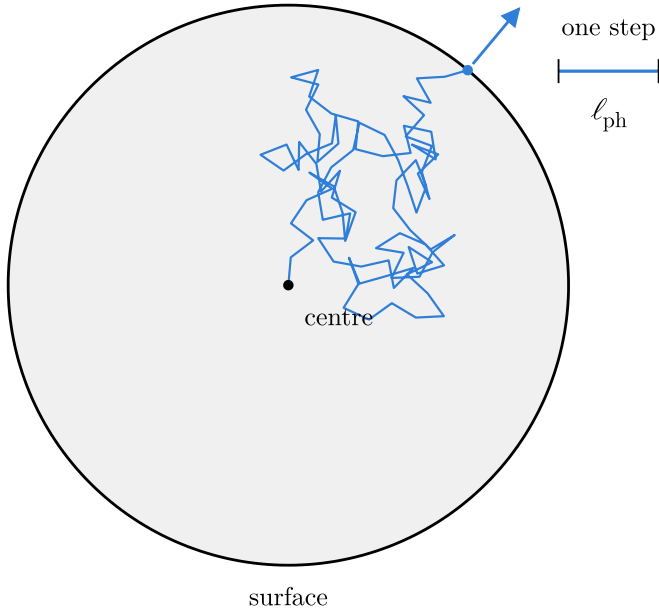


Figure 44. The path of a photon inside a star, drawn as a random walk of equal steps of length ℓ_{ph} in random directions, computed for a star whose radius is a small number of steps. After N steps the photon is typically a distance $\sqrt{N} \ell_{\text{ph}}$ from where it started, so it reaches the surface after about $(R/\ell_{\text{ph}})^2$ steps. The inset shows one step, enlarged.

Deep inside a star a photon travels only a short distance before it is absorbed, and Section 3.3 showed that this distance, the mean free path of a photon, is $\ell_{\text{ph}} = 1/\kappa\rho$, where κ is the opacity. In the interior of the Sun it is a small fraction of a millimetre. The absorbed energy is soon emitted again in a random direction, so the energy carried by radiation moves outwards by a random walk, as Figure 44 shows. In Section 4.1 of Vol. 2 a random walk of N steps of length ℓ along a line was found to carry the walker a mean square distance $N\ell^2$ from its starting point. The same result holds in three dimensions. Let the steps be vectors ℓ_j , each of length ℓ_{ph} , so that the displacement after N steps is their sum. Its square is the sum of the squares of the steps, which is $N\ell_{\text{ph}}^2$, plus the sum of the products $\ell_j \cdot \ell_k$ with $j \neq k$, each of which averages to zero,

since the directions of different steps are independent. The distance travelled after N steps is therefore typically $\sqrt{N} \ell_{\text{ph}}$, and energy starting at the centre reaches the surface after about $N = (R/\ell_{\text{ph}})^2$ steps, which take a time

$$t \approx \frac{N \ell_{\text{ph}}}{c} = \frac{R^2}{\ell_{\text{ph}} c}. \quad (4.27)$$

For the Sun, with mean free paths of a millimetre or less, this time is of order 10^4 years or more, compared with the two seconds or so that light would take to travel the radius of the Sun in empty space.

The random walk explains why the transport is slow, but the temperature gradient that it requires is most easily obtained from the momentum that the radiation carries. The radiation at any point inside the star is very nearly isotropic, and it is very nearly black-body radiation at the local temperature T , but slightly more of it travels outwards than inwards, and this small excess carries the flux $F = L_r/4\pi r^2$. Each photon carries a momentum equal to its energy divided by c , as shown in Section 1.2 of Vol. 3 of the Fermi Notes on Physics, so the flux F carries a momentum F/c outwards per unit area per unit time. A layer of thickness dr absorbs a fraction $\kappa\rho dr$ of the radiation crossing it, by the definition of the opacity in Section 3.3, and so it takes up a momentum $\kappa\rho F dr/c$ per unit area per unit time. The radiation pressure is the momentum carried by the radiation across unit area in unit time, so this momentum taken up by the layer must equal the drop in the radiation pressure across it,

$$P_{\text{rad}}(r) - P_{\text{rad}}(r + dr) = \frac{\kappa\rho F}{c} dr, \quad \text{so} \quad \frac{dP_{\text{rad}}}{dr} = -\frac{\kappa\rho}{c} \frac{L_r}{4\pi r^2}. \quad (4.28)$$

With $P_{\text{rad}} = \frac{1}{3}a_{\text{rad}}T^4$ from Section 4.2, the derivative on the left is $\frac{4}{3}a_{\text{rad}}T^3 dT/dr$, and using $a_{\text{rad}}c = 4\sigma_{\text{SB}}$ we find the temperature gradient

$$\frac{dT}{dr} = -\frac{3\kappa\rho L_r}{64\pi\sigma_{\text{SB}}r^2T^3}. \quad (4.29)$$

This relation gives the temperature gradient wherever energy is carried by radiation, and it holds wherever the mean free path of a photon is much shorter than the distance over which the temperature changes appreciably, since only then is the radiation close to blackbody radiation at the local temperature. A larger opacity or a larger luminosity requires a steeper fall of temperature. The relation can also be written as $F = -\frac{1}{3}c\ell_{\text{ph}} du/dr$, where $u = a_{\text{rad}}T^4$ is the energy density of the radiation. It has the same form as the diffusion of molecules in a gas in Section 1.4 of Vol. 2, where the diffusion coefficient is $\frac{1}{3}\bar{v}\ell_{\text{mol}}$ for molecules of mean speed $\bar{v}$ and mean free path ℓ_{mol} , which Vol. 2 writes as λ ; here the photons move at the speed c with the mean free path ℓ_{ph} , and the quantity that diffuses is the energy density of the radiation. For this reason the transport of energy by radiation in a stellar interior is called radiative diffusion.

The Opacity

The opacity κ_ν of Section 3.3 depends on frequency, and the κ in the radiative temperature gradient found above is a suitable average of it. Since the flux is carried most easily at the frequencies where the gas is most transparent, the average is taken of the mean free path $1/\kappa_\nu$ and not of κ_ν itself, and it is weighted by how strongly the Planck function $B_\nu(T)$ changes with temperature at each frequency, since that change is what drives the flux at that frequency. The result is

$$\frac{1}{\kappa} = \frac{\int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B_\nu}{\partial T} d\nu}{\int_0^\infty \frac{\partial B_\nu}{\partial T} d\nu}, \quad (4.30)$$

and this average is called the Rosseland mean opacity, after Svein Rosseland, who introduced it in 1924. Frequencies at which the gas is nearly transparent dominate the mean, in the same way that most

of the current through resistors joined in parallel flows through the one with the smallest resistance.

The opacity of a stellar interior comes from several processes. A photon can be scattered by a free electron, which is called electron scattering; its cross-section is the Thomson cross-section σ_T , the same at all frequencies, and it gives an opacity $\kappa = \sigma_T/\mu_e m_H$, where $\mu_e m_H$ is the mass of gas per free electron, which does not depend on the density or the temperature. A photon can also be absorbed by a free electron passing close to a nucleus, which is called free-free absorption, or by an ion that still holds an electron, which the photon then removes from it, and this is called bound-free absorption. Together these two absorption processes give a Rosseland mean opacity that is approximately

$$\kappa \approx \kappa_0 \rho T^{-7/2}, \quad (4.31)$$

where κ_0 is a constant that depends on the composition. This approximation is called Kramers' law, after Hendrik Kramers, who obtained this dependence on density and temperature in 1923. It shows that the opacity rises steeply as the temperature falls, until the temperature is low enough for most of the atoms to hold on to their electrons. In practice the opacities used in models of stars are computed in detail and tabulated, and Kramers' law and electron scattering are useful as guides to how they behave; electron scattering dominates in the hot interiors of massive stars, where the density is low and the temperature high.

Convection and the Schwarzschild Criterion

When the temperature gradient required by radiative diffusion is too steep, the gas begins to move, and hot gas rising while cool gas sinks carries the energy outwards. This process is called convection. To find when it happens, let us consider a small element of gas at radius r , displaced upwards by a small distance Δr . The element moves slowly enough for its pressure to stay equal to that of its surroundings, because pressure differences are evened out at the speed of sound, but quickly enough that it exchanges no heat with them, so its change of state is adiabatic. If after the displacement

the element is denser than its new surroundings, it sinks back, and the layer is stable. If it is less dense, it is pushed further upwards by buoyancy, as a bubble of air is in water, and the displacement grows, so the layer is unstable to convection.

The element and its surroundings are at the same pressure, and for an ideal gas of the same mean molecular weight the density at a given pressure is inversely proportional to the temperature. The element is therefore less dense than its surroundings if it is hotter. In an adiabatic change of an ideal gas, $P\mathcal{V}^\gamma$ and $T\mathcal{V}^{\gamma-1}$ are both constant, by Section 2.2 of Vol. 2, and eliminating the volume $\mathcal{V}$ between them gives $T \propto P^{(\gamma-1)/\gamma}$. Taking the logarithm of both sides and differentiating with respect to r gives the rate at which the temperature of the element changes as it rises through the star,

$$\left(\frac{dT}{dr}\right)_{\text{ad}} = \left(1 - \frac{1}{\gamma}\right) \frac{T}{P} \frac{dP}{dr}, \quad (4.32)$$

which is called the adiabatic temperature gradient. After a displacement Δr the temperature of the element is $T + (dT/dr)_{\text{ad}}\Delta r$, while that of the surroundings is $T + (dT/dr)\Delta r$, where dT/dr is the actual gradient in the star, which is the radiative gradient if no convection is taking place. Both gradients are negative, since the temperature falls outwards.

Figure 45 shows the two possibilities. In panel (a) the temperature of the surroundings falls with height more slowly than the adiabatic gradient, so the displaced element ends up cooler than its surroundings, and it sinks back. In panel (b) the temperature of the surroundings falls more steeply, the element ends up hotter than its surroundings, and it continues to rise. A layer is therefore unstable to convection when

$$\left|\frac{dT}{dr}\right|_{\text{rad}} > \left(1 - \frac{1}{\gamma}\right) \frac{T}{P} \left|\frac{dP}{dr}\right|, \quad (4.33)$$

a condition that is called the Schwarzschild criterion, after Karl Schwarzschild, who stated it in 1906. Dividing both sides by $(T/P)|dP/dr|$ puts the criterion in the form $d \ln T / d \ln P > 1 - 1/\gamma$,

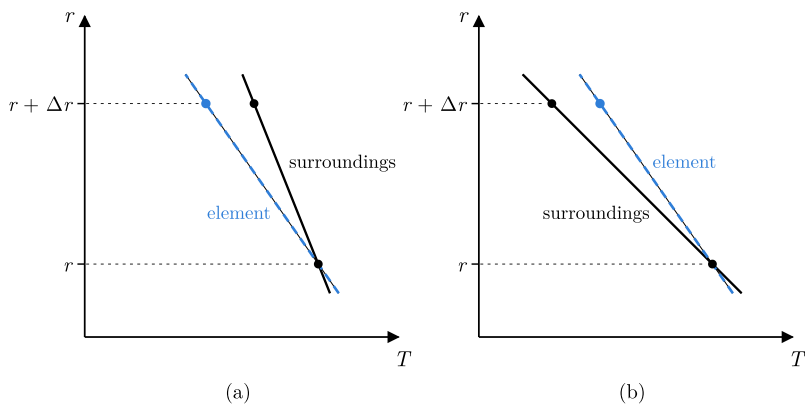


Figure 45. The temperature of an element of gas (accent, dashed) displaced upwards adiabatically from r to $r + \Delta r$, compared with the temperature of its surroundings (solid). In (a) the temperature of the surroundings falls more slowly with height than the adiabatic gradient, the displaced element is cooler and so denser than its surroundings, and it sinks back. In (b) the temperature of the surroundings falls more steeply than the adiabatic gradient, the element is hotter and so less dense than its surroundings, and it continues to rise.

where the derivative is taken along the radius of the star, and for a fully ionised gas with $\gamma = \frac{5}{3}$ the right-hand side is $\frac{2}{5}$.

Once convection begins it carries energy very efficiently, and a temperature gradient only slightly steeper than the adiabatic gradient is enough to carry the whole luminosity. In a convective region the temperature gradient is therefore very close to the adiabatic gradient, and in models of stars it is usually set equal to the adiabatic gradient. The detailed theory of the flow, which fixes how much steeper than adiabatic the gradient must be, is not treated in this course. Convection also mixes the gas, so that a convective region has a uniform composition.

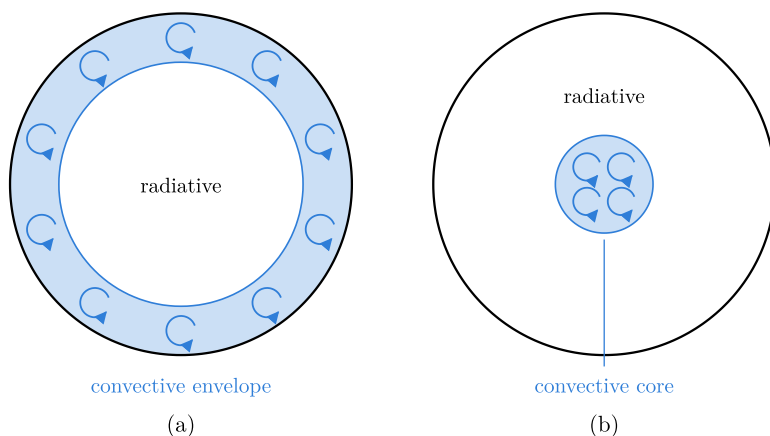


Figure 46. The regions in which energy is carried by convection (accent, with circulating arrows) and by radiation, in (a) a main-sequence star of about the mass of the Sun, with a radiative interior and a convective envelope, and (b) a main-sequence star several times more massive than the Sun, with a convective core and a radiative envelope. The boundaries are drawn at the radii given by detailed models.

The Schwarzschild criterion is met where the radiative gradient is large, which happens where the opacity is large or where the flux $L_r/4\pi r^2$ is large, or where γ is smaller than $\frac{5}{3}$, which lowers the adiabatic gradient. In the outer layers of a cool star the opacity is large, since Kramers' law makes it rise as the temperature falls, and in the layers where hydrogen is partly ionised γ is reduced, because much of the energy added to the gas goes into ionising it rather than

heating it. Stars like the Sun therefore have convective envelopes, and in the Sun the convective region extends from about $0.7 R_\odot$ to the surface, as panel (a) of Figure 46 shows. In main-sequence stars more massive than the Sun the nuclear energy is released in a small region at the centre, so $L_r/4\pi r^2$ is large there, and these stars have convective cores, as panel (b) shows. The least massive main-sequence stars, below roughly a third of the mass of the Sun, are convective throughout.

The Equations of Stellar Structure

The results of this section and of Section 4.2 form a set of four differential equations for the mass M_r , the pressure P , the luminosity L_r , and the temperature T as functions of r ,

$$\begin{aligned} \frac{dM_r}{dr} &= 4\pi r^2 \rho, & \frac{dP}{dr} &= -\frac{GM_r \rho}{r^2}, \\ \frac{dL_r}{dr} &= 4\pi r^2 \rho \epsilon, & \frac{dT}{dr} &= -\frac{3\kappa \rho L_r}{64\pi \sigma_{\text{SB}} r^2 T^3}, \end{aligned} \quad (4.34)$$

where the last equation holds in a radiative region, and in a convective region it is replaced by $dT/dr = (1 - 1/\gamma)(T/P) dP/dr$. These are called the equations of stellar structure. They are completed by three relations that describe the gas: the equation of state, which gives P in terms of ρ , T , and the composition; the opacity κ ; and the rate of energy generation ϵ , each as a function of the density, the temperature, and the composition.

The equations are of first order, so each of the four needs one boundary condition. At the centre, $M_r = 0$ and $L_r = 0$. At the surface, $M_r = M$, and the pressure and the temperature are both very small compared with their values inside; more exactly, the temperature at the photosphere is set by the effective temperature, as described in Section 3.4. That makes five conditions for four equations, and the extra condition fixes the radius R of the star, which is not known in advance. The mass and the composition of the star therefore determine its whole structure, including its radius and its luminosity, a statement called the Vogt–Russell theorem, although it is a statement about the equations rather than a theorem that has been proved. The equations cannot be solved in closed form for a

real star, and they are usually solved numerically, by integrating outwards from the centre and inwards from the surface and adjusting the unknown central and surface values until the two solutions meet.

Polytropes and the Lane–Emden Equation

The mechanical equations, those of mass continuity and hydrostatic equilibrium, involve the temperature only through the equation of state. If the pressure depends only on the density, they can be solved without the equations for L_r and T . Let us suppose that

$$P = K_p \rho^{1+1/n_p}, \quad (4.35)$$

where K_p and n_p are constants. A star of this kind is called a polytrope, and n_p is called its polytropic index. As an example, consider a star that is convective throughout. Its temperature gradient is adiabatic, so $P \propto \rho^\gamma$ everywhere, and with $\gamma = \frac{5}{3}$ it is a polytrope with $n_p = \frac{3}{2}$. You will show in Problem 2 that a star in which the pressure of radiation is the same fraction of the total pressure everywhere is a polytrope with $n_p = 3$; Eddington used this model in 1926, and it describes the Sun reasonably well.

Multiplying the equation of hydrostatic equilibrium by r^2/ρ gives $(r^2/\rho) dP/dr = -GM_r$, and differentiating with respect to r and using the equation of mass continuity eliminates M_r ,

$$\frac{1}{r^2} \frac{d}{dr} \left(\frac{r^2}{\rho} \frac{dP}{dr} \right) = -4\pi G \rho. \quad (4.36)$$

Let ρ_c be the central density, and let us write the density as $\rho = \rho_c w^{n_p}$, where w is a dimensionless function of r with $w = 1$ at the centre. Then $\rho^{1/n_p} = \rho_c^{1/n_p} w$, and differentiating $P = K_p \rho^{1+1/n_p}$ gives

$$\begin{aligned}
\frac{1}{\rho} \frac{dP}{dr} &= K_p \left(1 + \frac{1}{n_p} \right) \rho^{1/n_p-1} \frac{d\rho}{dr} \\
&= (n_p + 1) K_p \frac{d\rho^{1/n_p}}{dr} \\
&= (n_p + 1) K_p \rho_c^{1/n_p} \frac{dw}{dr}.
\end{aligned} \tag{4.37}$$

Finally, let $r = r_s \xi$, where ξ is dimensionless and the scale length r_s is chosen to absorb the constants,

$$r_s^2 = \frac{(n_p + 1) K_p \rho_c^{1/n_p-1}}{4\pi G}. \tag{4.38}$$

Substituting into the equation above and dividing by $4\pi G \rho_c$ gives

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{dw}{d\xi} \right) = -w^{n_p}, \tag{4.39}$$

which is called the Lane–Emden equation, after Jonathan Homer Lane and Robert Emden, who studied it in the late nineteenth and early twentieth centuries. The boundary conditions at the centre are $w = 1$ and $dw/d\xi = 0$, the second because the pressure gradient must vanish at the centre, where $M_r = 0$. The surface of the star is where the density first falls to zero, at the first zero ξ_1 of w , so the radius is $R = r_s \xi_1$.

For $n_p = 0$ the right-hand side is -1 , the density is uniform, and integrating twice with the conditions at the centre gives

$$w = 1 - \frac{\xi^2}{6}, \quad \xi_1 = \sqrt{6}. \tag{4.40}$$

For $n_p = 1$ the equation is linear, and the substitution $w = \Theta/\xi$ simplifies it. By the quotient rule, $\xi^2 dw/d\xi = \xi d\Theta/d\xi - \Theta$, and by the product rule the derivative of this is $\xi d^2\Theta/d\xi^2$, so the equation becomes $d^2\Theta/d\xi^2 = -\Theta$. Its solutions are $\sin \xi$ and $\cos \xi$, and only $\Theta = \sin \xi$ gives a w that is finite and equal to one at the centre, since $\sin \xi/\xi \rightarrow 1$ as $\xi \rightarrow 0$. Therefore

$$w = \frac{\sin \xi}{\xi}, \quad \xi_1 = \pi. \quad (4.41)$$

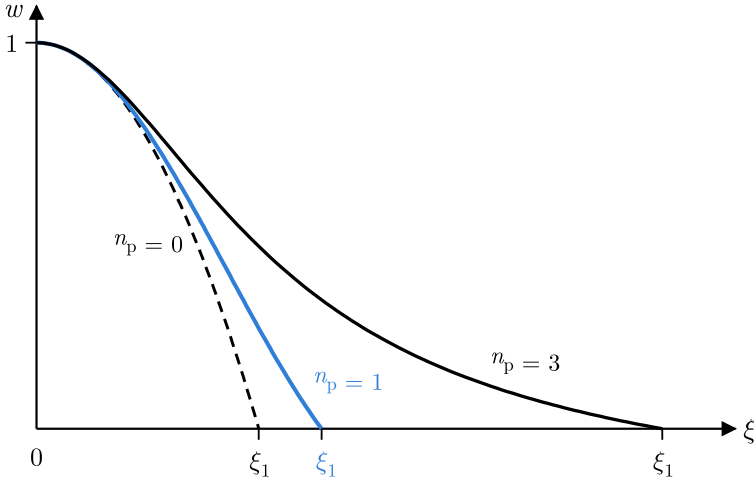


Figure 47. Solutions $w(\xi)$ of the Lane–Emden equation for $n_p = 0$ (dashed), $n_p = 1$ (accent), and $n_p = 3$, the last computed numerically. Each curve starts at $w = 1$ with zero slope and ends at its first zero ξ_1 , which marks the surface of the star. The density is $\rho_c w^{n_p}$, so a larger index gives a star more concentrated towards its centre.

For other values of n_p no solution in terms of familiar functions is known, apart from the case $n_p = 5$, and the equation is usually solved numerically, by stepping outwards from the centre in small increments of ξ . The first zero is at $\xi_1 \approx 3.65$ for $n_p = \frac{3}{2}$ and at $\xi_1 \approx 6.90$ for $n_p = 3$, and for $n_p = 5$ it moves off to infinity. Figure 47 compares the solutions for three indices. The mass of a polytrope follows from the Lane–Emden equation itself, since $\xi^2 w^{n_p}$ is minus the derivative of $\xi^2 dw/d\xi$,

$$M = \int_0^R 4\pi r^2 \rho dr = 4\pi r_s^3 \rho_c \int_0^{\xi_1} \xi^2 w^{n_p} d\xi = 4\pi r_s^3 \rho_c \xi_1^2 \left. \frac{dw}{d\xi} \right|_{\xi_1}. \quad (4.42)$$

With this result you will find in Problems 3 and 4 how the central density and the radius of a polytrope depend on its mass.

The Mass–Luminosity Relation from the Structure Equations

The equations of stellar structure also explain the mass–luminosity relation of main-sequence stars that was found by observation in Section 2.2. We estimate each quantity in the equations by its order of magnitude, replacing a derivative such as dP/dr by P/R and the density by M/R^3 , and dropping numerical factors of order one. The estimate assumes that stars of different masses have interiors of the same shape, differing only in scale, which is a reasonable approximation along the main sequence. Hydrostatic equilibrium gives $P \sim GM^2/R^4$, as in Section 4.2, and the equation of state of an ideal gas then gives

$$T \sim \frac{\mu m_{\text{H}} P}{\rho k_B} \sim \frac{\mu m_{\text{H}}}{k_B} \frac{GM}{R}. \quad (4.43)$$

Radiative diffusion, with $dT/dr \sim T/R$ and $L_r \sim L$, gives $T/R \sim \kappa \rho L / \sigma_{\text{SB}} R^2 T^3$, so

$$L \sim \frac{\sigma_{\text{SB}} R T^4}{\kappa \rho} \sim \frac{\sigma_{\text{SB}} R^4 T^4}{\kappa M}. \quad (4.44)$$

Substituting the temperature,

$$L \sim \frac{\sigma_{\text{SB}} G^4}{\kappa} \left(\frac{\mu m_{\text{H}}}{k_B} \right)^4 M^3. \quad (4.45)$$

The radius has cancelled, and the rate of energy generation never entered the estimate. The luminosity of a star in which energy is carried by radiation is set by how fast radiation can leak out of it, which depends on its mass, its composition, and its opacity, and the nuclear reactions at the centre adjust themselves to supply this luminosity, the star settling at the radius and the central temperature at which they do so.

If the opacity is due to electron scattering, κ is constant and $L \propto \mu^4 M^3$. If it follows Kramers' law, $\kappa \propto \rho T^{-7/2}$, then with $\rho \sim M/R^3$ and $T \propto M/R$ the opacity goes as $M^{-5/2} R^{1/2}$, and you will show in Problem 5 that $L \propto M^{11/2} R^{-1/2}$. The relation found by observation, roughly $L \propto M^{3.5}$ over much of the main sequence, lies between these two, and it becomes less steep for the most massive stars, in whose hot interiors electron scattering is the main source of opacity. The factor μ^4 shows that the luminosity of a star rises as its mean molecular weight increases, which it does as hydrogen is converted into helium in the core.

4.4 Thermonuclear Reactions in Stars

Section 4.2 showed that gravitational contraction could keep the Sun shining for only about 10^7 years, far less than the age of the solar system, and the source that keeps it shining for longer is nuclear fusion. In this section we first work out how much energy fusion can release, then explain how nuclei in a star manage to fuse even though their electric repulsion ought to keep them apart, and then calculate how fast they do so. Finally we describe the chains of reactions by which stars convert hydrogen into helium, and the way in which the rates of these reactions depend on temperature. The tunnelling of a particle through a barrier, on which the whole calculation rests, is treated in detail in Section 2.5 of Vol. 3.

Binding Energy and the Energy Released by Fusion

As you learned at school, the mass of a nucleus is less than the total mass of the protons and neutrons that make it up. For a nucleus of Z_n protons and N_n neutrons, with mass number $A = Z_n + N_n$ and mass m_{nuc} , the binding energy is

$$B = (Z_n m_p + N_n m_n - m_{\text{nuc}}) c^2, \quad (4.46)$$

which is the energy needed to pull the nucleus apart into its separate nucleons. Figure 48 shows the binding energy per nucleon, B/A . It rises steeply from the lightest nuclei, with helium-4 bound much more tightly than its neighbours, reaches a broad maximum

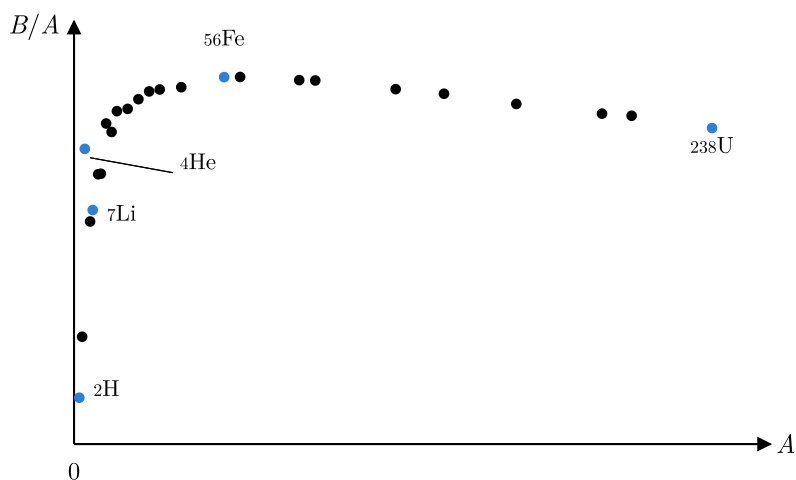


Figure 48. The binding energy per nucleon B/A of a selection of stable nuclei against the mass number A , plotted from measured nuclear masses. The nuclei labelled are drawn in the accent colour. The binding energy per nucleon rises steeply among the lightest nuclei, with helium-4 lying well above its neighbours, reaches a broad maximum near iron, and falls slowly towards uranium.

near iron and nickel, and then falls slowly towards the heaviest nuclei. Fusing light nuclei into heavier ones, up to the region of iron, therefore releases energy, because the nucleons end up more tightly bound.

The fusion that powers the Sun converts hydrogen into helium. The net result of the reactions by which this happens is that four hydrogen nuclei become one helium-4 nucleus, two of the protons turning into neutrons with the emission of two positrons and two neutrinos, and the positrons annihilate with two of the electrons in the gas. A neutrino is an uncharged particle of very small mass, which interacts so weakly with matter that it escapes from the star. When the electrons are counted in this way, the energy released is found from the masses of the atoms,

$$Q = (4m_{\text{H}} - m_{\text{He}}) c^2 \approx 26.7 \text{ MeV}, \quad (4.47)$$

where m_{He} is the mass of the helium-4 atom. This is about 0.7 per cent of the rest energy $4m_{\text{H}}c^2$ of the hydrogen. A star can burn in this way only the hydrogen near its centre, where the temperature is high enough, and for the Sun this is roughly a tenth of its mass. The time for which the Sun can shine at its present luminosity by burning this hydrogen is called the nuclear time scale,

$$t_{\text{nuc}} \approx \frac{\phi f_{\text{b}} M c^2}{L}, \quad (4.48)$$

where $\phi \approx 0.007$ is the fraction of the rest energy released and $f_{\text{b}} \approx 0.1$ the fraction of the mass of the star that is burned. For the Sun this comes to about 10^{10} years, comfortably longer than the 4.6×10^9 years for which it has been shining, and a thousand times the Kelvin–Helmholtz time scale.

The Coulomb Barrier

For two nuclei to fuse, they must come close enough for the strong nuclear force between them to act, which requires a separation of a few times 10^{-15} m. Let this separation be r_{n} . At larger separations the only force between the nuclei is the electric repulsion of their

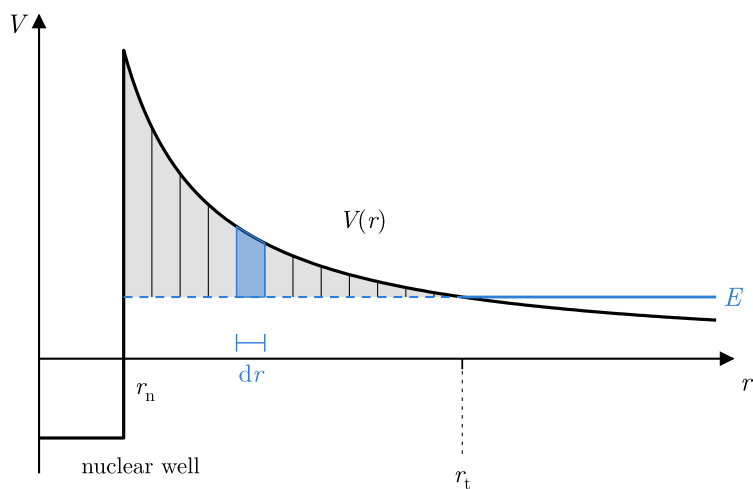


Figure 49. The potential energy $V(r)$ of two nuclei at a distance r apart, drawn schematically. Inside the range r_n of the nuclear force the nuclei attract, and outside it they repel each other electrically. A pair with relative energy E (accent) is classically stopped at the turning point r_t , and the shaded region between r_n and r_t is forbidden to it classically. The shaded region is cut into thin slices of width dr , one of which is drawn in the accent colour, for the calculation of the tunnelling probability. In a star r_t is very much larger than r_n .

charges. Let the charges be Z_1e and Z_2e . The potential energy of the pair at a separation r larger than r_n is then

$$V(r) = \frac{1}{4\pi\epsilon_0} \frac{Z_1Z_2e^2}{r}, \quad (4.49)$$

and inside r_n the nuclear attraction takes over, as Figure 49 shows. The peak of the potential energy at r_n is called the Coulomb barrier. For two protons at a separation of 10^{-15} m its height is about 1 MeV. At the centre of the Sun, where the temperature is about 1.6×10^7 K, the thermal energy $k_B T$ is only about 1.4 keV, roughly a thousand times less than the height of the barrier.

In classical physics a pair of protons can cross the barrier only if their relative energy exceeds its height. By the Boltzmann factor of Section 4.4 of Vol. 2, the fraction of pairs with so much energy is of order e^{-1000} , which is about 10^{-434} . The Sun contains about 10^{57} protons, so in classical physics not a single pair would ever fuse. Eddington suggested in 1920 that the Sun is powered by the conversion of hydrogen into helium, and he was confronted with exactly this difficulty. The answer came after Gamow's explanation of alpha decay in 1928 as the tunnelling of an alpha particle through the Coulomb barrier of a nucleus, when Atkinson and Houtermans applied the same idea to nuclei colliding in a star.

Tunnelling and the Gamow Factor

In Section 2.5 of Vol. 3 a particle of mass m_x and energy E meeting a rectangular barrier of height V_0 and width w_b was found to pass through it with a probability that, for a thick barrier, is close to $e^{-2\eta w_b}$ multiplied by a factor of order one, where $\eta = \sqrt{2m_x(V_0 - E)}/\hbar$; Vol. 3 writes this quantity as κ , and it is renamed here because κ is the opacity. The Coulomb barrier is not rectangular, but it can be cut into thin slices of width dr , as in Figure 49, across each of which the potential energy is nearly constant. Each slice reduces the probability by a factor $e^{-2\eta(r)dr}$, where

$$\eta(r) = \frac{\sqrt{2\mu_r[V(r) - E]}}{\hbar}, \quad (4.50)$$

and the mass is the reduced mass $\mu_r = m_1 m_2 / (m_1 + m_2)$ of the two nuclei, which is written μ in Section 5.3 of Vol. 1, since the collision of two nuclei is a two-body problem that reduces to the motion of a single particle of mass μ_r . Multiplying the factors for all the slices adds their exponents, and the probability of tunnelling through the whole barrier is

$$P_T \approx \exp \left(-2 \int_{r_n}^{r_t} \eta(r) dr \right), \quad (4.51)$$

where r_t is the classical turning point, at which $V(r_t) = E$. This approximation leaves out factors of order one, which change the result far less than the exponential does, and it is valid when η changes little over a distance $1/\eta$. A more careful method, called the WKB approximation, is not treated in this course; it gives the same exponent.

To evaluate the integral, let E be written as $V(r_t)$, so that $V(r) - E = E(r_t/r - 1)$ and $2\eta(r) = (2\sqrt{2\mu_r E}/\hbar)\sqrt{r_t/r - 1}$. In a star r_n is much smaller than r_t , and we set the lower limit of the integral to zero; you will find the correction that this introduces in Problem 2. With the substitution $r = r_t \sin^2 \theta$, for which $dr = 2r_t \sin \theta \cos \theta d\theta$ and $\sqrt{r_t/r - 1} = \cos \theta / \sin \theta$, the integral becomes

$$\begin{aligned} \int_0^{r_t} \sqrt{\frac{r_t}{r} - 1} dr &= r_t \int_0^{\pi/2} 2 \cos^2 \theta d\theta \\ &= r_t \int_0^{\pi/2} (1 + \cos 2\theta) d\theta \\ &= \frac{\pi r_t}{2}, \end{aligned} \quad (4.52)$$

where the identity $2 \cos^2 \theta = 1 + \cos 2\theta$ has been used. The exponent is therefore $\pi r_t \sqrt{2\mu_r E}/\hbar$, and with $r_t = Z_1 Z_2 e^2 / 4\pi\epsilon_0 E$ the tunnelling probability becomes

$$P_T \approx \exp \left(-\sqrt{\frac{E_G}{E}} \right), \quad E_G = 2\pi^2 \mu_r \left(\frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{\hbar} \right)^2. \quad (4.53)$$

The exponential is called the Gamow factor, and E_G is called the Gamow energy. For two protons E_G is about 0.49 MeV, and at an energy $E = k_B T$ at the centre of the Sun the exponent is about 19, so the tunnelling probability is of order 10^{-8} . That is small, but it is larger than the classical probability by a factor of more than 10^{400} . The Gamow energy goes as $\mu_r Z_1^2 Z_2^2$, so it grows rapidly with the charges of the nuclei. For a proton and a nitrogen-14 nucleus it is about 45 MeV, ninety times larger than for two protons, and reactions between heavier nuclei therefore need much higher temperatures.

The Reaction Rate and the Gamow Peak

Let the gas contain n_1 nuclei of one kind and n_2 of another per unit volume, and let σ be the cross-section for a reaction between them. A nucleus of the first kind moving at a relative speed v through the nuclei of the second kind sweeps out a volume $\sigma v dt$ in a time dt , and so meets $n_2 \sigma v dt$ of them. When this number is averaged over the relative speeds, the number of reactions per unit volume per unit time is

$$\mathcal{R}_{12} = n_1 n_2 \langle \sigma v \rangle, \quad (4.54)$$

where $\langle \sigma v \rangle$ is the mean of the product σv . If the two nuclei are of the same kind, as for two protons, the right-hand side is divided by two, so that each pair is counted once.

The cross-section contains the Gamow factor, and it is conventional to write it as

$$\sigma(E) = \frac{S(E)}{E} \exp \left(-\sqrt{\frac{E_G}{E}} \right). \quad (4.55)$$

The factor $1/E$ comes from the size of the target that a nucleus

presents in a quantum collision, which is of the order of the square of the reduced de Broglie wavelength $\hbar/p_{\text{rel}}$ of Section 1.3 of Vol. 3, where p_{rel} is the momentum of the relative motion. Since $p_{\text{rel}}^2 = 2\mu_r E$, this area goes as $1/E$. The remaining factor $S(E)$ is called the astrophysical S-factor, and it contains the nuclear physics of the reaction. For most reactions it changes slowly with energy, which allows it to be measured in the laboratory at energies higher than those in stars and extrapolated to them.

The relative velocities of the nuclei follow from the Maxwell distribution of Section 1.2 of Vol. 2. Section 5.3 of Vol. 1 showed that the kinetic energy of two bodies is the sum of $\frac{1}{2}M_{\text{pair}}v_{\text{cm}}^2$, for the motion of their centre of mass, and $\frac{1}{2}\mu_r v^2$, for their relative motion, where M_{pair} is the mass of the pair and v_{cm} the velocity of its centre of mass. The Boltzmann factor is therefore a product of a factor for v_{cm} and a factor for v , and the relative velocity has a Maxwell distribution with the mass replaced by μ_r . The Maxwell distribution of speeds is converted into a distribution of the relative energy $E = \frac{1}{2}\mu_r v^2$ by substituting $v = \sqrt{2E/\mu_r}$ and $dv = dE/\sqrt{2\mu_r E}$, which gives the fraction of pairs with relative energy between E and $E + dE$ as

$$f(E) dE = \frac{2}{\sqrt{\pi}} \frac{E^{1/2}}{(k_B T)^{3/2}} e^{-E/k_B T} dE. \quad (4.56)$$

Multiplying the cross-section by $v = \sqrt{2E/\mu_r}$ and by $f(E)$, and integrating over all energies, gives

$$\langle \sigma v \rangle = \sqrt{\frac{8}{\pi \mu_r}} \frac{1}{(k_B T)^{3/2}} \int_0^\infty S(E) \exp\left(-\frac{E}{k_B T} - \sqrt{\frac{E_G}{E}}\right) dE. \quad (4.57)$$

The integrand is the product of the Maxwell factor $e^{-E/k_B T}$, which falls as the energy rises, and the Gamow factor, which rises, so the product has a sharp peak, as Figure 50 shows. This peak is called the Gamow peak. Its position E_0 is where the exponent $E/k_B T + \sqrt{E_G/E}$ has its minimum. Setting the derivative of the exponent with respect to E to zero gives $1/k_B T = \frac{1}{2}E_G^{1/2} E^{-3/2}$, so

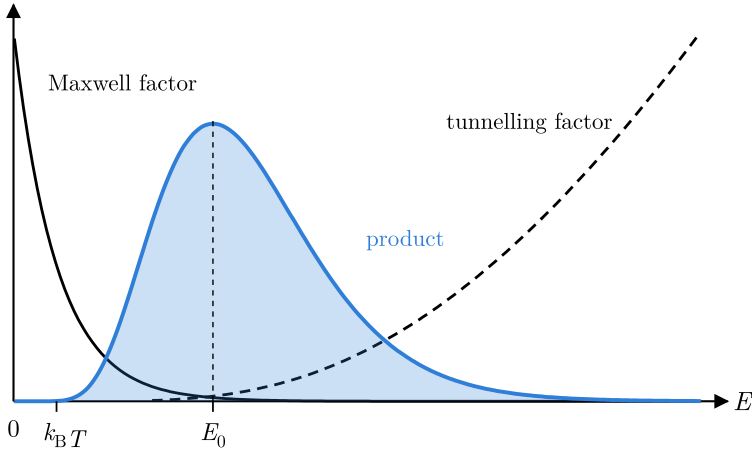


Figure 50. The two factors in the integrand for the reaction rate, each drawn to a separate vertical scale, against the relative energy E of a pair of nuclei. The Maxwell factor $e^{-E/k_B T}$ falls with energy, the tunnelling factor $\exp(-\sqrt{E_G/E})$ rises, and their product (accent, shaded) is the Gamow peak, centred on E_0 . The curves are drawn for a Gamow energy very much larger than $k_B T$, and the product is magnified by a large factor.

$$E_0 = \left(\frac{E_G (k_B T)^2}{4} \right)^{1/3}. \quad (4.58)$$

For two protons at the centre of the Sun, E_0 is about 6 keV, between four and five times $k_B T$. The reactions therefore take place among the small fraction of pairs whose energies are several times the mean thermal energy, and yet they tunnel through a barrier more than a hundred times higher than their energy.

The width of the peak follows from a Taylor expansion of the exponent about its minimum. The second derivative of $E/k_B T + \sqrt{E_G/E}$ is $\frac{3}{4} E_G^{1/2} E^{-5/2}$, and at E_0 , using $E_G^{1/2} = 2E_0^{3/2}/k_B T$ from the condition for the minimum, it equals $3/2 E_0 k_B T$. The value of the exponent at E_0 is $E_0/k_B T + 2E_0/k_B T$. Near the peak, therefore,

$$\exp \left(-\frac{E}{k_B T} - \sqrt{\frac{E_G}{E}} \right) \approx e^{-\tau_G} \exp \left(-\frac{3(E - E_0)^2}{4E_0 k_B T} \right), \quad (4.59)$$

$$\tau_G = \frac{3E_0}{k_B T} = 3 \left(\frac{E_G}{4k_B T} \right)^{1/3},$$

which is a Gaussian that falls to $1/e$ of its peak value at a distance $2\sqrt{E_0 k_B T/3}$ on either side of E_0 . At the centre of the Sun this half-width is a little over half of E_0 for two protons. Since the S-factor changes slowly, it can be taken outside the integral at the value $S(E_0)$, and the integral of the Gaussian is $\sqrt{4\pi E_0 k_B T/3}$, by the Gaussian integral of Section 1.2 of Vol. 2, with the lower limit extended from zero to minus infinity. This extension is a good approximation when the peak lies several half-widths above zero, which is the case when τ_G is large, since the ratio of E_0 to the half-width is $\sqrt{\tau_G}/2$. The result is

$$\langle \sigma v \rangle \approx \frac{4}{k_B T} \sqrt{\frac{2E_0}{3\mu_r}} S(E_0) e^{-\tau_G}. \quad (4.60)$$

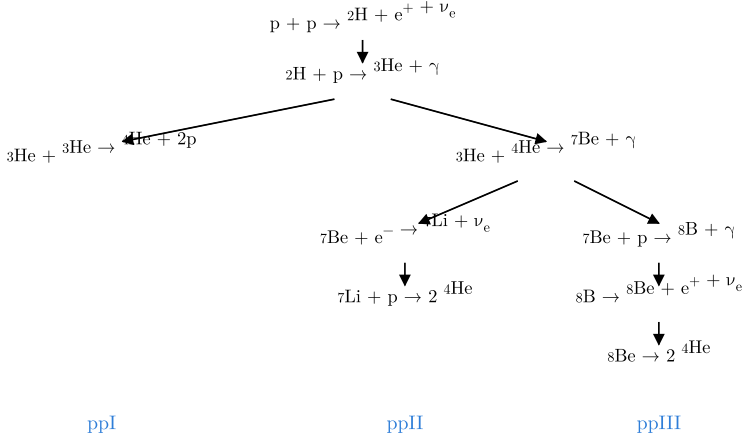


Figure 51. The proton–proton chains. The first two reactions are common to all three branches; ppI then ends with the fusion of two helium-3 nuclei, which requires the first two reactions to take place twice, while ppII and ppIII begin with the fusion of helium-3 and helium-4 and differ in the fate of the beryllium-7 nucleus that it forms.

The Proton–Proton Chains

The fusion of four protons into a helium nucleus does not happen in a single collision, since the simultaneous meeting of four nuclei is far too rare. It happens through a sequence of reactions between pairs of nuclei. In the Sun the principal set of reactions is called the proton–proton chain, and it has three branches, shown in Figure 51. The first reaction of every branch is

$$p + p \rightarrow {}^2\text{H} + e^+ + \nu_e, \quad (4.61)$$

in which two protons form a deuterium nucleus, ${}^2\text{H}$, with the emission of a positron and a neutrino. During the brief encounter one of the protons must turn into a neutron, by the process responsible for the beta-plus decay met at school, and that process is so slow that the S-factor of this reaction is extremely small. A proton at the centre of the Sun waits on average about 10^{10} years before it takes part in this reaction, which is why the Sun burns its hydrogen so slowly. The deuterium nucleus then captures another proton

within seconds, forming helium-3,



In the branch called ppI, two helium-3 nuclei, each formed in this way, fuse to give helium-4 and return two protons, $^3\text{He} + ^3\text{He} \rightarrow ^4\text{He} + 2\text{p}$. Alternatively, a helium-3 nucleus can fuse with a helium-4 nucleus already present, forming beryllium-7, and the beryllium-7 nucleus is then converted into two helium-4 nuclei in one of two ways. In the branch called ppII it captures an electron and becomes lithium-7, which captures a proton and splits into two helium-4 nuclei. In the branch called ppIII it captures a proton and becomes boron-8, which decays to beryllium-8 with the emission of a positron and a neutrino, and the beryllium-8 splits into two helium-4 nuclei. In the Sun most of the helium is made by the ppI branch, a smaller but substantial part by the ppII branch, and only a very small part by the ppIII branch. The net result of each branch is the conversion of four protons into a helium-4 nucleus with the release of 26.7 MeV, of which a few per cent is carried off by the neutrinos in ppI and ppII, and a larger fraction in ppIII, whose boron-8 decay emits neutrinos of high energy.

The CNO Cycle

In stars somewhat more massive than the Sun, hydrogen is converted into helium mainly by a different sequence of reactions, in which carbon, nitrogen, and oxygen act as catalysts. It was found independently by von Weizsäcker and by Bethe in 1938 and 1939, and it is called the CNO cycle. Nuclear physicists write a reaction such as $^{12}\text{C} + \text{p} \rightarrow ^{13}\text{N} + \gamma$ in the shorthand $^{12}\text{C}(\text{p}, \gamma)^{13}\text{N}$, with the particle absorbed and the particle emitted in the brackets. In this notation the main CNO cycle, drawn in Figure 52, is the following sequence of reactions:

1. $^{12}\text{C}(\text{p}, \gamma)^{13}\text{N}$;
2. the decay $^{13}\text{N} \rightarrow ^{13}\text{C} + \text{e}^+ + \nu_{\text{e}}$;
3. $^{13}\text{C}(\text{p}, \gamma)^{14}\text{N}$;
4. $^{14}\text{N}(\text{p}, \gamma)^{15}\text{O}$;

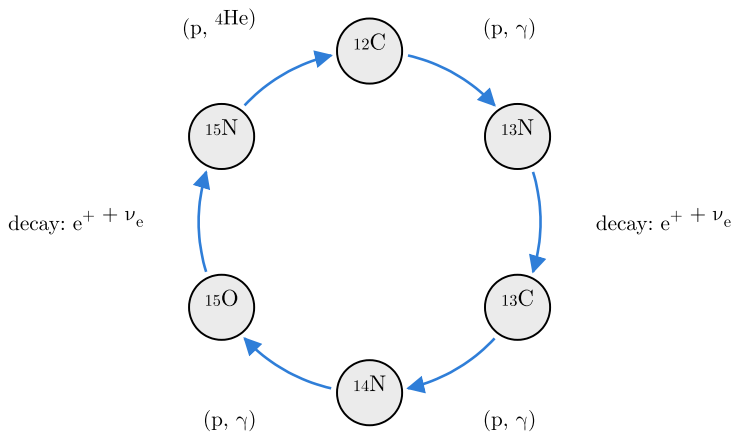


Figure 52. The CNO cycle. Each arrow is labelled with the reaction that carries one nucleus into the next, written as (particle absorbed, particle emitted) for a capture and as the particles emitted for a decay. In one passage round the cycle four protons are absorbed, a helium-4 nucleus, two positrons, and two neutrinos are emitted, and the carbon-12 nucleus is returned.

5. the decay $^{15}\text{O} \rightarrow ^{15}\text{N} + \text{e}^+ + \nu_{\text{e}}$;

6. $^{15}\text{N}(\text{p}, ^4\text{He})^{12}\text{C}$.

The carbon-12 nucleus with which the cycle began is returned at the end, having absorbed four protons and emitted a helium-4 nucleus, and so the net effect is again the conversion of four protons into helium, with the release of 26.7 MeV. This cycle is the main one of several; in a small fraction of passages the nitrogen-15 nucleus captures a proton and forms oxygen-16, which starts a second cycle through the isotopes of oxygen and fluorine that eventually returns nitrogen-14.

Every proton capture in the cycle has to overcome the Coulomb barrier of a nucleus with six or seven protons, so the Gamow energies are between about sixty-five and ninety times that of two protons, and the cycle needs a higher temperature than the proton-proton chain. The slowest step is the capture of a proton by nitrogen-14. In a steady state each step proceeds at the same rate, so the nuclei accumulate in the form that is slowest to react, and you will show in Problem 4 that most of the carbon, nitrogen, and oxygen that takes part in the cycle ends up as nitrogen-14. The rate of the cycle is proportional to the abundance of these catalysts, and so to the fraction Z of heavier elements in the star.

The Sensitivity of the Rates to Temperature

The rate of a reaction depends on temperature mainly through the factor $e^{-\tau_{\text{G}}}$ in $\langle\sigma v\rangle$, where $\tau_{\text{G}} = 3(E_{\text{G}}/4k_{\text{B}}T)^{1/3}$, and near a given temperature it is convenient to describe the dependence by a power law, $\langle\sigma v\rangle \propto T^{\zeta}$, with

$$\zeta = \frac{d \ln \langle\sigma v\rangle}{d \ln T}. \quad (4.63)$$

Since $E_0 \propto T^{2/3}$, the rate found above goes as $T^{-2/3}e^{-\tau_{\text{G}}}$ when the slowly changing S-factor is treated as constant, and you will show in Problem 3 that

$$\zeta = \frac{\tau_{\text{G}} - 2}{3}. \quad (4.64)$$

For two protons at the centre of the Sun τ_G is about 13, and ζ is about 4. For the capture of a proton by nitrogen-14 at the same temperature τ_G is about 60, and ζ is about 20. The rate of energy generation per unit mass is the number of reactions per unit volume per unit time multiplied by the energy released in each and divided by the density. Since n_1 and n_2 are each proportional to ρ multiplied by the mass fraction of the nuclei concerned, near the temperature of the centre of the Sun

$$\epsilon_{pp} \propto \rho X^2 T^4, \quad \epsilon_{CNO} \propto \rho X Z T^{20}, \quad (4.65)$$

for the proton–proton chain and the CNO cycle respectively. At the centre of the Sun the proton–proton chain supplies nearly all of the energy. The CNO cycle, with its much steeper dependence on temperature, overtakes it at a temperature of about 2×10^7 K, which is reached at the centres of main-sequence stars somewhat more massive than the Sun.

The steep dependence of the rates on temperature has several consequences. In a star burning hydrogen by the CNO cycle, the energy release is concentrated in a small region around the centre, where the temperature is highest, and this is why, as described in Section 4.3, main-sequence stars more massive than the Sun have convective cores. Nuclear reactions switch on over a narrow range of temperature, so the central temperatures of main-sequence stars of very different masses all lie within a factor of a few of 10^7 K. Finally, a small rise in the central temperature produces a large rise in the rate of energy generation, which, combined with the negative heat capacity of a star found in Section 4.2, makes the star expand and cool until the rate falls back, so that hydrogen burning on the main sequence is stable.

4.5 The Nucleosynthesis of the Elements

Section 4.4 followed the conversion of hydrogen into helium on the main sequence. When the hydrogen at the centre is used up, the core loses energy with nothing to replace it, and by the virial theorem of Section 4.2 it therefore contracts and grows hotter, until helium can burn. In this section we follow first the burning of he-

lium, then the later stages of burning in massive stars, and finally the capture of neutrons, which forms most of the nuclei heavier than iron. The abundances of the elements in the Sun, with which the section ends, are the record that these processes have left.

Helium Burning and the Triple-Alpha Process

The simplest reactions that helium might undergo produce no stable nucleus, because no nucleus with five or eight nucleons is stable. A pair of helium-4 nuclei, which are also called alpha particles, can combine to form a nucleus of beryllium-8, but its mass exceeds that of two alpha particles by Q_{Be}/c^2 , where

$$Q_{\text{Be}} = (m_{\text{Be}} - 2m_{\alpha})c^2 \approx 92 \text{ keV}, \quad (4.66)$$

with m_{Be} and m_{α} the masses of the two nuclei. A beryllium-8 nucleus therefore splits into two alpha particles, with a lifetime of about 10^{-16} s, and helium can burn only if a third alpha particle strikes it within this time, forming carbon-12. The net reaction



is called the triple-alpha process, and Salpeter proposed it in 1952. In 1953 Hoyle argued that the observed abundance of carbon required carbon-12 to have an excited state just above the energy of a beryllium-8 nucleus and an alpha particle at rest, so that the capture would be a resonance, a reaction whose cross-section is very large over a narrow range of energies. Fowler and his colleagues soon found the state at the energy Hoyle had predicted. The states involved are drawn in Figure 53.

We find the rate of the process in two steps, the first being to find how many beryllium-8 nuclei are present. At the temperatures of helium burning, of order 10^8 K, beryllium-8 nuclei form and break up so rapidly that their number takes the value set by the equilibrium $2\text{}^4\text{He} \rightleftharpoons \text{}^8\text{Be}$, which follows from the condition for chemical equilibrium of Section 3.4 of Vol. 2 of the Fermi Notes on Physics, applied as it was to the ionisation of hydrogen in Section 3.2. Let $\mu_{\text{ch},\alpha}$ and

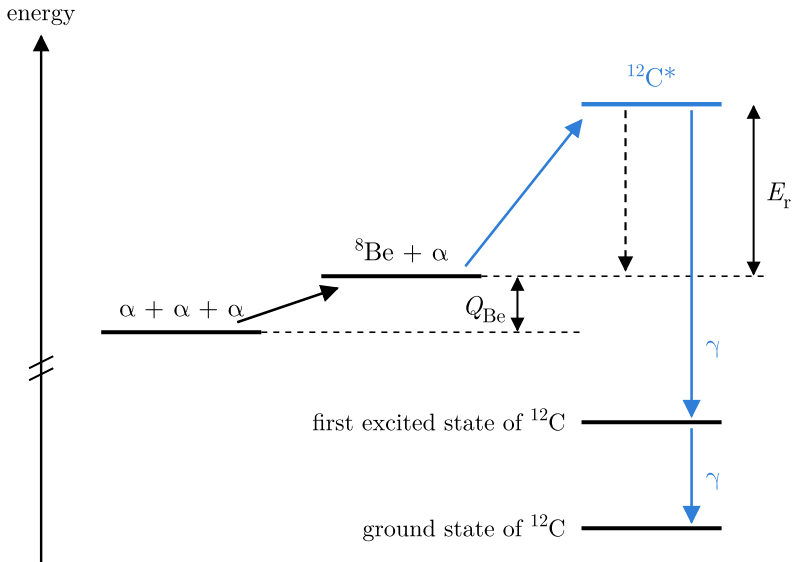


Figure 53. The energies of the states involved in the triple-alpha process. A beryllium-8 nucleus lies Q_{Be} above two separate alpha particles, and the excited state of carbon-12 lies E_r above a beryllium-8 nucleus and an alpha particle. Most nuclei formed in the excited state break up again (dashed), and a small fraction decay to the ground state of carbon-12 by emitting photons. The vertical scale is broken between the upper and the lower levels.

$\mu_{\text{ch,Be}}$ be the chemical potentials of helium-4 and beryllium-8, written μ_{ch} as in Section 3.2 because μ is the mean molecular weight in these notes. The condition reads $\mu_{\text{ch,Be}} = 2\mu_{\text{ch},\alpha}$. Helium-4 and beryllium-8 both have zero spin and so a single internal state, and when the rest energy of each nucleus is counted as part of its energy, as the ionisation energy was in Section 3.2, the chemical potential found there for a dilute gas of particles with lowest internal energy E_0 and internal partition function $\mathcal{Z}$, with $E_0 = m_x c^2$ and $\mathcal{Z} = 1$ for nuclei of mass m_x and number density n_x , becomes

$$\begin{aligned}\mu_{\text{ch},x} &= m_x c^2 + k_B T \ln \frac{n_x}{n_Q(m_x)}, \\ n_Q(m_x) &= \left(\frac{2\pi m_x k_B T}{h^2} \right)^{3/2} = \frac{1}{\lambda_{\text{th}}^3}.\end{aligned}\tag{4.68}$$

Substituting this expression for each species into $\mu_{\text{ch,Be}} = 2\mu_{\text{ch},\alpha}$, and taking the exponential of both sides, gives

$$n_{\text{Be}} = n_\alpha^2 \frac{n_Q(m_{\text{Be}})}{n_Q(m_\alpha)^2} e^{-Q_{\text{Be}}/k_B T} = n_\alpha^2 \left(\frac{h^2}{2\pi \mu_r k_B T} \right)^{3/2} e^{-Q_{\text{Be}}/k_B T}.\tag{4.69}$$

In the last expression m_{Be} has been set equal to $2m_\alpha$ in the factor in front of the exponential, which is accurate to about one part in 10^5 , and $\mu_r = m_\alpha/2$ is the reduced mass of two alpha particles, the quantity written μ in Section 5.3 of Vol. 1. At a temperature of 10^8 K and a density of 10^8 kg m^{-3} there is about one beryllium-8 nucleus for every 10^9 alpha particles.

The second step is the capture of an alpha particle by a beryllium-8 nucleus. As in Section 4.4, the number of captures per unit volume per unit time is $n_{\text{Be}} n_\alpha \langle \sigma v \rangle$, where $\sigma(E)$ is the cross-section at the energy E of relative motion and the average is over the Maxwell distribution of Section 1.2 of Vol. 2, applied to the relative velocity with the reduced mass μ_r of the pair in place of the mass of a molecule. Changing the variable from the relative speed v to $E = \frac{1}{2}\mu_r v^2$, the fraction of pairs with energy between E and $E + dE$ is

$$f(E) dE = \frac{2}{\sqrt{\pi}} \frac{E^{1/2}}{(k_B T)^{3/2}} e^{-E/k_B T} dE, \quad (4.70)$$

and $\langle \sigma v \rangle$ is the integral of $\sigma(E) v(E) f(E)$ over all energies, with $v(E) = (2E/\mu_r)^{1/2}$. At a resonance the cross-section is large only within a band of energies around the energy E_r of the excited state that is much narrower than $k_B T$, the scale over which $v(E)f(E)$ changes. That product can therefore be taken outside the integral at $E = E_r$, and

$$\begin{aligned} \langle \sigma v \rangle &\approx v(E_r) f(E_r) \int \sigma dE \\ &= \frac{2}{\sqrt{\pi}} \left(\frac{2}{\mu_r} \right)^{1/2} \frac{E_r}{(k_B T)^{3/2}} e^{-E_r/k_B T} \int \sigma dE. \end{aligned} \quad (4.71)$$

The integral of the cross-section over the resonance, which includes the probability of tunnelling through the Coulomb barrier treated in Section 4.4, is a property of the excited state and does not depend on the temperature. So is the small fraction of excited nuclei that decay to the ground state, which multiplies the rate by a constant.

Putting the two steps together, the rate at which carbon forms goes as $n_\alpha^3 T^{-3} e^{-Q_{3\alpha}/k_B T}$, where $Q_{3\alpha} = Q_{\text{Be}} + E_r \approx 0.38$ MeV is the energy of the excited state above three separate alpha particles. With $n_\alpha = \rho Y/m_\alpha$, and a fixed energy released by each carbon nucleus formed, the rate of energy generation per unit mass is

$$\epsilon_{3\alpha} \propto \rho^2 Y^3 T^{-3} e^{-Q_{3\alpha}/k_B T}. \quad (4.72)$$

The rate goes as the square of the density because three particles have to meet. You will show in Problem 1 that near a temperature T it goes as T^ζ with $\zeta = Q_{3\alpha}/k_B T - 3$, in the notation of Section 4.4, which is about 40 at 10^8 K, a far steeper dependence than that of the CNO cycle found in Section 4.4. Helium burning therefore begins over a narrow range of temperature close to 10^8 K, and it releases about a tenth as much energy per unit mass as hydrogen burning, as you will see in Problem 2.

As carbon builds up, some of it captures a further alpha particle in the reaction $^{12}\text{C} + ^4\text{He} \rightarrow ^{16}\text{O} + \gamma$. Oxygen-16 has no excited state at the energies reached in helium burning that would make this capture a resonance, so it is slow enough for much of the carbon to remain. Its rate at the energies of the Gamow peak of Section 4.4 is too small to measure in the laboratory and is known only by extrapolation, which makes it one of the least certain rates in calculations of stellar evolution. The capture of an alpha particle by oxygen-16 is slower still, since the Coulomb barrier is higher, and helium burning therefore leaves a core made mostly of carbon and oxygen.

The Later Burning Stages of Massive Stars

When the helium at the centre is used up, the core of carbon and oxygen contracts and grows hotter. Computed models show that only in stars that begin with more than about eight solar masses does it become hot enough for the next stage. In such a star each later stage repeats the pattern of the one before; a fuel burns in the core until it is exhausted, the core contracts and heats, and the next fuel ignites, while the previous one goes on burning in a shell. The stages, with the temperatures computed models give for them, are as follows:

1. Carbon burning, at about 6×10^8 K, mainly by $^{12}\text{C} + ^{12}\text{C} \rightarrow ^{20}\text{Ne} + ^4\text{He}$ and $^{12}\text{C} + ^{12}\text{C} \rightarrow ^{23}\text{Na} + \text{p}$, leaving mostly oxygen, neon, and magnesium.
2. Neon burning, at about 1.5×10^9 K, in which $^{20}\text{Ne} + \gamma \rightarrow ^{16}\text{O} + ^4\text{He}$ releases alpha particles that are captured in $^{20}\text{Ne} + ^4\text{He} \rightarrow ^{24}\text{Mg} + \gamma$, leaving mostly oxygen and magnesium.
3. Oxygen burning, at about 2×10^9 K, mainly by $^{16}\text{O} + ^{16}\text{O} \rightarrow ^{28}\text{Si} + ^4\text{He}$, leaving mostly silicon and sulphur.
4. Silicon burning, at about 3.5×10^9 K, which builds silicon up into iron, nickel, and their neighbours, which are called the iron group.

The first reaction of neon burning is an example of photodisintegration, the breaking apart of a nucleus by a photon from the tail of the Planck distribution of Section 5.6 of Vol. 2; the alpha particle

in neon-20 is bound less tightly than in oxygen-16, so neon is broken up first. Silicon burning proceeds in the same way, since two nuclei with fourteen protons each cannot cross their Coulomb barrier; photodisintegration frees protons, neutrons, and alpha particles from some nuclei, and other nuclei capture them, until the mixture approaches the equilibrium among all these reactions, which is called nuclear statistical equilibrium.

The temperature needed rises from stage to stage because the charges of the fusing nuclei rise. The exponential factor in the reaction rate of Section 4.4 depends on the temperature only through $E_G/k_B T$, and the Gamow energy E_G is proportional to $\mu_r Z_1^2 Z_2^2$, so the temperature needed rises roughly in proportion to E_G , as you will see in Problem 3.

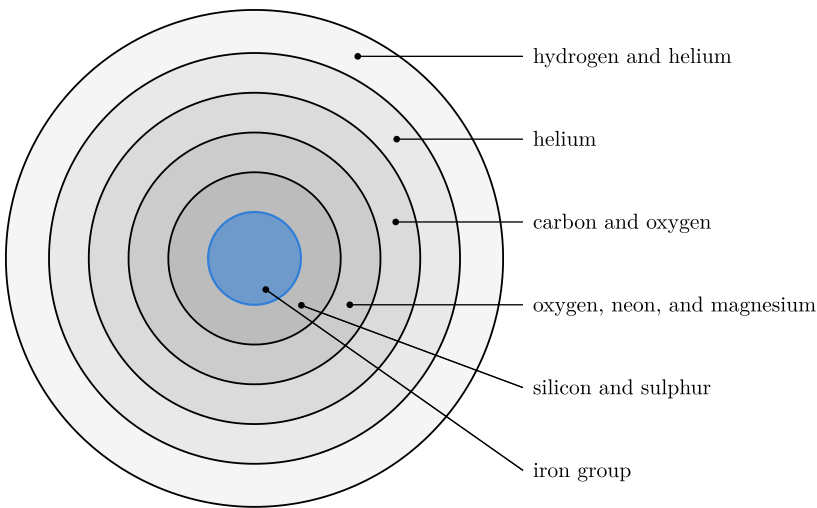


Figure 54. The layered structure of a massive star at the end of silicon burning, not drawn to scale, with each shell labelled by its main constituents; burning continues at the boundaries between the shells. The iron-group core is comparable in size to the Earth, while the envelope may reach as far as the orbit of Jupiter.

Since each fuel goes on burning in a shell after it has been exhausted in the core, a massive star at the end of silicon burning has the layered structure shown in Figure 54.

The Peak of the Binding Energy per Nucleon

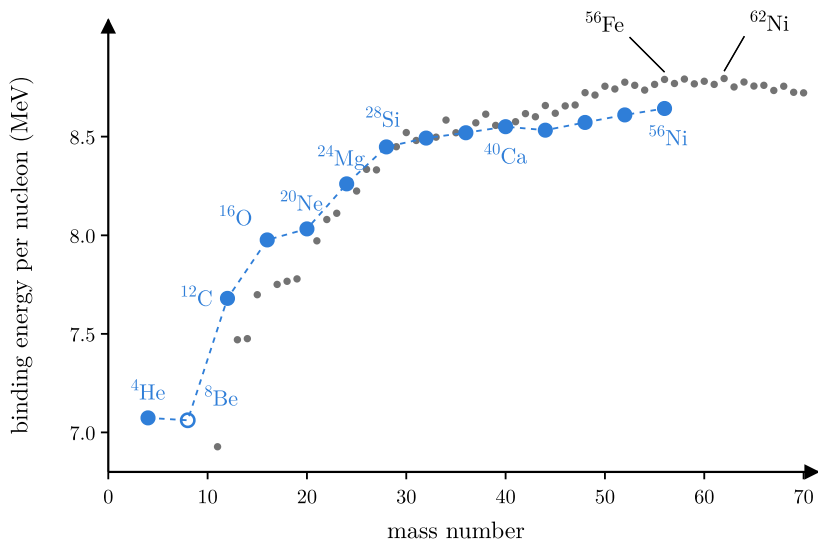


Figure 55. The measured binding energy per nucleon of the stable nuclei from the lightest that are bound nearly as tightly as helium-4 to a little beyond the iron group (grey), and of the nuclei made of whole numbers of alpha particles (accent), among which beryllium-8 (open circle) and several of the heavier ones are unstable. The vertical scale does not start at zero, so the lighter stable nuclei, which are bound less tightly than helium-4, lie below the bottom of the frame. The most tightly bound nuclei lie in the neighbourhood of iron and nickel.

The binding energy per nucleon of Section 4.4 is plotted in Figure 55 up to mass number seventy. The nuclei made of whole numbers of alpha particles lie above their neighbours, because the binding is strengthened when both the protons and the neutrons can be paired. Beryllium-8 lies slightly below helium-4, which is why it breaks up. The curve then flattens and reaches its peak near iron and nickel, where nickel-62, iron-58, and iron-56 differ by less than one part in a thousand.

The peak comes from a competition between the nuclear and the Coulomb forces. The nuclear force has a short range, so a nucleon at the surface, which has fewer neighbours, is less tightly bound than one inside, and the fraction of nucleons at the surface falls

as the nucleus grows. The Coulomb repulsion acts between every pair of protons, and in a nucleus of A nucleons, whose radius is proportional to $A^{1/3}$ and whose number of protons is roughly proportional to A , the Coulomb energy per nucleon goes as $A^2/A^{1/3}$ divided by A , that is as $A^{2/3}$. Beyond the iron group the growth of the Coulomb energy outweighs the diminishing effect of the surface, and the binding energy per nucleon falls.

The peak has two consequences for stars. The first is that fusing nuclei of the iron group into heavier ones absorbs energy, so silicon burning is the last stage that releases energy, and a core of iron-group nuclei has no further nuclear source. The second is that the later stages release little energy. Since a free proton has no binding energy and helium-4 is bound by about 7 MeV per nucleon, turning hydrogen into helium releases about 7 MeV per nucleon, while Figure 55 shows that all the later stages together release less than 2 MeV per nucleon.

The later stages are therefore brief. For a star of about twenty solar masses, computed models give about 10^7 years for hydrogen burning, 10^6 years for helium burning, 10^3 years for carbon burning, about a year each for neon and oxygen burning, and a few days for silicon burning. They are shortened further by neutrinos. Above about 5×10^8 K the most energetic photons can create pairs of electrons and positrons, and a pair occasionally annihilates into a neutrino and an antineutrino, which leave the star without interacting. From carbon burning onwards the core loses far more energy in this way than the surface radiates as light.

The Capture of Neutrons

Beyond the iron group, fusion absorbs energy and the Coulomb barriers are too high to cross. A neutron carries no charge, however, so it can be captured at low energies by a nucleus of any size, in a reaction such as $^{56}\text{Fe} + n \rightarrow ^{57}\text{Fe} + \gamma$, which raises the mass number by one and leaves the element unchanged. If the new nucleus has too many neutrons to be stable, one of them eventually turns into a proton by beta decay, $n \rightarrow p + e^- + \bar{\nu}_e$, which moves the nucleus one place up the table of elements at the same mass number.

The outcome depends on the competition between capture and de-

cay. Let n_n be the number density of free neutrons and σ the capture cross-section of a nucleus. A single nucleus captures neutrons at the rate $n_n \langle \sigma v \rangle$, so the mean time it waits for a capture is

$$t_{\text{cap}} = \frac{1}{n_n \langle \sigma v \rangle}. \quad (4.73)$$

Let t_β be the mean lifetime of an unstable nucleus against beta decay; there are two limiting cases.

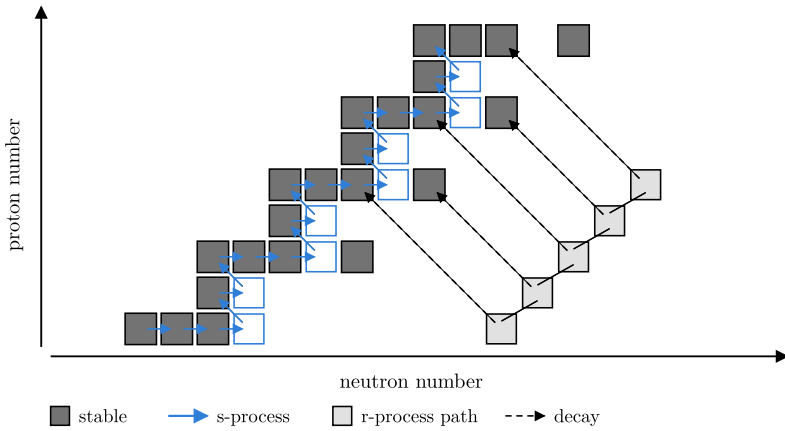


Figure 56. A schematic part of the chart of the nuclides. In the slow process (accent) an unstable nucleus decays (short diagonal steps) before it can capture another neutron, so the path keeps close to the stable nuclei (filled squares). In the rapid process the nuclei are driven far to the neutron-rich side (pale squares), and afterwards they decay along lines of constant mass number (dashed) to the first stable nucleus they meet.

When $t_{\text{cap}} \gg t_\beta$, an unstable nucleus almost always decays before its next capture, and the path of the nuclei runs along the stable nuclei, as Figure 56 shows. This is called the slow process, or s-process. It needs neutron densities of only 10^{13} to 10^{16} per cubic metre, for which t_{cap} ranges from months to many years, and the neutrons are released in the helium-burning layers of stars that have exhausted helium in their cores, by the reactions $^{13}\text{C} + ^4\text{He} \rightarrow ^{16}\text{O} + \text{n}$ and $^{22}\text{Ne} + ^4\text{He} \rightarrow ^{25}\text{Mg} + \text{n}$. The s-process ends at bismuth-209, since

a nucleus formed by capture beyond it soon emits an alpha particle and returns the path to lead.

When $t_{\text{cap}} \ll t_{\beta}$, a nucleus captures many neutrons before it decays and is driven far to the neutron-rich side, until the next neutron is so weakly bound that photodisintegration removes neutrons as fast as they are added. This is called the rapid process, or r-process, and it needs neutron densities above about 10^{26} per cubic metre. When the supply of neutrons ends, after a second or so, the nuclei decay along lines of constant mass number to stable nuclei, as Figure 56 shows. Some stable nuclei on the neutron-rich side of the s-process path, and the nuclei heavier than bismuth, such as thorium and uranium, can be reached only in this way. Where the r-process takes place is still debated; one site is the merger of two neutron stars, and in the light emitted by such a merger the formation of heavy elements was observed in 2017.

The capture cross-sections shape the abundances that the s-process makes. Let $\mathcal{N}_A$ be the number density of the nuclei of mass number A on the path, and σ_A their capture cross-section. Each nucleus of mass number A is formed when a nucleus of mass number $A - 1$ captures a neutron and destroyed when it captures one itself, so, taking the relative speed v to be the same for all the nuclei, since it is set mainly by the light neutrons,

$$\frac{d\mathcal{N}_A}{dt} = n_n v (\sigma_{A-1} \mathcal{N}_{A-1} - \sigma_A \mathcal{N}_A) . \quad (4.74)$$

Where the flow along the path is steady, $\sigma_A \mathcal{N}_A$ is the same for every mass number, and nuclei with small cross-sections accumulate. Nuclei with 50, 82, or 126 neutrons have closed shells of neutrons, as the atoms of the noble gases have closed shells of electrons, and their cross-sections are small, so the s-process makes peaks in the abundances at strontium, at barium, and at lead. The r-process reaches the same closed shells far from stability, at smaller proton numbers, and after the decays its peaks lie at smaller mass numbers, near tellurium and xenon and near platinum.

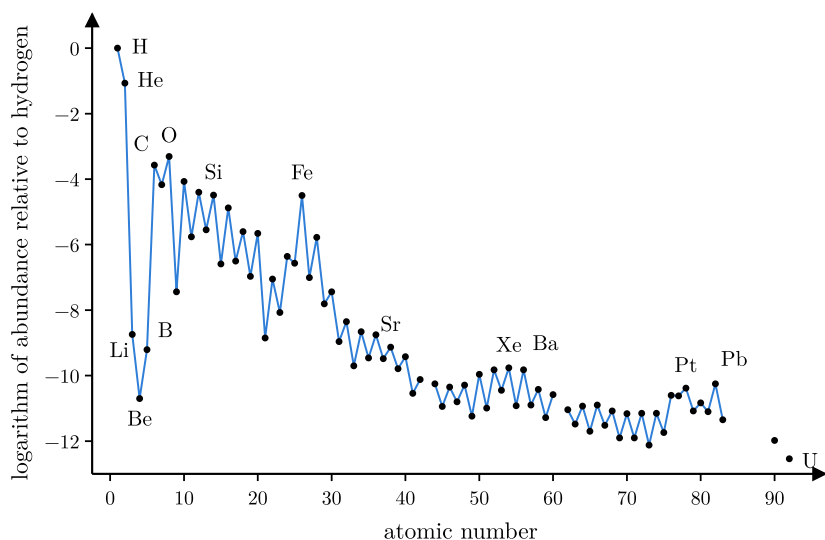


Figure 57. The abundances of the elements in the Sun, by number relative to hydrogen and on a logarithmic scale, from the solar spectrum and, for lithium, beryllium, and boron, from the most primitive meteorites. Elements with no stable isotopes and the rarest heavy elements are omitted.

The Abundances of the Elements

The abundances of the elements in the Sun are plotted in Figure 57. By mass the Sun is about seven tenths hydrogen and a little over a quarter helium, and all the other elements together make up between one and two per cent. The processes of this section account for the main features of the figure:

1. Hydrogen and most of the helium were formed in the early universe, whose thermal history is not treated in this course.
2. Lithium, beryllium, and boron are rare, because proton capture destroys them at a few million kelvin; the small amounts that exist were made mostly by collisions of fast particles with nuclei in interstellar space.
3. From carbon to calcium the most abundant nuclei are those made of whole numbers of alpha particles, the products of helium burning and the later stages.
4. Iron stands far above its neighbours, as the main product of silicon burning.
5. Beyond the iron group the abundances are many orders of magnitude lower, with the peaks of the s-process and the r-process, each of which made roughly half of the heavy nuclei, visible above a slow decline.

The list is not complete; some rare nuclei on the proton-rich side of the stable nuclei are made by processes not treated in this course. The elements made in stars reach interstellar space in stellar winds and explosions, and later generations of stars, the Sun among them, form from the enriched gas. The origin of the elements in stars was worked out in a paper of 1957 by Burbidge, Burbidge, Fowler, and Hoyle, and independently by Cameron.

4.6 The Sun

The Sun is a main-sequence star of spectral class G2, and it is the one star whose surface can be studied in detail and whose interior can be observed through the neutrinos it emits. Its mass, $M_{\odot} \approx 2.0 \times 10^{30}$ kg, follows from Kepler's third law of Section 2.1 applied to the planets, its radius, $R_{\odot} \approx 7.0 \times 10^8$ m, from its angular

diameter and distance, and its luminosity, $L_{\odot} \approx 3.8 \times 10^{26}$ W, from the flux measured above the atmosphere of the Earth. Its effective temperature is about 5800 K. Its age, about 4.6×10^9 years, is taken to be that of the oldest solids in meteorites, found by radioactive dating, as you will see in Problem 5. This section treats first the interior of the Sun and its neutrinos, then its atmosphere, and finally its magnetic activity.

The Standard Solar Model

A standard solar model is a solution of the equations of stellar structure of Section 4.3 for a star of one solar mass, followed in time from a star of uniform composition at the start of the main sequence to the present age, while the reactions of Section 4.4 turn hydrogen into helium at each radius. The model takes as given the mass, the age, and the ratio of heavy elements to hydrogen, measured from the solar spectrum and from meteorites. It also needs two quantities that cannot be found from first principles, the initial helium fraction Y and a parameter describing how efficiently convection carries energy, and these are adjusted until the model has the observed luminosity and radius at the present age. Everything else the model predicts, such as the flux of neutrinos, is then a test of it.

The general shape of the interior can be seen in a simpler model. Eddington found that the interior of the Sun is described fairly well by a polytrope of index three, one of the solutions of the Lane–Emden equation of Section 4.3 that is usually solved numerically. Figure 58 shows such a polytrope, with the luminosity computed for a rate of energy generation per unit mass that goes as ρT^4 , as the proton–proton chains do near the centre of the Sun. Half of the mass lies inside a little more than a quarter of the radius, and since the rate of energy generation depends on both the density and a high power of the temperature, nearly all of the luminosity is generated within the innermost quarter.

Standard solar models give a central temperature of about 1.6×10^7 K and a central density of about 1.5×10^5 kg m⁻³. Hydrogen makes up about seven tenths of the mass in the outer layers but only about a third at the centre, where roughly half of the original hydrogen has become helium. The proton–proton chains generate

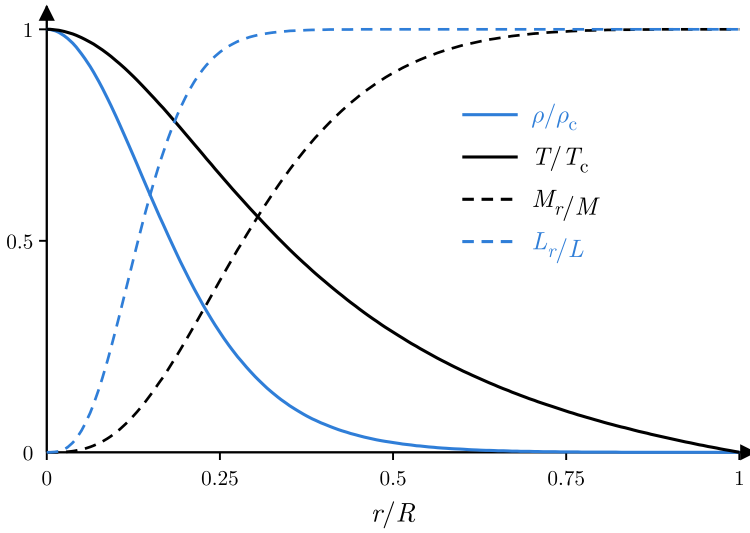


Figure 58. The density ρ , the temperature T , the mass M_r inside radius r , and the luminosity L_r crossing the sphere of radius r , each as a fraction of its central or total value, for a polytrope of the index that Eddington found to describe the Sun, with a rate of energy generation that goes as ρT^4 . The model is a rough approximation to the radiative interior of the Sun.

about ninety-nine per cent of the luminosity and the CNO cycle the rest. Radiation carries the energy out to about seven tenths of the radius, and above that, where the opacity is large enough for the Schwarzschild criterion of Section 4.3 to be met, convection carries it. The depth of the convection zone has been measured from the oscillations of the solar surface, a subject not treated in this course, and agrees with the models to about one per cent.

Solar Neutrinos

Every conversion of four protons into a helium nucleus releases two electron neutrinos, since two protons must become neutrons. Neutrinos interact so weakly with matter that almost all of them leave the Sun unabsorbed, so they show the core as it is now, whereas the energy carried by photons takes of the order of 10^5 years to diffuse out. Let Q be the energy released by each conversion, about 26.7 MeV, of which the neutrinos carry away only a few per cent. The number of conversions per second is then very nearly $L_\odot/Q$, and the flux of neutrinos at the distance d of the Earth from the Sun is

$$\Phi_{\nu_e} = \frac{2L_\odot}{4\pi d^2 Q}, \quad (4.75)$$

which is about 6×10^{14} neutrinos per square metre per second.

Different reactions produce neutrinos with different spectra, as Figure 59 shows. The first reaction of the chains, $p + p \rightarrow {}^2\text{H} + e^+ + \nu_e$, produces most of them, with energies below about 0.42 MeV, while the decay of boron-8, which occurs in only a small fraction of the chains, produces a few with energies up to about 15 MeV. Every detector responds only to neutrinos above a threshold energy, so each samples a different part of the spectrum.

The first detector, built by Davis deep in a gold mine in South Dakota to shield it from other particles, ran from the late 1960s. It was a tank of about six hundred tonnes of a cleaning fluid rich in chlorine, in which a neutrino with more than 0.81 MeV can turn chlorine-37 into argon-37 by $\nu_e + {}^{37}\text{Cl} \rightarrow {}^{37}\text{Ar} + e^-$. Every few months the argon atoms that had formed, a few tens at most, were

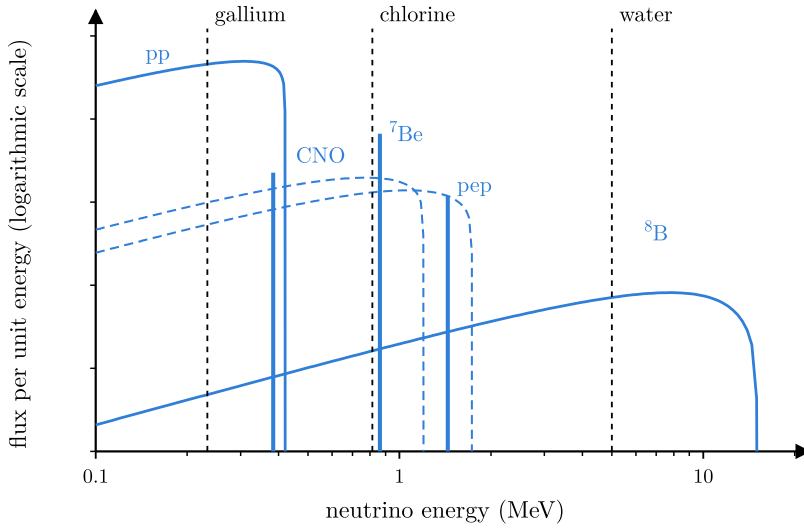


Figure 59. The spectrum of the solar neutrinos at the Earth, with the flux per unit energy on a logarithmic scale, computed from the shape of the spectrum of a beta decay and the fluxes of a standard solar model. The beryllium-7 and pep neutrinos have single energies and are drawn as vertical lines whose heights are their total fluxes. The dashed vertical lines mark the lowest energies to which three kinds of detector respond.

flushed out with helium and counted by their radioactive decays. Davis found only about a third of the rate that Bahcall had calculated from the standard solar model. From 1987 the Kamiokande detector in Japan observed solar neutrinos in a large tank of pure water. A neutrino striking an electron can knock it forward faster than light travels in water, and such an electron emits a cone of light, called Cherenkov radiation, much as a boat moving faster than the waves on a lake leaves a V-shaped wake. The light was recorded by photomultiplier tubes lining the walls; a photomultiplier tube is a vacuum tube in which a photon striking a metal surface releases an electron, which a series of electrodes multiplies into a pulse of current large enough to record. The direction of the cone showed that the neutrinos come from the Sun. Water detectors respond only to boron-8 neutrinos above about 5 MeV, and they found about half of the predicted rate. Detectors of gallium, whose threshold of 0.23 MeV admits the neutrinos of the first reaction of the chains, found a little more than half.

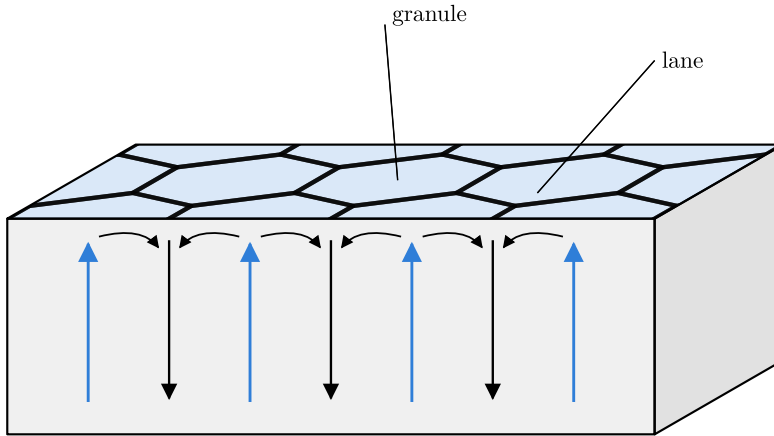
This deficit was called the solar neutrino problem. A slightly cooler core would reduce the boron-8 neutrinos strongly, since the reactions that make them depend on a high power of the temperature, but no change to the model could reproduce the deficits of all three kinds of detector together. Neutrinos come in three kinds, called flavours, the electron, muon, and tau neutrinos, and the chlorine and gallium detectors respond only to electron neutrinos. Pontecorvo had suggested that neutrinos with mass might change flavour in flight, a change called neutrino oscillation. A neutrino of definite flavour is a superposition of states of definite mass, which develop with different phases as it travels, so the probability of finding it in its original flavour oscillates with distance, much as the probability of finding the nitrogen atom of an ammonia molecule on one side of the plane of the hydrogen atoms oscillates in time in Section 4.2 of Vol. 3. In the dense interior of the Sun the change is enhanced by the scattering of the neutrinos from electrons. The theory of neutrino oscillations is not treated in this course.

The Sudbury Neutrino Observatory in Canada resolved the problem in 2001 and 2002. It held a thousand tonnes of heavy water, whose deuterium nuclei can be broken apart in two ways. The reaction $\nu_e + {}^2\text{H} \rightarrow \text{p} + \text{p} + \text{e}^-$ occurs only for electron neutrinos, while

the splitting of the deuterium nucleus into a proton and a neutron occurs equally for all three flavours. The rate of the second reaction agreed with the flux of boron-8 neutrinos predicted by the standard solar model, and the rate of the first was about a third of it. The Sun therefore produces the neutrinos the models predict, about two thirds of the high-energy ones arrive as muon or tau neutrinos, and neutrinos have mass.

The Photosphere, the Chromosphere, and the Corona

The photosphere, the layer from which most of the light escapes, was described in Section 3.4, where limb darkening was explained by the rise of temperature with depth. Seen at high resolution, the photosphere is covered with bright cells separated by darker lanes, a pattern called granulation.



hot gas rises at the centres and cooler gas sinks in the lanes

Figure 60. A block of the top of the convection zone, with the surface drawn in perspective. Hot gas rises at the centres of the cells (accent), spreads out and cools by radiating from the surface, and sinks in the darker lanes, giving the pattern of granulation.

Each bright cell, called a granule, is about a thousand kilometres across and lasts for five to ten minutes, and Doppler shifts show gas rising in the centres and sinking in the lanes at about a kilometre per second. Granules are the tops of convection cells, as Figure

60 shows, and the cooler gas in the lanes is darker by the Stefan–Boltzmann law.

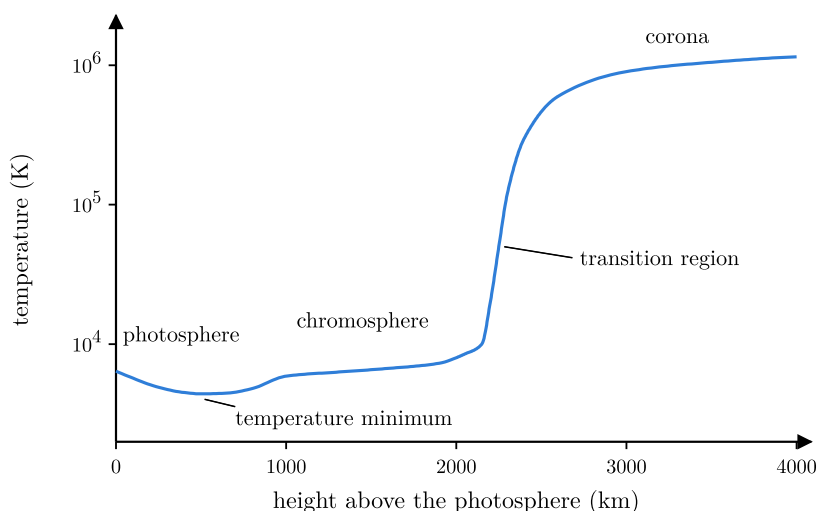


Figure 61. The temperature in the atmosphere of the Sun against height above the photosphere, drawn through the values of a representative model of the quiet Sun. The temperature falls to a minimum, rises slowly through the chromosphere, and rises steeply in the thin transition region to the temperature of the corona.

Above the photosphere the temperature falls, as it must for the absorption lines of Section 3.4 to form, to a minimum of about 4400 K a few hundred kilometres up, and then rises again, as Figure 61 shows. The layer in which it rises slowly, to about 10^4 K some two thousand kilometres up, is called the chromosphere. It is seen as a thin red ring at the edge of the disc during a total eclipse, when the Moon covers the photosphere, and its red colour comes mainly from the $H\alpha$ line of the Balmer series of Section 3.1. Above it, in a transition region a few hundred kilometres thick, the temperature climbs to between one and two million kelvin, the temperature of the corona, which extends for several solar radii and appears as a pale halo during an eclipse. It can be observed at other times with a coronagraph, a telescope in which a disc in the focal plane blocks the image of the photosphere.

The high temperature of the corona was inferred from its spectrum.

Emission lines seen during eclipses from 1869 onwards matched no known element, until around 1940 Grotrian and Edlén identified them as lines of iron and other elements that have lost many electrons, the brightest green line coming from iron that has lost thirteen of its electrons. By the Saha equation of Section 3.2, removing so many electrons requires temperatures of order 10^6 K, and the corona is also a strong source of X-rays, as gas at such temperatures must be. What heats the corona is not fully understood; the magnetic field is thought to carry energy up from the convection zone, and how that energy is released remains a subject of research.

Sunspots, the Magnetic Field, and the Solar Cycle

Sunspots are dark regions of the photosphere, each with a darker centre called the umbra and a lighter surrounding penumbra, from a thousand to many tens of thousands of kilometres across and lasting from hours to months. The umbra has a temperature of about 4000 K, so by the Stefan–Boltzmann law it emits about a quarter of the flux of the surrounding photosphere and looks dark only by contrast. In 1908 Hale measured the magnetic fields of sunspots with the Zeeman effect, in which a magnetic field splits a spectral line into components whose separations in frequency are proportional to the field, and found fields of a few tenths of a tesla.

To see why such a field matters, we work out the pressure that a magnetic field exerts on a gas. In a gas whose electrical resistance is negligible, as it nearly is in the Sun, the magnetic flux through any loop moving with the gas stays constant. This follows from Faraday’s law of Section 7.2 of Vol. 5, since a change of the flux would drive an electromotive force around the loop, and without resistance that electromotive force would drive an unlimited current. Let a bundle of field lines run straight through the gas, with a uniform field B across a cross-section of area S , so that the flux $\Phi_B = BS$ is constant. By Section 7.3 of Vol. 5 the field holds an energy $B^2/2\mu_0$ per unit volume, so the energy per unit length of the bundle is

$$U_B = \frac{B^2}{2\mu_0} S = \frac{\Phi_B^2}{2\mu_0 S}. \quad (4.76)$$

If the surrounding gas changes the area by dS , the energy changes by $dU_B = -(\Phi_B^2/2\mu_0 S^2) dS = -(B^2/2\mu_0) dS$. This equals the work $-P_B dS$ done on the bundle against a pressure P_B , so the field pushes outwards across its lines with the pressure

$$P_B = \frac{B^2}{2\mu_0}. \quad (4.77)$$

A sunspot is where a bundle of strong field lines crosses the photosphere. The total pressure must balance across its edge, so the gas pressure inside is lower than outside by $B^2/2\mu_0$, and a field of a few tenths of a tesla exerts a pressure comparable with the gas pressure of the photosphere. The field also resists the motion of the gas across it, since the gas must carry the field lines with it, and so it suppresses the convection that brings heat to the surface, which is why the surface inside the spot is cooler.

The number of sunspots rises and falls with a period of about eleven years, a variation called the solar cycle. Early in each cycle the spots appear at latitudes of thirty to forty degrees, and later ones appear closer to the equator. Spots usually come in pairs aligned roughly east and west, with fields of opposite signs, and Hale found that the leading spot of a pair has one sign in the northern hemisphere and the opposite sign in the southern. The signs reverse in the next cycle, so the full magnetic cycle lasts about twenty-two years.

The Sun does not rotate as a rigid body. The motion of sunspots across the disc shows that the equator rotates once in about 25 days and the regions near the poles once in about 35 days, a pattern called differential rotation. Babcock proposed in 1961 a description of the cycle based on it. Differential rotation stretches a field line that ran from pole to pole until it lies nearly parallel to the equator, as Figure 62 shows, and the field in the stretched bundles grows. A bundle then becomes buoyant, since by the argument above its gas has a lower pressure and so, at the same temperature, a lower density than the gas around it. Loops of it rise through the surface, and each loop crosses the photosphere twice, which accounts for the pairs of spots of opposite sign. The spreading of the fields of decaying spots then reverses the field that runs from pole to pole, and the cycle begins again with the signs reversed. The generation

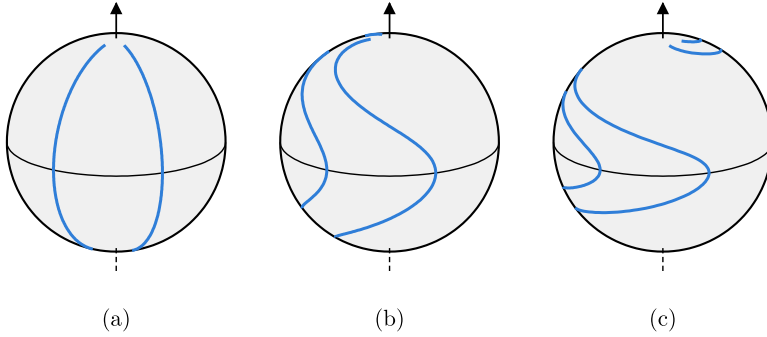


Figure 62. A magnetic field line on the surface of the Sun, drawn on a sphere whose north pole is tilted towards the viewer, wound up by differential rotation. The line runs initially from pole to pole (a); the faster rotation of the equator carries its middle ahead (b), and after several rotations it is stretched nearly parallel to the equator (c). The shapes are computed for an angular speed that goes as $1 - \delta \cos^2 \theta$, where θ is the angle from the north pole, and the part of the line on the far side is not drawn.

of a magnetic field by the motion of a conducting fluid is called a dynamo, and the theory of the solar dynamo is not treated in this course.

The Solar Wind

The corona expands continuously into interplanetary space as a flow of ionised gas called the solar wind. Biermann inferred it in 1951 from the ionised tails of comets, which always point away from the Sun, and Parker showed in 1958 that a hot corona cannot be in hydrostatic equilibrium out to large distances, for a reason you will see in Problem 2. Near the Earth the wind has a speed of about 400 km s^{-1} in its slow part and about 750 km s^{-1} in its fast part, and it holds a few million protons and electrons per cubic metre. The Sun loses about 10^9 kg each second in this way, which is too little to change its mass appreciably during its life on the main sequence. Where the wind meets the magnetic field of the Earth, charged particles guided down the field lines near the poles produce the aurorae.

4.7 Stellar Evolution on and beyond the Main Sequence

Section 4.1 recalled the life cycle of a star as it is taught at school, from a cloud of gas to a white dwarf, a neutron star, or a black hole. With the structure of a star from Sections 4.2 and 4.3 and its sources of energy from Sections 4.4 and 4.5, we can now follow that life cycle through its stages and give the reasons for them. We treat first the life of a star on the main sequence, then the evolution of a star of about one solar mass after it leaves the main sequence, and finally the different evolution of massive stars.

The Main-Sequence Lifetime

A star on the main sequence burns hydrogen in its core. Let q_{H} be the fraction of the rest energy of hydrogen that is released when it is turned into helium, $q_{\text{H}} = (4m_{\text{H}} - m_{\text{He}})/4m_{\text{H}} \approx 0.007$, where m_{He} is the mass of a helium atom. Computed models show that a star leaves the main sequence when it has burned the hydrogen in roughly the innermost tenth of its mass; let f_{c} be this fraction. The energy available is then $f_{\text{c}}XMq_{\text{H}}c^2$, and at a luminosity L the time spent on the main sequence is

$$t_{\text{MS}} = \frac{f_{\text{c}} q_{\text{H}} XM c^2}{L}. \quad (4.78)$$

For the Sun this is about 10^{10} years, so the Sun is a little under halfway through its life on the main sequence. The mass–luminosity relation found by observation in Section 2.2 gives $L \propto M^{3.5}$ roughly, over the middle of the main sequence, and if f_{c} and X are taken to be the same for all stars,

$$t_{\text{MS}} = t_{\text{MS},\odot} \left(\frac{M}{M_{\odot}} \right)^{-2.5}, \quad (4.79)$$

where $t_{\text{MS},\odot}$ is the main-sequence lifetime of the Sun. A star of ten solar masses spends only a few times 10^7 years on the main sequence, while a star of half a solar mass would spend several times 10^{10} years there. Stars of less than about $0.8 M_{\odot}$ have lifetimes longer than the

age of the universe, so none of them has yet left the main sequence. Below about $0.08 M_{\odot}$ the core never becomes hot enough to burn hydrogen, and such bodies are called brown dwarfs.

A star does not stay at one point on the main sequence. As hydrogen is turned into helium in the core, the number of particles per unit mass falls and the mean molecular weight rises. The gas pressure $P_{\text{gas}} = \rho k_B T / \mu m_H$ would then fall, and to keep the pressure that hydrostatic equilibrium requires, the core contracts slowly and grows hotter. The reaction rates rise with the temperature, and the luminosity grows. Models show that the luminosity of the Sun has increased by about thirty per cent since it arrived on the main sequence, and this slow drift is one reason why the main sequence observed in the Hertzsprung–Russell diagram is a band of some width rather than a line.

Leaving the Main Sequence

The later evolution of a star of about one solar mass is traced on the Hertzsprung–Russell diagram in Figure 63. In such a star energy is carried through the core by radiation, so the core is not mixed, and hydrogen is used up first at the centre, where the temperature is highest, and then in successively larger regions around it. When the hydrogen at the centre is exhausted, the helium core that remains has no source of energy, and hydrogen goes on burning in a shell around it. The star moves to the right across the diagram along a path that is called the subgiant branch, its envelope expanding and its surface cooling, while the helium made in the shell adds to the core.

The helium core contracts and grows hotter, and the shell around it, which lies at the edge of this hotter and more compact core, burns ever faster, so the luminosity rises steeply. At the same time the envelope swells to tens of times the radius of the Sun and becomes convective almost from the surface down to the shell, so that the star becomes a red giant, with the structure shown in Figure 64(a), and climbs the red-giant branch of Figure 63. The reason that the envelope expands while the core contracts is not a simple one, and computed models show the effect without any single argument accounting for it. When the convective envelope first reaches down

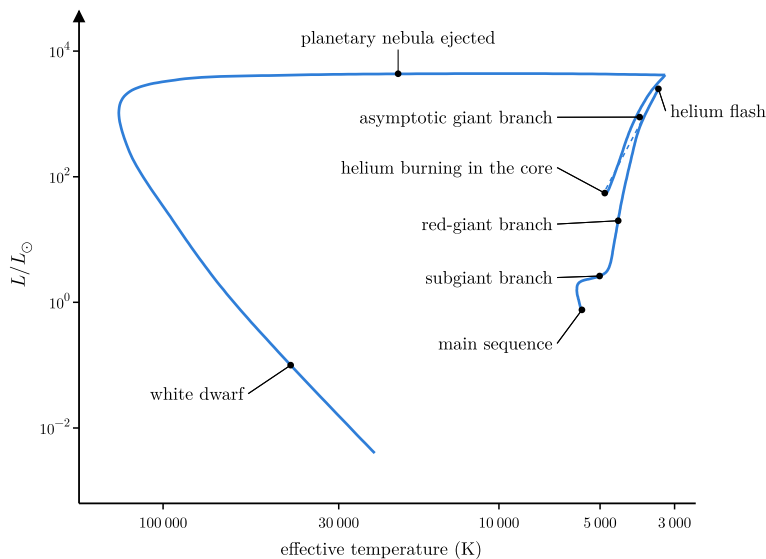


Figure 63. The evolutionary track of a star of about the mass of the Sun on the Hertzsprung–Russell diagram, drawn through representative points of computed stellar models. The dashed segment is the rapid move from the tip of the red-giant branch to the stage of helium burning in the core after the helium flash. The effective temperature increases to the left, as is usual for the diagram.

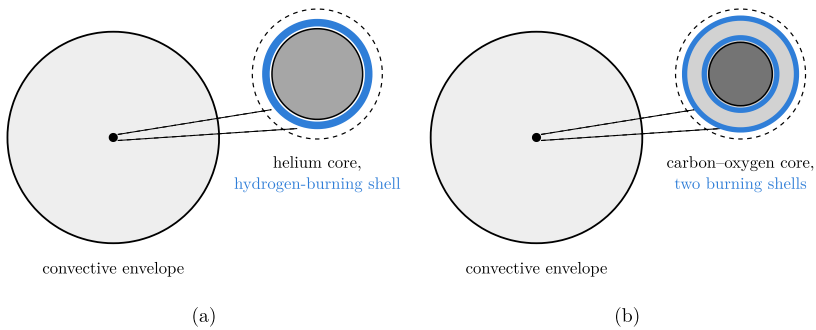


Figure 64. The structure of a red giant (a) and of a star on the asymptotic giant branch (b), not drawn to scale. In each the core, shown magnified in the inset, is a small fraction of the radius of the star and contains a large fraction of its mass. In (a) a shell of burning hydrogen (accent) surrounds a helium core. In (b) a carbon–oxygen core is surrounded by a shell of burning helium, a layer of helium, and a shell of burning hydrogen (accent).

into layers where the composition has been changed by nuclear reactions, it mixes some of the products to the surface, an event called the first dredge-up, and the surface abundances of some elements change measurably.

In a star of less than about two solar masses the helium core becomes so dense on the red-giant branch that its electrons can no longer be treated as an ideal gas. Their pressure then depends on the density but hardly at all on the temperature, a state of matter called degenerate, whose origin lies in quantum mechanics. By the time the core has grown to about $0.45 M_{\odot}$ and the star has reached a luminosity of a few thousand $L_{\odot}$ at the tip of the red-giant branch, the centre of the core has reached about 10^8 K, and helium ignites.

The Helium Flash and the Horizontal Branch

In a core of ideal gas the burning of a nuclear fuel regulates itself. Let E_{th} be the thermal energy of a core of monatomic ideal gas and E_{grav} its gravitational potential energy. By the virial theorem of Section 4.2, $2E_{\text{th}} + E_{\text{grav}} = 0$, so its total energy is $E_{\text{th}} + E_{\text{grav}} = -E_{\text{th}}$. If the reactions briefly release more energy than the core

loses, its total energy rises and its thermal energy therefore falls; the core expands and cools, and the reaction rate falls back. You will follow this argument in Problem 3.

In a degenerate core this regulation fails. The pressure does not rise when the temperature rises, so the core does not expand, and the energy released by the triple-alpha process only raises the temperature. The reaction rate, which goes as a power of the temperature of about 40, as you showed in Problem 1 of Section 4.5, rises steeply, and the burning runs away. For a few seconds the rate at which the core releases energy may reach $10^{11} L_{\odot}$, and this event is called the helium flash. The energy is absorbed in expanding the core and does not reach the surface, so the flash cannot be seen from outside. The runaway ends when the temperature has risen so far that the thermal pressure of the electrons again exceeds the pressure of degeneracy, and the core then expands and burns helium steadily.

After the flash the star settles quickly to a luminosity of about $50 L_{\odot}$, burning helium in its core and hydrogen in a shell. Stars of this kind lie on a sequence of roughly constant luminosity on the Hertzsprung–Russell diagram that is called the horizontal branch, and the position of a star along it depends on the mass of hydrogen left in its envelope, which in turn depends on how much mass the star lost on the red-giant branch and on its content of heavy elements. Stars with a composition like the Sun’s gather at the red end of the branch, in a group called the red clump, which is where the track of Figure 63 places the star. Helium burning releases about a tenth as much energy per unit mass as hydrogen burning, as you showed in Problem 2 of Section 4.5, and it proceeds at a higher luminosity, so it lasts only about 10^8 years, which is short compared with the main-sequence lifetime.

The Asymptotic Giant Branch and Planetary Nebulae

When the helium at the centre is exhausted, the core of carbon and oxygen contracts, and helium burns in a shell around it, with hydrogen burning in a second shell further out, as Figure 64(b) shows. The star climbs the Hertzsprung–Russell diagram a second time, along the asymptotic giant branch, so called because it approaches the red-giant branch from the left as it rises. The helium shell turns

out to be unstable, and it burns in a series of brief flashes, called thermal pulses, separated by intervals of thousands of years. The helium-burning layers between the shells are among the sites of the slow capture of neutrons of Section 4.5, and convection after each pulse can carry carbon and the products of the s-process up to the surface, so that some stars on the asymptotic giant branch show more carbon than oxygen in their spectra and are called carbon stars.

At this stage the star loses mass rapidly. Its envelope is so large and so weakly bound that pulsations and the pressure of radiation on dust grains that condense in its cool outer layers drive a slow, dense wind, and within some hundreds of thousands of years the envelope is almost entirely lost. The hot core that remains, still burning in its shells, moves to the left across the diagram at nearly constant luminosity, as Figure 63 shows, shrinking as its surface becomes hotter; you will see in Problem 4 how its radius depends on its temperature. When its surface temperature exceeds about 3×10^4 K, its ultraviolet radiation ionises the gas that was ejected, which then glows in the emission lines of hydrogen, oxygen, and other elements. This glowing shell is called a planetary nebula, a name given because many such nebulae look like the small discs of planets in a small telescope. The shell expands at a few tens of kilometres per second and fades within some tens of thousands of years, as it thins and the central star fades.

When the burning in the shells stops, the central star, a carbon–oxygen core of about $0.6 M_{\odot}$ for a star that began with one solar mass, cools and fades along the lower left of Figure 63 as a white dwarf. The star has lost a large fraction of its original mass. Stars that begin with up to about eight solar masses end in the same way; those with more than about two solar masses ignite helium in cores that are not degenerate, so they have no helium flash, and they leave white dwarfs of greater mass.

The Evolution of Massive Stars

Stars of more than about eight solar masses differ from the star of one solar mass followed above in three respects. Like all stars of more than about 1.2 solar masses, they burn hydrogen mainly

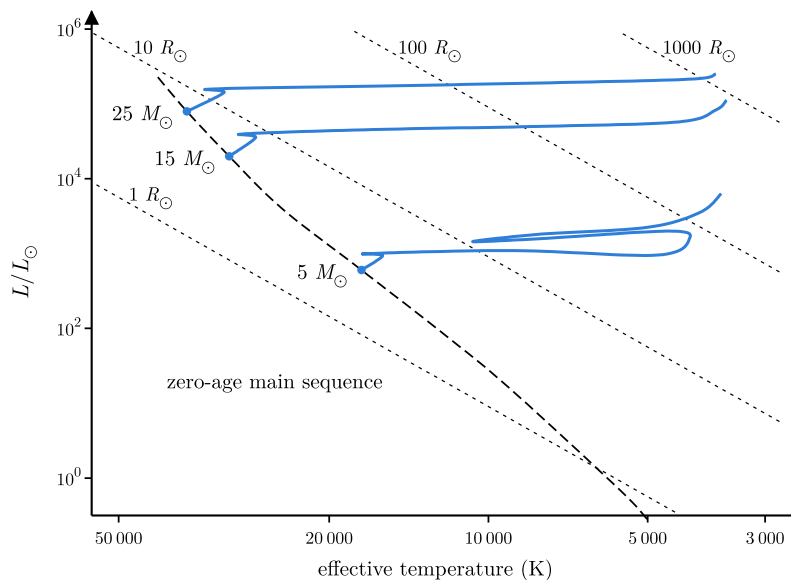


Figure 65. Evolutionary tracks of three stars of intermediate and high mass, labelled by their masses in units of the mass of the Sun, drawn through representative points of computed stellar models, together with the zero-age main sequence (dashed) and lines of constant radius (dotted). After leaving the main sequence the massive stars cross the diagram at nearly constant luminosity to become red supergiants. The track of the least massive of the three stars makes a loop towards higher effective temperatures during its burning of helium.

by the CNO cycle, whose steep dependence on temperature, found in Section 4.4, concentrates the energy generation so strongly at the centre that the core is convective and well mixed. When the hydrogen of the core is exhausted, the whole core runs out of fuel at the same time, and after a brief contraction, which appears as a small hook at the top of the main sequence, the star burns hydrogen in a shell. The cores of these massive stars are never degenerate while they burn helium, so they have no helium flash. And their cores go on to burn carbon, neon, oxygen, and silicon in turn, as described in Section 4.5.

Figure 65 shows the tracks of massive stars. After leaving the main sequence, each crosses the diagram from left to right at nearly constant luminosity, since the luminosity is set by the core while the envelope expands, and the star becomes a red supergiant with a radius of several hundred to a thousand solar radii. Stars of intermediate mass make loops towards higher effective temperatures, to the left of the diagram, and back again while they burn helium, as the track of five solar masses does in Figure 65, and these loops carry them across the region of the diagram where the Cepheid variables of Section 1.1 are found. The loss of mass in winds becomes more important as the mass of the star rises. Stars that begin with more than about twenty-five solar masses can lose their entire hydrogen envelope, exposing layers rich in helium, carbon, or nitrogen, and such stars, whose spectra show broad emission lines of these elements, are called Wolf-Rayet stars.

At the end of silicon burning a massive star has the layered structure of Figure 54, with a core of iron-group nuclei that can release no further energy by fusion. As Section 4.1 recalled from school, such a star ends its life in the explosion called a supernova, leaving a neutron star or a black hole.

4.8 Star Clusters and Their Ages

The evolution described in Section 4.7 takes millions to thousands of millions of years, far too long to be followed in any one star. A star cluster, a group of stars that formed together from the same cloud of gas, provides the test that a single star cannot. Its stars have, to a good approximation, the same age, the same initial composition,

and the same distance from us, and they differ only in their masses. The Hertzsprung–Russell diagram of a cluster therefore shows stars of many masses at one age, and comparing clusters of different ages shows how the diagram changes with time. Below we describe first the two main kinds of cluster, then find the age of a cluster from its diagram, and finally find its distance by main-sequence fitting.

Open and Globular Clusters

Clusters come in two main kinds. An open cluster contains from a hundred to a few thousand stars, loosely grouped over a few parsecs, and it lies in the disc of the Galaxy, the flattened system of stars and gas in which the Sun lies. Open clusters have compositions similar to that of the Sun, and most are young, with ages from a few million to a few thousand million years; the Pleiades and the Hyades are two of the nearest. Open clusters do not last long as groups, because the gravity of passing clouds of gas and of the Galaxy as a whole gradually pulls them apart. A globular cluster contains from tens of thousands to about a million stars, packed into a nearly spherical group a few tens of parsecs across, and the globular clusters of the Galaxy are spread through a roughly spherical region centred on the centre of the Galaxy, extending far above and below the disc. Globular clusters are old, and most of them contain between about a hundredth and a third of the proportion of heavy elements found in the Sun.

The diagram of a cluster is usually measured as the colour–magnitude diagram introduced in Section 1.1, in which the apparent magnitude m_V is plotted against the $B - V$ colour index $m_B - m_V$. The colour index is a measure of the effective temperature, and since all the stars are at the same distance, their apparent magnitudes differ from their absolute magnitudes by the same amount. The colour–magnitude diagram of a cluster is therefore its Hertzsprung–Russell diagram with the scales changed.

Isochrones and the Turn-Off Point

A curve on the Hertzsprung–Russell diagram occupied by stars of the same age and composition but different masses is called an isochrone. The diagram of a cluster traces out one isochrone, and

we can work out the main features of an isochrone from the lifetimes of Section 4.7. At the age t of the cluster, every star whose main-sequence lifetime is shorter than t has left the main sequence, while every star whose lifetime is longer is still on it. By the lifetime of Section 4.7, $t_{\text{MS}} = t_{\text{MS},\odot}(M/M_{\odot})^{-2.5}$, the stars leaving the main sequence at the age t have the mass

$$M_{\text{to}} = M_{\odot} \left(\frac{t}{t_{\text{MS},\odot}} \right)^{-0.4}, \quad (4.80)$$

and the main sequence of the cluster ends at the point occupied by stars of this mass, which is called the turn-off point. With $L \propto M^{3.5}$, the luminosity of the stars at the turn-off is

$$L_{\text{to}} = L_{\odot} \left(\frac{t}{t_{\text{MS},\odot}} \right)^{-1.4}. \quad (4.81)$$

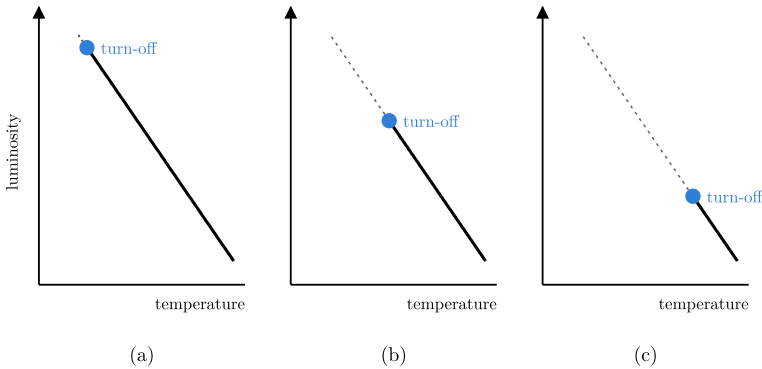


Figure 66. The main sequences of clusters of three ages, increasing from (a) to (c), drawn from $L \propto M^{3.5}$ and $R \propto M^{0.8}$ with the effective temperature increasing to the left. Stars more massive than the turn-off mass have left the main sequence (dotted), and the turn-off point (accent) moves down the main sequence as the cluster ages. The giant branches are omitted.

As the cluster ages, the turn-off point moves down the main sequence, as Figure 66 shows. The stars above the turn-off, which

have left the main sequence, lie on the subgiant and giant branches of Section 4.7. Since these later stages are brief compared with the life on the main sequence, the stars now on the giant branches have masses only slightly greater than the turn-off mass, as you will show in Problem 3, and the part of the diagram above the turn-off is populated by stars of almost the same mass, each at a different point of an evolutionary track of the kind traced in Figure 63.

The lower end of the main sequence of a young cluster is also incomplete, for a different reason. A newly formed star contracts towards the main sequence on the Kelvin–Helmholtz time scale of Section 4.2, $t_{\text{KH}} \approx 3GM^2/10RL$, until its centre is hot enough to burn hydrogen. With $L \propto M^{3.5}$ and $R \propto M^{0.8}$, this time goes as $M^{2-0.8-3.5} = M^{-2.3}$, so it is longest for the stars of lowest mass. In a cluster a few million years old, the stars of low mass are still contracting and lie above and to the right of the main sequence, while the massive stars have already reached it. The point at which the stars join the main sequence moves to lower masses as the cluster ages, and it gives a second measure of the age of a young cluster.

Ages of Clusters

Inverting the relation for L_{to} gives the age of a cluster from the luminosity of its turn-off,

$$t = t_{\text{MS},\odot} \left(\frac{L_{\text{to}}}{L_{\odot}} \right)^{-5/7}. \quad (4.82)$$

This relation shows the idea of the method, but it rests on the approximate relations $L \propto M^{3.5}$ and $t_{\text{MS}} \propto M/L$, which ignore the dependence of the luminosity on composition and the variation of the burned fraction f_c with mass. In practice astronomers usually find the age of a cluster by comparing its colour–magnitude diagram with isochrones computed from stellar models of the same composition, and choosing the age whose isochrone best matches the turn-off and the subgiant branch. The ages found in this way range from a few million years for the youngest open clusters, whose diagrams still contain hot, luminous stars of spectral class O and B at the top of the main sequence, through about 10^8 years for

the Pleiades and about 6×10^8 years for the Hyades, to several thousand million years for the oldest open clusters.

The age found from the turn-off depends on the luminosity of the turn-off, and so on the distance of the cluster, since the luminosity is found from the apparent brightness by the inverse-square law. An error in the distance modulus $m - \mathcal{M}$ of Section 1.1 therefore leads to an error in the age; you will show in Problem 2 that an error δ in the distance modulus changes the age found by a fraction $-(2 \ln 10/7)\delta$, so that an error of a tenth of a magnitude changes the age by several per cent. The distance of the cluster must be found first.

An Example: The Age from the Turn-Off Magnitude

As an example, let us combine the relation for the age with the magnitude scale of Section 1.1, for a cluster whose distance modulus $m - \mathcal{M}$ is known. Let m_{to} be the apparent bolometric magnitude of the turn-off, and $\mathcal{M}_{\odot}$ the absolute bolometric magnitude of the Sun. The absolute magnitude of the turn-off is $\mathcal{M}_{\text{to}} = m_{\text{to}} - (m - \mathcal{M})$, and since a difference of five magnitudes is a factor of a hundred in luminosity, as Section 1.1 showed,

$$\frac{L_{\text{to}}}{L_{\odot}} = 10^{-0.4(\mathcal{M}_{\text{to}} - \mathcal{M}_{\odot})}. \quad (4.83)$$

Substituting this ratio into $t = t_{\text{MS},\odot}(L_{\text{to}}/L_{\odot})^{-5/7}$, and using $0.4 \times 5/7 = 2/7$, gives

$$t = t_{\text{MS},\odot} 10^{2[m_{\text{to}} - (m - \mathcal{M}) - \mathcal{M}_{\odot}]/7}. \quad (4.84)$$

A turn-off one magnitude fainter therefore corresponds to an age larger by a factor $10^{2/7}$, a little less than two. The age is thus a steep function of the turn-off magnitude, and a fixed error in the magnitude, whether it comes from the photometry or from the distance, gives a fixed fractional error in the age, which is the result you will find in Problem 2.

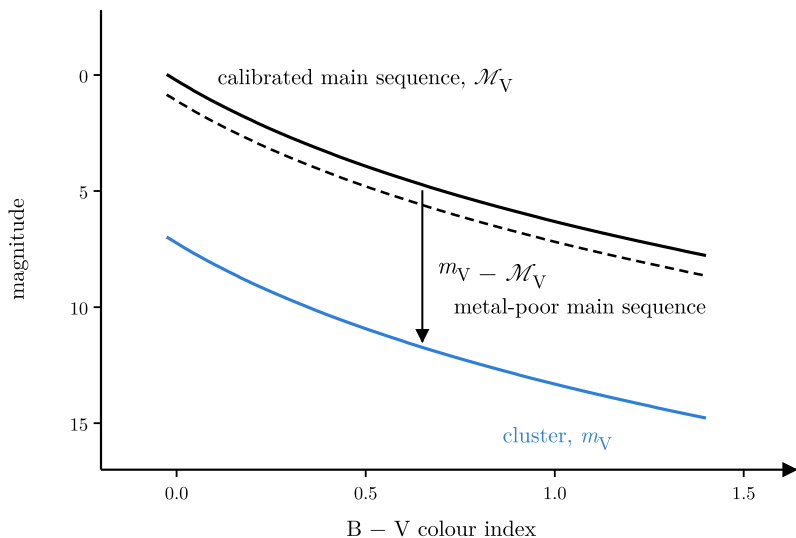


Figure 67. Main-sequence fitting in a colour-magnitude diagram. The calibrated main sequence gives the absolute magnitude $\mathcal{M}_V$ of stars of each colour, and the main sequence of a cluster of the same composition, plotted in apparent magnitude m_V , has the same shape shifted downwards. The shift at any colour is the distance modulus. The dashed curve is the calibrated main sequence of stars poorer in heavy elements, which lies below the solar one at every colour; fitting a metal-poor cluster to the solar sequence would overestimate its distance. All three sequences are drawn from $L \propto M^{3.5}$ and $R \propto M^{0.8}$, with a relation between colour and effective temperature, and with the bolometric correction taken to be zero.

Main-Sequence Fitting

Section 1.1 described main-sequence fitting as a rung of the distance ladder and drew it in Figure 4, and we return to it here to say how the calibrated main sequence is obtained and what limits the method. Main-sequence stars of the same colour and composition have the same luminosity, and so the same absolute magnitude. A calibrated main sequence, which gives the absolute magnitude $\mathcal{M}_V$ of main-sequence stars of each colour, is found from nearby stars whose distances are known from their parallaxes. The Gaia space telescope, which has measured the positions of more than a thousand million stars repeatedly over several years, has provided such parallaxes for many thousands of nearby main-sequence stars and for the stars of the nearest clusters, such as the Hyades.

For a distant cluster, the main sequence in its colour–magnitude diagram has the same shape as the calibrated main sequence, and at every colour its apparent magnitude exceeds the absolute magnitude by the same amount, as Figure 67 shows. That amount is the distance modulus of Section 1.1,

$$m_V - \mathcal{M}_V = 5 \log_{10} \frac{d}{10 \text{ pc}}, \quad (4.85)$$

and the distance of the cluster follows from it. Fitting the whole sequence, rather than comparing a single star, averages over the scatter of the individual stars, some of which are unresolved binaries that appear brighter than single stars of the same colour. The method assumes that the cluster and the calibrating stars have the same composition. At a given colour, a star with fewer heavy elements is fainter than a star of the Sun’s composition, and a metal-poor globular cluster must therefore be fitted to a main sequence of metal-poor nearby stars, of which there are relatively few. The method also assumes that no dust lies between the cluster and us to dim and redden its light. Once the distance is known, the absolute magnitude of the turn-off gives its luminosity, and the age follows.

Globular Clusters and the Age of the Galaxy

The colour–magnitude diagrams of globular clusters show every stage of Section 4.7 for stars of about $0.8 M_{\odot}$: a main sequence ending at a faint turn-off, a subgiant branch, a red-giant branch, a horizontal branch, and an asymptotic giant branch. The horizontal branches of many globular clusters extend far towards higher effective temperatures, to the left of the diagram, because their stars, poor in heavy elements and having lost mass on the red-giant branch, have thin envelopes of hydrogen. When the turn-offs are compared with isochrones computed for the composition of each cluster, the oldest globular clusters are found to be between about 1.2×10^{10} and 1.3×10^{10} years old, with an uncertainty of about a thousand million years, which comes partly from the distances and partly from the stellar models.

These are the oldest stars in the Galaxy whose ages can be measured with this precision, and the Galaxy can be no younger than the oldest stars in it. The Galaxy may be somewhat older than its oldest globular clusters, since the gas from which they formed first had to gather in the gravitational field of the forming Galaxy, and the ages of the globular clusters therefore set a lower limit on the age of the Galaxy.

Problems

§4.1 Stars and the Hertzsprung–Russell Diagram at School

1. The absolute bolometric magnitude of a star is $\mathcal{M}_{\text{bol}} = \mathcal{M}_{\text{bol},\odot} - 2.5 \log_{10}(L/L_{\odot})$. Show that on a diagram of $\mathcal{M}_{\text{bol}}$ against $\log_{10} T_{\text{eff}}$ the lines of constant radius are straight, with slope -10 , and find the change in $\mathcal{M}_{\text{bol}}$ when the radius of a star is multiplied by a factor f at a fixed effective temperature.
2. A white dwarf and a red giant have the same luminosity, and the effective temperature of the white dwarf is q times that of the giant. Find the ratio of their radii, and the ratio of their mean densities if their masses are equal.
3. Show that the lines of constant radius in Figure 39 are separated

vertically by two decades in luminosity for each decade in radius, and horizontally by half a decade in effective temperature for each decade in radius.

4. Along the main sequence let $L \propto M^a$ and $R \propto M^b$. Show that the mean density of a main-sequence star goes as M^{1-3b} and the acceleration due to gravity at its surface as M^{1-2b} , and say how each changes up the main sequence when b lies between one half and one.

§4.2 Hydrostatic Equilibrium and the Virial Theorem

1. Consider a star of uniform density ρ , mass M , and radius R . Find the pressure $P(r)$ inside it, and show that its central pressure is $3GM^2/8\pi R^4$. If the gas is ideal with a uniform mean molecular weight, find how its temperature depends on r .
2. The density of a model star falls linearly from ρ_c at the centre to zero at the surface, $\rho = \rho_c(1 - r/R)$. Find its mass M in terms of ρ_c and R , and show that its central pressure is $5GM^2/4\pi R^4$, which is 10/3 times that of the uniform star of Problem 1 with the same mass and radius. Figure 42 compares the pressures inside these two stars.
3. For the uniform star of Problem 1, show that the ratio of the pressure of radiation to the pressure of the gas at the centre is $(\pi a_{\text{rad}} G^3/18)(\mu m_{\text{H}}/k_B)^4 M^2$.
4. Find the mean molecular weight of a fully ionised gas of pure helium, and of a neutral gas of hydrogen and helium atoms with mass fractions X and $Y = 1 - X$.
5. A uniform sphere of gas of mass M and radius R is initially at rest, and its pressure is suddenly removed. Show that each shell of the sphere keeps the same mass inside it as it falls, and use Kepler's third law for a very eccentric orbit to show that the whole sphere collapses to the centre in the time $(\pi/2)\sqrt{R^3/2GM}$.

§4.3 Energy Transport and the Equations of Stellar Structure

1. Show that in a region of a star in which the gas is ideal, with a uniform mean molecular weight, the adiabatic temperature gradient is $(dT/dr)_{\text{ad}} = -(1 - 1/\gamma) \mu m_{\text{H}} GM_r / k_B r^2$.
2. Suppose that everywhere in a star the pressure of the gas is a fixed fraction β of the total pressure P , so that $P_{\text{rad}} = (1 - \beta)P$. Show that $P = K_{\text{p}} \rho^{4/3}$, so that the star is a polytrope with $n_{\text{p}} = 3$, and find K_{p} in terms of β , μ , and the constants.
3. For the polytrope with $n_{\text{p}} = 1$, show that the radius $R = \pi \sqrt{K_{\text{p}}/2\pi G}$ does not depend on the central density or the mass, and that the central density is $\pi^2/3$ times the mean density.
4. Show that for a polytrope of index n_{p} with a given constant K_{p} , the radius and the mass are related by $R \propto M^{(1-n_{\text{p}})/(3-n_{\text{p}})}$, and find how the radius of a fully convective star with a given K_{p} changes with its mass.
5. Repeat the estimate of the mass-luminosity relation with Kramers' opacity $\kappa = \kappa_0 \rho T^{-7/2}$, and show that $L \propto \mu^{15/2} M^{11/2} R^{-1/2}$.

§4.4 Thermonuclear Reactions in Stars

1. Taking the mass of a nucleus of mass number A to be Am_{p} , find the ratio of the Gamow energy for a proton and a carbon-12 nucleus to that for two protons. By what factor is the energy E_0 of the Gamow peak larger for the first reaction at the same temperature?
2. Keep the lower limit r_{n} in the integral for the tunnelling exponent. Show that for $r_{\text{n}} \ll r_{\text{t}}$ the exponent is $\sqrt{E_{\text{G}}/E} [1 - (4/\pi) \sqrt{r_{\text{n}}/r_{\text{t}}}]$ to first order in $\sqrt{r_{\text{n}}/r_{\text{t}}}$, and explain why the correction increases the tunnelling probability.
3. Using $\langle \sigma v \rangle \propto T^{-2/3} e^{-\tau_{\text{G}}}$ with $\tau_{\text{G}} \propto T^{-1/3}$, show that $\zeta = (\tau_{\text{G}} - 2)/3$.
4. In the CNO cycle of Figure 52, suppose that the cycle runs in a steady state, so that each nucleus is destroyed at the same

rate as it is formed. Show that the numbers of carbon-12 and nitrogen-14 nuclei per unit volume are in the ratio $\langle\sigma v\rangle_{14}/\langle\sigma v\rangle_{12}$, where the subscripts label the proton captures by nitrogen-14 and carbon-12, and explain why most of the catalyst ends up as nitrogen-14.

5. Show that at the energy E_0 of the Gamow peak the Maxwell factor is $e^{-E_0/k_B T}$ and the tunnelling factor is $e^{-2E_0/k_B T}$, so that two thirds of the exponent τ_G comes from tunnelling. Show also that the mean relative kinetic energy of the pairs is $\frac{3}{2}k_B T$, and compare it with E_0 .

§4.5 The Nucleosynthesis of the Elements

1. Starting from $\epsilon_{3\alpha} \propto \rho^2 Y^3 T^{-3} e^{-Q_{3\alpha}/k_B T}$, show that $d \ln \epsilon_{3\alpha} / d \ln T = Q_{3\alpha}/k_B T - 3$, and hence that a small fractional rise δ in the temperature raises the rate by a fraction of approximately $(Q_{3\alpha}/k_B T - 3)\delta$.
2. Let q_H be the fraction of the rest energy released when four hydrogen atoms become one helium atom, and q_{He} the fraction released when three alpha particles become one carbon-12 nucleus. Write each in terms of the masses m_H and m_{He} of the hydrogen and helium atoms and m_α and m_C of the helium and carbon nuclei, and show that the energy released per unit mass of fuel in helium burning is smaller than in hydrogen burning by the factor q_{He}/q_H .
3. Taking the mass of a nucleus of mass number A to be $A m_u$, and using $E_G \propto \mu_r Z_1^2 Z_2^2$, show that at a fixed value of $E_G/k_B T$ the temperature needed for oxygen burning is $(4/3)^5$ times that needed for carbon burning.
4. Show that where the s-process flow is steady, $\sigma_A \mathcal{N}_A$ is the same for all mass numbers on the path. A nucleus with a closed shell of neutrons has a cross-section smaller by a factor s than those of its neighbours on the path. By what factor is its abundance larger?
5. An unstable nucleus on the path has lifetime t_β against beta decay and captures neutrons at the rate $n_n \langle\sigma v\rangle$. Find the neutron density at which capture and decay are equally likely, and show

that the probability that the nucleus decays before it captures a neutron is $1/(1 + t_\beta/t_{\text{cap}})$.

§4.6 The Sun

1. Explain the factor of two in $\Phi_{\nu_e} = 2L_\odot/4\pi d^2 Q$. If a fraction f_{loss} of the energy released by each conversion is carried away by the neutrinos, and so does not contribute to $L_\odot$, show that the flux becomes $2L_\odot/[4\pi d^2 Q(1 - f_{\text{loss}})]$.
2. The corona is an isothermal ideal gas at temperature T with mean molecular weight μ , held by the gravity of the Sun alone. Using the equation of hydrostatic equilibrium, show that its pressure is $P(r) = P_0 \exp[(GM_\odot \mu m_H / k_B T)(1/r - 1/r_0)]$, where P_0 is the pressure at the base of the corona at radius r_0 . Find the pressure as $r \rightarrow \infty$, and explain why a static corona is impossible if this limit is much larger than the pressure of the interstellar gas.
3. A sunspot has the same temperature as the photosphere around it, where the gas pressure is P_{out} . Show that the density inside is smaller than outside by the factor $1 - B^2/2\mu_0 P_{\text{out}}$, and find the largest field for which pressure balance is possible.
4. The angular speed of the solar surface at angle θ from the north pole is $\Omega(\theta) = \Omega_{\text{eq}}(1 - \delta \cos^2 \theta)$, as in Figure 62. Find the time after which the equator has made one more rotation than the gas at angle θ , and show that near the pole this time is $2\pi/\delta\Omega_{\text{eq}}$.
5. A sample contains N_p nuclei of a radioactive parent with half-life $t_{1/2}$ and N_d nuclei of its daughter, none of which were present when the sample formed. Show that the age of the sample is $t = (t_{1/2}/\ln 2) \ln(1 + N_d/N_p)$.

§4.7 Stellar Evolution on and beyond the Main Sequence

1. Show that with $L \propto M^{3.5}$ the main-sequence lifetime goes as $M^{-2.5}$. Find the mass of the stars whose main-sequence lifetime equals a time t , in terms of $M_\odot$, t , and $t_{\text{MS},\odot}$.
2. A red giant of luminosity L burns hydrogen in a shell at the edge of a helium core of mass M_c , and the gas above the shell

has hydrogen fraction X . Show that the core grows at the rate $dM_c/dt = L/(Xq_H c^2)$, and find the time for the core to grow from M_1 to M_2 at constant luminosity.

3. For a core of monatomic ideal gas in which the nuclear reactions release energy at the rate L_{nuc} while the core loses energy at the rate L , use the virial theorem to show that $dE_{\text{th}}/dt = L - L_{\text{nuc}}$. Explain why this makes the burning stable when the rate of the reactions rises with the temperature.
4. After leaving the asymptotic giant branch, the remaining core of a star moves across the Hertzsprung–Russell diagram at constant luminosity. Show that its radius goes as T_{eff}^{-2} , and find by what factor the radius falls while the effective temperature rises by a factor k .
5. A planetary nebula is a shell of mass M_s and constant thickness Δr , whose radius grows as $v_s t$. Show that its number density goes as t^{-2} , and write it in terms of M_s , Δr , v_s , t , and the mean molecular weight μ .

§4.8 Star Clusters and Their Ages

1. Derive $t = t_{\text{MS},\odot}(L_{\text{to}}/L_{\odot})^{-5/7}$ from $t_{\text{MS}} \propto M^{-2.5}$ and $L \propto M^{3.5}$, and find how the effective temperature of the turn-off depends on t if $R \propto M^{0.8}$.
2. The luminosity of the turn-off of a cluster is found from its apparent brightness and a distance modulus that is in error by a small amount δ . Show that the luminosity found is in error by the fraction $0.4 \ln 10 \delta$, and that the age found is in error by the fraction $-(2 \ln 10/7)\delta$.
3. The stars of a cluster leave the main sequence at the age t with mass M_{to} , and after leaving it each spends a time t_g on the giant branches, with $t_g \ll t$. Show that the stars now on the giant branches have masses between M_{to} and $M_{\text{to}}(1 + 0.4 t_g/t)$.
4. The main sequence of a cluster lies at apparent magnitude m_V at a colour where the calibrated main sequence gives $\mathcal{M}_V$. Find the distance of the cluster. If the stars of the cluster are poorer in heavy elements than the calibrating stars, so that at this colour

their true absolute magnitude is larger by Δ , show that the distance found is too large by the factor $10^{\Delta/5}$.

Stellar Death and Compact Objects

5.1 Degenerate Matter and White Dwarfs

White dwarfs were introduced in Section 4.1 as the faint, hot stars that lie below the main sequence. The best studied of them is Sirius B, the faint companion of Sirius, whose mass, found from the orbit of the binary as in Section 2.2, is about $1.0 M_{\odot}$, and whose radius, found from $L = 4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4$, is about $0.008 R_{\odot}$, a little less than the radius of the Earth. Its mean density is therefore about $3 \times 10^9 \text{ kg m}^{-3}$.

A white dwarf has no nuclear source of energy, and it radiates the heat stored in its interior. By the virial theorem of Section 4.2, a star supported by the pressure of an ideal gas contracts as it loses energy, but white dwarfs of the same mass are observed to have the same radius at every temperature, so the pressure that supports them does not fall as they cool. In this section we show that the pressure of the electrons at such densities depends hardly at all on the temperature, because of the Pauli exclusion principle, and we then use it to find the radius of a white dwarf, the largest mass that a white dwarf can have, and the rate at which a white dwarf cools.

The Pauli Exclusion Principle

Electrons, protons, and neutrons each have spin one-half, as described in Section 4.2 of Vol. 3 of the Fermi Notes on Physics, so that S_z has the two values $\pm\frac{1}{2}\hbar$. The Pauli exclusion principle states that no two identical particles of spin one-half can occupy the same quantum state. A state here means a state of motion in space together with a value of the spin quantum number $m_s = \pm\frac{1}{2}$ that fixes $S_z = m_s\hbar$, so each state of motion in space can hold at most two electrons, one with spin up and one with spin down. Pauli proposed the principle in 1925 to account for the arrangement of the electrons in atoms; as an example, the two electrons of a helium atom in its ground state occupy the 1s level, in the notation of Section 5.2 of Vol. 3, with opposite spins, and the third electron of a lithium atom must occupy a level with $n = 2$. The connection of the principle with the quantum mechanics of identical particles is not treated in this course.

In a classical gas at low temperature every particle would settle into the state of lowest energy. In a gas of electrons that state can hold only two of them, so the electrons fill the states in pairs from the lowest upwards, and even at zero temperature most of them have large energies. A gas in this condition is called degenerate.

Counting the States of Electrons in a Box

Let N_e electrons be held in a cube of side ℓ_b and volume $V = \ell_b^3$, with the forces between them neglected, and let us find the state of lowest energy of the whole gas, which is its state at zero temperature. The states of one electron in the box were found in Section 5.1 of Vol. 3. Each is labelled by three positive integers, which Vol. 3 writes n_1 , n_2 , and n_3 and which we write q_1 , q_2 , and q_3 , since n is a number density in these notes. Along each axis the state is a standing wave with a whole number of half-wavelengths across the box, and by the de Broglie relation $p = h/\lambda$ of Section 1.1 of Vol. 3 the component of the momentum along that axis has the magnitude $\pi\hbar q_j/\ell_b$. The magnitude of the momentum is therefore

$$p = \frac{\pi\hbar}{\ell_b} q, \quad q = \sqrt{q_1^2 + q_2^2 + q_3^2}, \quad (5.1)$$

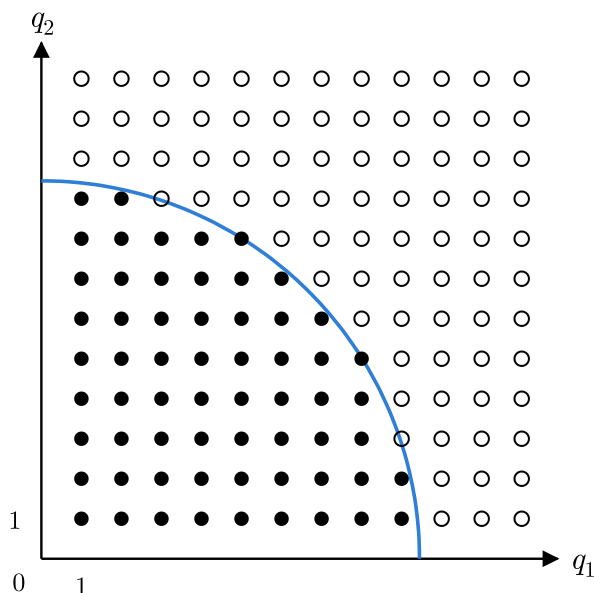


Figure 68. The states of an electron in a cubical box, drawn in one plane of constant q_3 in the space of the integers q_1 , q_2 , and q_3 . Each dot is one state of motion in space, which holds two electrons of opposite spin. In the ground state of the gas the states inside the sphere of radius q_F are occupied (filled dots) and those outside are empty (open dots); the arc is the circle in which the sphere meets the plane.

and the energy, for an electron moving slowly compared with light, is $p^2/2m_e$, as found in Section 5.1 of Vol. 3.

We place the states as points in a space whose Cartesian coordinates are q_1 , q_2 , and q_3 , as was done for a gas in Section 5.1 of Vol. 2 of the Fermi Notes on Physics. The states sit at the corners of every unit cell of a cubic lattice of unit spacing, so that there is one state per unit cell, and they fill one octant, since the integers are positive. In the ground state the electrons occupy, two to a state, all the states with q less than some value q_F , as Figure 68 shows. These states lie inside one eighth of a sphere of radius q_F , and so

$$N_e = 2 \times \frac{1}{8} \times \frac{4\pi}{3} q_F^3 = \frac{\pi q_F^3}{3}. \quad (5.2)$$

The sphere of radius q_F is called the Fermi sphere. The momentum $p_F = \pi\hbar q_F/\ell_b$ of the most energetic electrons is called the Fermi momentum, and with $n_e = N_e/\ell_b^3$ it is

$$p_F = \hbar (3\pi^2 n_e)^{1/3}, \quad (5.3)$$

which depends on the number density of the electrons and not on the size of the box. The energy of the most energetic electrons is called the Fermi energy, and for electrons moving slowly compared with light it is

$$E_F = \frac{p_F^2}{2m_e} = \frac{\hbar^2}{2m_e} (3\pi^2 n_e)^{2/3}. \quad (5.4)$$

You will check in Problem 1 that the same count follows from the result of Section 4.3 of Vol. 2, that the number of states of one particle in a volume V with momentum in a region of volume d^3p is $V d^3p/h^3$. The Fermi energy is inversely proportional to the mass of the particle, so the nuclei in a star, which are thousands of times heavier than an electron, remain an ideal gas at densities at which the electrons are degenerate.

The Pressure of a Degenerate Electron Gas

The occupied states with q between q and $q + dq$ fill one eighth of a spherical shell of volume $4\pi q^2 dq$, and they hold $\pi q^2 dq$ electrons. Each has the energy $\pi^2 \hbar^2 q^2 / 2m_e \ell_b^2$, so the total kinetic energy of the gas is

$$E_K = \int_0^{q_F} \frac{\pi^2 \hbar^2 q^2}{2m_e \ell_b^2} \pi q^2 dq = \frac{\pi^3 \hbar^2 q_F^5}{10m_e \ell_b^2} = \frac{3}{5} N_e E_F, \quad (5.5)$$

where the last step uses $N_e = \pi q_F^3 / 3$ and $E_F = \pi^2 \hbar^2 q_F^2 / 2m_e \ell_b^2$.

To find the pressure, let us change the volume of the box slowly at fixed N_e . The energy of every state goes as $1/\ell_b^2$, which is $V^{-2/3}$, so the levels move by the same factor and keep their order, and the electrons stay in their states. The gas remains in its ground state and takes in no heat, and by the first law in the form $dU = -P dV$, given in Section 2.2 of Vol. 2 for a change with no heat exchanged, the pressure is $-dE_K/dV$. Since E_K is proportional to $V^{-2/3}$,

$$P_{\text{deg}} = -\frac{dE_K}{dV} = \frac{2}{3} \frac{E_K}{V} = \frac{2}{5} n_e E_F = \frac{(3\pi^2)^{2/3}}{5} \frac{\hbar^2}{m_e} n_e^{5/3}. \quad (5.6)$$

This pressure is called the degeneracy pressure. Let μ_e be the mass per electron in units of m_H , so that $n_e = \rho / \mu_e m_H$; for fully ionised hydrogen $\mu_e = 1$, and for fully ionised helium, carbon, or oxygen $\mu_e = 2$, since their nuclei hold equal numbers of protons and neutrons. The degeneracy pressure goes as $\rho^{5/3}$, the equation of state of a polytrope with $n_p = \frac{3}{2}$ in the notation of Section 4.3, and it contains no temperature.

The calculation assumed $p_F \ll m_e c$. At higher densities most of the electrons move at speeds close to c , and their energy is pc to a good approximation, a result of special relativity, which is not treated in this course, and the relation used for the levels of an ultra-relativistic gas in Problem 2 of Section 5.1 of Vol. 2. The gas is then called ultra-relativistic. The de Broglie relation holds at any speed, so the count of states is unchanged, while the energy of a state becomes $\pi \hbar c q / \ell_b$. The total kinetic energy is

$$E_K = \int_0^{q_F} \frac{\pi \hbar c q}{\ell_b} \pi q^2 dq = \frac{\pi^2 \hbar c q_F^4}{4 \ell_b} = \frac{3}{4} N_e p_F c. \quad (5.7)$$

Every level now goes as $V^{-1/3}$, and the same argument gives

$$P_{\text{deg}} = \frac{1}{3} \frac{E_K}{V} = \frac{1}{4} n_e p_F c = \frac{(3\pi^2)^{1/3}}{4} \hbar c n_e^{4/3}, \quad (5.8)$$

the relation between pressure and energy density found for radiation in Section 5.6 of Vol. 2. The pressure goes as $\rho^{4/3}$, the equation of state of a polytrope with $n_p = 3$. Between the non-relativistic and the ultra-relativistic limits the energy of an electron is given by special relativity as $\sqrt{p^2 c^2 + m_e^2 c^4} - m_e c^2$, and the pressure passes smoothly from the first form to the second. For $\mu_e = 2$ the Fermi momentum equals $m_e c$ at a density of about $2 \times 10^9 \text{ kg m}^{-3}$.

The Condition for Degeneracy

At a temperature T above zero the electrons carry thermal energies of the order of $k_B T$. If $k_B T$ is much less than E_F , only the electrons near the top of the Fermi sphere can move into empty states, since an electron deep inside the sphere finds every state within $k_B T$ of its energy already occupied. The gas is then very close to its ground state, and its pressure is the degeneracy pressure. If $k_B T$ is much greater than E_F , the mean number of electrons in each state is much less than one, the exclusion principle has almost no effect, and the gas is ideal. The electrons are therefore degenerate when

$$T \ll T_F = \frac{E_F}{k_B} = \frac{\hbar^2}{2m_e k_B} \left(\frac{3\pi^2 \rho}{\mu_e m_H} \right)^{2/3}, \quad (5.9)$$

where T_F is called the Fermi temperature. You will show in Problem 2 that the condition is equivalent to $n_e \gg n_Q$, where $n_Q = (2\pi m_e k_B T / \hbar^2)^{3/2}$ is the quantity that appeared in the chemical potential of an ideal gas in Problem 1 of Section 4.3 of Vol. 2, and which is called the quantum concentration.

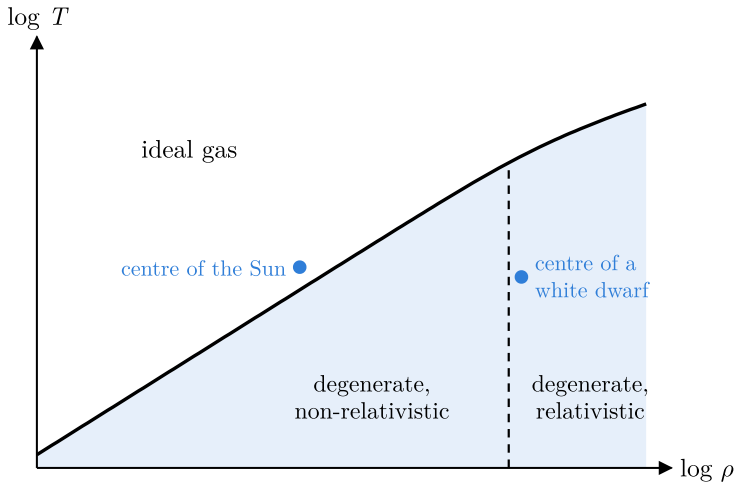


Figure 69. The plane of density and temperature for matter with $\mu_e = 2$, with logarithmic axes. Along the solid curve $k_B T$ equals the Fermi energy of the electrons, and in the shaded region below it the electrons are degenerate. The dashed line marks the density at which $p_F = m_e c$, to the right of which the degenerate electrons are relativistic. The dots mark the centre of the Sun and the centre of a typical white dwarf.

The degenerate and the ideal regions are drawn in Figure 69, where the solid curve is $T = T_F$, computed with the energy of an electron at any speed. At the centre of the Sun T_F is about 5×10^6 K, a few times less than the central temperature, so the gas there is ideal. At the centre of a white dwarf the density is 10^9 to 10^{10} kg m $^{-3}$, which gives a Fermi temperature of the order of 10^9 K, about a hundred times the central temperature, and the electrons are strongly degenerate.

The Mass–Radius Relation

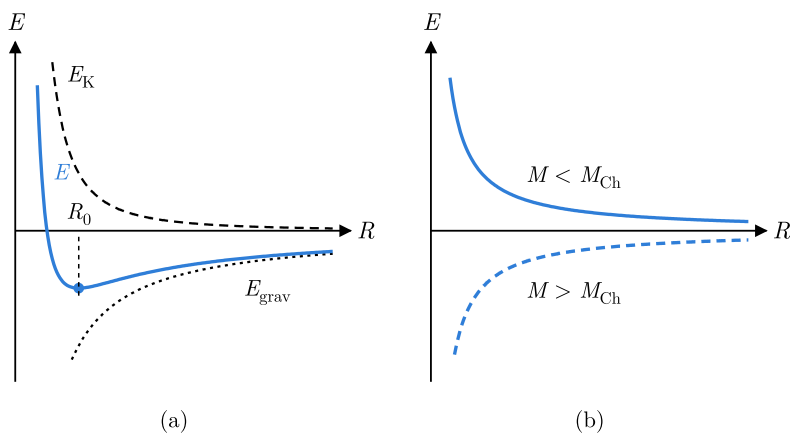


Figure 70. (a) The kinetic energy E_K of the degenerate electrons (dashed), the gravitational potential energy E_{grav} (dotted), and their sum E (accent) for a white dwarf of fixed mass whose electrons are non-relativistic, as functions of the radius. The sum has its minimum at the radius R_0 . (b) The total energy when the electrons are ultra-relativistic, which goes as $1/R$ and is positive for $M < M_{\text{Ch}}$ (solid) and negative for $M > M_{\text{Ch}}$ (dashed).

To find the radius of a white dwarf of mass M , we estimate its energy as a function of its radius and look for the radius at which the energy is least, which is a stable equilibrium. We treat the star as a uniform sphere, which is a valid approximation for an estimate, although the density of a real white dwarf rises towards the centre. The star holds $N_e = M/\mu_e m_H$ electrons in the volume $\frac{4}{3}\pi R^3$, and the kinetic energy $\frac{3}{5}N_e E_F$ of the electrons becomes

$$E_K = \frac{3}{10} \left(\frac{9\pi}{4} \right)^{2/3} \frac{\hbar^2 N_e^{5/3}}{m_e R^2} = \frac{C_1}{R^2}, \quad (5.10)$$

which defines the constant C_1 . The gravitational potential energy of a uniform sphere, found in Section 4.2, is

$$E_{\text{grav}} = -\frac{3}{5} \frac{GM^2}{R} = -\frac{C_2}{R}. \quad (5.11)$$

The thermal energy of the nuclei is small by comparison and is left out, so the total energy is $E = C_1/R^2 - C_2/R$. The kinetic and the gravitational terms go as different powers of R , and the sum has a minimum, as Figure 70(a) shows. Setting $dE/dR = -2C_1/R^3 + C_2/R^2$ to zero gives $R = 2C_1/C_2$, and with $N_e = M/\mu_e m_H$,

$$R = \left(\frac{9\pi}{4} \right)^{2/3} \frac{\hbar^2}{Gm_e (\mu_e m_H)^{5/3}} M^{-1/3}. \quad (5.12)$$

The radius of a white dwarf decreases as its mass grows, since a stronger gravitational field squeezes the electrons to a higher Fermi energy before their pressure balances it, and MR^3 is the same for all white dwarfs of the same composition. At the minimum $2E_K + E_{\text{grav}} = 0$, as you will check in Problem 3, which is the virial theorem of Section 4.2, since for a gas with $P = \frac{2}{3}E_K/V$ the product $3PV$ is $2E_K$. For $M = 0.6 M_\odot$, close to the most common mass of a white dwarf, and $\mu_e = 2$, the formula gives a radius of about 8.5×10^6 m, a little larger than the Earth, and for $M = 1.0 M_\odot$ it gives about 7×10^6 m, close to the radius of Sirius B.

The Chandrasekhar Mass

As the mass of a white dwarf grows, its mean density $3M/4\pi R^3$ grows as M^2 , and you will show in Problem 4 that for a mass of the order of the Sun it reaches the density at which $p_F = m_e c$. Let us take the limit in which all of the electrons are ultra-relativistic. Their kinetic energy $\frac{3}{4}N_e p_F c$ becomes

$$E_K = \frac{3}{4} \left(\frac{9\pi}{4} \right)^{1/3} \frac{\hbar c N_e^{4/3}}{R} = \frac{C_3}{R}, \quad (5.13)$$

which goes as $1/R$, like the gravitational energy. The total energy $E = (C_3 - C_2)/R$ has no minimum, as Figure 70(b) shows. If $C_3 > C_2$, the energy falls as the radius grows, so the star expands until its electrons are no longer ultra-relativistic and settles on the non-relativistic branch. If $C_3 < C_2$, the energy falls without limit as the radius shrinks, and no pressure of the electrons can stop the star from collapsing. Since C_3 goes as $M^{4/3}$ and C_2 as M^2 , the boundary $C_3 = C_2$ fixes a mass,

$$M_{\text{Ch}} = \frac{15\sqrt{5}\pi}{16} \left(\frac{\hbar c}{G} \right)^{3/2} \frac{1}{(\mu_e m_H)^2}, \quad (5.14)$$

which is called the Chandrasekhar mass, after Chandrasekhar, who found it in 1931. No white dwarf can have a greater mass. The electron mass does not appear, since the energy of an ultra-relativistic electron does not depend on it. For $\mu_e = 2$ the estimate gives about $1.7 M_\odot$. The exact value comes from the Lane–Emden equation of Section 4.3 with $n_p = 3$, solved numerically, and is $M_{\text{Ch}} \approx 5.8 M_\odot / \mu_e^2$, or about $1.4 M_\odot$ for $\mu_e = 2$. The masses of the white dwarfs measured in binaries are all below this value.

The non-relativistic and the ultra-relativistic limits are joined by carrying out the same minimisation with the energy of an electron at any speed, which requires an integral over the Fermi sphere that is usually evaluated numerically. The resulting radius is drawn against the mass in Figure 71. At small masses it follows $R \propto M^{-1/3}$, and as the mass approaches M_{Ch} the radius falls to zero, because the electrons become ultra-relativistic throughout the star.

The Cooling of White Dwarfs

The carbon and oxygen of a typical white dwarf would fuse only at the temperatures of several hundred million kelvin described in Section 4.5, so a white dwarf shines by losing its stored heat. The degenerate electrons already occupy the lowest states open to them

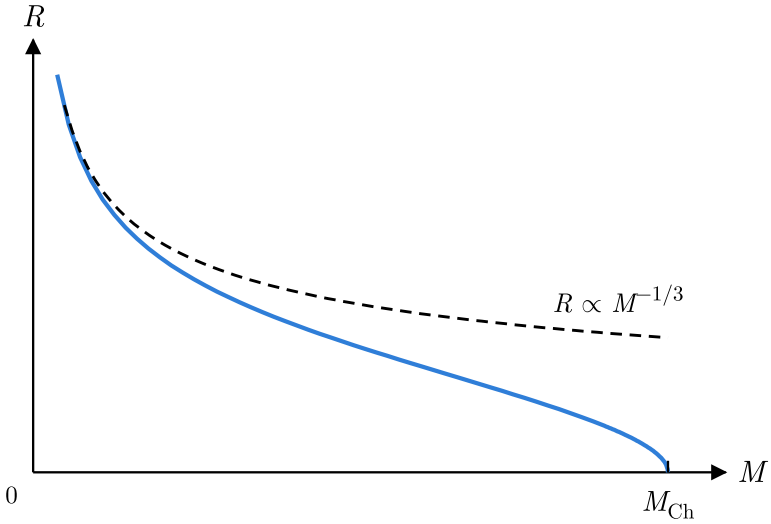


Figure 71. The radius of a white dwarf as a function of its mass (accent), found by minimising the total energy of a uniform sphere of degenerate electrons with the energy $\sqrt{p^2 c^2 + m_e^2 c^4} - m_e c^2$ of an electron at any speed. The dashed curve is the non-relativistic relation $R \propto M^{-1/3}$. The radius falls to zero as the mass approaches the Chandrasekhar mass of the estimate.

and cannot give up their kinetic energy, and the heat that is lost is the thermal energy of the nuclei, which form an ideal gas. For a white dwarf of mass M made of nuclei of mass number A at the temperature T_c , this energy is

$$U = \frac{3}{2} \frac{M}{Am_u} k_B T_c. \quad (5.15)$$

The interior is at nearly the same temperature throughout, because degenerate electrons conduct heat very well, as the electrons in a metal do; an electron can be scattered only into an empty state, and few empty states are accessible to most electrons, so they travel far between collisions. The heat escapes by radiative diffusion through a thin surface layer in which the electrons are not degenerate, and you will show in Problem 5, with the Kramers opacity of Section 4.3, that the luminosity is

$$L = C_L M T_c^{7/2}, \quad (5.16)$$

where C_L is a constant that depends on the composition of the layer. Setting $dU/dt = -L$ gives

$$\frac{3k_B M}{2Am_u} \frac{dT_c}{dt} = -C_L M T_c^{7/2}, \quad (5.17)$$

in which the mass cancels. Separating the variables and integrating $T_c^{-7/2} dT_c$ from the initial temperature T_0 gives $T_c^{-5/2} - T_0^{-5/2} = (t/t_0) T_0^{-5/2}$, with

$$t_0 = \frac{3k_B}{5Am_u C_L T_0^{5/2}} = \frac{2}{5} \frac{U_0}{L_0}, \quad (5.18)$$

where U_0 and L_0 are the thermal energy and the luminosity at $t = 0$. Since L goes as $T_c^{7/2}$, the luminosity is

$$L = L_0 \left(1 + \frac{t}{t_0}\right)^{-7/5}. \quad (5.19)$$

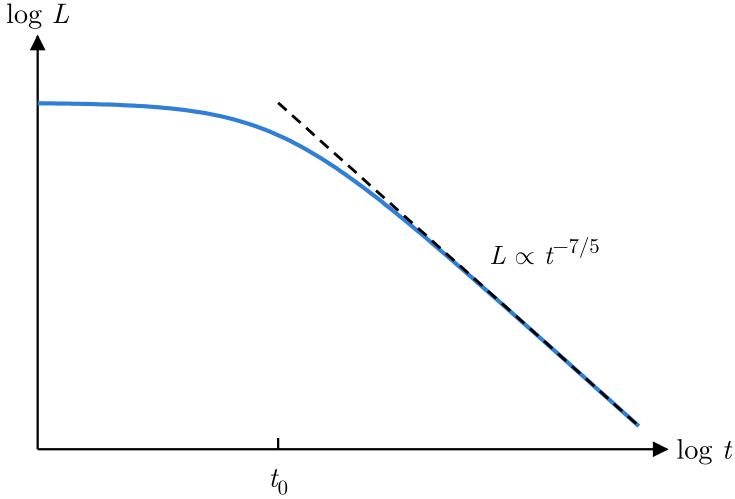


Figure 72. The luminosity of a cooling white dwarf as a function of time (accent), with logarithmic axes. After a time long compared with t_0 the luminosity goes as $t^{-7/5}$, which is the dashed straight line.

The luminosity is drawn in Figure 72. For $t \gg t_0$ it goes as $t^{-7/5}$, and the age of the white dwarf is $t \approx \frac{2}{5}U/L$, independent of its initial temperature. The faintest white dwarfs in the disc of the Galaxy have cooling ages of about 10^{10} years, and the number of white dwarfs observed at each luminosity falls sharply below the luminosity of the faintest, since no white dwarf in the disc has had time to cool further. This gives an age for the disc of the Galaxy that can be compared with the ages of the oldest star clusters found in Section 4.8.

5.2 Core Collapse and Supernovae

A supernova is a star that brightens within days to a luminosity that can rival that of a whole galaxy, and then fades over months. In this section we follow the core of a massive star from the end of its nuclear burning to its collapse, then describe how supernovae are classified by their spectra, and finally show how the light of a supernova in the months after the explosion is supplied by radioactive decay.

The Iron Core of a Massive Star

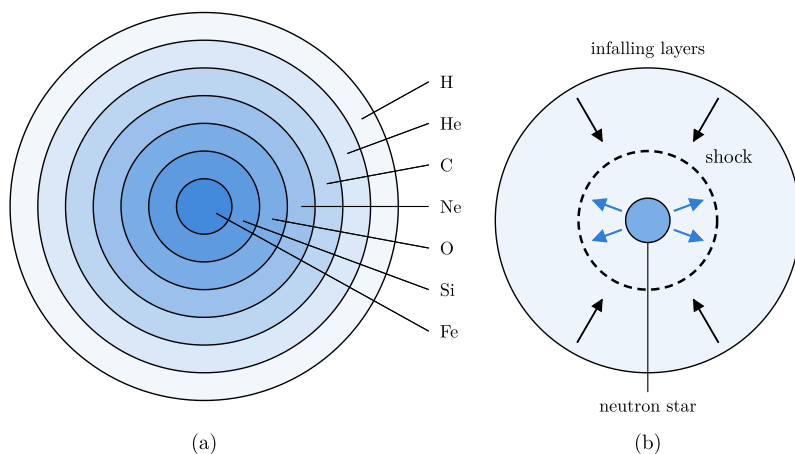


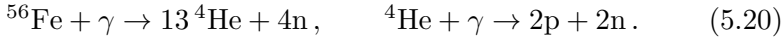
Figure 73. (a) The layers of a massive star at the end of its nuclear burning, drawn schematically and not to scale, each labelled by the most abundant element it contains. (b) The core shortly after it has collapsed. The neutron star at the centre (dark shading) has bounced, and a shock (dashed circle) moves outwards through the layers that are still falling in.

Section 4.7 described how a star of more than about eight solar masses burns hydrogen, then helium, and then heavier nuclei in its core, each stage hotter and shorter than the one before, and Section 4.5 described the burning stages themselves. Each stage leaves a shell of its ashes behind it, and a burning shell continues to consume the fuel at its base, so that the star ends with the layered structure shown in Figure 73(a). The last stage, the burning of silicon, lasts about a day and builds a core of iron and of the nuclei close to it in mass number. Since the binding energy per nucleon is greatest near iron, as Section 4.5 showed, the fusion of iron into heavier nuclei would absorb energy, and the core has no further source of nuclear energy.

The iron core is supported by the pressure of its degenerate electrons, which at its density of 10^{12} to $10^{13} \text{ kg m}^{-3}$ are relativistic, and the burning shell of silicon around it keeps adding iron to it. Section 5.1 showed that such a core cannot have a mass greater than M_{Ch} , and when its mass approaches this limit, which for the hot core is of the order of $1.5 M_{\odot}$, the core becomes unstable.

Why the Core Collapses

There are two processes that remove the pressure that supports the core. The first is the break-up of nuclei by photons, called photodisintegration. At a temperature of about 5×10^9 K the photons in the high-energy tail of the Planck spectrum can break nuclei apart, through the reactions



These reactions reverse the fusion that built the iron and absorb the energy that it released, about 124 MeV for each iron nucleus broken into helium and a further 28 MeV for each helium nucleus broken into nucleons, as you will find in Problem 1. The energy is taken from the thermal energy of the gas, so the pressure falls.

The second process is the capture of electrons by protons, both free and inside nuclei,



in which ν_{e} is an electron neutrino. The capture requires an electron with a kinetic energy of at least 0.78 MeV, the amount by which the mass of a neutron exceeds the combined mass of a proton and an electron, and at the density of the core the Fermi energy of the electrons is several MeV. Each capture removes an electron and so lowers the degeneracy pressure, and the neutrino escapes and carries its energy away.

The core then collapses in a time of the order of the dynamical time scale $t_{\text{dyn}} \sim (G\rho)^{-1/2}$ of Section 4.2, a small fraction of a second, with the inner core falling inwards at up to about a quarter of the speed of light, and as the density rises the captures turn it into matter made mostly of neutrons.

The Bounce and the Neutrinos

The collapse of the inner core stops when its density reaches that of an atomic nucleus, about $2 \times 10^{17} \text{ kg m}^{-3}$. The nucleons are then packed as closely as in a nucleus, and the strong nuclear force

between them, which is attractive at larger separations, becomes strongly repulsive, so the matter becomes very hard to compress. The inner core overshoots slightly and rebounds, and the rebound, called the bounce, sends a shock wave outwards into the outer core, which is still falling in, as in Figure 73(b). The shock loses energy by breaking up the iron nuclei it passes through and stalls, and it is thought to be revived by the neutrinos from the hot interior, a small fraction of which are absorbed behind it; how the explosion succeeds in detail is the subject of current research. The propagation of shock waves is not treated in this course.

The energy released by the collapse is the gravitational potential energy of the neutron star that is formed. With the potential energy of a uniform sphere found in Section 4.2, a neutron star of mass M and radius R has released the energy

$$E_{\text{grav}} \approx \frac{3}{5} \frac{GM^2}{R}, \quad (5.22)$$

since the core before the collapse was several hundred times larger and its potential energy correspondingly smaller in magnitude. For $M = 1.4 M_{\odot}$ and $R = 12$ km this is about 3×10^{46} J. Almost all of it is carried away by neutrinos and antineutrinos, whose mean free path in the hot, dense core is shorter than its radius, so that, like the photons in a star described in Section 4.3, they escape by a random walk, over about ten seconds. The kinetic energy of the ejected layers is about one per cent of the total, and the light of the supernova about one hundredth of that.

The Neutrinos from SN 1987A

On 23 February 1987 a supernova, called SN 1987A, was seen in the Large Magellanic Cloud, a small galaxy close to the Milky Way at a distance of about 50 kpc; the star that exploded had been catalogued beforehand as a blue supergiant. A few hours before the light was first seen, two neutrino detectors recorded a burst of events within about thirteen seconds, eleven at Kamiokande II in Japan and eight at IMB in the United States. The neutrinos arrived first because the shock took several hours to reach the surface of the star.

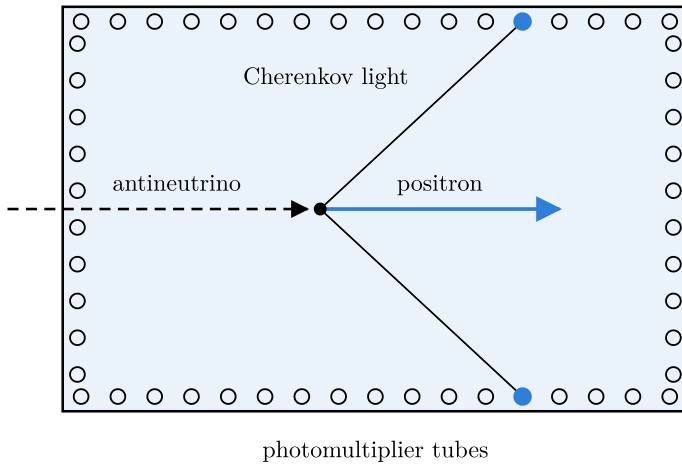


Figure 74. A water Cherenkov detector of the kind that recorded the neutrinos from SN 1987A. An antineutrino (dashed) is absorbed by a proton in the water and produces a positron (accent), which moves faster than light travels in the water and emits a cone of Cherenkov light; the photomultiplier tubes lining the tank (small circles) record where the cone meets the wall.

Each detector was a tank holding thousands of tonnes of pure water, lined with photomultiplier tubes, which are devices that turn a faint flash of light into a measurable electric pulse, as Figure 74 shows. An electron antineutrino absorbed by a proton in the water, in the reaction $\bar{\nu}_e + p \rightarrow n + e^+$, produces a positron with most of its energy. The positron moves faster than light travels in water, which is about $c/1.33$, and a charged particle doing so emits a cone of faint light called Cherenkov light, from which the tubes find the energy and the direction of the positron.

The Number of Neutrinos Detected

As an example of what the detection tells us, consider the number of events expected in a detector at distance d from a supernova. Let the supernova release the energy E_{grav} in neutrinos, of which a fraction f_a is carried by electron antineutrinos, and let the mean energy of an antineutrino be $\bar{E}$. The number of antineutrinos emitted is $f_a E_{\text{grav}}/\bar{E}$, and since they are emitted equally in all directions, the number crossing each unit area at the distance d is

$$\mathcal{F} = \frac{f_a E_{\text{grav}}}{4\pi d^2 \bar{E}}. \quad (5.23)$$

Let the detector hold N_p free protons, the hydrogen nuclei of the water, and let σ be the cross-section for the absorption of an antineutrino by a proton. A single proton absorbs, on average, $\sigma\mathcal{F}$ antineutrinos, just as a target of area σ intercepts the particles of a uniform stream that fall on that area, so the expected number of events is

$$N_{\text{det}} = \frac{f_a E_{\text{grav}} \sigma N_p}{4\pi d^2 \bar{E}}. \quad (5.24)$$

The cross-section grows with the energy of the antineutrino, and at about 15 MeV it is of the order of 10^{-45} m^2 . With $E_{\text{grav}} \approx 3 \times 10^{46} \text{ J}$, f_a about one sixth, a few thousand tonnes of water, and the distance of the Large Magellanic Cloud, the formula gives about twenty events, of the same order as the eleven recorded at Kamiokande II. The energies of the detected positrons, up to about

40 MeV, and the duration of the burst agree with the random walk of neutrinos out of a hot core, and the detection confirmed that core collapse releases an energy of the order of the gravitational energy of a neutron star, almost all of it as neutrinos.

The Classification of Supernovae

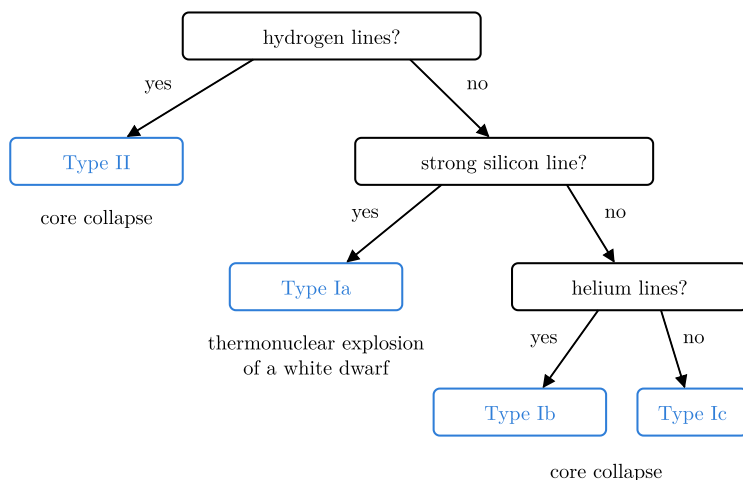


Figure 75. The classification of supernovae by the absorption lines in their spectra near maximum light. The accent boxes are the classes, and the words beneath them name the mechanism by which each class is thought to explode.

Supernovae are classified by the lines in their spectra near maximum light, as Figure 75 shows. A supernova whose spectrum shows no lines of hydrogen is called Type I, and one whose spectrum shows them is called Type II. Type I is divided further. A Type I supernova with a strong absorption line of singly ionised silicon is called Type Ia; one without the silicon line but with lines of helium is called Type Ib; and one with neither is called Type Ic. There are further subclasses, based on the shape of the light curve or on details of the spectrum, which are not treated in this course.

The classification corresponds largely to two different mechanisms. Supernovae of Types II, Ib, and Ic are found where stars are forming, such as the arms of spiral galaxies, and almost never in elliptical

galaxies, which have formed no stars for billions of years. Since a massive star lives only a few tens of millions of years, as Section 4.7 showed, these supernovae are the core collapses of massive stars. A star that keeps its hydrogen envelope becomes a Type II supernova, one that has lost it to a strong wind or to a companion becomes a Type Ib, and one that has lost its helium layer as well becomes a Type Ic.

Supernovae of Type Ia are found in galaxies of every kind, including elliptical galaxies, so they come from stars that live a long time, and they are thought to be the thermonuclear explosions of white dwarfs made of carbon and oxygen. A white dwarf in a close binary can gain mass from its companion, or two white dwarfs can merge; which route is the more common is not yet known, and accretion and the evolution of close binaries are not treated in this course. As the mass approaches M_{Ch} , the central density rises, and carbon begins to fuse at the centre. In a star supported by an ideal gas, such as the Sun, a rise of temperature raises the pressure, the core expands and cools, and the rate of burning falls again, as the virial theorem of Section 4.2 requires. In a white dwarf the pressure is the degeneracy pressure of Section 5.1, which does not depend on the temperature, so the heated matter does not expand. The temperature rises, the rate of fusion rises steeply with it, as Section 4.4 showed, and the burning runs away, converting much of the star into nickel-56 and other nuclei near iron. The energy released exceeds the gravitational binding energy of the white dwarf when a large enough fraction of its mass is burned, as you will show in Problem 3, so the star is blown apart and leaves no remnant. Since every such white dwarf explodes at nearly the same mass, supernovae of Type Ia have similar peak luminosities, and after a correction found by observation, that the brighter ones fade more slowly, they are the standard candles described in Section 1.1.

Light Curves Powered by Radioactive Decay

The light of a supernova in the months after its peak cannot be the heat of the explosion, since the ejected gas expands and cools within days, and it is supplied by the radioactive nuclei made in the explosion. The burning is too rapid to change the equal numbers of protons and neutrons in the fuel, and it forms large amounts of

nickel-56, the most tightly bound nucleus with equal numbers of each: several tenths of a solar mass in a Type Ia supernova and less in a core collapse. Nickel-56 decays by capturing an electron into cobalt-56, with a mean life of 8.8 days, and cobalt-56 decays into stable iron-56, with a mean life of 111 days. The gamma rays and positrons from the decays heat the expanding gas, which radiates the heat as light.

Let t_{Ni} and t_{Co} be the mean lives of nickel-56 and cobalt-56, let N_0 be the number of nickel-56 nuclei made in the explosion at $t = 0$, and let $N_{\text{Ni}}(t)$ and $N_{\text{Co}}(t)$ be the numbers present at time t . By the law of radioactive decay, $N_{\text{Ni}} = N_0 e^{-t/t_{\text{Ni}}}$. Cobalt is made at the rate $N_{\text{Ni}}/t_{\text{Ni}}$ at which nickel decays, and decays at the rate $N_{\text{Co}}/t_{\text{Co}}$, so

$$\frac{dN_{\text{Co}}}{dt} + \frac{N_{\text{Co}}}{t_{\text{Co}}} = \frac{N_0}{t_{\text{Ni}}} e^{-t/t_{\text{Ni}}} . \quad (5.25)$$

This is a linear first-order equation, which we solve with the integrating factor $e^{t/t_{\text{Co}}}$, as in Section 5.1 of Vol. 4 of the Fermi Notes on Physics. Multiplying by the integrating factor makes the left-hand side the derivative of $N_{\text{Co}} e^{t/t_{\text{Co}}}$, and integrating from $t = 0$, when there is no cobalt, gives

$$N_{\text{Co}}(t) = \frac{N_0 t_{\text{Co}}}{t_{\text{Co}} - t_{\text{Ni}}} (e^{-t/t_{\text{Co}}} - e^{-t/t_{\text{Ni}}}) . \quad (5.26)$$

Let Q_{Ni} and Q_{Co} be the energies deposited in the gas by one decay of each kind. If all of the energy is absorbed and radiated promptly, the luminosity is

$$L(t) = \frac{Q_{\text{Ni}} N_{\text{Ni}}}{t_{\text{Ni}}} + \frac{Q_{\text{Co}} N_{\text{Co}}}{t_{\text{Co}}} . \quad (5.27)$$

The luminosity is drawn in Figure 76. The decay of nickel supplies the luminosity in the first weeks, and after a few mean lives of nickel the decay of cobalt takes over. For $t \gg t_{\text{Ni}}$,

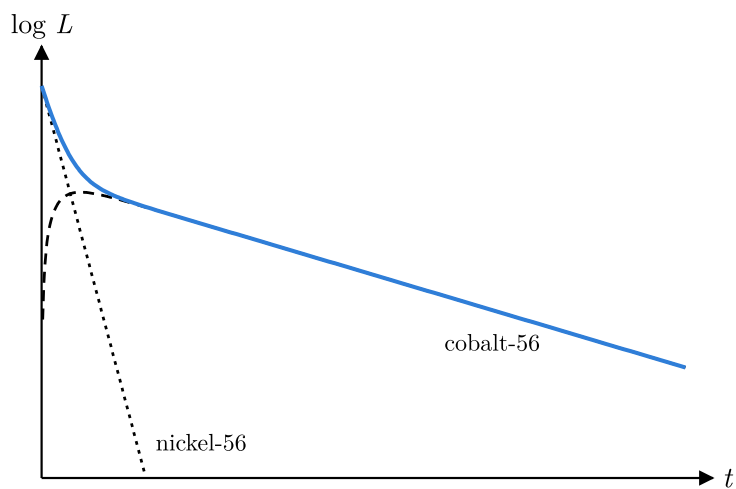


Figure 76. The luminosity supplied by the decay of nickel-56 and of the cobalt-56 made from it (accent), with the luminosity on a logarithmic scale. The dotted line is the contribution of the decay of nickel, and the dashed curve that of the decay of cobalt. At late times the luminosity falls as $e^{-t/t_{Co}}$, which is a straight line on these axes.

$$L(t) \approx \frac{Q_{\text{Co}} N_0}{t_{\text{Co}} - t_{\text{Ni}}} e^{-t/t_{\text{Co}}}, \quad (5.28)$$

so the logarithm of the luminosity falls linearly with time, and you will show in Problem 5 that the magnitude increases at the constant rate $2.5 \log_{10} e/t_{\text{Co}}$, about one magnitude in a hundred days. The late light curve of SN 1987A fell at this rate for about two years, and the luminosity on the straight part of the curve fixes N_0 , which corresponds to about $0.07 M_{\odot}$ of nickel. The decline later becomes steeper, as the expanding gas thins out and more of the gamma rays escape unabsorbed. Near the peak the light curve also differs from Figure 76, because the heat deposited in the thick gas takes days to diffuse out, and the transport of radiation through the expanding gas is not treated in this course.

Supernova Remnants

The gas ejected by a supernova moves outwards at about 10^7 m s^{-1} , sweeps the surrounding interstellar gas into an expanding shell, and heats it to temperatures at which it emits X-rays, and the shell remains visible for tens of thousands of years as a supernova remnant. Examples include the Crab Nebula, the remnant of a supernova recorded by Chinese astronomers in 1054, which holds the pulsar described in Section 5.3, and the remnants of the supernovae seen by Tycho in 1572 and by Kepler in 1604. Supernovae return the elements made inside stars, as described in Section 4.5, to the interstellar gas from which later stars form.

5.3 Neutron Stars and Pulsars

In 1934, two years after Chadwick discovered the neutron, Baade and Zwicky proposed that a supernova turns an ordinary star into a star made of neutrons, and the collapse described in Section 5.2 makes such a star from the iron core of a massive star. In this section we first estimate the radius of a neutron star with the methods of Section 5.1, then find how fast it spins and how strong its magnetic field is, and then show that the pulsars are rotating neutron stars and work out what their slowing down reveals. Finally we

describe the black holes that are left when the collapsing core is too massive to become a neutron star.

Degenerate Neutrons

A free neutron decays, with a mean life of about fifteen minutes, into a proton, an electron, and an electron antineutrino. Inside a neutron star the decay is blocked by the exclusion principle, since the electrons present are degenerate, with a Fermi energy greater than the energy the decay would give to an electron, so every state that the new electron could occupy is filled. The matter is therefore made mostly of neutrons, with a small fraction of protons and electrons.

Neutrons have spin one-half, as stated in Section 4.2 of Vol. 3, and the count of states in Section 5.1 applies to them with the neutron mass m_n in place of m_e and the number density $n_n = \rho/m_n$ in place of n_e , so a gas of neutrons that are not relativistic has the pressure

$$P_{\text{deg}} = \frac{(3\pi^2)^{2/3}}{5} \frac{\hbar^2}{m_n} n_n^{5/3}. \quad (5.29)$$

The minimisation of the energy in Section 5.1 then gives the radius of a neutron star of mass M , with the difference that the particles that provide the pressure also provide the mass, so that m_e and $\mu_e m_H$ are both replaced by m_n :

$$R = \left(\frac{9\pi}{4}\right)^{2/3} \frac{\hbar^2}{G m_n^{8/3}} M^{-1/3}. \quad (5.30)$$

For $M = 1.4 M_\odot$ this gives about 11 km, and you will show in Problem 1 that a neutron star is smaller than a white dwarf of the same mass by a factor of several hundred, mostly because the neutron is about two thousand times heavier than the electron. The mean density of a neutron star is a few times $10^{17} \text{ kg m}^{-3}$, similar to the density inside an atomic nucleus. Measurements of the radii of neutron stars, from the X-rays emitted by their hot surfaces, give values between about 10 and 14 km.

The argument of Section 5.1 also gives a largest mass for a star of degenerate neutrons, but its value is not reliable, for two reasons. At nuclear densities the neutrons are not an ideal gas, since the strong nuclear force between them contributes to the pressure, and this force is known only approximately at such densities. And the radius of a neutron star is only a few times its Schwarzschild radius, which is introduced below, so its gravity must be described by general relativity, which is not treated in this course. Calculations that include both the nuclear force and general relativity give a maximum mass between about 2 and $3 M_{\odot}$, and the heaviest neutron stars measured in binaries have masses of about $2 M_{\odot}$.

The Rotation and Magnetic Field of a Neutron Star

A neutron star is born from the collapse of a core that was rotating and carried a magnetic field, and both the rotation and the field are greatly amplified by the collapse. Let the core have radius R_i , angular speed Ω_i , and magnetic field B_i , and let the neutron star have radius R . We treat the core and the neutron star as uniform spheres, with the moment of inertia $I_{\text{rot}} = \frac{2}{5}MR^2$ found in Section 3.3 of Vol. 1 of the Fermi Notes on Physics. No external torque acts during the collapse, so the angular momentum $J = I_{\text{rot}}\Omega$ is conserved, and

$$\Omega = \Omega_i \left(\frac{R_i}{R} \right)^2. \quad (5.31)$$

The matter of a star is ionised and conducts electricity very well. By the flux rule of Section 7.2 of Vol. 5 of the Fermi Notes on Physics, the electromotive force round any loop that moves with the matter, even one that changes its shape, equals minus the rate of change of the magnetic flux through it. In a perfect conductor a non-zero electromotive force round a loop would drive an unlimited current, so the flux through every loop that moves with the matter stays constant, which is a valid approximation for a highly conducting gas over times short compared with the time its currents take to decay. The field is then said to be frozen into the matter. The flux through the star is of the order of $B\pi R^2$, and if it is constant,

$$B = B_i \left(\frac{R_i}{R} \right)^2. \quad (5.32)$$

As an example, consider a core the size of a white dwarf, with R_i of a few thousand kilometres, collapsing to a radius of 10 km, so that $(R_i/R)^2$ is of the order of 10^5 . White dwarfs rotate with periods of the order of hours, and those that are magnetic have fields of 10^2 to 10^4 T, so a neutron star made from such a core would spin with a period of a fraction of a second and have a field of 10^7 to 10^9 T. These estimates are only indicative, since the rotation and the field of a collapsing core are not measured directly, but they agree in order of magnitude with the pulsars described below.

A star cannot spin arbitrarily fast. At the equator the gravitational acceleration GM/R^2 must at least supply the centripetal acceleration $\Omega^2 R$, or matter would be flung off, so $\Omega^2 \leq GM/R^3$. With $M = \frac{4}{3}\pi R^3 \bar{\rho}$, where $\bar{\rho}$ is the mean density, the shortest period is

$$P_{s,\min} = 2\pi \left(\frac{R^3}{GM} \right)^{1/2} = \left(\frac{3\pi}{G\bar{\rho}} \right)^{1/2}, \quad (5.33)$$

which depends only on the mean density. For a white dwarf of $0.6 M_\odot$ it is about twenty seconds, and for a neutron star it is less than a millisecond.

Pulsars

In 1967 Jocelyn Bell, a research student working with Antony Hewish at Cambridge, found a radio source that emitted a short pulse every 1.34 seconds, with a regularity that no known astronomical source could match. Many more such sources, called pulsars, were soon found, with periods from about a millisecond to several seconds, each of which increases slowly with time. In 1968 a pulsar with a period of 33 milliseconds was found at the centre of the Crab Nebula, the remnant of the supernova of 1054, which linked pulsars to supernovae.

A pulsar cannot be a white dwarf. A pulse lasting a time Δt cannot come from a region larger than about $c\Delta t$, since light from different

parts of a larger region would arrive at different times and smear the pulse out, and the pulses of the Crab pulsar last about a millisecond, so the source is at most a few hundred kilometres across. And a white dwarf spinning with the period of the Crab pulsar would fly apart, since its shortest period, found above, is about twenty seconds. A neutron star is small enough to satisfy the first condition and dense enough to satisfy the second. Pulsation of the star was proposed as well, but the pulsation of stars is not treated in this course, and the pulses are now explained by the rotation of a neutron star.

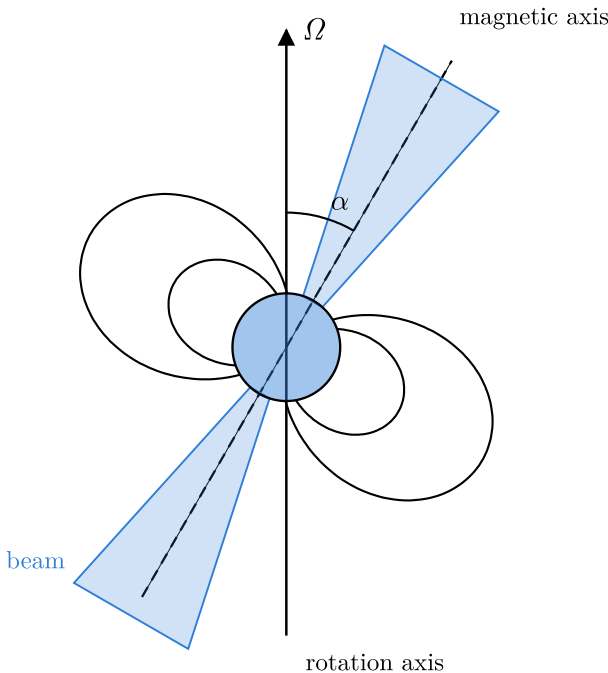


Figure 77. The lighthouse model of a pulsar. The neutron star (shaded) rotates with angular speed Ω about the vertical axis, and its magnetic axis (dashed) is inclined to the rotation axis at the angle α . The solid curves are field lines of the dipole field. Beams of radio waves (accent) are emitted along the magnetic axis from above the two magnetic poles and are carried round by the rotation.

This explanation is called the lighthouse model, and it is drawn

in Figure 77. A neutron star has a dipole magnetic field whose axis, the magnetic axis, is inclined to the rotation axis at an angle α , as the magnetic axis of the Earth is inclined to its rotation axis. Charged particles are pulled out of the surface by the strong electric fields produced by the rotating magnetic field, and near the magnetic poles they are accelerated along the field lines and emit radio waves in narrow beams along the magnetic axis. The rotation carries the beams round the rotation axis, and an observer whose line of sight is crossed by a beam sees a pulse once in each rotation, as a ship sees the beam of a lighthouse. The mechanism by which the particles emit the radio waves is not well understood, and the processes of high-energy radiation are not treated in this course.

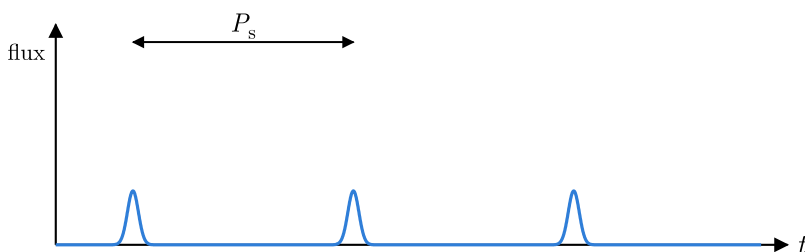


Figure 78. The radio flux received from a pulsar in the lighthouse model, computed for a beam whose intensity falls off with the angle from the magnetic axis and a line of sight that the beam crosses once in each rotation. A pulse arrives once in every spin period P_s .

The flux received by such an observer is drawn in Figure 78. The pulse is short compared with the period, since the beam is narrow and sweeps past the line of sight quickly, and the interval between pulses is the spin period of the star. An observer whose line of sight is never crossed by a beam does not see the star as a pulsar, so only a fraction of the neutron stars in the Galaxy can be seen in this way.

The Slowing Down of a Pulsar

The energy of a pulsar comes from its rotation. The rotational kinetic energy is $E_{\text{rot}} = \frac{1}{2} I_{\text{rot}} \Omega^2$, and differentiating it with respect

to time gives the rate at which the rotation loses energy,

$$-\frac{dE_{\text{rot}}}{dt} = -I_{\text{rot}}\Omega\dot{\Omega} = \frac{4\pi^2 I_{\text{rot}} \dot{P}_s}{P_s^3}, \quad (5.34)$$

where the second form follows from $\Omega = 2\pi/P_s$, whose derivative is $\dot{\Omega} = -2\pi\dot{P}_s/P_s^2$. For the Crab pulsar, with $I_{\text{rot}} \approx 10^{38} \text{ kg m}^2$ and the measured period and its rate of change, this is about $5 \times 10^{31} \text{ W}$, which is about a hundred thousand times the luminosity of the Sun and is enough to supply the energy radiated by the whole Crab Nebula.

Let μ_m be the magnitude of the magnetic dipole moment of the star, defined as in Section 5.2 of Vol. 5, which writes it μ ; here μ is the mean molecular weight. A magnetic dipole whose moment rotates about an axis inclined to it at the angle α radiates electromagnetic waves at the frequency of the rotation, and the power it radiates is

$$L_{\text{dip}} = \frac{\mu_0 \mu_m^2 \Omega^4 \sin^2 \alpha}{6\pi c^3}, \quad (5.35)$$

a result of the theory of radiation from moving charges and currents, which is beyond this course. The lighthouse beams carry only a small part of the energy, and in this model the rotation loses its energy mainly by dipole radiation. Setting $-dE_{\text{rot}}/dt = L_{\text{dip}}$ gives

$$\dot{\Omega} = -K_{\Omega}\Omega^3, \quad K_{\Omega} = \frac{\mu_0 \mu_m^2 \sin^2 \alpha}{6\pi c^3 I_{\text{rot}}}, \quad (5.36)$$

so a pulsar slows down at a rate proportional to the cube of its angular speed.

The equation can be solved for Ω by separating the variables. Let Ω_0 be the angular speed at the birth of the pulsar, at $t = 0$. Integrating $\Omega^{-3} d\Omega = -K_{\Omega} dt$ gives $\Omega^{-2} - \Omega_0^{-2} = 2K_{\Omega}t$. If the pulsar was born spinning much faster than it spins now, the term Ω_0^{-2} can be neglected, and the age is $t = 1/2K_{\Omega}\Omega^2$. Replacing K_{Ω} by $-\dot{\Omega}/\Omega^3$ expresses the age in terms of the measured period and its rate of change,

$$t_{\text{char}} = -\frac{\Omega}{2\dot{\Omega}} = \frac{P_s}{2\dot{P}_s}, \quad (5.37)$$

which is called the characteristic age of the pulsar. For the Crab pulsar it is about 1,250 years, while the supernova was seen about 970 years ago. The agreement is close, and the difference arises partly because the birth period of the Crab pulsar was not much shorter than its present period, which you will allow for in Problem 4.

The Magnetic Field of a Pulsar

The magnetic moment of a pulsar follows from the same measurements. From the definition of K_Ω , $\mu_m^2 \sin^2 \alpha = 6\pi c^3 I_{\text{rot}} K_\Omega / \mu_0$, and $K_\Omega = -\dot{\Omega} / \Omega^3 = P_s \dot{P}_s / 4\pi^2$, so

$$\mu_m \sin \alpha = \left(\frac{3c^3 I_{\text{rot}} P_s \dot{P}_s}{2\pi \mu_0} \right)^{1/2}. \quad (5.38)$$

In Section 5.2 of Vol. 5 the field on the axis of a magnetic dipole was found to be $\mu_0 \mu_m / 2\pi r^3$ at a distance r from it along its axis. We take the field outside the star to be that of a dipole at its centre, which is exact outside a uniformly magnetised sphere, a result whose proof is beyond this course. The field at a magnetic pole of a neutron star of radius R is $B_p = \mu_0 \mu_m / 2\pi R^3$, and

$$B_p \sin \alpha = \frac{1}{R^3} \left(\frac{3\mu_0 c^3 I_{\text{rot}} P_s \dot{P}_s}{8\pi^3} \right)^{1/2}. \quad (5.39)$$

The field goes as $(P_s \dot{P}_s)^{1/2}$. For the Crab pulsar, with $R = 10$ km and $I_{\text{rot}} = 10^{38}$ kg m², the formula gives a field of about 8×10^8 T at the pole, which agrees in order of magnitude with the estimate from the frozen-in flux above, and most pulsars have fields between about 10^7 and 10^9 T.

Pulsars are often displayed on a plane with $\log P_s$ and $\log \dot{P}_s$ as axes, as in Figure 79. Since t_{char} goes as $P_s / \dot{P}_s$ and B_p as $(P_s \dot{P}_s)^{1/2}$, lines

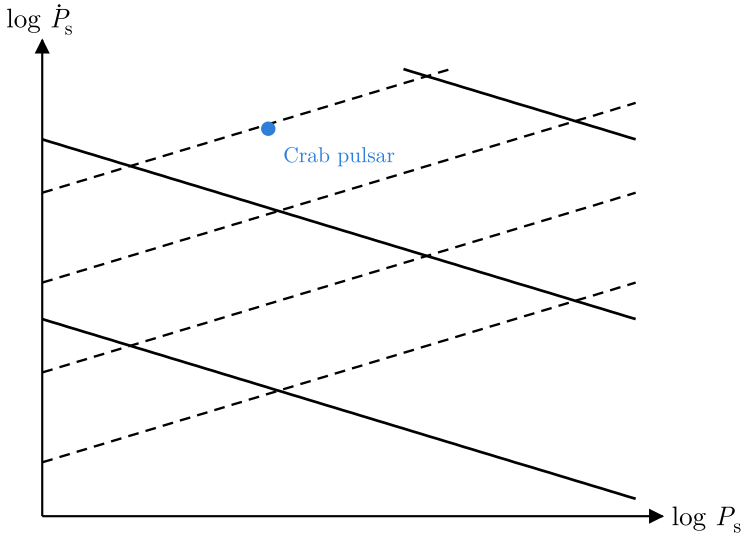


Figure 79. The plane of the spin period P_s and its rate of change $\dot{P}_s$, with logarithmic axes. The dashed lines are lines of constant characteristic age $P_s/2\dot{P}_s$, and the age increases from the upper left to the lower right. The solid lines are lines of constant magnetic field, with the field increasing towards the upper right. The Crab pulsar is marked.

of constant characteristic age have slope one on these axes and lines of constant field have slope minus one, and a pulsar with a constant field moves down a line of slope minus one as it slows. The Crab pulsar lies near the upper left, young and fast. The pulsars with periods of a few milliseconds lie near the lower left, with fields of only 10^4 to 10^5 T and ages of billions of years; they are thought to be old neutron stars spun up by matter falling onto them from a companion, a process of accretion that is not treated in this course.

Black Holes

When the core left by a collapse has a mass greater than the maximum mass of a neutron star, no known pressure can support it, and it collapses to a black hole. Which stars end in this way is not yet well established, although stars that begin with more than about twenty to twenty-five solar masses are thought to be the likeliest to do so.

A black hole is an object from which not even light can escape. You will have met at school its Schwarzschild radius,

$$R_S = \frac{2GM}{c^2}, \quad (5.40)$$

which is the radius at which the escape speed $\sqrt{2GM/R}$ of Newtonian gravity would equal c . The correct description is given by general relativity, which gives the same expression for the radius of the surface from within which nothing can escape, and general relativity is not treated in this course. For the Sun the Schwarzschild radius is about 3 km, and for a neutron star of $1.4 M_\odot$ it is about 4 km, only a few times smaller than the radius of the star, which is why general relativity is needed to describe neutron stars.

An isolated black hole emits almost no light, but a black hole in a binary can be found from the motion of its companion. The radial velocity of the visible star gives the mass function of Section 2.2,

$$f_m = \frac{M_2^3 \sin^3 i}{(M_1 + M_2)^2}, \quad (5.41)$$

where M_1 is the mass of the visible star, M_2 that of the unseen companion, and i the inclination of the orbit. The mass function is found from the period and the semi-amplitude of the radial velocity alone, but i and M_1 are usually not known. You will show in Problem 5 that the mass of the companion is never less than f_m , for every value of i and M_1 , as Figure 80 illustrates.

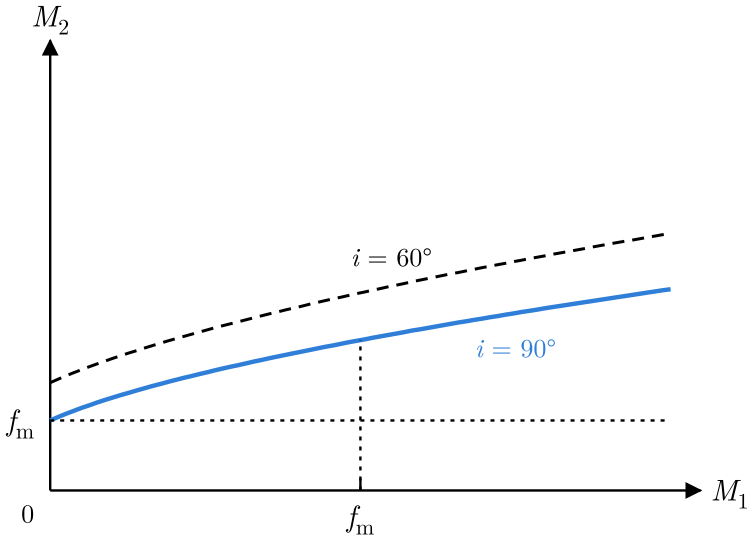


Figure 80. The mass M_2 of the unseen companion in a binary with mass function f_m , as a function of the mass M_1 of the visible star, for an orbit seen edge on (solid) and for an orbit of smaller inclination (dashed). The dotted line is $M_2 = f_m$, and no curve lies below it. The vertical dotted line marks $M_1 = f_m$.

If the mass function itself exceeds the maximum mass of a neutron star, the companion must be a black hole. As an example, the binary V404 Cygni has a mass function of about $6 M_\odot$, so its unseen companion has a mass greater than $6 M_\odot$, well above the largest mass of a neutron star. About twenty such binaries are known in the Galaxy, and in many of them gas drawn from the visible star is heated until it emits X-rays as it falls towards the black hole; the discs of gas that form around compact objects in this way are not treated in this course. Since 2015 black holes have also been detected through the gravitational waves emitted when two of them

spiral together and merge.

Problems

§5.1 Degenerate Matter and White Dwarfs

1. Use the result of Section 4.3 of Vol. 2, that a particle in a volume V with momentum in a region of volume d^3p has $V d^3p/h^3$ states of motion, to count the states with momentum less than p_F , and show that it gives the same Fermi momentum as the count of Figure 68.
2. The quantum concentration of electrons at temperature T is $n_Q = (2\pi m_e k_B T/h^2)^{3/2}$. Show that $E_F/k_B T = \frac{1}{4} 3^{2/3} \pi^{1/3} (n_e/n_Q)^{2/3}$, so that the electrons are degenerate when $n_e \gg n_Q$.
3. For the non-relativistic white dwarf with $E = C_1/R^2 - C_2/R$, show that at the radius of least energy $2E_K + E_{\text{grav}} = 0$, and that the total energy there is $-E_K$. How does this energy depend on the mass of the white dwarf?
4. Find the density ρ_{rel} at which $p_F = m_e c$. Using the non-relativistic mass-radius relation, find the mass at which the mean density of a white dwarf equals ρ_{rel} , and show that it is $(4/5)^{3/2} M_{\text{Ch}}$.
5. In the non-degenerate surface layer of a white dwarf, the pressure obeys the equation of hydrostatic equilibrium $dP/dr = -GM\rho/r^2$, the temperature obeys the equation of radiative diffusion $dT/dr = -3\kappa\rho L/64\pi\sigma_{\text{SB}}r^2T^3$ of Section 4.3, the gas is ideal with $P = \rho k_B T/\mu m_H$, and the opacity has the Kramers form $\kappa = \kappa_0 \rho T^{-7/2}$. Show that ρ goes as $(M/L)^{1/2} T^{13/4}$ in the layer, taking the pressure and temperature to be small at the surface. The layer ends where the electrons become degenerate, at a density proportional to $T^{3/2}$. Show that the luminosity is proportional to $MT_c^{7/2}$, where T_c is the temperature at the base of the layer, which is the temperature of the interior.

§5.2 Core Collapse and Supernovae

1. Let $E_b(^{56}\text{Fe})$ and $E_b(^4\text{He})$ be the binding energies of the nuclei of iron-56 and helium-4. Show that the energy absorbed in breaking one nucleus of iron-56 into thirteen nuclei of helium-4 and four neutrons is $E_b(^{56}\text{Fe}) - 13E_b(^4\text{He})$, and that the energy absorbed in breaking one nucleus of helium-4 into nucleons is $E_b(^4\text{He})$. Find the total energy absorbed per unit mass when iron is broken completely into nucleons, and compare it with the thermal energy per unit mass, $\frac{3}{2}k_B T/56m_u$, of a gas of iron nuclei at temperature T .
2. A detector holding N_p free protons records the antineutrinos from a supernova at distance d , with the quantities defined in the text. Find the distance beyond which the expected number of events is less than one, and show that doubling the mass of the detector increases this distance by a factor of $\sqrt{2}$.
3. A white dwarf of mass M and radius R is made of carbon and oxygen, and in an explosion a mass M_b of it is burned to nickel-56, releasing the energy ΔE_b for each nucleon. Treating the white dwarf as a uniform sphere, find the condition that the energy released be greater than the gravitational binding energy of the white dwarf, and express it as a condition on M_b/M .
4. Show that the number of cobalt-56 nuclei is greatest at the time $t_{\text{Ni}} t_{\text{Co}} \ln(t_{\text{Co}}/t_{\text{Ni}})/(t_{\text{Co}} - t_{\text{Ni}})$.
5. The apparent magnitude of a supernova is $m = -2.5 \log_{10} F + \text{constant}$, where F is the flux received. Show that on the late part of the light curve $dm/dt = 2.5 \log_{10} e/t_{\text{Co}}$, and explain why this rate does not depend on the distance to the supernova or on the amount of nickel made in it.

§5.3 Neutron Stars and Pulsars

1. Using the mass-radius relations of Sections 5.1 and 5.3, show that the ratio of the radius of a neutron star to that of a white dwarf of the same mass is $(m_e/m_n)(\mu_e m_H/m_n)^{5/3}$, and find its approximate value for $\mu_e = 2$, taking $m_n \approx m_H$ and $m_n/m_e \approx 1840$.

2. The core of a star has radius R_i , spin period P_i , and magnetic field B_i , and it collapses to a uniform sphere of radius R with the same mass. Find the ratio of the rotational kinetic energy after the collapse to that before it, and the ratio of the magnetic energy, taking the energy to be of the order of $B^2/2\mu_0$ times the volume of the star, as in Section 7.3 of Vol. 5. Where does the extra energy come from?
3. A pulsar has spin period P_s and rate of change $\dot{P}_s$. Assuming that it slows down by magnetic dipole radiation with a constant field, find $\ddot{P}_s$ in terms of P_s and $\dot{P}_s$. The quantity $\Omega\ddot{\Omega}/\dot{\Omega}^2$ is called the braking index; show that it equals three for dipole radiation.
4. A pulsar was born with period P_0 and has slowed down by magnetic dipole radiation with a constant field. Show that its true age is $t = (P_s/2\dot{P}_s)[1 - (P_0/P_s)^2]$, and hence that the characteristic age exceeds the true age.
5. Show from the mass function that $M_2 \geq f_m$ for every value of i and of M_1 . For an orbit seen edge on, find M_2 when $M_1 = 0$ and when $M_1 = f_m$, and check the second value against Figure 80.

The Interstellar Medium

6.1 Gas, Dust, and the Phases of the Interstellar Medium

The gas and dust between the stars are called the interstellar medium. The gas is mostly hydrogen and helium, with a mean density near the Sun of about 10^6 m^{-3} , or one atom in every cubic centimetre, and the dust, in small solid grains, makes up about one per cent of the mass. The total mass of the medium is roughly a tenth of the mass of the stars of the Galaxy. It lies between us and distant objects in the plane of the Galaxy, so its effect on starlight must be removed before the magnitudes of Section 1.1 give distances, and it is the material from which stars form and to which dying stars return the elements made in them, as described in Sections 4.5 and 5.2.

Extinction and Reddening

In 1930 Trumpler found that the distances of open clusters obtained from the magnitudes of their stars, with the distance modulus of Section 1.1, were systematically larger than those obtained from their angular diameters, and increasingly so with distance. The stars were fainter than the inverse-square law allowed, and they were also redder than their spectral classes implied. The spectral class of a star, fixed by the relative strengths of its absorption lines

as in Section 3.1, sets its intrinsic colour, and these ratios of line strengths are not changed by anything that dims the whole spectrum, so a star whose $B - V$ colour index is larger than the value for its class has lost more of its blue light than of its yellow light.

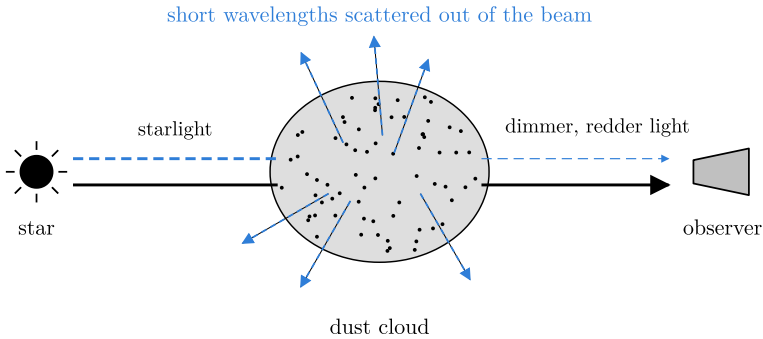


Figure 81. Light from a star passing through a cloud of dust grains. The grains scatter and absorb short wavelengths more strongly than long ones, so the light that reaches the observer is fainter and redder than the light that left the star. The scattered light leaves the beam in other directions.

The dimming, which is called interstellar extinction, and the change of colour, which is called interstellar reddening, are both caused by grains of dust, which absorb light and scatter it out of the line of sight more efficiently at short wavelengths than at long ones, as Figure 81 shows.

Let $F_{\lambda,0}$ be the flux per unit wavelength that would arrive from a star if there were no dust, and let F_{λ} be the flux that arrives. The dust neither creates nor adds light at the wavelengths of starlight, so along the line of sight the formal solution of the equation of radiative transfer in Section 3.3, with the source function set to zero, gives

$$F_{\lambda} = F_{\lambda,0} e^{-\tau_{\lambda}}, \quad (6.1)$$

where τ_{λ} is the optical depth of the dust along the line of sight. Let n_d be the number density of grains and σ_d the cross-section with which a grain removes light from the beam, by absorption and by scattering together. By the argument that gave the mean free path

of a molecule in Section 1.3 of Vol. 2 of the Fermi Notes on Physics, and the mean free path of a photon $1/n\sigma_\nu$ in Section 3.3, a photon travels a mean distance $1/n_d\sigma_d$ before it meets a grain, and over a path of length x through a uniform cloud $\tau_\lambda = n_d\sigma_d x$, the product $n_d\sigma_d$ taking the place of the product $\kappa_\lambda\rho$ of Section 3.3.

The number of magnitudes by which the dust makes the star fainter at the wavelength λ is called the extinction A_λ . With the definition of the magnitude scale in Section 1.1, $m_1 - m_2 = -2.5 \log_{10}(F_1/F_2)$, this is

$$A_\lambda = -2.5 \log_{10} \frac{F_\lambda}{F_{\lambda,0}} = 2.5 \tau_\lambda \log_{10} e \approx 1.086 \tau_\lambda, \quad (6.2)$$

where the second step uses $\log_{10} e^{-\tau} = -\tau \log_{10} e$. An extinction of one magnitude therefore corresponds to an optical depth of about one. The distance modulus of Section 1.1 gains a term, and in the V band it becomes

$$m_V - \mathcal{M}_V = 5 \log_{10} \frac{d}{10 \text{ pc}} + A_V, \quad (6.3)$$

where $\mathcal{M}_V$ is the absolute magnitude in the V band. If the extinction is ignored, the distance obtained is too large by the factor $10^{A_V/5}$, which is what Trumpler found. In the plane of the Galaxy near the Sun the extinction averages about one magnitude for every kiloparsec of path, although it is very patchy, and towards the Galactic centre A_V is about 30 magnitudes, so that the centre is hidden from optical telescopes.

The dependence of the extinction on wavelength, found by comparing the spectra of two stars of the same spectral class, one behind dust and one not, is called the extinction curve, and its average form is drawn in Figure 82.

Since A_λ is larger at the B band than at the V band, the colour index of a reddened star exceeds its intrinsic value. Let $(m_B - m_V)_0$ be the intrinsic B – V colour index, which is known from the spectral class. The difference

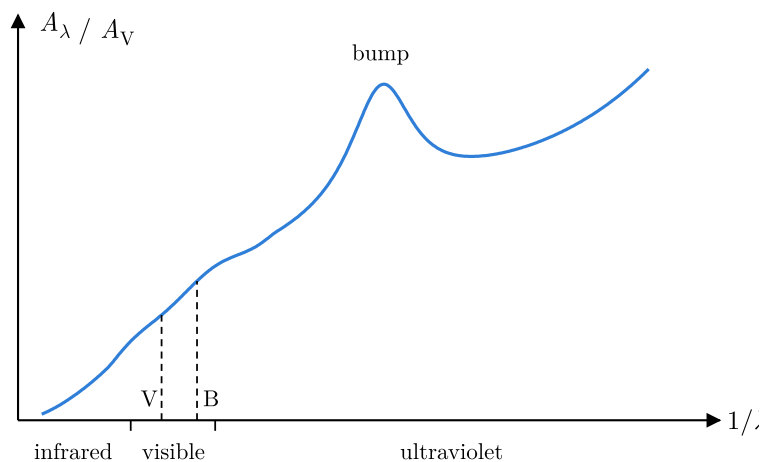


Figure 82. The mean interstellar extinction curve, A_λ/A_V plotted against $1/\lambda$, computed from a standard fit to the extinction measured along many lines of sight in the diffuse interstellar medium. The dashed lines mark the V and B bands, and the ticks on the axis mark the ends of the visible range. The extinction rises roughly in proportion to $1/\lambda$ through the visible, and has a broad bump in the ultraviolet.

$$E_{B-V} = (m_B - m_V) - (m_B - m_V)_0 = A_B - A_V \quad (6.4)$$

is called the colour excess, and the second equality follows because the extinction adds A_B to m_B and A_V to m_V . The colour excess can be measured for any star whose spectral class is known, while the extinction itself cannot be measured without knowing the distance. The colour excess and the extinction are connected by the ratio

$$\mathcal{R}_V = \frac{A_V}{E_{B-V}}, \quad (6.5)$$

which is called the ratio of total to selective extinction. It is found by observation to be close to 3.1 along most lines of sight through the diffuse medium, and larger, up to about 5, through dense clouds. The ratio depends only on the shape of the extinction curve, since it is $A_V/(A_B - A_V)$, and you will show in Problem 1 how it follows from an extinction proportional to $1/\lambda$.

Correcting a Distance for Extinction

As an example, consider a main-sequence star of known spectral class, lying in the plane of the Galaxy behind an unknown amount of dust. Nearby stars of the same class, whose distances are known from their parallaxes, give its absolute magnitude $\mathcal{M}_V$ and its intrinsic colour $(m_B - m_V)_0$. The measured magnitudes m_B and m_V then give the colour excess, and with the ratio $\mathcal{R}_V$ the extinction is $A_V = \mathcal{R}_V E_{B-V}$. Solving the distance modulus above for d ,

$$d = 10 \text{ pc} \times 10^{(m_V - \mathcal{M}_V - \mathcal{R}_V E_{B-V})/5}. \quad (6.6)$$

The correction is large. An extinction of one magnitude, which is typical of a path of a kiloparsec in the plane, would make the uncorrected distance too large by a factor $10^{0.2}$, or by about 60 per cent. The method relies on $\mathcal{R}_V$ being the same along the path as along the lines of sight from which it was measured, which is a valid approximation in the diffuse medium and less reliable through dense clouds.

The Dust Grains

The shape of the extinction curve indicates the size of the grains. A particle much smaller than the wavelength scatters light in proportion to λ^{-4} , which is why the molecules of the air make the sky blue, and a particle much larger than the wavelength removes light equally at all wavelengths, which is why the water droplets of a cloud look white. The dependence on $1/\lambda$ between these limits is the behaviour of grains whose size is comparable to the wavelength divided by 2π , from a few nanometres up to about $0.3 \mu\text{m}$. The theory of scattering by small spheres is not treated in this course.

Absorption bands in the infrared show that the grains are made of silicates, the minerals of ordinary rock, and of carbon, to which the ultraviolet bump of Figure 82 is attributed, with mantles of ice in cold clouds. The energy that the grains absorb from starlight is radiated again in the far infrared, since the grains are cold, and you will estimate their temperature in Problem 2.

The dust is well mixed with the gas. The number of hydrogen atoms per unit area along a line of sight, called the column density N_{H} , can be measured from the absorption lines of hydrogen in the ultraviolet spectra of stars, and it is found by observation to be proportional to the colour excess, with

$$\frac{N_{\text{H}}}{E_{\text{B-V}}} \approx 5.8 \times 10^{25} \text{ m}^{-2} \text{ mag}^{-1} \quad (6.7)$$

along most lines of sight. Combining this ratio with the sizes and densities of the grains shows that the dust makes up about one per cent of the mass of the interstellar medium.

The 21-Centimetre Line of Neutral Hydrogen

Most of the hydrogen in the interstellar medium is atomic, neutral, and in the ground state. It cannot absorb the Balmer lines of Section 3.1, which need atoms in the level $n = 2$, and the Lyman lines that it does absorb lie in the ultraviolet, which does not pass through the atmosphere. The ground state is itself split into two levels, however, and a transition between them gives a line in the radio band.

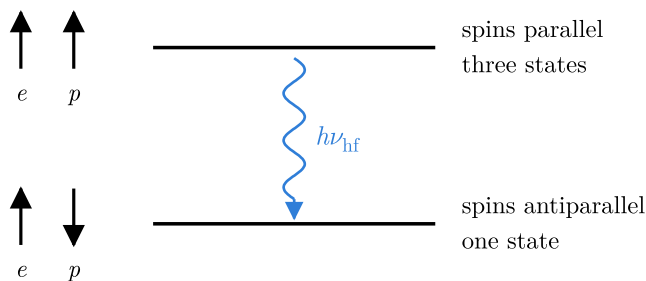


Figure 83. The splitting of the ground state of hydrogen by the interaction of the spins of the electron (e) and the proton (p). The upper level, in which the spins are parallel, has three states and the lower level, in which they are antiparallel, has one. A transition from the upper level to the lower emits a photon of energy $h\nu_{\text{hf}}$, whose wavelength is about 21 centimetres.

The electron and the proton each have spin one-half, and each carries a magnetic moment proportional to its spin, $\mu = \gamma_{\text{g}}\mathbf{S}$, as in Section 4.2 of Vol. 3 of the Fermi Notes on Physics, where the constant of proportionality is called the gyromagnetic ratio and is written γ ; in these notes γ is the ratio of heat capacities, so we write γ_{g} . Each moment feels the magnetic field of the other, so the energy of the atom depends on the relative orientation of the two spins. There are four spin states of the pair, since each spin has two states along any axis. They form a level of three states, in which the spins are said to be parallel, and a level of one state, in which they are said to be antiparallel, as shown in Figure 83; the grouping of the four states into levels of three and one follows from the rules for adding angular momenta, which are beyond this course. The level with the spins parallel lies higher, by $h\nu_{\text{hf}}$ with

$$\nu_{\text{hf}} \approx 1420.4 \text{ MHz}, \quad \lambda_{\text{hf}} = \frac{c}{\nu_{\text{hf}}} \approx 21.1 \text{ cm}. \quad (6.8)$$

This splitting is called the hyperfine splitting of the ground state, and the line is called the 21-centimetre line. Its energy, about $5.9 \times$

10^{-6} eV, is smaller than the ionisation energy of hydrogen, $\chi_{\text{H}} \approx 13.6$ eV, by a factor of more than a million, for two reasons. The energy of two magnetic moments a Bohr radius apart is smaller than the electrostatic energy of the electron and the proton at that distance by a factor of order $(v/c)^2$, where v is the speed of the electron in the Bohr orbit and v/c is the fine-structure constant, about $1/137$, as you found in Problem 1 of Section 1.3 of Vol. 3; and the gyromagnetic ratio of a particle is of order its charge divided by its mass, as Section 4.2 of Vol. 3 showed for the electron, so the moment of the proton is smaller than that of the electron by a further factor of the order of m_e/m_p .

The populations of the two levels follow from the Boltzmann factor with statistical weights, as in Section 3.2. With the statistical weights $g_{\text{u}} = 3$ for the upper level and $g_{\text{l}} = 1$ for the lower, the ratio of the number density n_{u} in the upper level to the number density n_{l} in the lower level is

$$\frac{n_{\text{u}}}{n_{\text{l}}} = 3 e^{-h\nu_{\text{hf}}/k_{\text{B}}T}. \quad (6.9)$$

The energy $h\nu_{\text{hf}}$ corresponds to a temperature $h\nu_{\text{hf}}/k_{\text{B}} \approx 0.068$ K, far below any temperature found in the interstellar medium, so the exponential is very close to one and three quarters of the atoms are in the upper level.

An atom in the upper level returns to the lower by emitting a photon with a probability per unit time $\Gamma_{\text{sp}} \approx 2.9 \times 10^{-15} \text{ s}^{-1}$, so an isolated atom would remain in the upper level for about eleven million years. In cold gas an atom collides with another every few hundred years, so collisions keep the populations in the ratio above, and the populations follow the temperature of the gas. The line is nevertheless strong, because a line of sight through the Galaxy passes through an enormous number of atoms.

The strength of the line gives the column density directly. Consider a line of sight through a cloud, and let N_{u} be the column density of atoms in the upper level, that is, the number in a column of unit cross-section along the line of sight. Each of these atoms emits photons of energy $h\nu_{\text{hf}}$ at the rate Γ_{sp} , equally in all directions,

so the energy emitted per unit time from unit area of the column into the solid angle $d\Omega$ is $N_u \Gamma_{\text{sp}} h\nu_{\text{hf}} d\Omega/4\pi$. Provided the gas is optically thin at the frequency of the line, so that absorption along the line of sight can be neglected, the equation of radiative transfer of Section 3.3 reduces to the sum of the emission along the ray, and the specific intensity integrated over the width of the line is

$$\int I_\nu d\nu = \frac{h\nu_{\text{hf}}}{4\pi} \Gamma_{\text{sp}} N_u = \frac{3h\nu_{\text{hf}}}{16\pi} \Gamma_{\text{sp}} N_H, \quad (6.10)$$

where the second step uses $N_u = \frac{3}{4} N_H$. The result does not depend on the temperature of the gas, since the fraction of atoms in the upper level is three quarters at every interstellar temperature, and a measurement of the integrated intensity therefore gives the column density of neutral hydrogen.

The line was predicted by van de Hulst in 1944 and first detected by Ewen and Purcell in 1951. Its wavelength is so much larger than the grains that the dust is transparent to it, so the neutral gas can be observed throughout the Galaxy, and the Doppler shift of the line gives the radial velocity of the gas that emits it.

Emission Nebulae

The O and early B stars of Section 3.1 are hot enough to emit a large number of photons with energies above χ_H , which ionise the hydrogen around them. A star of this kind sits inside a cloud of ionised gas, which is called an H II region, since astronomers usually write H I for neutral hydrogen and H II for ionised hydrogen; an example is the Orion Nebula. Protons and electrons in the cloud recombine, usually into an excited level, and the atom drops to the ground state in a series of transitions. Many of the photons emitted are Balmer photons, and the red line $H\alpha$, from $n = 3$ to $n = 2$, gives these nebulae their colour in photographs. The balance between heating by the photoelectrons and cooling by lines of oxygen and other ions excited in collisions holds the temperature near 10^4 K. The glowing clouds of ionised gas are called emission nebulae, and H II regions are one kind; others include the planetary nebulae of Section 4.7 and the remnants of the supernovae of Section 5.2.

The Strömgren Sphere

The size of an H II region follows from the balance between ionisation and recombination. Let $\dot{N}_{\text{ion}}$ be the number of photons with energies above χ_{H} that the star emits per unit time, and let the star sit in a uniform cloud of pure hydrogen with n_{H} atoms, neutral or ionised, per unit volume. The rate at which a given proton captures an electron is proportional to the number density of electrons, for the same reason that the collision rate of a molecule in Section 1.3 of Vol. 2 is proportional to the number density of the molecules it can strike, so the number of recombinations per unit volume per unit time is

$$n_{\text{e}} n_{\text{p}} \alpha_{\text{rec}} , \quad (6.11)$$

where the constant α_{rec} is called the recombination coefficient. It depends weakly on the temperature and is about $2.6 \times 10^{-19} \text{ m}^3 \text{ s}^{-1}$ at 10^4 K . A recombination directly into the ground state emits a photon that can itself ionise an atom, and such photons are absorbed close to where they are made, so this value of α_{rec} counts only recombinations into the excited levels.

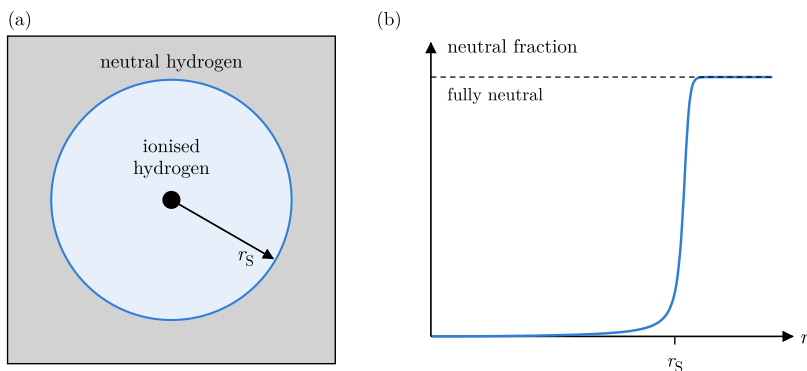


Figure 84. (a) A hot star inside a uniform cloud of hydrogen ionises a sphere of radius r_{S} , the Strömgren sphere, whose edge is sharp. (b) The fraction of the hydrogen that is neutral, plotted against the distance r from the star, computed from the balance of ionisation and recombination at each radius for a gas in which the mean free path of an ionising photon in neutral hydrogen is much smaller than r_{S} .

In a steady state every ionising photon from the star is used up in balancing one recombination. Inside the ionised region the gas is almost fully ionised, with $n_e = n_p = n_H$, and if the region is a sphere of radius r_S , as drawn in Figure 84(a), the balance gives

$$\dot{N}_{\text{ion}} = \frac{4\pi}{3} r_S^3 n_H^2 \alpha_{\text{rec}}, \quad r_S = \left(\frac{3\dot{N}_{\text{ion}}}{4\pi n_H^2 \alpha_{\text{rec}}} \right)^{1/3}. \quad (6.12)$$

The sphere is called the Strömgren sphere, after Strömgren, who worked it out in 1939, and r_S is called the Strömgren radius. An O star emits of order 10^{49} ionising photons per second, and in gas with $n_H \approx 10^8 \text{ m}^{-3}$ its Strömgren radius is a few parsecs. The steady state is a valid approximation because the time $1/n_e \alpha_{\text{rec}}$ that a proton waits to capture an electron, about a thousand years at this density, is much shorter than the lifetime of an O star.

The edge of the sphere is sharp. Inside it, each neutral atom is ionised so often by the photons from the star that only a small fraction of a per cent of the gas is neutral, as follows from balancing ionisations against recombinations at each radius. Near r_S the photons have almost all been used up, and those that remain are absorbed in the neutral gas beyond within a few mean free paths $1/n_H \sigma_{\text{ion}}$, where $\sigma_{\text{ion}} \approx 6 \times 10^{-22} \text{ m}^2$ is the cross-section for photoionisation near the threshold. For the values above this mean free path is about 10^{-4} of r_S , so the neutral fraction rises from nearly zero to one in a thin shell, called the ionisation front, as Figure 84(b) shows.

The number $\dot{N}_{\text{ion}}$ depends very steeply on the temperature of the star, because only photons far out in the tail of the Planck spectrum have energies above χ_H . You will show in Problem 3 that a star twice as hot as another of the same radius emits more ionising photons by a factor of more than a hundred, when the hotter star is an O star. This is why the large H II regions surround O stars, and why stars cooler than the early B stars ionise very little gas.

The Phases of the Medium

The interstellar gas is found in a small number of distinct conditions, called phases, each with a typical temperature and density. The main phases found by observation are the following, and the list is not complete, since the dense molecular clouds and the H II regions around hot stars form further phases:

1. the cold neutral medium, at $T \approx 80$ K with $n \approx 3 \times 10^7 \text{ m}^{-3}$, in clouds and sheets seen in the 21-centimetre line;
2. the warm medium, at $T \approx 8000$ K with $n \approx 3 \times 10^5 \text{ m}^{-3}$, partly neutral and partly ionised;
3. the hot ionised medium, at $T \approx 10^6$ K with $n \approx 3 \times 10^3 \text{ m}^{-3}$, seen in soft X-rays.

The molecular clouds, at 10 to 20 kelvin and densities above about 10^9 m^{-3} , hold their hydrogen as the molecule H_2 and are held together by their gravity; the formation of stars in them is not treated in this course.

The products nT of the three phases in the list are all close to $3 \times 10^9 \text{ K m}^{-3}$, so their pressures $P = nk_B T$ are nearly the same, about $4 \times 10^{-14} \text{ Pa}$. The phases are therefore in approximate pressure equilibrium with one another. Pressure equilibrium is to be expected between regions that touch, since a region at higher pressure than its surroundings expands and pushes the surrounding gas back, in about the time a sound wave takes to cross it, until the pressures are equal.

The cold and warm phases can coexist at one pressure because of the way the gas is heated and cooled. It is heated mainly by electrons that starlight knocks out of the dust grains, and cooled when collisions excite atoms and ions that then radiate. The cooling rises steeply with temperature near a few tens of kelvin, where a far-infrared line of singly ionised carbon is excited, and again near 8000 K, where collisions begin to excite hydrogen, and it changes slowly between. Over a range of pressures, heating and cooling therefore balance at two stable temperatures, and gas at an intermediate temperature heats or cools until it reaches one of them. The two phases resemble a liquid and its vapour coexisting at one pressure, as in Section 6.1 of Vol. 2, although each phase here is

maintained by a steady balance of heating and cooling. The calculation of these rates is not treated in this course.

The hot ionised medium cools so slowly, at its low density, that the blast waves of supernovae, occurring every few decades somewhere in the Galaxy, reheat it before it can cool. It is believed to fill a large fraction of the volume of the disc, perhaps a half, while holding only a small fraction of the mass. The dynamics of blast waves is not treated in this course.

Problems

§6.1 Gas, Dust, and the Phases of the Interstellar Medium

1. Suppose that the extinction is exactly proportional to $1/\lambda$, and let the B and V bands have the effective wavelengths λ_B and λ_V . Show that $\mathcal{R}_V = \lambda_B/(\lambda_V - \lambda_B)$.
2. A spherical grain of radius a_g lies at a distance r from a star of luminosity L , far enough away that the star can be treated as a point. Suppose that the grain absorbs all the starlight falling on its cross-section, and radiates as a blackbody at a uniform temperature T_g . Show that $T_g = (L/16\pi\sigma_{\text{SB}}r^2)^{1/4}$, and that it does not depend on a_g . Real grains radiate less efficiently than a blackbody at wavelengths much larger than their size. Explain whether this makes them hotter or cooler than T_g .
3. Treat a star of radius R as a blackbody at temperature T_{eff} , with $x_0 = \chi_H/k_B T_{\text{eff}}$ much larger than one. Using the relation between intensity and flux of Section 3.3, and dropping the -1 in the denominator of the Planck function, show that the number of ionising photons it emits per unit time is $\dot{N}_{\text{ion}} \approx (8\pi^2 R^2/c^2)(k_B T_{\text{eff}}/h)^3(x_0^2 + 2x_0 + 2)e^{-x_0}$. Find the ratio of $\dot{N}_{\text{ion}}$ for two stars of the same radius whose temperatures differ by a factor of two, in terms of x_0 for the hotter star.
4. An H II region of radius r_S surrounds a star in gas of density n_H . Find how r_S , the total mass of ionised gas, and the ratio of the thickness of the ionisation front to r_S change if the same star sits in gas of density $2n_H$.

5. A cold cloud at temperature T_{cold} is in pressure equilibrium with warm gas at temperature T_{warm} . Find the ratio of their densities. If the cold and warm phases contain equal masses of gas, find the ratio of the volumes they fill.

The Milky Way

7.1 The Milky Way

The faint band of light called the Milky Way, which crosses the night sky, is the disc of our Galaxy seen from inside. Galileo resolved it into stars in 1610, and the Herschels, counting stars in different directions in the 1780s, concluded that the stars form a flattened system. The word Galaxy, with a capital letter, is used for this system, and galaxy, with a small letter, for the many others like it. The Galaxy contains a few hundred billion stars, with a total mass in stars of about $5 \times 10^{10} M_{\odot}$, together with the gas and dust of the previous part.

The Disc, the Bulge, and the Halo

The stars of the Galaxy belong to a small number of components, which are drawn in Figure 85. The positions of stars in the disc are described with the cylindrical coordinates (s, ϕ, y) , which are those of Section 3.4 of Vol. 4 of the Fermi Notes on Physics with the height written y in place of z , since z is kept for the redshift. The origin is at the Galactic centre and the y -axis lies along the axis of rotation of the disc, so that s is the distance from the axis and y the height above the plane. The main components are the following:

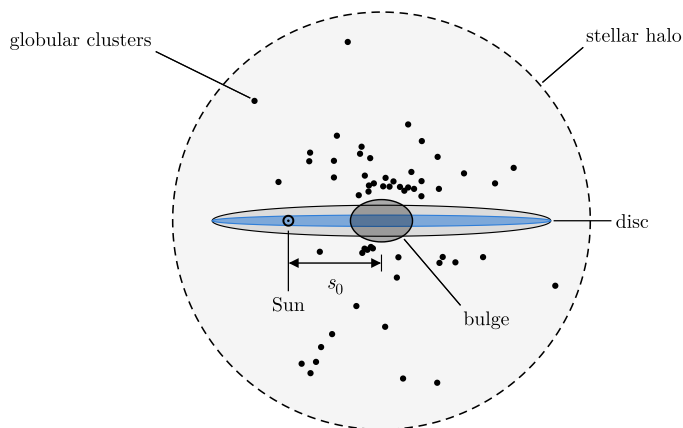


Figure 85. The Galaxy seen edge on. The thin disc (accent) lies inside the thicker disc (shaded), with the bulge at the centre, and the whole is surrounded by the stellar halo (dashed), in which the globular clusters (dots) move. The Sun lies in the disc at a distance s_0 from the centre.

1. the thin disc, which holds most of the stars, all of the gas and dust, and the spiral arms, has a radius of about 15 kpc, and has a thickness described by a scale height of about 300 pc, over which the density of stars falls by a factor e ;
2. the thick disc, with a scale height of about 1 kpc, whose stars are older than those of the thin disc and contain a smaller fraction of heavy elements;
3. the bulge, a concentration of old stars, rich in heavy elements, in the central few kiloparsecs, which counts of stars in the infrared show to be elongated into a bar;
4. the stellar halo, a roughly spherical distribution of old stars that are poor in heavy elements, extending well beyond the disc, which contains most of the globular clusters of the Galaxy.

The Sun lies in the thin disc, a distance s_0 from the centre, close to the plane.

In 1944 Baade, studying the stars of the Andromeda galaxy, divided stars into two groups, which are called stellar populations. Stars of Population I are young, contain heavy elements in proportions close to those of the Sun, with $Z \approx 0.02$, and move on nearly

circular orbits in the disc; examples include the O and B stars, the Cepheid variables of Section 1.1, and the stars of open clusters. Stars of Population II are old, contain much smaller fractions of heavy elements, down to Z of about 10^{-4} or less, and move on orbits that are often elongated and inclined at large angles to the disc; examples include the stars of the halo and of the globular clusters. The two populations differ in composition because each generation of stars returns to the interstellar medium some of the heavy elements made in it, by the processes of Sections 4.5 and 5.2, so that stars formed later are richer in heavy elements. The ages of the globular clusters, found in Section 4.8, are more than about twelve billion years, so the halo formed early in the history of the Galaxy. The division into two populations is a simplification, and the old stars of the bulge, which are rich in heavy elements, fit neither group well.

The Distance to the Galactic Centre

In 1918 Shapley measured the distances of the globular clusters, using variable stars in them whose mean luminosities are nearly all the same, a fact found by observation, in the way the Cepheids of Section 1.1 are used. He found that the clusters form a roughly spherical system whose centre lies far from the Sun, in the direction of the constellation Sagittarius. The globular clusters orbit the centre of mass of the Galaxy, so the centre of their distribution marks the centre of the Galaxy, and Shapley's result showed for the first time that the Sun is far from that centre. His distance, about 15 kpc, was too large, because interstellar extinction, described in the previous part, was not known until 1930 and he did not allow for it. Modern measurements, among them one made from the orbits of stars close to the centre, give

$$s_0 \approx 8.2 \text{ kpc} . \quad (7.1)$$

Directions in the plane of the Galaxy are given by the angle measured at the Sun from the direction of the Galactic centre, increasing towards the direction in which the Sun moves, which is called the Galactic longitude l , so that the Sun moves towards $l = 90^\circ$.

The angle of a direction above or below the plane is called the Galactic latitude.

Differential Rotation

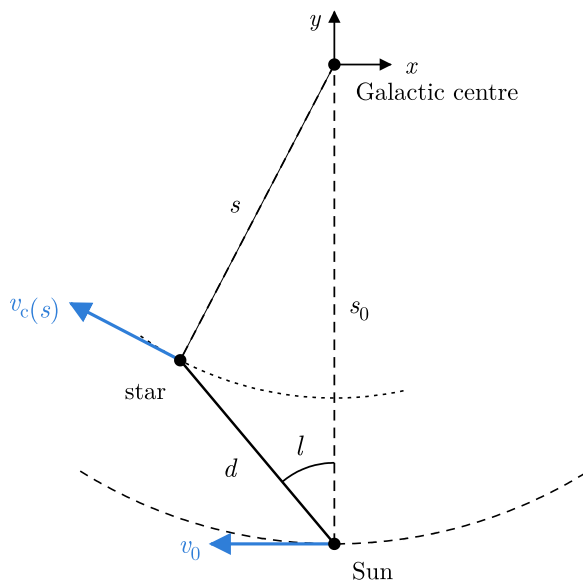


Figure 86. The Galactic plane seen from the side on which the rotation appears clockwise. The Sun is at a distance s_0 from the centre and a star at a distance d from the Sun, in the direction of longitude l , is at a distance s from the centre. Each moves on a circle about the centre, with the circular speed at its radius, v_0 for the Sun and $v_c(s)$ for the star.

The stars and gas of the disc move round the centre on nearly circular orbits. Let $v_c(s)$ be the speed of a circular orbit at radius s , which is called the circular speed, and let

$$\omega(s) = \frac{v_c(s)}{s} \quad (7.2)$$

be the angular speed of that orbit. If the disc rotated as a rigid body, ω would be the same at every radius and the stars would keep fixed positions relative to one another. The disc does not rotate in this way; ω depends on s , and the rotation is called differential.

The speed of the Sun is $v_0 = v_c(s_0) \approx 230 \text{ km s}^{-1}$, and its angular speed is $\omega_0 = v_0/s_0$. An orbit of the Sun takes $2\pi s_0/v_0$, about 220 million years.

What can be measured for another star is its velocity relative to the Sun, and we now find this velocity from the rotation. We set up coordinates in the plane such that the origin is at the Galactic centre, the x -axis points to the right and the y -axis points from the Sun towards the centre, as in Figure 86, which shows the plane from the side on which the rotation appears clockwise. The Sun is at $(0, -s_0)$. The unit vector along the line of sight towards longitude l , and the unit vector across the line of sight in the direction of increasing l , are

$$\hat{\mathbf{n}} = (-\sin l, \cos l), \quad \hat{\mathbf{t}} = (-\cos l, -\sin l), \quad (7.3)$$

and a star at distance d from the Sun is at $(-d \sin l, -s_0 + d \cos l)$. A body at position (x, y) moving clockwise on a circle about the origin with angular speed ω has velocity $\omega(y, -x)$, since this vector is perpendicular to the position, has magnitude ωs , and at the point $(s, 0)$ points in the direction of $-y$, which is the clockwise sense. The velocity of the star is therefore $\omega(s)(-s_0 + d \cos l, d \sin l)$, and that of the Sun is $\omega_0(-s_0, 0)$.

The radial velocity of the star relative to the Sun is the component of the difference of these velocities along $\hat{\mathbf{n}}$. Taking the scalar products,

$$\mathbf{v} \cdot \hat{\mathbf{n}} = \omega(s) [(-s_0 + d \cos l)(-\sin l) + d \sin l \cos l] = \omega(s) s_0 \sin l, \quad (7.4)$$

and $\mathbf{v}_0 \cdot \hat{\mathbf{n}} = \omega_0 s_0 \sin l$, so

$$v_r = [\omega(s) - \omega_0] s_0 \sin l. \quad (7.5)$$

In the same way, $\mathbf{v} \cdot \hat{\mathbf{t}} = \omega(s) [(-s_0 + d \cos l)(-\cos l) - d \sin^2 l] = \omega(s)(s_0 \cos l - d)$ and $\mathbf{v}_0 \cdot \hat{\mathbf{t}} = \omega_0 s_0 \cos l$, so the velocity across the line of sight is

$$v_t = [\omega(s) - \omega_0] s_0 \cos l - \omega(s) d. \quad (7.6)$$

The radial velocity is measured from the Doppler shift of the lines of the spectrum, and v_t from the angular rate v_t/d at which the star moves across the sky, which is called its proper motion, once its distance is known.

The Oort Constants

For stars near the Sun, with $d \ll s_0$, these results can be simplified. The distance of the star from the centre satisfies

$$s^2 = (d \sin l)^2 + (s_0 - d \cos l)^2 = s_0^2 - 2s_0 d \cos l + d^2, \quad (7.7)$$

and to first order in d/s_0 the binomial expansion of the square root gives $s \approx s_0 - d \cos l$. Expanding $\omega(s)$ to first order about s_0 ,

$$\omega(s) - \omega_0 \approx \left(\frac{d\omega}{ds} \right)_0 (s - s_0) \approx - \left(\frac{d\omega}{ds} \right)_0 d \cos l, \quad (7.8)$$

where the subscript 0 means that the derivative is evaluated at s_0 . Substituting into v_r and using $2 \sin l \cos l = \sin 2l$,

$$v_r \approx \mathcal{A} d \sin 2l, \quad \mathcal{A} = -\frac{1}{2} s_0 \left(\frac{d\omega}{ds} \right)_0. \quad (7.9)$$

For v_t , the last term is $-\omega_0 d$ to first order in d , and the first term becomes $-s_0 (d\omega/ds)_0 d \cos^2 l = \mathcal{A} d (1 + \cos 2l)$, using $2 \cos^2 l = 1 + \cos 2l$. Therefore

$$v_t \approx \mathcal{A} d \cos 2l + \mathcal{B} d, \quad \mathcal{B} = \mathcal{A} - \omega_0. \quad (7.10)$$

The constants $\mathcal{A}$ and $\mathcal{B}$ are called the Oort constants, after Oort, who derived these relations in 1927 and used them to show that the Galaxy rotates differentially. Differentiating $\omega = v_c/s$ by the

quotient rule gives $d\omega/ds = (dv_c/ds)/s - v_c/s^2$, and in terms of the circular speed the constants are

$$\mathcal{A} = \frac{1}{2} \left[\frac{v_0}{s_0} - \left(\frac{dv_c}{ds} \right)_0 \right], \quad \mathcal{B} = -\frac{1}{2} \left[\frac{v_0}{s_0} + \left(\frac{dv_c}{ds} \right)_0 \right]. \quad (7.11)$$

The radial velocities of stars at a given distance therefore vary with longitude as $\sin 2l$. Stars towards $l = 45^\circ$ and $l = 225^\circ$ recede and stars towards $l = 135^\circ$ and $l = 315^\circ$ approach, because stars closer to the centre than the Sun move round faster in angle and pull ahead of it, while stars further out fall behind. The amplitude of this pattern, divided by d , gives $\mathcal{A}$, and the proper motions of the stars give $\mathcal{B}$. The measured values are $\mathcal{A} \approx 15 \text{ km s}^{-1} \text{ kpc}^{-1}$ and $\mathcal{B} \approx -12 \text{ km s}^{-1} \text{ kpc}^{-1}$. From them, $\omega_0 = \mathcal{A} - \mathcal{B} \approx 27 \text{ km s}^{-1} \text{ kpc}^{-1}$, which with s_0 gives the value of v_0 quoted above, and $(dv_c/ds)_0 = -(\mathcal{A} + \mathcal{B})$ is small, so the circular speed changes little with radius near the Sun.

The Rotation Curve and the Tangent-Point Method

The circular speed as a function of radius, $v_c(s)$, is called the rotation curve. Inside the orbit of the Sun it is measured with the 21-centimetre line of Section 6.1, which reaches across the whole disc through the dust. Along a line of sight with $0 < l < 90^\circ$, the neutral gas lies at many distances, each at a different radius s , and the line is made up of emission at many radial velocities.

The radial velocity depends on the position of the gas only through $\omega(s)$. In the Galaxy ω decreases outwards, so along the line of sight v_r is largest where s is smallest. By the expression for s^2 above, s^2 is smallest where its derivative with respect to d , $2d - 2s_0 \cos l$, vanishes, which is at $d = s_0 \cos l$, and there

$$s = s_0 \sin l. \quad (7.12)$$

This point is called the tangent point, because the line of sight there touches the circle of radius $s_0 \sin l$, as Figure 87 shows, and the gas there moves directly along the line of sight. The largest radial velocity in the 21-centimetre line towards the longitude l , $v_{r,\max}$,

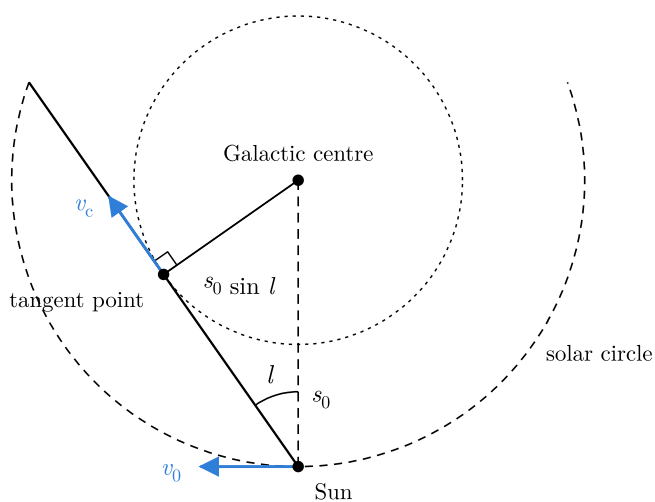


Figure 87. The tangent point. The line of sight from the Sun at longitude l passes closest to the Galactic centre at the point where it touches the circle of radius $s_0 \sin l$, and there the gas moves along the line of sight. The dashed circle is the orbit of the Sun, called the solar circle.

therefore comes from gas at the tangent point, and the expression for v_r with $\omega = v_c/s$ and $s = s_0 \sin l$ gives $v_{r,\max} = v_c(s_0 \sin l) - v_0 \sin l$, or

$$v_c(s_0 \sin l) = v_{r,\max}(l) + v_0 \sin l. \quad (7.13)$$

Measuring the largest radial velocity at each longitude from 0 to 90° gives the rotation curve at every radius from the centre out to s_0 . This is called the tangent-point method. Outside the solar circle the line of sight has no tangent point, and the rotation curve there is measured from objects whose distances are known independently, such as Cepheids and the hot stars in H II regions, using the expression for v_r with s found from d and l .

The Mass inside a Radius

The rotation curve gives the mass of the Galaxy. Suppose, as an approximation, that the mass is distributed with spherical symmetry, and let $M(s)$ be the mass inside a sphere of radius s about the centre. By the shell theorem of Section 5.2 of Vol. 1 of the Fermi Notes on Physics, the shells of mass outside the radius s exert no net force on a star at that radius, and the shells inside act as a point mass $M(s)$ at the centre. For a circular orbit the gravitational force supplies the centripetal acceleration,

$$\frac{v_c^2(s)}{s} = \frac{GM(s)}{s^2}, \quad M(s) = \frac{s v_c^2(s)}{G}. \quad (7.14)$$

For a flattened distribution such as the disc, the shell theorem does not apply and the relation holds only approximately, but it gives the right order of magnitude. At the radius of the Sun, $M(s_0)$ is about $10^{11} M_\odot$.

The Flat Rotation Curve

Measurements from the 21-centimetre line, and at larger radii from stars, clusters, and satellite galaxies, show that the rotation curve of the Galaxy stays nearly flat, with v_c close to v_0 , from a few kiloparsecs out to beyond 20 kpc. The light of the disc falls off

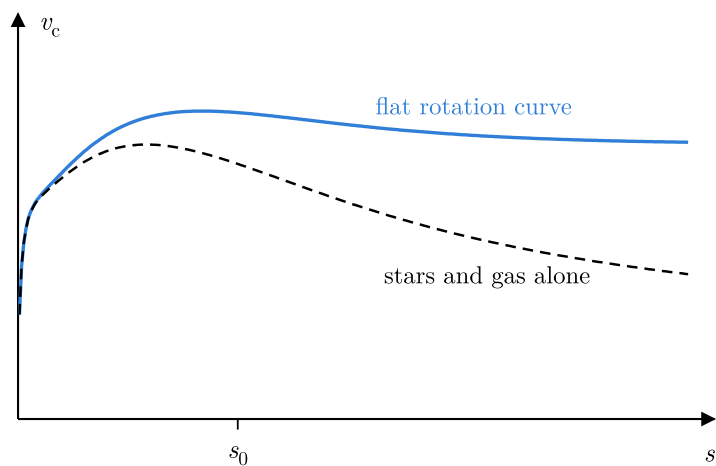


Figure 88. The rotation curve of a model of the Galaxy. The dashed curve is the circular speed that the stars and gas of the disc and bulge would produce alone, which falls at large radii. The actual curve is a flat rotation curve of the kind measured, which requires additional mass at large radii. The tick marks the radius s_0 of the Sun.

with radius, and if the mass followed the light, then beyond most of the disc $M(s)$ would be nearly constant and $v_c = \sqrt{GM/s}$ would fall as $s^{-1/2}$, as it does for the planets of the solar system. Figure 88 compares a rotation curve that falls in this way with the flat curve that is measured.

A flat rotation curve means instead that $M(s) = sv_c^2/G$ grows in proportion to s . For a spherical distribution the mass in a shell between s and $s + ds$ is $4\pi s^2 \rho ds$, so $dM/ds = 4\pi s^2 \rho$, and with $dM/ds = v_c^2/G$ the density is

$$\rho(s) = \frac{v_c^2}{4\pi G s^2}. \quad (7.15)$$

The mass required at large radii is not seen in stars or gas, and it is called dark matter. Its distribution around the Galaxy, which is roughly spherical and extends far beyond the disc, is called the dark halo. Its density cannot go on falling as $1/s^2$ forever, since the total mass would then be infinite, and the motions of satellite galaxies and distant halo stars suggest a total mass for the Galaxy of about $10^{12} M_\odot$, many times the mass in stars and gas.

The Black Hole at the Galactic Centre

At the centre of the Galaxy lies a compact source of radio waves, discovered in 1974 and called Sagittarius A*. The centre is hidden at visible wavelengths by about 30 magnitudes of extinction, but in the infrared, where the extinction is much smaller, individual stars close to the source can be seen. Their positions have been measured since the early 1990s with telescopes using adaptive optics, a technique in which a mirror in the light path changes its shape many times a second to undo the blurring by the atmosphere described in Section 1.2, the distortion being measured on the image of a bright reference star. The stars move on Keplerian ellipses about a common focus at the position of the radio source.

The star called S2 has an orbital period P_{orb} of about 16 years and an eccentricity of about 0.88, and its orbit is drawn in Figure 89. Let M be the mass of the central object. The mass of the star is negligible by comparison, so Kepler's third law, as reviewed in

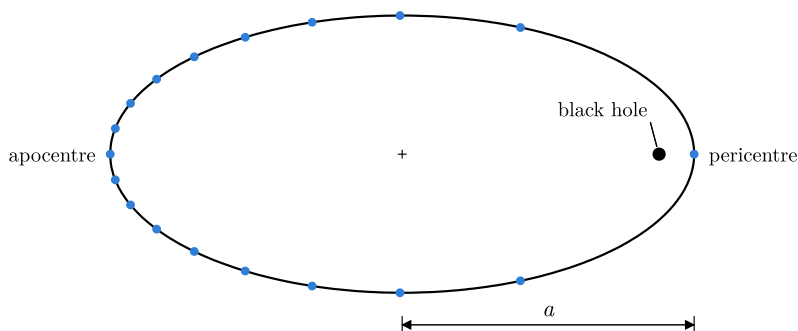


Figure 89. The orbit of a star about the black hole at the Galactic centre, drawn for the eccentricity of the star S2. The black hole lies at a focus of the ellipse of semi-major axis a . The dots mark the positions of the star at equal intervals of time, computed from Kepler's equation, and crowd together near apocentre, where the star moves slowly.

Section 2.1 and derived in Section 5.4 of Vol. 1, gives

$$P_{\text{orb}}^2 = \frac{4\pi^2 a^3}{GM}. \quad (7.16)$$

The orbit on the sky is the true ellipse seen in projection, and fitting the measured positions with the orbital elements of Section 2.1 gives the inclination, the eccentricity, the period, and the semi-major axis as an angle θ_a . Since $a = \theta_a d$, where d is the distance to the centre,

$$M = \frac{4\pi^2 \theta_a^3 d^3}{GP_{\text{orb}}^2}. \quad (7.17)$$

The mass depends on the cube of the distance, so the distance has to be known as well. It is found from the same star. The orbital fit gives the velocity of the star in angle per unit time, and the radial velocity measured from its spectrum gives one component of the velocity in kilometres per second. Since the physical velocity is the angular velocity multiplied by d , fitting the positions and the radial velocities together gives d , and this is how one of the modern

values of s_0 quoted earlier in this section was obtained. You will work out a simple case in Problem 4. The results are

$$M \approx 4 \times 10^6 M_\odot, \quad d = s_0 \approx 8.2 \text{ kpc}. \quad (7.18)$$

At closest approach S2 is only about 120 astronomical units from the centre, so this mass lies within a sphere not much larger than the solar system. The Schwarzschild radius of Section 5.3, $2GM/c^2$, is for this mass about 1.2×10^{10} m, less than a tenth of an astronomical unit, and the closest approach of S2 is about 1500 times larger. No cluster of stars or of stellar remnants of this mass could remain packed into so small a region for long, and the only known object that fits the measurements is a black hole. The arguments about the stability of such clusters belong to stellar dynamics, which is not treated in this course. In 2022 the Event Horizon Telescope, an array of radio telescopes across the Earth combined as an interferometer of the kind described in Section 1.2, whose resolution is set by the separation of its dishes and therefore by the diameter of the Earth, produced an image of the region around Sagittarius A* on the scale of a few Schwarzschild radii.

Problems

§7.1 The Milky Way

1. Find the Oort constants in terms of v_0 and s_0 for a flat rotation curve, for a Keplerian rotation curve with $v_c \propto s^{-1/2}$, and for rigid rotation with $v_c \propto s$. In which of these three cases is the radial velocity of every nearby star zero, and why?
2. The dark halo of a galaxy is modelled as a sphere of density $\rho = v_c^2/4\pi G s^2$ out to a radius s_h , with no mass beyond. Find the mass of the halo, and the circular speed at radii larger than s_h .
3. Show that for $90^\circ < l < 180^\circ$ and a flat rotation curve, the radial velocity of every point on the line of sight is negative, so that the line of sight has no tangent point. Use the geometry of Figure 87 to explain why.

4. A star moves on a circular orbit of radius a about a black hole of mass M at distance d , with the plane of the orbit containing the line of sight. The star is seen to move back and forth on the sky, with the largest angular distance from the black hole θ_{m} , and the semi-amplitude of its radial velocity is K . Find d and M in terms of θ_{m} , K , and P_{orb} .
5. Let $\bar{\rho}$ be the mean density of the mass inside the radius s , treated as a sphere. Show that the period of a circular orbit at radius s is $\sqrt{3\pi/G\bar{\rho}}$ for any form of the rotation curve.

Galaxies

8.1 Galaxies and Their Scaling Relations

Faint patches of light that are not stars, called nebulae, were catalogued in large numbers in the eighteenth and nineteenth centuries, and some of them, such as the Andromeda nebula, showed a spiral form. Whether they were clouds within the Galaxy or systems of stars as large as the Galaxy itself was debated until 1925, when Hubble found Cepheid variables in the Andromeda nebula and used the period–luminosity relation of Section 1.1 to show that it lies far outside the Galaxy. It is now called the Andromeda galaxy, at a distance of about 0.78 Mpc, and the number of galaxies that could be observed with present telescopes is of the order of 10^{11} .

The Hubble Sequence

In 1926 Hubble arranged the galaxies he could resolve into a sequence by their appearance, which is drawn in Figure 90 in the form of a tuning fork that he introduced in 1936. The classes are the following, and the list does not include every galaxy, since dwarf galaxies of low surface brightness and galaxies disturbed by collisions with their neighbours fit none of them well.

The elliptical galaxies have smooth images, elliptical in outline, with the brightness falling steadily from the centre outwards and no disc

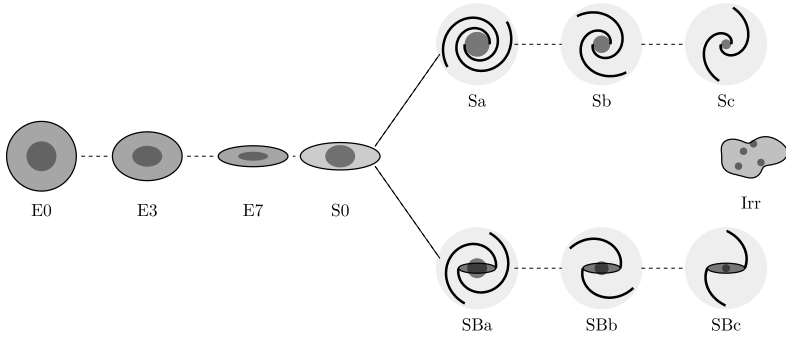


Figure 90. The Hubble sequence. The elliptical galaxies, from E0 to E7, are arranged by the flattening of their images, and the lenticular galaxies, S0, sit where the sequence divides into the ordinary spirals, Sa to Sc, and the barred spirals, SBa to SBc. From a to c the bulge becomes smaller and the arms more open. The irregular galaxies, Irr, lie outside the sequence.

or spiral arms. They are classed by the flattening of the image. Let q be the ratio of the minor axis to the major axis of the image; the class is written E followed by the integer nearest to $10(1 - q)$, so that a circular image is E0 and the flattest ellipticals observed are E7. The class describes the image, which depends on the direction from which the galaxy is seen, since a flattened galaxy seen along its shortest axis gives a circular image. The stars of an elliptical galaxy are old, there is little gas or dust, and the light is red. The stars move on orbits in all directions, and the galaxy is held up against its gravity mainly by the random motions of its stars, whose spread of speeds is measured by the velocity dispersion σ_v described below. Ellipticals range from dwarfs of about $10^7 L_\odot$ to giants of more than $10^{11} L_\odot$.

The spiral galaxies have a disc, with spiral arms in which young blue stars, gas, and dust are concentrated, and a central bulge. They are classed from Sa to Sc by three features that change together along the sequence: the bulge becomes smaller compared with the disc, the arms become more loosely wound, and the fraction of gas and young stars increases. In the barred spirals, classed from SBa to SBc, a bar of stars crosses the centre and the arms begin at its ends. Roughly half of spiral galaxies have bars, and the Galaxy,

with the bar in its bulge described in Section 7.1, is one of them. The disc of a spiral rotates, and the circular speed v_c of its gas and stars is typically between 100 and 300 km s⁻¹.

The lenticular galaxies, S0, have a disc and a bulge, like a spiral, but no spiral arms and little gas, and their stars are old. The irregular galaxies have no regular structure and are usually rich in gas and young stars; examples include the Large and Small Magellanic Clouds, which are companions of the Galaxy.

Hubble thought that the sequence might describe how galaxies change as they age, from left to right. That idea has been abandoned, but the names early type for the ellipticals and lenticulars and late type for Sc galaxies and irregulars are still in use. How galaxies form and why they take these forms is not treated in this course.

Surface Brightness

The light of a galaxy is spread over its image, and the distribution of the light is described by the luminosity emitted per unit area of the galaxy, as seen projected on the sky, which is called the surface brightness and written $\mathcal{I}$. It is a function of position in the image, and for a galaxy with circular symmetry on the sky it depends on the distance s from the centre measured in the plane of the sky, which is the cylindrical radius of Section 7.1 taken about the line of sight through the centre. For a disc seen face on, s is the galactocentric radius in the disc. Astronomers often measure $\mathcal{I}$ in units of $L_\odot \text{pc}^{-2}$.

The flux per unit solid angle received from a galaxy does not depend on its distance, and this is why the surface brightness is useful. A small patch of the galaxy of area dS at distance d emits the luminosity $\mathcal{I} dS$, so the flux arriving from it is $\mathcal{I} dS/4\pi d^2$, and the patch subtends the solid angle $d\Omega = dS/d^2$. The flux received per unit solid angle is therefore

$$\frac{dF}{d\Omega} = \frac{\mathcal{I}}{4\pi}, \quad (8.1)$$

in which d has cancelled; the flux falls as $1/d^2$, and so does the solid

angle over which it is spread. This holds for galaxies near enough that the redshift of their light can be neglected. The flux per unit solid angle is usually expressed in magnitudes per square arcsecond, $m_{\text{sb}} = -2.5 \log_{10} \mathcal{I} + \text{constant}$, the constant depending on the photometric band, so that a larger m_{sb} means a fainter surface. The night sky itself has a surface brightness of about 22 mag arcsec⁻² in the V band at a dark site, and the faint outer parts of galaxies are measured against this background.

The Exponential Disc and the de Vaucouleurs Law

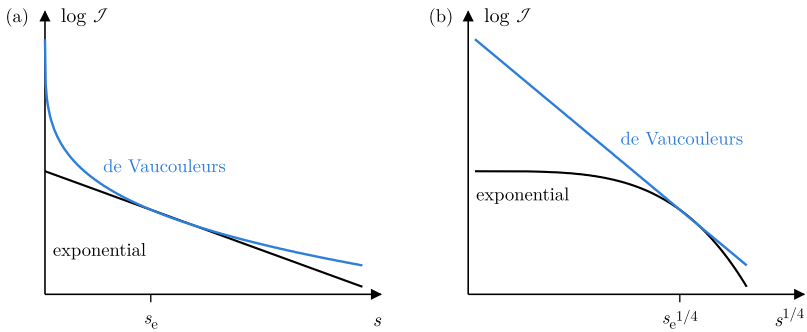


Figure 91. The surface-brightness profiles of an exponential disc (solid) and of the de Vaucouleurs law (accent), drawn with the same effective radius s_e and the same surface brightness there, on a logarithmic scale of $\mathcal{I}$. (a) Against the radius s , the exponential disc is a straight line. (b) Against $s^{1/4}$, the de Vaucouleurs law is a straight line.

The surface brightness of the disc of a spiral galaxy is found by observation to fall exponentially with radius,

$$\mathcal{I}(s) = \mathcal{I}_0 e^{-s/s_d}, \quad (8.2)$$

where $\mathcal{I}_0$ is the central surface brightness and s_d is called the scale length of the disc. For the Galaxy s_d is about 2.6 kpc. On a logarithmic scale the profile is a straight line against s , as in Figure 91(a), and in magnitudes it becomes $m_{\text{sb}}(s) = m_{\text{sb}}(0) + 1.086 s/s_d$, using $2.5 \log_{10} e \approx 1.086$ as in Section 6.1. The luminosity of the

disc is found by adding up rings of radius s and width ds , each of area $2\pi s ds$,

$$L = \int_0^\infty J_0 e^{-s/s_d} 2\pi s ds = 2\pi J_0 s_d^2, \quad (8.3)$$

where the integral $\int_0^\infty s e^{-s/s_d} ds = s_d^2$ follows by integrating by parts, differentiating s and integrating the exponential. In 1970 Freeman found that the central surface brightness of the discs of many large spirals is close to $21.7 \text{ mag arcsec}^{-2}$ in the B band. This result, called Freeman's law, is partly a consequence of how galaxies are selected, since discs of much lower surface brightness are hard to see against the sky, and galaxies of low surface brightness have since been found in large numbers.

The surface brightness of an elliptical galaxy, and of the bulge of a spiral, is found by observation to follow

$$J(s) = J_e \exp \left\{ -7.67 \left[\left(\frac{s}{s_e} \right)^{1/4} - 1 \right] \right\}, \quad (8.4)$$

which is called the de Vaucouleurs law, after de Vaucouleurs, who proposed it in 1948. The radius s_e is called the effective radius, and it is the radius of the circle on the sky that contains half the light of the galaxy; the constant 7.67 is the number that makes this so, and J_e is the surface brightness at s_e . Against $s^{1/4}$ the logarithm of the surface brightness is a straight line, as in Figure 91(b), and against s it is steep at the centre and falls slowly at large radii, as in Figure 91(a). You will show in Problem 3 that the total luminosity is about $7.22\pi s_e^2 J_e$. The profile of a spiral is well described as the sum of a de Vaucouleurs bulge and an exponential disc, and the ratio of the luminosity of the bulge to that of the disc is one measure of the position of the galaxy along the sequence from Sa to Sc.

The Mass-to-Light Ratio

The ratio of the mass of a galaxy, or of a part of it, to its luminosity,

$$\Upsilon = \frac{M}{L}, \quad (8.5)$$

is called the mass-to-light ratio, and it is usually given in units of $M_{\odot}/L_{\odot}$, so that $\Upsilon = 1$ for the Sun. It depends on the photometric band in which L is measured. For a population of stars it depends on which stars dominate the mass and which the light. By the mass–luminosity relation of Section 2.2, found by observation, a main-sequence star of mass M has $L \propto M^{3.5}$ roughly, so the few massive stars of a young population give most of its light while holding little of its mass, and Υ is below one. In an old population the massive stars have died, the light comes from giants and from the stars near the turn-off of Section 4.8, and most of the mass is in faint stars of low mass, so Υ is a few in visible light.

The mass of a spiral galaxy can be found from its rotation curve with $M(s) = sv_c^2/G$ of Section 7.1, and dividing by the luminosity inside the same radius gives the mass-to-light ratio there. Inside the visible disc the values are typically a few times those expected from the stars alone. The rotation curves of spirals, like that of the Galaxy, stay flat out to radii where the light of the disc has fallen to a small fraction of its central value, so the mass inside s keeps growing in proportion to s while the luminosity inside s approaches a constant, and the mass-to-light ratio inside s rises with s . You will find how it rises in Problem 5.

The Virial Theorem for a Galaxy

The virial theorem, derived for a star in Section 4.2 and for a pair of bodies in Section 5.3 of Vol. 1 of the Fermi Notes on Physics, applies to any bound system of bodies moving under their mutual gravity whose motion has reached a steady state. For a galaxy of stars it states that the kinetic energy E_{kin} of the stars and the gravitational potential energy E_{grav} of the galaxy satisfy

$$2E_{\text{kin}} + E_{\text{grav}} = 0, \quad (8.6)$$

with each energy averaged over time. Let M be the mass of the galaxy, R a radius that describes its size, and $\langle v^2 \rangle$ the mean square

speed of its stars, so that $E_{\text{kin}} = \frac{1}{2}M\langle v^2 \rangle$. The potential energy can be written

$$E_{\text{grav}} = -\eta \frac{GM^2}{R}, \quad (8.7)$$

where η is a number of order one that depends on how the mass is distributed, and for a sphere of uniform density of radius R it is $\frac{3}{5}$, as was found in Section 4.2 by assembling the sphere from thin shells. Substituting into the virial theorem gives

$$M = \frac{\langle v^2 \rangle R}{\eta G}. \quad (8.8)$$

For a spiral galaxy, whose stars and gas move on nearly circular orbits at the speed v_c , the mean square speed is $\langle v^2 \rangle = v_c^2$. For an elliptical galaxy the stars move in random directions. The velocity along the line of sight of a star picked at random has a mean of zero, relative to the centre of the galaxy, and a spread, called the velocity dispersion σ_v , which is its standard deviation. If the motions are the same in every direction, each of the three components of the velocity has the variance σ_v^2 , and adding them gives $\langle v^2 \rangle = 3\sigma_v^2$.

The velocity dispersion is measured from the spectrum of the galaxy, which is the sum of the spectra of its stars. Each star contributes its spectrum shifted by the Doppler shift of its radial velocity, so the absorption lines in the combined spectrum are broadened by the spread of radial velocities. The broadened spectrum is compared with the spectrum of a single star of similar type, broadened by a range of trial dispersions, and the dispersion that matches is taken as σ_v . For large ellipticals σ_v is between about 100 and 300 km s⁻¹.

The Tully–Fisher Relation

To apply these results to spiral galaxies we need the circular speed, which is measured from the 21-centimetre line of the neutral gas in the disc. Let the axis of the disc make the angle i with the line of sight, which is the inclination of the disc, and let ϕ be the azimuth of a point of the disc, measured in the disc from the major

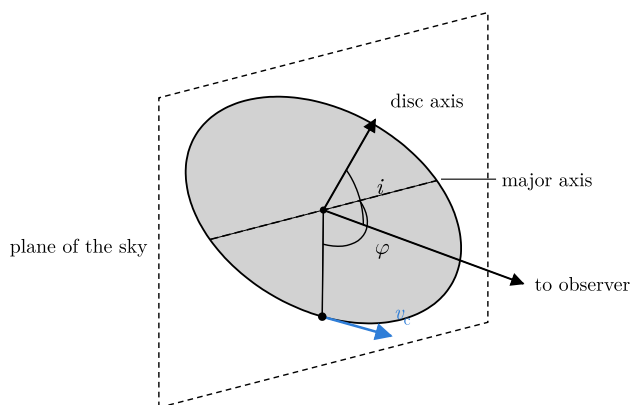


Figure 92. A disc galaxy whose axis makes an angle i with the line of sight to the observer, drawn in perspective. The disc is tilted out of the plane of the sky about its major axis. Gas at the azimuth ϕ from the major axis moves with the circular speed v_c along the ring.

axis of the image, which is the line along which the disc meets the plane of the sky, as in Figure 92. Let $\hat{\mathbf{e}}_1$ be the unit vector along the major axis and $\hat{\mathbf{e}}_2$ the unit vector in the disc perpendicular to it. The gas at azimuth ϕ moves along the ring with velocity $v_c(-\sin \phi \hat{\mathbf{e}}_1 + \cos \phi \hat{\mathbf{e}}_2)$. The vector $\hat{\mathbf{e}}_1$ lies in the plane of the sky and has no component along the line of sight, while $\hat{\mathbf{e}}_2$ is tilted out of the plane of the sky by the angle i and has a component of magnitude $\sin i$ along it. Choosing the sense of $\hat{\mathbf{e}}_2$ so that this component is positive, the radial velocity of the gas relative to the centre of the galaxy is therefore

$$v_r = v_c \sin i \cos \phi. \quad (8.9)$$

The line from the whole disc is the sum of the emission at all these radial velocities, and its shape is drawn in Figure 93. For a flat rotation curve and gas spread evenly in azimuth, the number of elements of gas per unit interval of v_r is proportional to $1/|dv_r/d\phi| = 1/(v_c \sin i |\sin \phi|)$, which becomes large near $\phi = 0$ and $\phi = \pi$, where v_r is close to $\pm v_c \sin i$. The line therefore has two

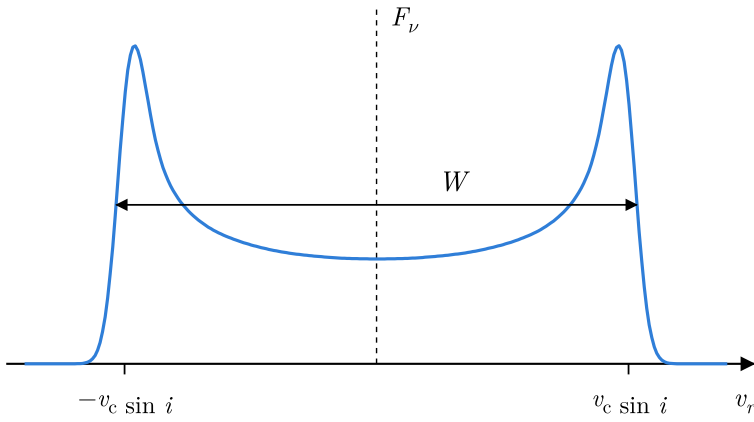


Figure 93. The 21-centimetre line of an inclined disc galaxy, computed for gas spread uniformly over a disc whose rotation curve rises near the centre and is flat beyond, with a small random motion of the gas added. The flux per unit frequency, F_ν , plotted against the radial velocity relative to the centre, has two horns near $\pm v_c \sin i$, and the width W of the line is measured between the points where the flux falls to half its peak.

peaks, called horns, at its edges, and its full width is

$$W \approx 2v_c \sin i. \quad (8.10)$$

The inclination is found from the shape of the image, since a thin circular disc seen at inclination i appears as an ellipse whose axis ratio is $\cos i$, as you will show in Problem 2, and the circular speed is then $v_c = W/2 \sin i$.

In 1977 Tully and Fisher found by observation that the luminosities of spiral galaxies are closely related to their circular speeds, with

$$L \propto v_c^4 \quad (8.11)$$

approximately. The relation is called the Tully–Fisher relation, and it is tightest when the luminosity is measured in the near infrared, where the extinction by the dust of the galaxy itself is small. Its form can be understood from the virial theorem with two assumptions. The first is that all spirals have the same mass-to-light ratio Υ , so that $L = M/\Upsilon$. The second is that they all have the same mean surface brightness inside the radius R , defined as $\langle J \rangle = L/\pi R^2$, as Freeman’s law suggests. With $\langle v^2 \rangle = v_c^2$, the virial mass gives

$$L = \frac{M}{\Upsilon} = \frac{v_c^2 R}{\eta G \Upsilon}, \quad R = \left(\frac{L}{\pi \langle J \rangle} \right)^{1/2}, \quad (8.12)$$

and substituting the second expression into the first, dividing by $L^{1/2}$, and squaring gives

$$L = \frac{v_c^4}{\pi \eta^2 G^2 \Upsilon^2 \langle J \rangle}. \quad (8.13)$$

The derivation gives the exponent found by observation, and it shows that the luminosity rises as the fourth power of the speed because a faster rotation requires a larger mass, and a larger mass, at fixed surface brightness, is spread over a larger radius. Neither

assumption is exact, and the mass inside R includes the dark matter of Section 7.1 as well as the stars, so the tightness of the observed relation suggests that the dark and visible matter of spirals are closely linked.

Distances from the Tully–Fisher Relation

The relation gives the distances of spiral galaxies, and it is one of the rungs of the distance ladder of Section 1.1 beyond the Cepheids. As an example, consider a spiral galaxy at an unknown distance d . The width W of its 21-centimetre line and the axis ratio of its image give v_c . Galaxies near enough for their distances to be found from Cepheids give the constant of proportionality in the relation, which we write $L = C_{\text{TF}} v_c^4$, so the luminosity of the galaxy is known. The measured flux $F = L/4\pi d^2$ then gives

$$d = \left(\frac{L}{4\pi F} \right)^{1/2} = v_c^2 \left(\frac{C_{\text{TF}}}{4\pi F} \right)^{1/2}. \quad (8.14)$$

In magnitudes, since $\mathcal{M} = -2.5 \log_{10} L + \text{constant}$, the relation becomes $\mathcal{M} = -10 \log_{10} v_c + \text{constant}$, and the distance follows from the distance modulus of Section 1.1. The scatter of real galaxies about the relation gives an uncertainty of about 20 per cent in the distance of a single galaxy, and the method has been applied out to distances of the order of 100 Mpc.

The Faber–Jackson Relation and the Fundamental Plane

In 1976 Faber and Jackson found by observation that the luminosities of elliptical galaxies are related to their velocity dispersions, with

$$L \propto \sigma_v^4 \quad (8.15)$$

approximately. The relation is called the Faber–Jackson relation, and it follows from the virial theorem in the same way as the Tully–Fisher relation, with the same two assumptions. With $\langle v^2 \rangle = 3\sigma_v^2$ the virial mass is $M = 3\sigma_v^2 R/\eta G$, and the steps above give

$$L = \frac{9\sigma_v^4}{\pi\eta^2 G^2 \Upsilon^2 \langle \mathcal{J} \rangle}. \quad (8.16)$$

The Faber–Jackson relation has a larger scatter than the Tully–Fisher relation, because the mean surface brightnesses of ellipticals vary considerably from one galaxy to another. When the effective radius, the velocity dispersion, and the mean surface brightness inside the effective radius are all measured, the ellipticals are found to lie close to a plane in the space of the logarithms of these three quantities, which is called the fundamental plane, and the Faber–Jackson relation is the projection of that plane onto the plane of luminosity and velocity dispersion.

Problems

§8.1 Galaxies and Their Scaling Relations

1. For an exponential disc, find the luminosity inside the radius s , and show that the radius $s_{1/2}$ containing half the light satisfies $(1+x)e^{-x} = \frac{1}{2}$ with $x = s_{1/2}/s_d$.
2. A thin circular disc of radius R is seen with its axis at an angle i to the line of sight, as in Figure 92. Show that its image is an ellipse with axis ratio $q = \cos i$, and hence that $v_c = W/2\sqrt{1-q^2}$.
3. Write the de Vaucouleurs law as $\mathcal{J} = \mathcal{J}_e \exp\{-k_V[(s/s_e)^{1/4} - 1]\}$, with the constant k_V left general. Using the substitution $u = k_V(s/s_e)^{1/4}$ and the integral $\int_0^\infty u^N e^{-u} du = N!$, which is worked out in Section 5.2 of Vol. 3 of the Fermi Notes on Physics, show that the total luminosity is $L = 8! \pi e^{k_V} k_V^{-8} \mathcal{J}_e s_e^2$.
4. Consider two spiral galaxies with the same circular speed, the apparent magnitude of the second being larger than that of the first by Δm . Assuming that both obey the same Tully–Fisher relation and that extinction can be neglected, find the ratio of their distances.
5. A spiral galaxy has an exponential disc and a flat rotation curve with circular speed v_c . Find the mass-to-light ratio $\Upsilon(s)$ of the part of the galaxy inside the radius s , and show that for $s \gg s_d$ it grows in proportion to s .

Cosmological Models, Parameters, and the Dark Sector

9.1 The Expanding Universe

In this part we turn from single galaxies to the universe as a whole, and to the matter it contains that emits no light. This section first reviews the expansion of the universe as it is taught at school, then shows that a uniform expansion must follow a linear relation between velocity and distance, and finally uses the rate of expansion to estimate the age of the universe and to explain why the sky is dark at night. Section 9.2 sets out the evidence that most of the mass in galaxies and clusters of galaxies emits no detectable light, and Section 9.3 describes the candidates proposed for this matter and the experiments that search for it.

Redshift and Hubble's Law

The expansion of the universe is part of the school course, and we recall it only briefly before taking it further. The light from a distant galaxy arrives with every line of its spectrum shifted to a longer wavelength. Let λ_0 be the wavelength of a line measured in the laboratory, and λ its wavelength in the spectrum of the galaxy.

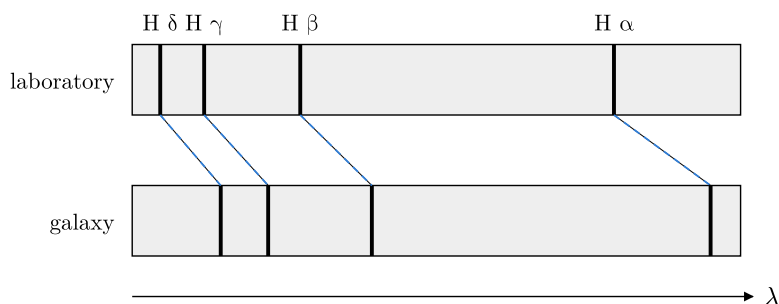


Figure 94. The Balmer lines of hydrogen at the wavelengths measured in the laboratory (top) and in the spectrum of a distant galaxy (bottom). Every wavelength is multiplied by the same factor $1 + z$, so the shift $\lambda - \lambda_0 = z\lambda_0$ is largest for the line of longest wavelength.

The redshift of the galaxy is defined as

$$z = \frac{\lambda - \lambda_0}{\lambda_0}, \quad (9.1)$$

and it is found to be the same for every line, as Figure 94 shows for the Balmer lines of Section 3.1. For a source moving directly away from us with a speed v_r much less than c , the Doppler shift gives

$$z \approx \frac{v_r}{c}, \quad (9.2)$$

and the redshift of a nearby galaxy is interpreted as a speed of recession.

In 1929 Hubble compared the redshifts of nearby galaxies, most of them measured by Slipher, with distances found from Cepheid variables (Section 1.1) in a few of them and from the brightest stars in the others, and found that the speed of recession grows in proportion to the distance. The relation has since been confirmed to far greater distances with supernovae of Type Ia (Section 1.1). This relation is called Hubble's law, and it is found by observation to take the form

$$v_r = H_0 d, \quad (9.3)$$

where H_0 is called the Hubble constant. Measurements by different methods currently give values between about 67 and $73 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and we shall use $H_0 \approx 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The reasons for the spread belong to the measurement of cosmological parameters, which is not treated in this course.

Hubble's law describes the average motion of the galaxies. Each galaxy also moves relative to the expansion, with a velocity called its peculiar velocity, typically a few hundred kilometres per second, produced by the gravitational pull of its neighbours, so the law is a useful measure of distance only beyond a few tens of megaparsecs. At the other extreme, $z \approx v_r/c$ holds only for $z \ll 1$. For larger redshifts the relation between redshift and distance depends on the metric of an expanding universe, which is not treated in this course, and in this part we keep to redshifts much less than one.

The Evidence for the Big Bang

If the galaxies are moving apart now, they were closer together in the past, and the matter of the universe was then denser and hotter, as a gas compressed without any exchange of heat becomes hotter (Section 2.2 of Vol. 2 of the Fermi Notes on Physics). The theory that the universe expanded from a hot, dense state is called the Big Bang theory. At school it is supported chiefly by the following observations:

1. the expansion itself, described by Hubble's law;
2. the cosmic microwave background, radiation arriving from every direction with the spectrum of a blackbody (Section 5.6 of Vol. 2) at about 2.7 K , interpreted as the radiation of the hot early universe, cooled by the expansion;
3. the abundance of helium, which makes up about a quarter of the mass of ordinary matter, in the oldest stars as well as in the Sun.

The hydrogen burning of Section 4.4 has made far less helium than this over the age of the Galaxy, as you will show in Problem 4,

and the rest was formed in the first few minutes of the expansion. The calculation of the helium abundance, and the physics of the microwave background, belong to the thermal history of the early universe, which is not treated in this course.

The Cosmological Principle

The assumption that on large enough scales the universe is homogeneous, meaning that it has the same properties at every place, and isotropic, meaning that it looks the same in every direction, is called the cosmological principle. On small scales the universe is neither; galaxies are gathered into groups and clusters, strung along filaments and sheets around large regions containing few galaxies, called voids. Surveys that have measured the redshifts, and hence the distances, of more than a million galaxies find that the number of galaxies in a volume more than about 100 Mpc across comes out much the same wherever the volume is placed. The evidence for isotropy is stronger; the temperature of the cosmic microwave background is the same in every direction to about one part in 10^5 , once the effect of the motion of the Earth has been removed, and faint radio sources, most of them distant galaxies, are spread evenly over the sky.

Isotropy can be observed only from where we are. If we add the assumption that our position in the universe is not special, so that an observer in any other galaxy would also find the universe isotropic, homogeneity follows. To see this, let P and Q be any two points, and let C be the point midway between them. A rotation through half a turn about a line through C perpendicular to PQ carries P to Q. The universe looks the same in every direction from C, so this rotation changes none of its average properties, and the mean density at P is therefore equal to the mean density at Q. Since P and Q were any two points, the mean density is the same everywhere.

Homogeneity can also be tested by counting galaxies, as Hubble did in the 1930s. If galaxies of luminosity L are spread uniformly with number density n , those brighter than a flux F are the ones nearer than $(L/4\pi F)^{1/2}$, and their number is $N(> F) = \frac{4}{3}\pi n(L/4\pi F)^{3/2}$. You will show in Problem 1 that the number goes as $F^{-3/2}$ for any mixture of luminosities, and hence as $10^{0.6m}$ for galaxies brighter

than apparent magnitude m . The counts of the brighter galaxies, whose redshifts are small, are found to follow this law approximately.

A Uniform Expansion Is a Linear Law

The linear form of Hubble's law follows from the cosmological principle, as we now show. We set up coordinates such that our galaxy, A, is at the origin, and let $\mathbf{v}(\mathbf{r})$ be the velocity that we measure for galaxies at position $\mathbf{r}$, averaged over their peculiar velocities. An observer in another galaxy, B, at position $\mathbf{r}_B$, moves with velocity $\mathbf{v}(\mathbf{r}_B)$, and sees a galaxy at position $\mathbf{r}$ at the relative position $\mathbf{r}' = \mathbf{r} - \mathbf{r}_B$. Since the speeds are much less than c , velocities are subtracted as vectors, and the observer at B measures the velocity of that galaxy as

$$\mathbf{v}_B(\mathbf{r}') = \mathbf{v}(\mathbf{r}_B + \mathbf{r}') - \mathbf{v}(\mathbf{r}_B). \quad (9.4)$$

If the universe is homogeneous, the observer at B sees the same pattern of motion that we do, so the velocity measured at B for a galaxy at relative position $\mathbf{r}'$ is the same function of $\mathbf{r}'$ as ours, $\mathbf{v}_B(\mathbf{r}') = \mathbf{v}(\mathbf{r}')$. Therefore

$$\mathbf{v}(\mathbf{r}_B + \mathbf{r}') = \mathbf{v}(\mathbf{r}_B) + \mathbf{v}(\mathbf{r}') \quad (9.5)$$

for every pair of positions $\mathbf{r}_B$ and $\mathbf{r}'$. A function with this property is linear. Setting $\mathbf{r}_B = \mathbf{r}' = \mathbf{r}$ gives $\mathbf{v}(2\mathbf{r}) = 2\mathbf{v}(\mathbf{r})$, and repeating the step gives $\mathbf{v}(q\mathbf{r}) = q\mathbf{v}(\mathbf{r})$ for every whole number q . Applying this result to the position $\mathbf{r}/q$ gives $\mathbf{v}(\mathbf{r}/q) = \mathbf{v}(\mathbf{r})/q$, so the relation holds for every rational multiple of $\mathbf{r}$, and, for a velocity that changes smoothly with position, for every real multiple. Each component of the velocity is therefore a linear combination of the components of the position,

$$v_j = \sum_{k=1}^3 H_{jk} r_k, \quad (9.6)$$

where the numbers H_{jk} form a three-by-three matrix $\mathbf{H}$ that is the same at every place, though it may change with time.

Isotropy fixes the matrix. Any matrix is the sum of a symmetric part, $\frac{1}{2}(H_{jk} + H_{kj})$, and an antisymmetric part, $\frac{1}{2}(H_{jk} - H_{kj})$. You will show in Problem 3 that the antisymmetric part describes a rotation of the whole pattern of galaxies about an axis, and an axis is a special direction, so isotropy requires the antisymmetric part to vanish. A real symmetric matrix has three perpendicular eigenvectors, as stated in Section 4.3 of Vol. 4, and along each of these directions the velocity is parallel to the position, with the eigenvalue as the ratio of speed to distance. If the three eigenvalues were different, the expansion would be faster along some directions than along others, which isotropy forbids, so they must all be equal, to H_0 say. Then $\mathbf{H}$ is H_0 times the identity matrix, and

$$\mathbf{v} = H_0 \mathbf{r}, \quad (9.7)$$

which is Hubble's law.

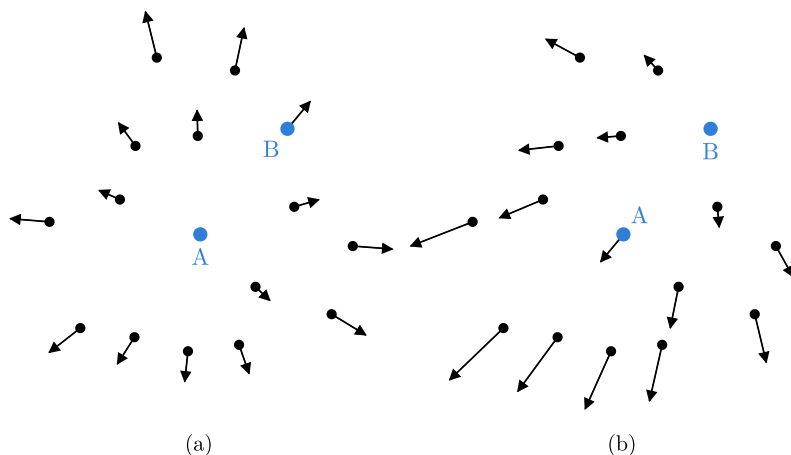


Figure 95. The velocities of a set of galaxies in a uniformly expanding universe, drawn as arrows in proportion to the velocities measured (a) by an observer in galaxy A and (b) by an observer in galaxy B. Each observer sees every other galaxy moving directly away with a speed proportional to its distance.

Figure 95 shows the pattern of velocities measured from two different galaxies. Each observer finds every other galaxy receding at a speed proportional to its distance, so each seems to be at the centre of the expansion, and the expansion therefore has no centre. Every separation grows at the same fractional rate, and the pattern of galaxies is magnified uniformly, as a photograph is when it is enlarged. The Hubble constant is the same everywhere but may change with time, and the subscript 0 marks its present value.

The Hubble Time

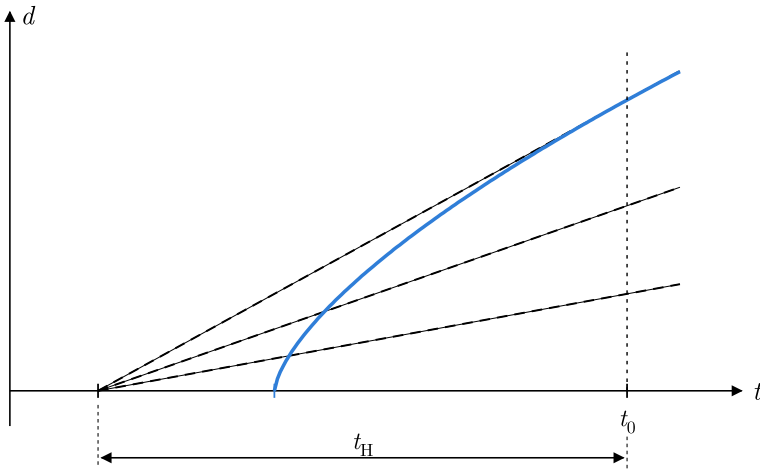


Figure 96. The separation d between our galaxy and three others, plotted against time t . The dashed lines show separations that grow at constant speed, which all reach zero a time t_H before the present time t_0 . The accent curve shows an expansion that slows down with time and has the same rate of expansion at t_0 as the top line; it lies below its tangent and reaches zero at a later time, marked on the axis.

Hubble's law gives an estimate of the age of the universe. If every galaxy had always moved away from us at its present speed, a galaxy now at distance d would have been at our position a time d/v_r ago, and by Hubble's law this time is

$$t_H = \frac{d}{H_0 d} = \frac{1}{H_0}, \quad (9.8)$$

the same for every galaxy, as the dashed lines of Figure 96 show. The quantity t_H is called the Hubble time, and with the measured value of H_0 it is about 14 Gyr.

The galaxies have not always moved at constant speed. Their mutual gravitational attraction slows the expansion, so if gravity were the only influence the separation of two galaxies, plotted against time, would follow a curve that bends downwards, like the accent curve of Figure 96. At the present time t_0 this curve has the same slope as the straight line for the same galaxy, since the curve and the line both describe the present speed $H_0 d$, and a curve that bends downwards lies below its tangent line at earlier times. The curve therefore reaches zero separation later than the straight line does, and the age of the universe, the time that has passed since the separation was zero, is less than t_H . An expansion that speeds up gives an age greater than t_H . How the rate of expansion changes with time depends on what the universe contains, and its calculation needs the equations of an expanding universe, which are not treated in this course; the measurements that use them give an age of about 13.8 Gyr, close to the Hubble time.

The age of the universe must exceed the age of the oldest stars in it. The ages of globular clusters found from the turn-off point of their main sequences in Section 4.8 are about 12 to 13 Gyr, which is consistent with a universe whose age is close to t_H . The value Hubble found, $H_0 \approx 500 \text{ km s}^{-1} \text{ Mpc}^{-1}$, gave a Hubble time of about 2 Gyr, less than the age of the Earth found from radioactive dating, then put at about 3 Gyr and now at about 4.5 Gyr. The conflict was eased in 1952, when Baade showed that the Cepheids used to set the scale of distances belonged to the two stellar populations of Section 7.1, which follow different period–luminosity relations, and the corrected distances were about twice as large.

Olbers' Paradox

The question of why the sky is dark at night is named after Olbers, who discussed it in 1823. Suppose that the universe is infinite, unchanging in time, and uniformly filled with stars, n per unit volume, each with luminosity L and radius R . Consider a thin spherical shell of radius r and thickness dr centred on the Earth,

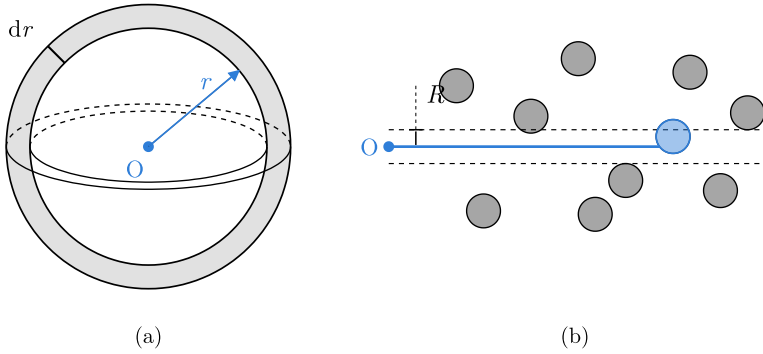


Figure 97. (a) A thin spherical shell of radius r and thickness dr centred on an observer O , drawn in perspective. (b) A line of sight from O through a field of stars of radius R , which ends on the first star whose centre lies within R of the line, that is, inside the cylinder bounded by the dashed lines.

as drawn in Figure 97(a). The shell contains $4\pi r^2 n dr$ stars, each of which delivers a flux $L/4\pi r^2$, so the shell contributes a flux

$$dF = 4\pi r^2 n dr \frac{L}{4\pi r^2} = nL dr, \quad (9.9)$$

which does not depend on r , so adding the shells out to a distance D gives a total flux nLD , which grows without limit as D increases.

This calculation ignores that nearer stars hide the stars behind them. A line of sight meets a star if the centre of the star lies within a distance R of the line, so the line of sight sweeps out a cylinder of radius R , drawn in Figure 97(b), and meets the first star whose centre lies inside the cylinder. The argument is the one used for the mean free path of a molecule in Section 1.3 of Vol. 2, with the line of sight in place of the molecule and the area πR^2 in place of the collision cross-section, and without the factor of $\sqrt{2}$ found there, which comes from the motion of the other molecules, since the stars move so much more slowly than light that they can be treated as at rest; this argument gives the mean distance that a line of sight travels before it meets a star,

$$\ell = \frac{1}{n\pi R^2}. \quad (9.10)$$

In an infinite universe every line of sight therefore ends on the surface of a star. The specific intensity of radiation does not change along a ray in empty space, as shown in Section 3.3, so the sky in every direction would be as bright as the surface of a star. The Earth would be bathed in radiation as intense as that at the surface of the Sun, and it would be heated to a temperature close to that of the solar surface. This conflict between a simple argument and the dark night sky is called Olbers' paradox.

The paradox is resolved by the finite age of the universe. The light arriving now left its source less than the age of the universe ago. Measuring time from the moment of zero separation, so that the age of the universe is the present time t_0 , we receive light only from stars nearer than about ct_0 ; the expansion changes this distance by a factor of order one, which does not matter here. Let $\mathcal{P}(D)$ be the probability that a line of sight travels a distance D without meeting a star. In a further distance dD it meets a star with probability dD/ℓ , so $d\mathcal{P} = -\mathcal{P} dD/\ell$, and integrating with $\mathcal{P}(0) = 1$ gives $\mathcal{P} = e^{-D/\ell}$. The fraction of the sky covered by stars nearer than ct_0 is then

$$1 - e^{-ct_0/\ell} \approx \frac{ct_0}{\ell}, \quad (9.11)$$

provided $ct_0 \ll \ell$, and the sky is darker than the surface of a star by this factor. The mean density of starlight in the universe corresponds to roughly 10^8 stars like the Sun in every cubic megaparsec, and with this density ℓ is of order 10^{15} times ct_H . The stars therefore cover only about one part in 10^{15} of the sky.

The stars could not fill space with so much radiation in any case, since you will show in Problem 5 that they would have to shine for a time ℓ/c , of order 10^{25} years, whereas a star like the Sun shines for about 10^{10} years. The redshift of the light from distant sources reduces the brightness of the sky further, by a smaller factor.

9.2 The Evidence for Dark Matter

The mass of a galaxy or a cluster of galaxies can be found in two independent ways, by adding up the stars and gas that can be seen, with the mass-to-light ratio of Section 8.1, and by measuring the gravitational effects of the mass on the motions of stars, gas, and galaxies and on rays of light. In the galaxies and clusters where both methods have been applied to a large enough region, the second gives much more mass than the first. Matter that is detected only through its gravity, and that emits and absorbs too little radiation to be seen, is the dark matter named in Section 7.1, where the flat rotation curve of the Galaxy first required it. This section first presents the evidence from the rotation of spiral galaxies, then from the motions of galaxies and the hot gas in clusters, then from gravitational lensing, and finally from a pair of colliding clusters in which the mass and the gas are found in different places.

Flat Rotation Curves

Section 7.1 showed that, for matter distributed with spherical symmetry, the circular speed v_c at galactocentric radius s measures the mass $M(s)$ inside that radius,

$$v_c^2 = \frac{GM(s)}{s}. \quad (9.12)$$

For a flattened distribution, such as a disc, the relation between v_c and the mass inside the orbit differs from this one by a factor of order one, and the rotation curve of a disc is usually computed numerically.

Spiral galaxies other than the Milky Way are seen from outside. Let the disc of a spiral be inclined at an angle i to the plane of the sky, and let v_{sys} be the radial velocity of the galaxy as a whole. Gas on a circular orbit, where the orbit crosses the major axis of the image of the disc, has radial velocity $v_r = v_{\text{sys}} \pm v_c \sin i$, the sign depending on the side of the centre. In the 1970s Rubin and Ford measured rotation curves in this way with a spectrograph, the instrument of Section 2.2 that spreads the light passing through a narrow slit into a spectrum; they laid the slit along the major axis

and measured the Doppler shifts of the emission lines of ionised gas at each point along it. The curves were extended to larger radii with the 21-centimetre line of Section 6.1, since the neutral hydrogen of a spiral usually reaches several times further out than its stars.

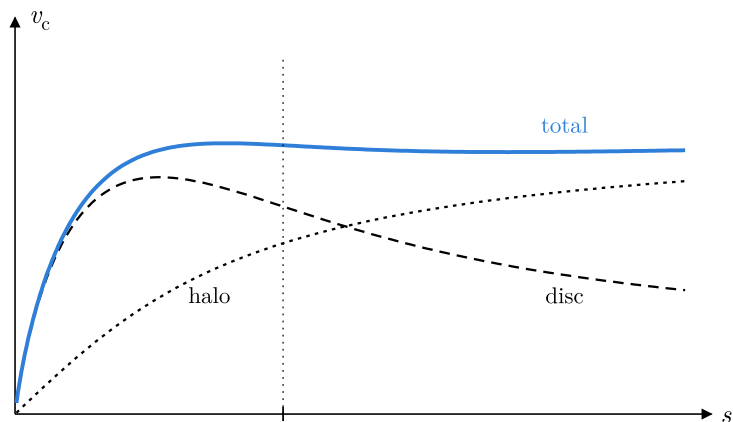


Figure 98. The rotation curve of a model spiral galaxy. The dashed curve is the circular speed that the exponential disc of stars and gas would produce alone, the dotted curve is the contribution of a spherical halo of dark matter, and the accent curve is the total, which is nearly flat at large radii, as the observed rotation curves are. The dotted vertical line marks the radius beyond which little starlight is seen.

If the mass followed the light, the rotation curve would fall at large radii. The surface brightness of a spiral disc falls exponentially with distance from the centre (Section 8.1), so beyond a few scale lengths almost all the light, and on this assumption almost all the mass, lies inside the orbit, and $v_c = (GM/s)^{1/2}$ falls as $s^{-1/2}$, as the speeds of the planets do with distance from the Sun; the dashed curve of Figure 98 shows this for the disc alone. In nearly every spiral measured far enough out, the observed curve instead rises in the inner part and then stays nearly flat to the largest radius reached, like the accent curve.

A flat rotation curve, $v_c = \text{constant}$, requires

$$M(s) = \frac{v_c^2 s}{G}, \quad (9.13)$$

so the mass inside radius s goes on growing in proportion to s where there is little light. If the extra mass is spread in a spherical halo, its density is, by the argument given for the Galaxy in Section 7.1,

$$\rho(s) = \frac{v_c^2}{4\pi G s^2}. \quad (9.14)$$

A distribution with this density is called a singular isothermal sphere. The name comes from the result, which you will show in Problem 1, that it is the density of a self-gravitating ball of gas, or of particles, whose velocity dispersion σ_v is the same at every radius, with $v_c^2 = 2\sigma_v^2$.

As Section 8.1 described, the light from an exponential disc converges to a finite total while $M(s)$ grows in proportion to s , so the mass-to-light ratio Υ inside radius s grows roughly in proportion to s beyond the stellar disc, from a few in solar units, which the stars can account for, to a few tens at the largest radii measured. Since the rotation curves do not turn down where the measurements end, the total mass of a spiral is at least several times the mass of its stars and gas.

The Velocities of Galaxies in Clusters

In 1933 Zwicky found that the radial velocities of the galaxies in the Coma cluster are spread over a range far wider than the mass of the visible galaxies could hold together, and he turned the spread into a mass with the virial theorem. Section 5.3 of Vol. 1 of the Fermi Notes on Physics derived the theorem for two bodies, and Section 4.2 obtained the same relation for the gas of a star; for a cluster we extend the argument of Vol. 1 to many bodies.

Let the cluster consist of bodies of masses m_j at positions $\mathbf{r}_j$, and let $\mathbf{F}_{jk}$ be the gravitational force on body j due to body k . Following Vol. 1, we differentiate the quantity $Q = \sum_j m_j \dot{\mathbf{r}}_j \cdot \mathbf{r}_j$ with respect to time,

$$\frac{dQ}{dt} = \sum_j m_j \dot{\mathbf{r}}_j \cdot \dot{\mathbf{r}}_j + \sum_j \mathbf{r}_j \cdot m_j \ddot{\mathbf{r}}_j = 2E_{\text{kin}} + \sum_j \sum_{k \neq j} \mathbf{r}_j \cdot \mathbf{F}_{jk}, \quad (9.15)$$

where E_{kin} is the total kinetic energy, which Vol. 1 writes T . In the double sum, the terms for a given pair of bodies can be combined, since $\mathbf{F}_{kj} = -\mathbf{F}_{jk}$ by Newton's third law, which gives

$$\mathbf{r}_j \cdot \mathbf{F}_{jk} + \mathbf{r}_k \cdot \mathbf{F}_{kj} = (\mathbf{r}_j - \mathbf{r}_k) \cdot \mathbf{F}_{jk} = -\frac{Gm_j m_k}{|\mathbf{r}_j - \mathbf{r}_k|^3}, \quad (9.16)$$

because $\mathbf{F}_{jk} = -Gm_j m_k (\mathbf{r}_j - \mathbf{r}_k) / |\mathbf{r}_j - \mathbf{r}_k|^3$. The right-hand side is the gravitational potential energy of the pair, so the double sum is the total gravitational potential energy E_{grav} , which Vol. 1 writes U . For a bound cluster the positions and velocities stay within finite limits, so Q stays bounded, and its rate of change averages to zero over a long time, as in Vol. 1. The time averages then satisfy

$$2\langle E_{\text{kin}} \rangle + \langle E_{\text{grav}} \rangle = 0. \quad (9.17)$$

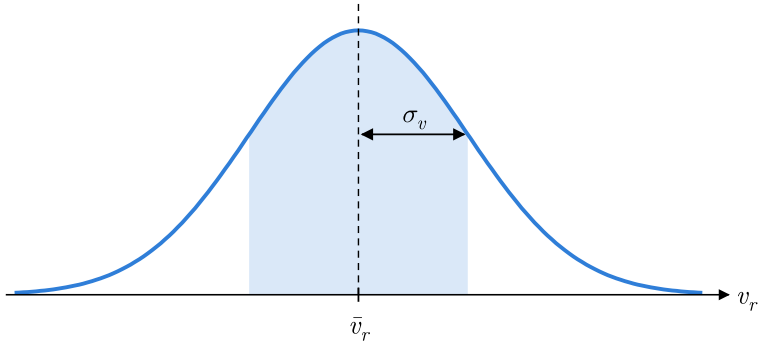


Figure 99. The distribution of the radial velocities v_r of the galaxies in a cluster, which is observed to be close to a Gaussian about the mean $\bar{v}_r$. The shaded band lies within one velocity dispersion σ_v of the mean.

Only the radial velocities of the galaxies can be measured, and they are observed to be spread about their mean $\bar{v}_r$, the velocity of the cluster as a whole, in a distribution close to a Gaussian, drawn in Figure 99, like the velocity components of the molecules of a gas in Section 1.2 of Vol. 2. The spread is measured by the velocity dispersion σ_v of Section 8.1, the root mean square of $v_r - \bar{v}_r$. If the motions are the same in every direction, each component of the velocity relative to the cluster has mean square σ_v^2 , as in Section 8.1, and the kinetic energy of a cluster of total mass M is

$$E_{\text{kin}} = \frac{1}{2}M \cdot 3\sigma_v^2. \quad (9.18)$$

The potential energy depends on how the mass is distributed. For a uniform sphere of mass M and radius R it is $-3GM^2/5R$, which is the energy of a uniformly charged sphere worked out in Section 1.4 of Vol. 5 with the Coulomb constant replaced by $-G$ and the charge by the mass. We write $E_{\text{grav}} = -\alpha_v GM^2/R$, with α_v a number of order one that depends on the distribution, and which equals $3/5$ for a uniform sphere. The virial theorem gives $3M\sigma_v^2 = \alpha_v GM^2/R$, and therefore

$$M = \frac{3\sigma_v^2 R}{\alpha_v G} = \frac{5\sigma_v^2 R}{G}, \quad (9.19)$$

the second form holding for a uniform sphere. For the Coma cluster, with $\sigma_v \approx 1000 \text{ km s}^{-1}$ and a radius of a megaparsec or two, this gives a mass of order $10^{15} M_\odot$, while the luminosity of its galaxies is of order $10^{13} L_\odot$, so the mass-to-light ratio of the cluster is some hundreds in solar units, compared with a few for the stars.

The estimate assumes that the cluster is bound and has been undisturbed long enough for the time averages to apply, that the motions are the same in every direction, and that the galaxies are distributed like the mass. Each assumption could change the mass by a factor of order one, but none can account for a factor of a hundred. The mass-to-light ratio Zwicky found was larger still, since you will show in Problem 2 that the ratio is proportional to the Hubble constant, and the value he used was about eight times the present one; even

so corrected, his result required far more mass than the galaxies contain.

Hot Gas in Clusters

X-ray telescopes in orbit (Section 1.2) show that clusters of galaxies are filled with fully ionised gas at temperatures of 10^7 to 10^8 K, which emits X-rays mainly by the radiation of electrons deflected as they pass the ions, a process called bremsstrahlung; high-energy radiation processes of this kind are not treated in this course. The temperature of the gas is found from the shape of its X-ray spectrum, and its density from the X-ray brightness across the image of the cluster, and the gas has several times the mass of the stars in the galaxies.

The time a sound wave takes to cross a cluster, which is its radius divided by the speed of sound, of order $(P/\rho)^{1/2} = (k_B T / \mu m_H)^{1/2}$ as in Section 4.2, is less than a gigayear, much shorter than the age of the cluster, so the gas has had time to settle into hydrostatic equilibrium in the gravitational field of the cluster. For a spherical cluster the equation of hydrostatic equilibrium of Section 4.2 applies,

$$\frac{dP}{dr} = -\frac{GM(r)\rho}{r^2}, \quad (9.20)$$

where ρ is the density of the gas and $M(r)$ is the total mass inside radius r , including the galaxies, the gas, and anything else. With the ideal gas law $P = \rho k_B T / \mu m_H$, where $\mu \approx 0.6$ for fully ionised gas of the composition of the Sun, as found in Section 4.2, the derivative of the pressure is found by the product rule,

$$\frac{dP}{dr} = \frac{k_B}{\mu m_H} \left(T \frac{d\rho}{dr} + \rho \frac{dT}{dr} \right), \quad (9.21)$$

and equating the two expressions for dP/dr and solving for the mass gives

$$M(r) = -\frac{k_B T r}{G \mu m_H} \left(\frac{d \ln \rho}{d \ln r} + \frac{d \ln T}{d \ln r} \right), \quad (9.22)$$

where we have used $(r/\rho) d\rho/dr = d \ln \rho / d \ln r$, and similarly for T . The temperature and the two logarithmic derivatives are measured in X-rays.

An Isothermal Cluster

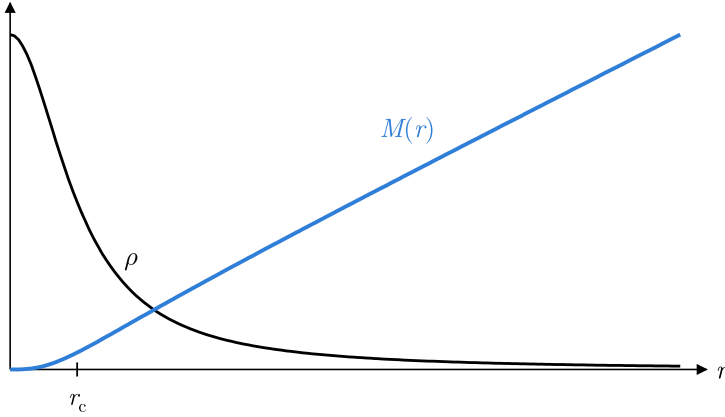


Figure 100. The density ρ of the hot gas of an isothermal cluster with the profile given in the text, and the total mass $M(r)$ inside radius r that holds the gas in hydrostatic equilibrium, each scaled to its largest value in the range shown. Beyond the core radius r_c the density falls steeply while the mass goes on growing in proportion to r .

As an example, consider a cluster in which the gas has the same temperature everywhere, and its density follows the profile

$$\rho(r) = \rho_c \left(1 + \frac{r^2}{r_c^2} \right)^{-3\beta_g/2}, \quad (9.23)$$

which is found to fit the X-ray images of many clusters. Here ρ_c is the central density, r_c is called the core radius, and β_g is a constant found from the fit, typically about $2/3$. Taking the logarithm and differentiating with respect to $\ln r$ gives

$$\frac{d \ln \rho}{d \ln r} = -\frac{3\beta_g}{2} \frac{r}{1 + r^2/r_c^2} \frac{2r}{r_c^2} = -\frac{3\beta_g r^2}{r_c^2 + r^2}, \quad (9.24)$$

and since $d \ln T / d \ln r = 0$, the mass inside radius r is

$$M(r) = \frac{3\beta_g k_B T}{G\mu m_H} \frac{r^3}{r_c^2 + r^2}. \quad (9.25)$$

Figure 100 shows the density and the mass for $\beta_g = 2/3$. Well outside the core the mass grows in proportion to r , as it does for the flat rotation curves of spirals, while the density of the gas falls as $r^{-3\beta_g}$. For a rich cluster at about 10^8 K the mass inside a megaparsec is several times $10^{14} M_\odot$, in agreement with the virial estimates, of which the hot gas makes up only about a tenth and the stars a few per cent. If the gas and the galaxies move in the same gravitational field, you will show in Problem 5 that $k_B T \sim \mu m_H \sigma_v^2$, and this relation is found to hold approximately.

Gravitational Lensing

A ray of light passing a body of mass M at a distance b , called the impact parameter, is bent towards the body through an angle

$$\hat{\alpha} = \frac{4GM}{c^2 b}, \quad (9.26)$$

provided b is much larger than the Schwarzschild radius $2GM/c^2$ of Section 5.3. This result comes from general relativity, which is not treated in this course. A calculation that treats light as a stream of particles moving at speed c and uses Newtonian gravity gives half this angle, as you will show in Problem 3. In 1919 Eddington and his colleagues photographed stars close to the Sun during a total eclipse and found them displaced from their usual positions, away from the Sun, by amounts consistent with a deflection of $4GM_\odot/c^2 R_\odot \approx 1.75$ seconds of arc for a ray grazing the surface of the Sun, which is the relativistic value and twice the Newtonian one.

A body that bends light in this way is called a gravitational lens. Consider a source S at distance D_S from the observer O, and a lens

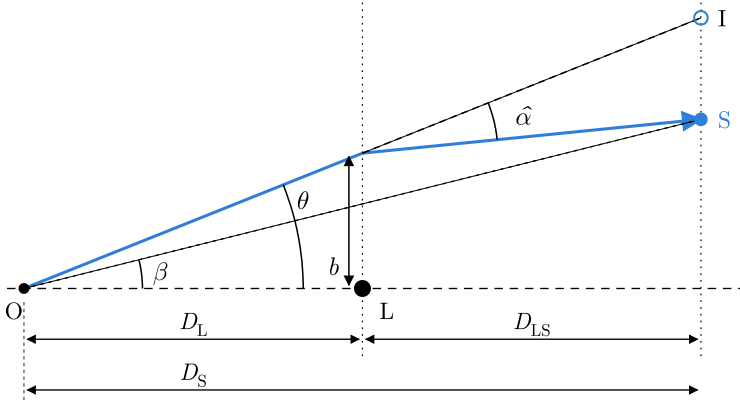


Figure 101. The geometry of a gravitational lens, with the angles greatly exaggerated. A ray from the source S passes the lens L at impact parameter b , is bent through the angle $\hat{\alpha}$, and reaches the observer O , who sees an image of the source at I , in the direction θ from the line OL , while the source itself lies in the direction β .

L of mass M at distance D_L , with $D_{LS} = D_S - D_L$ the distance from the lens to the source, as drawn in Figure 101. All the angles are small. Let β be the angle between the line OL and the direction of the source, and θ the angle between OL and the direction in which the observer sees the image. The ray that reaches the observer passes the lens at $b = D_L \theta$ and is bent through $\hat{\alpha}$. Measured at the distance of the source, the height of the image I above the line OL is θD_S , the height of the source is βD_S , and these two heights differ by the distance $\hat{\alpha} D_{LS}$ by which the bending moves the ray over the path from the lens to the source, so that $\theta D_S = \beta D_S + \hat{\alpha} D_{LS}$. Dividing by D_S and substituting $\hat{\alpha} = 4GM/c^2 D_L \theta$ gives

$$\beta = \theta - \frac{\theta_E^2}{\theta}, \quad \theta_E^2 = \frac{4GM}{c^2} \frac{D_{LS}}{D_L D_S}, \quad (9.27)$$

and the first of these relations is called the lens equation. Where the lens and the source are at redshifts that are not small, the distances in it must be defined with the metric of an expanding universe, which is not treated in this course, and here we take them

to be ordinary distances.

If the source lies directly behind the lens, $\beta = 0$, and by symmetry the image is a ring of angular radius θ_E about the lens, called an Einstein ring; θ_E is called the Einstein radius. For $\beta \neq 0$, multiplying the lens equation by θ gives the quadratic $\theta^2 - \beta\theta - \theta_E^2 = 0$, whose roots are

$$\theta_{\pm} = \frac{1}{2} \left(\beta \pm \sqrt{\beta^2 + 4\theta_E^2} \right). \quad (9.28)$$

There are two images, one outside the Einstein ring on the same side of the lens as the source, and one inside the ring on the opposite side, and you will find some of their properties in Problem 4.

For a lens whose projected mass is distributed with circular symmetry about the line of sight, the deflection at impact parameter b is $4GM(< b)/c^2b$, where $M(< b)$ is the mass inside a cylinder of radius b about the line of sight, in the same way that the field of a spherical body at radius r depends only on the mass inside r . An Einstein ring of measured angular radius θ_E therefore gives the mass inside the cylinder of radius $D_L\theta_E$,

$$M(< D_L\theta_E) = \frac{c^2}{4G} \frac{D_LD_S}{D_{LS}} \theta_E^2, \quad (9.29)$$

without any assumption about how the matter moves or whether it is in equilibrium. Behind rich clusters, background galaxies are seen stretched into thin arcs along the Einstein ring of the cluster, of angular radius tens of seconds of arc. The masses found inside the arcs, and farther out from the small distortions of the shapes of many fainter background galaxies, agree with the virial and X-ray masses to within a factor of order one.

Colliding Clusters

The cluster 1E 0657–56, known as the Bullet Cluster, is a pair of clusters that passed through each other about 10^8 years ago at several thousand kilometres per second. The galaxies are small compared with the distances between them and passed through

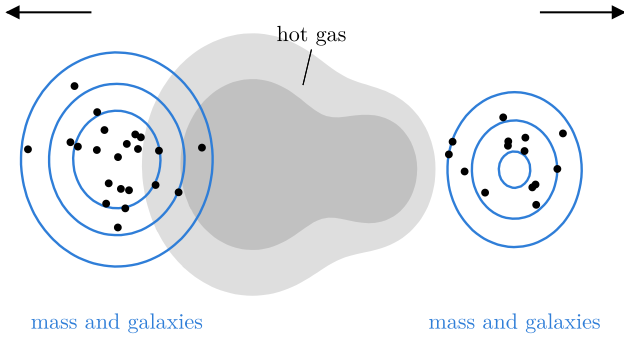


Figure 102. A schematic drawing of a pair of clusters that have passed through each other, with the directions of motion marked by the arrows at the top. The shaded regions show the hot gas, seen in X-rays, which has been slowed and left between the clusters. The accent contours show the mass found by gravitational lensing, which is centred on the galaxies, drawn as dots.

almost without colliding, but the two clouds of hot gas, each filling its cluster, collided, were slowed by each other's pressure, and were left between the two groups of galaxies, as drawn in Figure 102. Since the gas has several times the mass of the stars, most of the ordinary matter now lies between the groups of galaxies.

In 2006 Clowe and his colleagues mapped the mass from the distortions of background galaxies and found it in two clumps centred on the groups of galaxies, well separated from the gas. If the mass were mostly ordinary matter it would follow the gas, so most of it is in a form that, like the galaxies, passed through with few collisions, as expected of dark matter that interacts only weakly with itself and with ordinary matter.

The separation also limits how often particles of dark matter collide with one another. Let σ be the cross-section for a collision between two particles of dark matter of mass m_{DM} , and let a particle of one cluster cross a length ℓ_c of the other, where the dark matter has mean density ρ_{DM} . As in the mean-free-path argument of Section 1.3 of Vol. 2, but without the factor of $\sqrt{2}$ found there, since

the clusters meet at a speed much greater than the speeds of the particles within either of them, the mean number of collisions that the particle makes is

$$\frac{\rho_{\text{DM}}}{m_{\text{DM}}} \sigma \ell_c = \frac{\sigma}{m_{\text{DM}}} \rho_{\text{DM}} \ell_c. \quad (9.30)$$

If this number were larger than about one, the dark matter would have been slowed like the gas, and with the mass per unit area $\rho_{\text{DM}} \ell_c$ measured by the lensing, this requires σ/m_{DM} to be less than about $0.1 \text{ m}^2 \text{ kg}^{-1}$.

Proposals that explain flat rotation curves by changing the law of gravity at small accelerations, in place of adding unseen mass, are not treated in this course; in such a proposal the lensing would be expected to follow the ordinary matter, most of which is in the gas.

9.3 Dark Matter Candidates and Their Detection

The evidence of Section 9.2 shows how much dark matter there is and where it lies, but not what it is. This section first describes some of the candidates that have been proposed, namely ordinary matter in compact bodies too faint to see, neutrinos, weakly interacting massive particles, and axions, and then the ways in which particles of dark matter in the halo of the Milky Way might be detected.

The Density of Dark Matter near the Sun

Any search in the laboratory depends on the density of dark matter where the Earth is. The circular speed of the Milky Way at the distance s_0 of the Sun from the Galactic centre is about 230 km s^{-1} (Section 7.1), and the rotation curve is close to flat. If all the mass were in a singular isothermal sphere, the density at the Sun would be, by Section 9.2,

$$\rho_{\text{DM}} = \frac{v_c^2}{4\pi G s_0^2}, \quad (9.31)$$

which with the measured values is about $10^{-21} \text{ kg m}^{-3}$. Part of the mass inside the orbit of the Sun is in the disc, and more careful models give about 0.3 to $0.4 \text{ GeV } c^{-2} \text{ cm}^{-3}$, roughly the mass of one proton in every three cubic centimetres. In such a halo the particles move in random directions with a velocity dispersion $\sigma_v = v_c/\sqrt{2}$, as you showed in Problem 1 of Section 9.2.

Compact Objects in the Halo

The dark matter might be ordinary matter in bodies too faint to be seen in the halo, which are called massive compact halo objects; examples include faint stars of low mass, brown dwarfs, which have the composition of stars but are too light, below about $0.08 M_\odot$, to fuse hydrogen, and the white dwarfs, neutron stars, and black holes of Sections 5.1 to 5.3. In 1986 Paczyński pointed out that such a body passing close to the line of sight to a background star would act as a gravitational lens and brighten the star for a while. For a lens in the halo and a star in the Large Magellanic Cloud, a small galaxy near the Milky Way like the Small Magellanic Cloud of Section 1.1, the Einstein radius of Section 9.2 is less than a millisecond of arc, too small for the two images to be told apart, and what is observed is their total brightness. Lensing in which the images are not resolved is called microlensing, and it was met in Section 2.3 as a way of detecting planets.

The magnification follows from the image positions of Section 9.2. Let $\eta = \beta/\theta_E$ be the angular distance of the source from the lens in units of the Einstein radius, and let $y_\pm = \theta_\pm/\theta_E$, so that

$$y_\pm = \frac{1}{2} (\eta \pm \sqrt{\eta^2 + 4}) . \quad (9.32)$$

The specific intensity is constant along a ray in empty space (Section 3.3), so each image has the surface brightness of the source, and its brightness is proportional to its solid angle. Round the lens, an arc of the source at angular distance η becomes an arc of the image at distance y subtending the same angle at the lens, so lengths in this direction are multiplied by y/η , and in the direction away from the lens they are multiplied by $dy/d\eta$. The magnification of each image is therefore

$$A_{\pm} = \left| \frac{y_{\pm}}{\eta} \frac{dy_{\pm}}{d\eta} \right|. \quad (9.33)$$

Differentiating $y_{\pm}$ gives $dy_{\pm}/d\eta = \frac{1}{2}(1 \pm \eta/\sqrt{\eta^2 + 4})$, which can be rewritten as $\pm y_{\pm}/\sqrt{\eta^2 + 4}$, so that $A_{\pm} = y_{\pm}^2/\eta\sqrt{\eta^2 + 4}$. The two image positions satisfy $y_+ + y_- = \eta$ and $y_+ y_- = -1$, so $y_+^2 + y_-^2 = (y_+ + y_-)^2 - 2y_+ y_- = \eta^2 + 2$, and the total magnification is

$$A = A_+ + A_- = \frac{\eta^2 + 2}{\eta\sqrt{\eta^2 + 4}}. \quad (9.34)$$

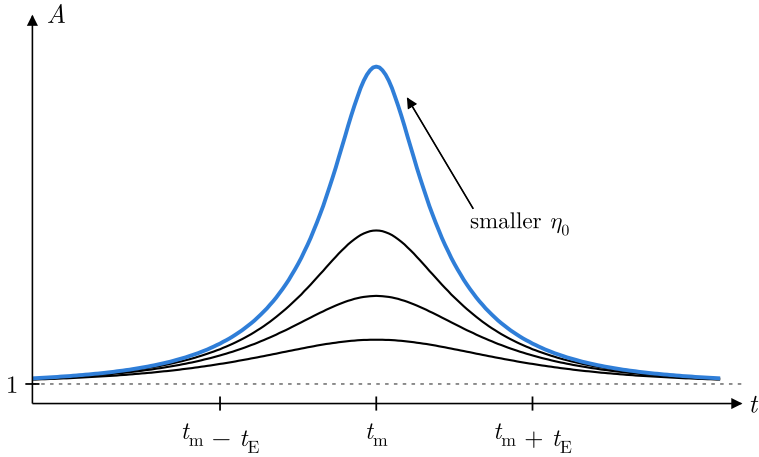


Figure 103. The magnification A of a background star by a compact lens moving across the line of sight, plotted against time t for several values of the closest approach η_0 , in units of the Einstein radius. The curves are symmetric about the time t_m of closest approach, and the peak is higher for smaller η_0 .

A lens moving across the line of sight with transverse speed $v_{\perp}$ crosses the Einstein radius, whose length in the plane of the lens is $D_L \theta_E$, in a time

$$t_E = \frac{D_L \theta_E}{v_{\perp}}, \quad (9.35)$$

called the Einstein crossing time. If the lens passes closest to the line of sight at time t_m , at a distance η_0 in units of the Einstein radius, then by Pythagoras' theorem $\eta(t) = [\eta_0^2 + (t - t_m)^2/t_E^2]^{1/2}$. Substituting this into $A(\eta)$ gives the light curves of Figure 103, which are symmetric in time, the same at every wavelength, since the bending of light does not depend on the wavelength, and not repeated, which distinguishes them from the brightening of variable stars. Since θ_E grows as $M^{1/2}$, the crossing time measures the mass of the lens statistically, D_L and $v_\perp$ being unknown for a single event; for a lens of half a solar mass in the halo it is of order a month.

The chance that a given star is being lensed at any moment is small. Consider the lenses between distances D_L and $D_L + dD_L$, of mass M and number density ρ_{DM}/M . A star is appreciably magnified when it lies within the Einstein radius of one of them, that is, when a lens lies within a disc of area $\pi D_L^2 \theta_E^2$ about the line of sight. The fraction of the sky covered by these discs, added up along the line of sight, is

$$\tau_{ml} = \int_0^{D_S} \frac{\rho_{DM}}{M} \pi D_L^2 \theta_E^2 dD_L = \frac{4\pi G}{c^2} \int_0^{D_S} \rho_{DM} \frac{D_L(D_S - D_L)}{D_S} dD_L, \quad (9.36)$$

which is called the optical depth for microlensing, by analogy with the optical depth of Section 3.3. The mass of the lenses has cancelled, so the optical depth depends only on the density of the halo. For a density of order $v_c^2/4\pi G s_0^2$ along a line of sight of length comparable with s_0 , you will find in Problem 2 that τ_{ml} is of order v_c^2/c^2 , which is less than 10^{-6} . If the whole halo were made of compact objects, fewer than one star in a million in the Large Magellanic Cloud would be appreciably magnified at any moment.

The MACHO, EROS, and OGLE surveys of the 1990s and 2000s followed the brightness of tens of millions of stars in the Magellanic Clouds for several years. They found far fewer events than a halo made of compact objects would produce, and concluded that compact objects with masses between roughly 10^{-7} and $10 M_\odot$ make up at most a small fraction of the halo. Evidence from the abundances of the light elements and from the microwave background indicates that ordinary matter makes up only about a sixth of all matter, but

that evidence belongs to the thermal history of the early universe, which is not treated in this course.

Massive Neutrinos

Neutrinos interact with matter only through the weak interaction and gravity, the oscillations described in Section 4.6 show that they have mass, and the Big Bang left a background of them with a number density comparable to that of the photons of the microwave background. Their masses are small; experiments on the beta decay of tritium, a radioactive isotope of hydrogen, measure how far the largest energy of the emitted electron falls short of the energy released, the difference being the rest energy of the neutrino, and find the mass to be less than about $1 \text{ eV } c^{-2}$. With so small a mass the background neutrinos could make up only a part of the dark matter, and the argument that follows rules them out as the dark matter of galaxies.

An argument due to Tremaine and Gunn rules out light neutrinos as the dark matter of galaxies, using the Pauli exclusion principle of Section 5.1, since neutrinos, like electrons, have spin one-half. Consider the halo of a galaxy, with density ρ of neutrinos of mass m_ν , and suppose that no neutrino bound to the halo moves faster than some speed v_{max} , of the order of the escape speed. The states of a particle in a box of volume V with momentum less than p number $g_\nu V \cdot \frac{4}{3}\pi p^3/h^3$, as counted in Section 5.1, where g_ν is the number of spin states for each momentum, and $g_\nu = 2$ once the neutrinos and their antiparticles are both counted. The number density of neutrinos with speeds below v_{max} can therefore be at most

$$n_{\text{max}} = \frac{4\pi g_\nu}{3} \frac{(m_\nu v_{\text{max}})^3}{h^3}, \quad (9.37)$$

which is reached when every state below the momentum $m_\nu v_{\text{max}}$ is filled. The number density required is ρ/m_ν , and requiring it to be no more than n_{max} gives

$$m_\nu^4 \geq \frac{3\rho h^3}{4\pi g_\nu v_{\max}^3} . \quad (9.38)$$

The bound is strongest where the density is high and the speeds are low, which is the case in the smallest galaxies. You will show in Problem 4 that for an isothermal halo with $v_{\max} = 2\sigma_v$ the bound becomes $m_\nu^4 \geq 3h^3/64\pi^2 g_\nu Gr^2\sigma_v$ at radius r . For the dwarf galaxies that orbit the Milky Way, with σ_v of order 10 km s^{-1} at radii of order 1 kpc, it requires m_ν to exceed about $100 \text{ eV } c^{-2}$, far above the measured limit, so these galaxies cannot be held together by neutrinos.

Weakly Interacting Massive Particles

A weakly interacting massive particle is a hypothetical stable particle with a mass perhaps between a few $\text{GeV } c^{-2}$ and a few $\text{TeV } c^{-2}$, which interacts with ordinary matter through the weak interaction or something of similar strength. No such particle is known, but several extensions of the present theory of the elementary particles predict one. Much of the interest in them comes from the calculation, which belongs to the thermal history of the early universe and is not treated in this course, that a cross-section typical of the weak interaction would leave about the observed density of them over from the hot early universe. In the halo they would move at a few hundred kilometres per second, much slower than light, and dark matter of this kind is called cold dark matter.

Axions

An axion is a hypothetical particle of very small mass, perhaps between 10^{-6} and $10^{-3} \text{ eV } c^{-2}$, proposed in the late 1970s to explain a property of the strong interaction between quarks, which is not treated in this course. Axions made in the early universe would move slowly despite their small mass, so they too would behave as cold dark matter. In a strong magnetic field an axion can turn into a photon of energy $m_a c^2$, the rest energy of the axion, which lies in the microwave region. The searches place a microwave cavity, a closed metal box inside which standing electromagnetic waves exist only at sharply defined frequencies, like the modes of the cavity of

Section 5.6 of Vol. 2, in a strong magnet, and look for a very faint signal when the frequency of a mode matches $m_a c^2/h$, changing the size of the cavity step by step to scan the possible masses.

Direct Detection

A particle of dark matter passing through a detector will occasionally collide with a nucleus in it, and the energy of the recoil follows from the elastic collisions of Section 2.4 of Vol. 1. Let a particle of mass m_{DM} and speed v strike a nucleus of mass m_{nuc} at rest, and let

$$\mu_r = \frac{m_{\text{DM}} m_{\text{nuc}}}{m_{\text{DM}} + m_{\text{nuc}}} \quad (9.39)$$

be the reduced mass of the two, which Vol. 1 writes μ . The centre of mass moves with speed $V = m_{\text{DM}} v / (m_{\text{DM}} + m_{\text{nuc}})$, and in the frame of the centre of mass the nucleus moves initially with velocity $-\mathbf{V}$. As shown in Vol. 1, an elastic collision leaves each speed in the centre-of-mass frame unchanged and turns each velocity through the scattering angle, which Vol. 1 writes θ^* and we write ϑ . After the collision the nucleus therefore moves in the centre-of-mass frame with speed V in a direction at the angle ϑ to $-\mathbf{V}$, and adding $\mathbf{V}$ gives its velocity in the laboratory.

In Figure 104 the velocity of the nucleus in the laboratory is OP , the third side of a triangle whose sides OC and CP both have length V and meet at the angle ϑ , so the cosine rule gives $v_{\text{nuc}}^2 = 2V^2(1 - \cos \vartheta)$, and the recoil energy is

$$E_{\text{R}} = \frac{1}{2} m_{\text{nuc}} v_{\text{nuc}}^2 = \frac{\mu_r^2 v^2}{m_{\text{nuc}}} (1 - \cos \vartheta), \quad (9.40)$$

where we have used $m_{\text{nuc}} V = \mu_r v$. The recoil is largest for a head-on collision, $\vartheta = \pi$, when P lies at Q , and then $E_{\text{R}} = 2\mu_r^2 v^2 / m_{\text{nuc}}$, which is the fraction $4m_{\text{DM}} m_{\text{nuc}} / (m_{\text{DM}} + m_{\text{nuc}})^2$ of the kinetic energy of the particle, as you showed in Problem 3 of Section 2.4 of Vol. 1. For a particle and a nucleus both of mass about $100 \text{ GeV } c^{-2}$, moving at a few hundred kilometres per second, the largest recoil

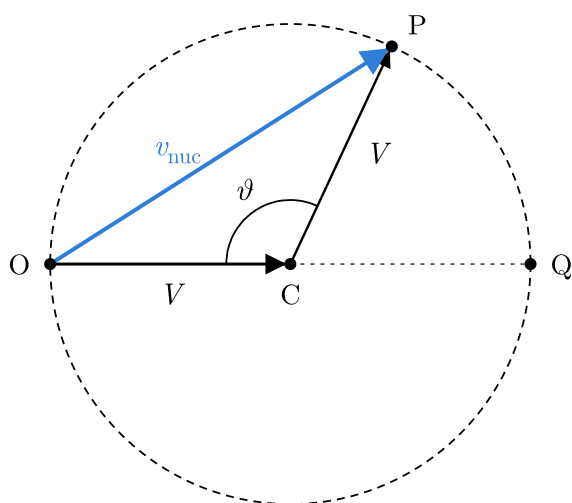


Figure 104. The recoil velocity of a nucleus struck by a particle of dark matter. The velocity of the centre of mass is $\vec{OC}$, of length V , and after the collision the nucleus moves relative to the centre of mass with the velocity $\vec{CP}$, also of length V , turned through the angle ϑ from the direction $\vec{CO}$. Its velocity in the laboratory is $\vec{OP}$ (accent), which is longest when P is at Q .

energy is a few tens of kiloelectronvolts. As you also showed in Problem 3 of Section 2.4 of Vol. 1, for a given particle the largest recoil is greatest when the nucleus has the same mass as the particle, which is one reason nuclei of large mass, such as those of xenon and germanium, are often chosen as targets.

The rate of collisions is found as in Section 1.3 of Vol. 2. A nucleus meets the particles of dark matter that lie in a cylinder of cross-section σ swept out by their relative motion, so it is struck $n_{\text{DM}}\sigma\langle v \rangle$ times per unit time, where $n_{\text{DM}} = \rho_{\text{DM}}/m_{\text{DM}}$ is the number density of the particles and $\langle v \rangle$ their mean speed relative to the detector. A detector of mass M_{det} contains $M_{\text{det}}/m_{\text{nuc}}$ nuclei, so the rate of events is

$$\Gamma = \frac{M_{\text{det}}}{m_{\text{nuc}}} \frac{\rho_{\text{DM}}}{m_{\text{DM}}} \sigma \langle v \rangle. \quad (9.41)$$

For the cross-sections now being tested this is fewer than one event per tonne of target per year, so the detectors are placed deep underground, where the rock stops most of the particles produced by cosmic rays, and are built of materials of low radioactivity. Some of the largest hold several tonnes of liquid xenon. A recoiling nucleus produces a flash of light in the liquid, recorded by photomultiplier tubes, which are devices that turn a faint flash of light into a measurable pulse of current, and it frees electrons, which an electric field drifts to the surface, where they produce a second flash. The ratio of the two flashes tells a recoiling nucleus from an electron struck by a gamma ray. No experiment of this kind has yet found a confirmed signal, and the results are upper limits on σ for each value of m_{DM} .

Annual Modulation

The motion of the Earth gives the signal a distinctive variation through the year. If the halo does not rotate, the Sun moves through it with the circular speed, $v_{\odot} \approx 230 \text{ km s}^{-1}$, and the Earth moves round the Sun at $v_{\oplus} \approx 30 \text{ km s}^{-1}$ in a plane that makes an angle $\psi \approx 60^\circ$ with the direction of the motion of the Sun, as drawn in Figure 105(a). Let $\mathbf{w}$ be the velocity of the Earth relative to the Sun, so that the velocity of the Earth relative to the halo is $\mathbf{v}_{\odot} + \mathbf{w}$,

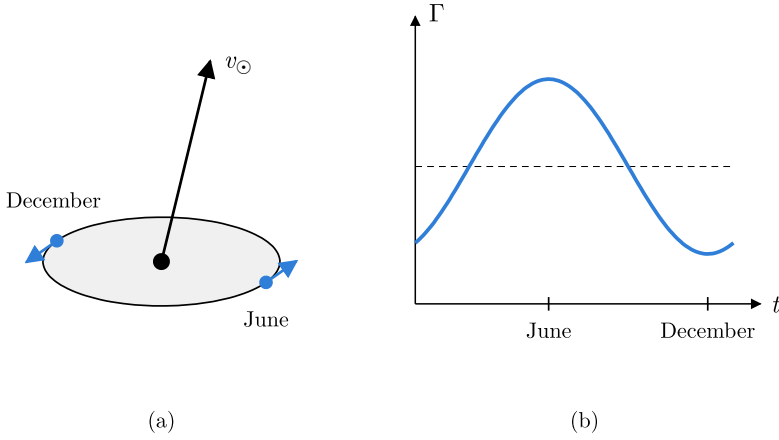


Figure 105. (a) The Sun moves through the halo with velocity $\mathbf{v}_\odot$, and the Earth moves round the Sun in an orbit whose plane is inclined to $\mathbf{v}_\odot$, so that the Earth moves partly with the Sun in June and partly against it in December. (b) The resulting rate Γ of collisions in a detector through the year, with the size of the variation exaggerated.

and its square is $v_\odot^2 + 2\mathbf{v}_\odot \cdot \mathbf{w} + w^2$. Since $w \ll v_\odot$, the binomial expansion gives the speed v_h of the Earth through the halo as

$$v_h \approx v_\odot + \hat{\mathbf{v}}_\odot \cdot \mathbf{w} = v_\odot + v_\oplus \cos \psi \cos \frac{2\pi(t - t_J)}{P_{\text{orb}}}, \quad (9.42)$$

where $\hat{\mathbf{v}}_\odot$ is the unit vector along $\mathbf{v}_\odot$, P_{orb} is one year, and t_J is the time, near the beginning of June, at which the component of $\mathbf{w}$ along $\mathbf{v}_\odot$ is largest; the component of $\mathbf{w}$ perpendicular to $\mathbf{v}_\odot$ changes the speed only at second order, through the term w^2 . With $\cos \psi \approx \frac{1}{2}$, the speed of the Earth through the halo varies by about $\pm 15 \text{ km s}^{-1}$ over the year.

The rate of events depends on this speed, because a faster motion through the halo raises the mean relative speed $\langle v \rangle$, and it also raises the number of particles fast enough to produce a recoil above the smallest energy the detector can register. The rate therefore varies as

$$\Gamma(t) \approx \Gamma_0 \left[1 + q_m \frac{v_{\oplus} \cos \psi}{v_{\odot}} \cos \frac{2\pi(t - t_J)}{P_{\text{orb}}} \right], \quad (9.43)$$

where Γ_0 is the mean rate and $q_m = d \ln \Gamma / d \ln v_h$ measures how strongly the rate depends on the speed. The coefficient q_m is of order one, and depends on the velocity distribution of the particles and on the smallest recoil energy detected. The rate therefore changes by a few per cent over the year, with a maximum near the beginning of June, as drawn in Figure 105(b), and since few other sources of events would share this period and phase, the variation is a useful test of a claimed signal. The DAMA experiment, with crystals of sodium iodide in the Gran Sasso laboratory in Italy, has reported such a modulation for many years, but other experiments with the same target material have not found it, and the interpretation of the result is disputed.

Indirect Detection

If a particle of dark matter is identical with its antiparticle, two that meet can annihilate into gamma rays, into pairs of particles and antiparticles such as electrons and positrons, or into neutrinos. The rate of annihilation per unit volume goes as the square of the number density, since each particle meets others at a rate proportional to the number density, so the searches look where dark matter is densest, towards the centre of the Milky Way, towards its dwarf satellite galaxies, and towards the centre of the Sun, where particles would collect after losing energy in collisions. Gamma-ray telescopes in space look for the gamma rays, and neutrino detectors, such as a cubic kilometre of Antarctic ice watched by thousands of sensors that record the faint light from the fast charged particles a neutrino produces when it interacts, look for the neutrinos. An excess of gamma rays from around the Galactic centre has been reported, but it may come from a population of pulsars there, and no signal has yet been accepted as coming from dark matter.

Problems

§9.1 The Expanding Universe

- Galaxies are spread uniformly through space, and a fraction f_q of them have luminosity L_q , where q labels the different luminosities. Show that the number of galaxies brighter than flux F is proportional to $F^{-3/2}$, and hence that the number brighter than apparent magnitude m goes as $10^{0.6m}$. Show also that the total flux received from all the galaxies with fluxes greater than F grows as $F^{-1/2}$ as F is made smaller, and explain how this result is related to Olbers' paradox.
- Suppose that every separation between galaxies grows in proportion to $(t - t_b)^\zeta$, where t_b is the time at which the separations were zero and ζ is a positive constant. Find the Hubble constant at time t , and show that the present age of the universe is $t_0 - t_b = \zeta t_H$. Which values of ζ describe an expansion that slows down?
- Let ω be a constant vector, and consider the velocity field $\mathbf{v} = \omega \times \mathbf{r}$. Show that it has the linear form $v_j = \sum_k H_{jk} r_k$ with an antisymmetric matrix $\mathbf{H}$, and that every antisymmetric matrix arises in this way from some ω . Show also that an observer in any galaxy sees the other galaxies circulating about an axis through that galaxy parallel to ω , and that no galaxy has a radial velocity relative to any other.
- A galaxy of mass M has shone with a constant luminosity L for a time t_H , all of it produced by the fusion of hydrogen into helium, which releases a fraction f_{He} of the rest energy of the hydrogen consumed. Show that the fraction of the mass of the galaxy turned into helium is $\Delta Y = Lt_H/f_{\text{He}}Mc^2$, and express it in terms of the mass-to-light ratio $\Upsilon = M/L$ of Section 8.1.
- Stars of radius R and effective temperature T_{eff} , with number density n , shine into a universe that holds no radiation to begin with, and none of the radiation is absorbed. The energy density of blackbody radiation at the temperature T_{eff} , which is the energy density everywhere once every line of sight ends on the surface of a star, is $u_s = 4\sigma_{\text{SB}}T_{\text{eff}}^4/c$ (Section 5.6 of Vol. 2). Show that the time the stars need to raise the energy density

everywhere to u_s is ℓ/c , where ℓ is the mean distance a line of sight travels before meeting a star.

§9.2 The Evidence for Dark Matter

1. A spherical ball of gas has the same velocity dispersion σ_v everywhere, so that its pressure is $P = \rho\sigma_v^2$. Using the equation of hydrostatic equilibrium of Section 4.2 together with $dM/dr = 4\pi r^2\rho$, show that $\rho = \sigma_v^2/2\pi Gr^2$ is a solution, and that the circular speed in this ball is $v_c = \sqrt{2}\sigma_v$ at every radius.
2. The distance d to a cluster is found from its mean redshift with Hubble's law. Its radius R is found from its angular radius, and its luminosity from the flux received from its galaxies. Show that the mass found from the virial theorem is proportional to $1/H_0$, that the luminosity is proportional to $1/H_0^2$, and hence that the mass-to-light ratio is proportional to H_0 .
3. A particle moving at speed c passes a body of mass M with impact parameter b , and is deflected through a small angle. Treating its path as a straight line, show that the component of the gravitational acceleration perpendicular to the path, integrated over the time of passage, gives a change of velocity $2GM/bc$, and hence a deflection $2GM/c^2b$. The integral $\int_{-\infty}^{\infty} b(b^2 + x^2)^{-3/2} dx$ can be done with the substitution $x = b \tan \phi$.
4. For a point lens, show that the two images satisfy $\theta_+\theta_- = -\theta_E^2$, that their angular separation $\theta_+ - \theta_-$ is never less than $2\theta_E$, and that when $\beta \gg \theta_E$ the outer image lies close to the direction of the source and the inner image close to the lens.
5. An isothermal cluster of radius R has gas with the profile of the text, with $r_c \ll R$, and galaxies with velocity dispersion σ_v . Equate the mass inside R found from the gas with the virial mass of a uniform sphere, and show that $k_B T = 5\mu m_H \sigma_v^2 / 3\beta_g$.

§9.3 Dark Matter Candidates and Their Detection

1. Show that the total magnification of a point lens is approximately $1/\eta$ when $\eta \ll 1$, that it equals $3/\sqrt{5}$ when the source lies on the Einstein radius, and that $A - 1 \approx 2/\eta^4$ when $\eta \gg 1$.

2. Lenses of total mass density ρ_{DM} , uniform along the line of sight, lie between the observer and a source at distance D_{S} . Show that the optical depth for microlensing is $\tau_{\text{ml}} = 2\pi G\rho_{\text{DM}}D_{\text{S}}^2/3c^2$. Taking $\rho_{\text{DM}} = v_{\text{c}}^2/4\pi Gs_0^2$ and $D_{\text{S}} = s_0$, show that $\tau_{\text{ml}} = v_{\text{c}}^2/6c^2$. Show also that for a fixed source the Einstein radius measured in the plane of the lens, $D_{\text{L}}\theta_{\text{E}}$, is largest for a lens halfway to the source.
3. A particle of mass m_{DM} and speed v strikes a nucleus of mass m_{nuc} at rest. Show that to produce a recoil of energy E_{R} the particle must have a speed of at least $(m_{\text{nuc}}E_{\text{R}}/2\mu_{\text{r}}^2)^{1/2}$, and find this smallest speed for a nucleus much heavier than the particle.
4. For the singular isothermal sphere of Problem 1 of Section 9.2, with $\rho = \sigma_v^2/2\pi Gr^2$, and with $v_{\text{max}} = 2\sigma_v$, show that the bound of Tremaine and Gunn becomes $m_{\nu}^4 \geq 3h^3/64\pi^2 g_{\nu} Gr^2 \sigma_v$. Explain why the bound is strongest for small galaxies.
5. A detector of mass M_{det} registers every collision, so that its rate of events is proportional to the speed v_{h} of the Earth through the halo. Show that the rate varies through the year with a fractional amplitude $v_{\oplus} \cos \psi / v_{\odot}$, and that the component of the orbital velocity perpendicular to $\mathbf{v}_{\odot}$ changes the rate by a fraction of order $v_{\oplus}^2 / v_{\odot}^2$.

Solutions

Chapter 1. Observational Foundations and Instrumentation

§1.1 Magnitudes, Colours, and the Distance Ladder

1. The distances are $1/p_1$ and $1/p_2$ in parsecs, so the distance moduli give $m_1 - m_2 = 5 \log_{10}(d_1/d_2) = 5 \log_{10}(p_2/p_1)$. The ratio of the fluxes is $F_1/F_2 = (d_2/d_1)^2 = (p_1/p_2)^2$.
2. When $x = hc/\lambda k_B T$ is small, the Taylor series gives $e^x - 1 \approx x$, so $F_{s,\lambda} \approx 2\pi c k_B T / \lambda^4$. The temperature then cancels from the ratio, which becomes $(\Delta\lambda_B/\Delta\lambda_V)(\lambda_V/\lambda_B)^4$, and the colour index tends to $-2.5 \log_{10}$ of this ratio plus C_{BV} , a constant. In Figure 3 the accent curve flattens towards the dotted line at high temperature, so a small change in the colour index corresponds to a large change in temperature.
3. The fraction is $F_{s,\lambda}(\lambda_V) \Delta\lambda/\sigma_{SB} T^4$. With $T = hc/(x\lambda_V k_B)$, the factor $1/T^4$ is proportional to x^4 , and the fraction is proportional to $x^4/(e^x - 1)$. By the quotient rule its derivative vanishes when $4x^3(e^x - 1) = x^4 e^x$, and dividing by $x^3 e^x$ gives $x = 4(1 - e^{-x})$, whose root is a little below four. The V band then collects the largest fraction of the flux at $T = hc/(x\lambda_V k_B)$ with that root, so the bolometric correction is closest to zero near this temperature, which is somewhat hotter than the Sun, and more negative for hotter and cooler bodies.

4. With $T_{\text{eff}} = T_{\text{eff},\odot}$, the relation between magnitude, radius, and temperature gives $-\Delta = -5 \log_{10}(R/R_{\odot})$, so $R = 10^{\Delta/5} R_{\odot}$.
5. The two Cepheids have the same period and hence the same absolute magnitude $\mathcal{M}_V$. The distance to the Milky Way Cepheid is $d_2 = 1/p$ in parsecs. Subtracting the two distance moduli gives $m_1 - m_2 = 5 \log_{10}(d_1/d_2)$, so the distance to the Small Magellanic Cloud is $d_1 = 10^{(m_1 - m_2)/5}/p$ in parsecs, with p in seconds of arc.

§1.2 Telescopes, Detectors, and Observing across the Spectrum

1. For a bright star $S/N = \sqrt{\dot{N}_{\star} t}$, so $t = \mathcal{S}^2 / \dot{N}_{\star}$. With $\dot{N}_{\star}$ from the text, $t = 4\mathcal{S}^2 hc 10^{0.4m_V} / (\eta \pi D^2 \lambda_V \Delta \lambda_V F_{\lambda,0})$, which goes as $10^{0.4m_V} / D^2$.
2. From $m = -2.5 \log_{10}(F/F_0) = -(2.5/\ln 10) \ln(F/F_0)$, differentiating gives $dm = -(2.5/\ln 10) dF/F$, so the size of the uncertainty is $\delta m = (2.5/\ln 10) \delta F/F$. The fractional uncertainty in the flux is the inverse of the signal-to-noise ratio, and $2.5/\ln 10 \approx 1.09$.
3. The photons from the star arrive at the rate $\dot{N}_{\star} \propto D^2$. The sky photons arrive at a rate proportional to the collecting area and to the solid angle of the seeing disc, so $\dot{N}_{\text{sky}} \propto D^2 \theta_s^2$. For a faint star $t \propto \dot{N}_{\text{sky}} / \dot{N}_{\star}^2 \propto D^2 \theta_s^2 / D^4 = \theta_s^2 / D^2$, so halving θ_s reduces the time by the same factor as doubling D .
4. The phase $2\pi \ell \sin \theta / \lambda$ changes by 2π from one fringe to the next. Near θ_1 , the Taylor series gives $\sin \theta \approx \sin \theta_1 + (\theta - \theta_1) \cos \theta_1$, so the phase changes by 2π when $\theta - \theta_1 = \lambda / (\ell \cos \theta_1)$. The length $\ell \cos \theta_1$ is the component of the baseline at right angles to the direction of the source, which is the length of the baseline as it appears from the source.
5. By Wien's law, $\lambda_{\text{max}} = hc / (\zeta_p k_B T_t)$, where ζ_p is the number close to five of Section 1.1. The total flux goes as T_t^4 , a power of the temperature. On the short-wavelength side of the peak, where $hc / \lambda_o k_B T_t \gg 1$, the Wien approximation gives $F_{s,\lambda} \approx (2\pi hc^2 / \lambda_o^5) e^{-hc / \lambda_o k_B T_t}$, which falls exponentially as T_t

is lowered. Cooling the telescope moves the peak to longer wavelengths, further from λ_o , and the exponential factor makes the glow at λ_o fall far faster than the total.

Chapter 2. Celestial Mechanics and Planetary Systems

§2.1 Kepler's Laws and Orbital Elements

1. From the coordinates of B, $\cos \theta = (\cos \psi - e)/(1 - e \cos \psi)$, so $1 - \cos \theta = (1 + e)(1 - \cos \psi)/(1 - e \cos \psi)$ and $1 + \cos \theta = (1 - e)(1 + \cos \psi)/(1 - e \cos \psi)$. Dividing the first by the second and using $\tan^2(x/2) = (1 - \cos x)/(1 + \cos x)$ gives $\tan^2(\theta/2) = [(1 + e)/(1 - e)] \tan^2(\psi/2)$, and the positive root is taken because θ and ψ lie in the same half of the orbit.
2. Kepler's equation reads $\psi = \eta + e \sin \psi$, and replacing ψ by η inside the sine makes an error of order e^2 , so $\psi \approx \eta + e \sin \eta$. To first order $\sqrt{(1 + e)/(1 - e)} \approx 1 + e$, so $\tan(\theta/2) \approx \tan(\psi/2) + e \tan(\psi/2)$. The Taylor expansion of $\tan$ about $\psi/2$ gives $\theta/2 \approx \psi/2 + e \tan(\psi/2) \cos^2(\psi/2) = \psi/2 + \frac{1}{2}e \sin \psi$, so $\theta \approx \psi + e \sin \psi \approx \eta + 2e \sin \eta$.
3. Since η grows uniformly, the time average is $(1/2\pi) \int_0^{2\pi} r \, d\eta$. Differentiating Kepler's equation gives $d\eta = (1 - e \cos \psi) \, d\psi$, so the average is $(a/2\pi) \int_0^{2\pi} (1 - e \cos \psi)^2 \, d\psi = (a/2\pi)(2\pi + \pi e^2) = a(1 + \frac{1}{2}e^2)$, the integral of $\cos \psi$ vanishing and that of $\cos^2 \psi$ being π .
4. With $r = a(1 - e \cos \psi) = a$ we need $\cos \psi = 0$, so $\psi = \pi/2$, and Kepler's equation gives $\eta = \pi/2 - e$. The fraction of the period is $\eta/2\pi = \frac{1}{4} - e/2\pi$.
5. For a circular orbit $r = a$ and the body moves at constant angular speed, so the angle from the ascending node is $2\pi(t - t_0)/P_{\text{orb}}$, where t_0 is an instant at which the body is at the ascending node. Then $\zeta = a \sin i \sin[2\pi(t - t_0)/P_{\text{orb}}]$, which is sinusoidal with amplitude $a \sin i$, and the largest value of $d\zeta/dt$ is $2\pi a \sin i / P_{\text{orb}}$, reached as the body passes the ascending node.

§2.2 Binary Stars and the Masses and Radii of Stars

- Directions with inclination between i and $i + di$ fill a band of the unit sphere of area $2\pi \sin i di$, and the hemisphere $0 \leq i \leq \pi/2$ has area 2π , so the probability is $\sin i di$. The mean of $\sin^3 i$ is $\int_0^{\pi/2} \sin^4 i di$. Writing $\sin^4 i = (1 - \cos 2i)^2/4 = \frac{3}{8} - \frac{1}{2} \cos 2i + \frac{1}{8} \cos 4i$ and integrating, the cosines give zero and the result is $\frac{3}{8} \cdot \frac{\pi}{2} = 3\pi/16$.
- Let $h = r^2 d\theta/dt$ and $\ell = a(1 - e^2)$. Differentiating $r = \ell/(1 + e \cos \theta)$ gives $dr/dt = (r^2 e \sin \theta/\ell) d\theta/dt = h e \sin \theta/\ell$, and $r d\theta/dt = h/r = h(1 + e \cos \theta)/\ell$. Then $d\zeta/dt = (h \sin i/\ell)[e \sin \theta \sin(\omega + \theta) + (1 + e \cos \theta) \cos(\omega + \theta)]$, and the terms in e combine to $e \cos(\omega + \theta - \theta) = e \cos \omega$. The area swept per unit time is $\frac{1}{2}h = \pi a^2 \sqrt{1 - e^2}/P_{\text{orb}}$, so $h/\ell = 2\pi a/P_{\text{orb}} \sqrt{1 - e^2}$. For star 1, a is replaced by a_1 , so $a_1 \sin i = P_{\text{orb}} K_1 \sqrt{1 - e^2}/2\pi$, and repeating the calculation of the mass function with this value gives the extra factor $(1 - e^2)^{3/2}$.
- In coordinates in the plane of the sky with the x -axis along the line of nodes, the relative position is $a(\cos \phi, \sin \phi \cos i)$, where ϕ is the angle from the ascending node, so the separation squared is $a^2(\cos^2 \phi + \sin^2 \phi \cos^2 i)$, which is smallest, $a^2 \cos^2 i$, at $\phi = \pm\pi/2$. The discs overlap only if this minimum is less than $(R_1 + R_2)^2$. For random orientations $\cos i$ is uniformly distributed between zero and one, since the probability $\sin i di$ of Problem 1 is $|d(\cos i)|$, so the probability that $\cos i < (R_1 + R_2)/a$ is $(R_1 + R_2)/a$.
- The semi-major axis of the orbit of the visible star is $a_1 = \alpha_1/p$ astronomical units, and $a_1 = aM_2/(M_1 + M_2)$, with $a^3 = (M_1 + M_2)P_{\text{orb}}^2$ in solar units. Eliminating a gives $(\alpha_1/p)^3/P_{\text{orb}}^2 = M_2^3/(M_1 + M_2)^2$. The right-hand side is the mass function with $\sin i = 1$, since the fit to the apparent orbit gives the true α_1 , and the equation can be solved for M_2 once M_1 is known.
- The smaller star moves through a distance equal to its diameter between t_a and t_b and the diameter of the larger star between t_a and t_c , at the same speed, so $R_s/R_1 = (t_b - t_a)/(t_c - t_a)$. Between t_a and t_d it moves through $2(R_s + R_1)$ at the speed $2\pi a/P_{\text{orb}}$, so $(R_s + R_1)/a = \pi(t_d - t_a)/P_{\text{orb}}$.

§2.3 Exoplanets and the Formation of Planetary Systems

1. From the mass function of Section 2.2 with $M_2 = M_{\text{pl}}$, $K^3 = 2\pi GM_{\text{pl}}^3 \sin^3 i / P_{\text{orb}}(M + M_{\text{pl}})^2$, and taking the cube root gives the result. For equal masses about the same star, $K_1/K_2 = (P_2/P_1)^{1/3}$.
2. The planet crosses the disc of the star when their separation on the sky, whose smallest value is $a \cos i$, is less than $R + R_{\text{pl}}$. As you showed in Problem 3 of Section 2.2, $\cos i$ is uniformly distributed between zero and one for random orientations, so the probability is $(R + R_{\text{pl}})/a$, which is about R/a since $R_{\text{pl}} \ll R$.
3. The planet moves at the speed $2\pi a/P_{\text{orb}}$ and crosses a diameter $2R$ of the star, so $t_{\text{T}} = 2R/(2\pi a/P_{\text{orb}}) = P_{\text{orb}}R/\pi a$. Then $a/R = P_{\text{orb}}/\pi t_{\text{T}}$, and Kepler's third law gives $M = 4\pi^2 a^3/GP_{\text{orb}}^2$. The mean density is $3M/4\pi R^3 = (3\pi/GP_{\text{orb}}^2)(a/R)^3 = (3\pi/GP_{\text{orb}}^2)(P_{\text{orb}}/\pi t_{\text{T}})^3 = 3P_{\text{orb}}/\pi^2 G t_{\text{T}}^3$.
4. The planet absorbs $(1 - A_{\text{b}})\pi R_{\text{pl}}^2 L/4\pi a^2$ and radiates $4\pi R_{\text{pl}}^2 \sigma_{\text{SB}} T^4$, so $T^4 = (1 - A_{\text{b}})L/16\pi \sigma_{\text{SB}} a^2$, and $L = 4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4$ gives the result. A planet radiating from one hemisphere, of area $2\pi R_{\text{pl}}^2$, has half the radiating area, so T^4 is doubled and T is larger by the factor $2^{1/4}$.
5. With weak focusing dR_{pl}/dt is the same constant for both bodies, so $R_{\text{A}} - R_{\text{B}}$ is constant while both radii grow, and $M_{\text{A}}/M_{\text{B}} = (R_{\text{A}}/R_{\text{B}})^3 = [1 + (R_{\text{A}} - R_{\text{B}})/R_{\text{B}}]^3$ decreases towards one. With strong focusing $dM_{\text{pl}}/dt = CM_{\text{pl}}^{4/3}$ with the same constant C for both, so $d \ln(M_{\text{A}}/M_{\text{B}})/dt = C(M_{\text{A}}^{1/3} - M_{\text{B}}^{1/3})$, which is positive, and the ratio increases.

Chapter 3. Radiation, Radiative Transfer, and Stellar Atmospheres

§3.1 Stellar Spectra and Their Classification

1. For $\text{H}\alpha$, $1/\lambda = \mathcal{R}_{\text{H}}(1/4 - 1/9) = 5\mathcal{R}_{\text{H}}/36$, and for $\text{H}\beta$, $1/\lambda = \mathcal{R}_{\text{H}}(1/4 - 1/16) = 3\mathcal{R}_{\text{H}}/16$. The ratio of the wavelengths is $(3/16)(36/5) = 27/20$. The Balmer limit has $1/\lambda = \mathcal{R}_{\text{H}}/4$, so its wavelength divided by that of $\text{H}\alpha$ is $(5/36)/(1/4) = 5/9$.

2. Since $\mathcal{M}_{\text{bol}} = -2.5 \log_{10} L$ plus a constant, the ratio of the luminosities of the giant and the main-sequence star is $10^{0.4\Delta\mathcal{M}}$. At equal effective temperatures $R \propto L^{1/2}$, so the ratio of the radii is $10^{0.2\Delta\mathcal{M}}$. With equal masses $g \propto R^{-2}$, so the surface gravity of the main-sequence star is $10^{0.4\Delta\mathcal{M}}$ times that of the giant.
3. The fraction of atoms with velocity between v_x and $v_x + dv_x$ is $\sqrt{m_{\text{atom}}/2\pi k_B T} e^{-m_{\text{atom}} v_x^2/2k_B T} dv_x$. An atom with velocity v_x absorbs at the shift $\Delta\lambda = \lambda v_x/c$, so $v_x = c\Delta\lambda/\lambda$, and when each atom absorbs independently the light removed at the shift $\Delta\lambda$ is proportional to the number of atoms absorbing there, which goes as $\exp[-m_{\text{atom}} c^2 (\Delta\lambda)^2 / 2k_B T \lambda^2]$. This is a Gaussian whose variance is $\lambda^2 k_B T / m_{\text{atom}} c^2$, so the standard deviation is $(\lambda/c) \sqrt{k_B T / m_{\text{atom}}}$. The width goes as $m_{\text{atom}}^{-1/2}$, and the ratio of the widths of the hydrogen and iron lines is $\sqrt{56} = 2\sqrt{14}$.
4. From the distance modulus, $d = (10 \text{ pc}) \times 10^{(m-\mathcal{M})/5}$, so $\ln d = \ln(10 \text{ pc}) + (m-\mathcal{M})(\ln 10)/5$. Differentiating with respect to $\mathcal{M}$ at fixed m gives $\delta d/d = (\ln 10/5) \delta\mathcal{M}$ in magnitude.
5. At the Balmer limit $1/\lambda = \mathcal{R}_{\text{H}}/4$, so the photon energy is $hc\mathcal{R}_{\text{H}}/4 = \chi_{\text{H}}/4$, which is the energy needed to take the electron from $E_2 = -\chi_{\text{H}}/4$ to zero. The level $k = 3$ has energy $-\chi_{\text{H}}/9$, so a photon can ionise the atom if $hc/\lambda \geq \chi_{\text{H}}/9$, that is if $\lambda \leq 9/\mathcal{R}_{\text{H}}$. Dividing by the wavelength $4/\mathcal{R}_{\text{H}}$ of the Balmer limit gives $9/4$.

§3.2 The Boltzmann and Saha Equations

1. The levels are equally populated when $4e^{-3\chi_{\text{H}}/4k_B T} = 1$, that is when $3\chi_{\text{H}}/4k_B T = \ln 4 = 2 \ln 2$, so $T = 3\chi_{\text{H}}/(8k_B \ln 2)$.
2. With $n_{\text{II}} = n_e = xn_{\text{H}}$ and $n_{\text{I}} = (1-x)n_{\text{H}}$, the left-hand side of the Saha equation is $x^2 n_{\text{H}}^2 / [(1-x)n_{\text{H}}] = n_{\text{H}} x^2 / (1-x)$, and the factor $2\mathcal{Z}_{\text{II}}/\mathcal{Z}_{\text{I}}$ is one. Dividing both sides by n_{H} gives the result.
3. Taking the logarithm of the Saha equation in terms of P_e gives $\ln(n_{\text{II}}/n_{\text{I}}) = \frac{5}{2} \ln T - \chi/k_B T$ plus terms that do not depend on T . Differentiating with respect to $\ln T$, and using $d(1/T)/d \ln T = -1/T$, gives $5/2 + \chi/k_B T$. When $\chi \gg k_B T$ this logarithmic derivative is large, so a small fractional change in T changes

$n_{\text{II}}/n_{\text{I}}$ by a large factor, and the ratio passes from much less than one to much more than one over a small range of T .

4. When $n_2 \ll n_1$, $n_2/n_{\text{I}} \approx n_2/n_1 = 4e^{-3\chi_{\text{H}}/4k_{\text{B}}T}$, and when the hydrogen is nearly all ionised, $n_{\text{I}}/(n_{\text{I}} + n_{\text{II}}) \approx n_{\text{I}}/n_{\text{II}}$, which by the Saha equation with $\mathcal{Z}_{\text{I}} = 2$ and $\mathcal{Z}_{\text{II}} = 1$ is $(P_e/k_{\text{B}}T)(h^2/2\pi m_e k_{\text{B}}T)^{3/2}e^{\chi_{\text{H}}/k_{\text{B}}T}$. The product is the stated expression, since $e^{-3\chi_{\text{H}}/4k_{\text{B}}T}e^{\chi_{\text{H}}/k_{\text{B}}T} = e^{\chi_{\text{H}}/4k_{\text{B}}T}$. At fixed P_e it goes as $T^{-5/2}e^{\chi_{\text{H}}/4k_{\text{B}}T}$, and both factors decrease as T rises.
5. The term for the level k is $2k^2e^{-\chi_{\text{H}}(1-1/k^2)/k_{\text{B}}T}$. For small $k \geq 2$ the exponent is at least $\frac{3}{4}\chi_{\text{H}}/k_{\text{B}}T$, which is large, so each of these terms is much less than the first term, 2. For large k the term tends to $2k^2e^{-\chi_{\text{H}}/k_{\text{B}}T}$, and the sum of these terms is about $e^{-\chi_{\text{H}}/k_{\text{B}}T} \sum_k 2k^2 \approx e^{-\chi_{\text{H}}/k_{\text{B}}T} \int_0^{k_{\text{max}}} 2k^2 dk = \frac{2}{3}k_{\text{max}}^3 e^{-\chi_{\text{H}}/k_{\text{B}}T}$. With $k_{\text{max}} = (n^{-1/3}/a_0)^{1/2}$ we have $k_{\text{max}}^3 = (na_0^3)^{-1/2}$, and the sum is small compared with 2 when $\frac{1}{3}(na_0^3)^{-1/2}e^{-\chi_{\text{H}}/k_{\text{B}}T} \ll 1$.

§3.3 Radiative Transfer

1. At the distance d the sphere fills a cone of half-angle θ_s about the direction of its centre, with $\sin \theta_s = R/d$, and by the constancy of the specific intensity along a ray the specific intensity is I within the cone and zero outside it. Taking θ from the direction of the centre, the flux is $\int_0^{2\pi} \int_0^{\theta_s} I \cos \theta \sin \theta d\theta d\phi = 2\pi I \cdot \frac{1}{2} \sin^2 \theta_s = \pi I (R/d)^2$. At the surface, $d = R$, this is the outward flux πI of the text.
2. Within the small solid angle $\cos \theta \approx 1$, so $F_\nu = \int I_\nu \cos \theta d\Omega \approx I_\nu \Delta\Omega$, $u_\nu = (1/c) \int I_\nu d\Omega = I_\nu \Delta\Omega/c$, and $P_{\text{rad},\nu} = (1/c) \int I_\nu \cos^2 \theta d\Omega \approx I_\nu \Delta\Omega/c$. The ratio of the pressure to the energy density is one, while for isotropic radiation it is $1/3$, which is the mean of $\cos^2 \theta$ over all directions.
3. Half the photons travel the distance $\ell_{1/2}$ given by $e^{-\ell_{1/2}/\ell_{\text{ph}}} = 1/2$, so $\ell_{1/2} = \ell_{\text{ph}} \ln 2$. The mean of ℓ^2 is $\int_0^\infty \ell^2 e^{-\ell/\ell_{\text{ph}}} d\ell / \ell_{\text{ph}}$, and with $x = \ell/\ell_{\text{ph}}$ this is $\ell_{\text{ph}}^2 \int_0^\infty x^2 e^{-x} dx = 2\ell_{\text{ph}}^2$, the integral being done by parts twice.
4. With $I_\nu(0) = 0$ the formal solution gives $I_\nu = S_\nu(1 - e^{-\tau_\nu})$. When both optical depths are small, $1 - e^{-\tau} \approx \tau$, and the ratio

of the intensity at the centre of the line to that in the continuum is τ_l/τ_c , so the line appears in emission. When both are large, both intensities tend to S_ν , the ratio tends to one, and the line disappears.

5. With $S_\nu = B_\nu(T_g)$ uniform, the result follows from the formal solution with a uniform source function. The emerging intensity minus $B_\nu(T_s)$ is $[B_\nu(T_g) - B_\nu(T_s)](1 - e^{-\tau_\nu})$, and since $1 - e^{-\tau_\nu} > 0$ it is negative if and only if $B_\nu(T_g) < B_\nu(T_s)$. The Planck function depends on T through $1/(e^{h\nu/k_B T} - 1)$, and as T rises the exponent $h\nu/k_B T$ falls, the denominator falls, and B_ν rises. The inequality between the Planck functions therefore holds if and only if $T_g < T_s$.

§3.4 Stellar Atmospheres and the Formation of Lines

1. The condition is $\frac{3}{4}(\tau + \frac{2}{3}) = 2$, so $\tau + \frac{2}{3} = \frac{8}{3}$ and $\tau = 2$. At the top $T^4 = \frac{1}{2}T_{\text{eff}}^4$, so the ratio of the fourth powers of the temperatures is 4, and the ratio of the temperatures is $\sqrt{2}$.
2. A ring of the disc at the impact parameter b and of width db has area $2\pi b db$. With $b = R \sin \theta$, $b db = R^2 \sin \theta \cos \theta d\theta$, and θ runs from 0 at the centre to $\pi/2$ at the limb. The mean over the area πR^2 is $2 \int_0^{\pi/2} [(2 + 3 \cos \theta)/5] \cos \theta \sin \theta d\theta$. With $x = \cos \theta$ this is $(2/5) \int_0^1 (2x + 3x^2) dx = (2/5)(1 + 1) = 4/5$ of $I(0)$.
3. With $g = GM/R^2$, $H_{\text{sc}}/R = k_B T/(m_{\text{av}} g R) = k_B T R/GM m_{\text{av}}$. At the same mass and temperature the ratio is proportional to R , so it is larger for the giant by the ratio of the radii. The atmosphere of a giant is therefore a larger fraction of its radius, and the plane-parallel approximation is less accurate for giants, and much less accurate for supergiants, than for main-sequence stars.
4. The optical depth of the layer at the wavelength λ is $\tau_\lambda = N\sigma_\lambda$, the number of atoms in a column of unit cross-section multiplied by the cross-section of each. By the formal solution with no emission from the thin layer, $F_\lambda = F_c e^{-\tau_\lambda} \approx F_c(1 - \tau_\lambda)$, so $(F_c - F_\lambda)/F_c \approx N\sigma_\lambda$, and integrating over wavelength gives $W = N \int \sigma_\lambda d\lambda$. Only the integral of the cross-section enters, so the shape of the profile does not.

5. With $x = \tau_\nu / \cos \theta$, the emerging specific intensity is $\int_0^\infty (S_0 + S_1 x \cos \theta + S_2 x^2 \cos^2 \theta) e^{-x} dx$, and with $\int_0^\infty x^j e^{-x} dx = j!$ for $j = 0, 1$, and 2 this is $S_0 + S_1 \cos \theta + 2S_2 \cos^2 \theta$. The Eddington–Barbier relation gives $S_\nu(\cos \theta) = S_0 + S_1 \cos \theta + S_2 \cos^2 \theta$, which is smaller by $S_2 \cos^2 \theta$.

Chapter 4. Stellar Structure, Energy Generation, and Evolution

§4.1 Stars and the Hertzsprung–Russell Diagram at School

1. Taking the logarithm of $L/L_\odot = (R/R_\odot)^2 (T_{\text{eff}}/T_{\text{eff},\odot})^4$ and multiplying by -2.5 gives $\mathcal{M}_{\text{bol}} = \mathcal{M}_{\text{bol},\odot} - 10 \log_{10}(T_{\text{eff}}/T_{\text{eff},\odot}) - 5 \log_{10}(R/R_\odot)$. At fixed R this is a linear function of $\log_{10} T_{\text{eff}}$ with slope -10 . Multiplying R by f at fixed T_{eff} changes $\mathcal{M}_{\text{bol}}$ by $-5 \log_{10} f$, so a larger star has a smaller, that is a brighter, magnitude.
2. With $L = 4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4$ the same for the white dwarf and the giant, $R \propto T_{\text{eff}}^{-2}$, so $R_{\text{wd}}/R_{\text{g}} = q^{-2}$. The mean density goes as M/R^3 , so for equal masses $\bar{\rho}_{\text{wd}}/\bar{\rho}_{\text{g}} = (R_{\text{g}}/R_{\text{wd}})^3 = q^6$.
3. From $\log_{10}(L/L_\odot) = 4 \log_{10}(T_{\text{eff}}/T_{\text{eff},\odot}) + 2 \log_{10}(R/R_\odot)$, raising $\log_{10} R$ by one at fixed T_{eff} raises $\log_{10} L$ by two, which is the vertical separation. At fixed L the same change must be balanced by a fall of one half in $\log_{10} T_{\text{eff}}$, which is the horizontal separation.
4. The mean density is $3M/4\pi R^3 \propto M^{1-3b}$, and the surface gravity is $GM/R^2 \propto M^{1-2b}$. For $\frac{1}{2} < b < 1$ the first exponent is negative and the second is negative as well, so both the mean density and the surface gravity decrease up the main sequence, the density more quickly than the gravity.

§4.2 Hydrostatic Equilibrium and the Virial Theorem

1. With $M_r = Mr^3/R^3$, the equation of hydrostatic equilibrium gives $dP/dr = -GM\rho r/R^3$, and integrating inwards from $P(R) = 0$ gives $P = (GM\rho/2R^3)(R^2 - r^2)$. With $\rho = 3M/4\pi R^3$ this is $P = (3GM^2/8\pi R^4)(1 - r^2/R^2)$, whose value at $r = 0$

- is the stated central pressure. Since $T = \mu m_H P / \rho k_B$ with ρ and μ constant, the temperature goes as $1 - r^2/R^2$, falling from $T_c = \mu m_H GM / 2k_B R$ at the centre to zero at the surface.
- The mass inside r is $M_r = 4\pi\rho_c(r^3/3 - r^4/4R)$, so $M = \pi\rho_c R^3/3$. Then $P_c = \int_0^R GM_r \rho dr / r^2 = 4\pi G \rho_c^2 \int_0^R (r/3 - r^2/4R)(1 - r/R) dr = 4\pi G \rho_c^2 R^2 (\frac{1}{6} - \frac{1}{12} - \frac{1}{9} + \frac{1}{16}) = 5\pi G \rho_c^2 R^2 / 36$. Substituting $\rho_c = 3M/\pi R^3$ gives $5GM^2/4\pi R^4$, and dividing by $3GM^2/8\pi R^4$ gives $10/3$.
 - At the centre $P_{\text{rad}}/P_{\text{gas}} = (a_{\text{rad}}T_c^4/3)(\mu m_H/\rho k_B T_c) = a_{\text{rad}}\mu m_H T_c^3/3\rho k_B$. When $T_c = \mu m_H GM/2k_B R$ and $\rho = 3M/4\pi R^3$ are substituted, the powers of R cancel and the result is $(4\pi a_{\text{rad}}/9)(\mu m_H/k_B)^4 (G^3/8)M^2$, which simplifies to the stated form.
 - Fully ionised helium has three particles, a nucleus and two electrons, for each mass $4m_H$, so $\mu = \frac{4}{3}$. A neutral gas has one particle for each atom, so a unit mass contains $X/m_H + Y/4m_H$ particles and $1/\mu = X + Y/4$, which gives $\mu = 1$ for pure hydrogen and $\mu = 4$ for pure helium.
 - Uniform density means that the initial acceleration $-GM_r/r^2 = -(GM/R^3)r$ is proportional to r , so every shell falls through the same fraction of its radius in the same time and the shells never cross, which keeps M_r fixed for each. A shell starting at rest at radius r_0 therefore moves like a body falling towards a point mass M_{r_0} , which is a Kepler orbit of eccentricity close to one with semi-major axis $r_0/2$. The fall to the centre takes half the period, $\frac{1}{2} \times 2\pi \sqrt{(r_0/2)^3/GM_{r_0}}$, and with $M_{r_0} = Mr_0^3/R^3$ this is $(\pi/2)\sqrt{R^3/2GM}$, the same for every shell.

§4.3 Energy Transport and the Equations of Stellar Structure

- Hydrostatic equilibrium gives $dP/dr = -GM_r\rho/r^2$, and the equation of state gives $T/P = \mu m_H/\rho k_B$. Substituting these two relations into $(dT/dr)_{\text{ad}} = (1 - 1/\gamma)(T/P)dP/dr$, the density cancels and the stated result follows.
- The gas pressure is $\beta P = \rho k_B T / \mu m_H$ and the radiation

pressure is $(1 - \beta)P = a_{\text{rad}}T^4/3$. The first gives $T = \beta\mu m_{\text{H}}P/\rho k_B$, and substituting it into the second gives $(1 - \beta)P = (a_{\text{rad}}/3)(\beta\mu m_{\text{H}}/k_B)^4 P^4/\rho^4$. Solving for P gives $P = K_{\text{p}}\rho^{4/3}$ with $K_{\text{p}} = [3(1 - \beta)/a_{\text{rad}}]^{1/3}(k_B/\beta\mu m_{\text{H}})^{4/3}$, which is constant because β and μ are.

3. With $n_{\text{p}} = 1$ the scale length is $r_{\text{s}}^2 = 2K_{\text{p}}/4\pi G = K_{\text{p}}/2\pi G$, which does not involve ρ_{c} , and $R = \pi r_{\text{s}}$. At $\xi_1 = \pi$ the derivative of $\sin \xi/\xi$ is $(\pi \cos \pi - \sin \pi)/\pi^2 = -1/\pi$, so $M = 4\pi r_{\text{s}}^3 \rho_{\text{c}} \pi^2/\pi = 4\pi^2 r_{\text{s}}^3 \rho_{\text{c}}$. The mean density is $3M/4\pi R^3 = 3\rho_{\text{c}}/\pi^2$, so $\rho_{\text{c}} = (\pi^2/3)\bar{\rho}$.
4. From the definition of r_{s} , $R \propto r_{\text{s}} \propto \rho_{\text{c}}^{(1-n_{\text{p}})/2n_{\text{p}}}$, and from the formula for the mass, $M \propto r_{\text{s}}^3 \rho_{\text{c}} \propto \rho_{\text{c}}^{(3-n_{\text{p}})/2n_{\text{p}}}$, since ξ_1 and the slope there depend only on n_{p} . Eliminating ρ_{c} gives $R \propto M^{(1-n_{\text{p}})/(3-n_{\text{p}})}$. A fully convective star has $n_{\text{p}} = \frac{3}{2}$, so $R \propto M^{-1/3}$, and with a given K_{p} the more massive star is the smaller.
5. With $\kappa = \kappa_0 \rho T^{-7/2}$ and $\rho \sim M/R^3$, the estimate $L \sim \sigma_{\text{SB}} R^4 T^4/\kappa M$ becomes $L \sim (\sigma_{\text{SB}}/\kappa_0) R^7 T^{15/2}/M^2$. Substituting $T \sim (\mu m_{\text{H}} G/k_B) M/R$ gives

$$L \sim (\sigma_{\text{SB}}/\kappa_0)(\mu m_{\text{H}} G/k_B)^{15/2} M^{11/2} R^{-1/2} \quad (\text{S.1})$$

, which is the stated dependence.

§4.4 Thermonuclear Reactions in Stars

1. The Gamow energy goes as $\mu_{\text{r}} Z_1^2 Z_2^2$. For two protons $\mu_{\text{r}} = m_{\text{p}}/2$ and $Z_1 Z_2 = 1$; for a proton and carbon-12, $\mu_{\text{r}} = 12m_{\text{p}}/13$ and $Z_1 Z_2 = 6$. The ratio is $36 \times (12/13)/(1/2) = 864/13$. Since $E_0 \propto E_{\text{G}}^{1/3}$ at fixed temperature, E_0 is larger by the factor $(864/13)^{1/3}$, which is about four.
2. With the substitution $r = r_{\text{t}} \sin^2 \theta$, the integral from r_{n} to r_{t} is $r_{\text{t}}[\theta + \sin \theta \cos \theta]$ evaluated between $\theta_{\text{n}} = \arcsin \sqrt{r_{\text{n}}/r_{\text{t}}}$ and $\pi/2$, which is $r_{\text{t}}[\pi/2 - \theta_{\text{n}} - \sin \theta_{\text{n}} \cos \theta_{\text{n}}]$. For small θ_{n} , both θ_{n} and $\sin \theta_{\text{n}} \cos \theta_{\text{n}}$ are close to $\sqrt{r_{\text{n}}/r_{\text{t}}}$, so the integral is $(\pi r_{\text{t}}/2)[1 - (4/\pi)\sqrt{r_{\text{n}}/r_{\text{t}}}]$, and the exponent is multiplied by the same bracket. The correction makes the exponent smaller, because the part of the barrier inside r_{n} , which is the highest part, is not there to be tunnelled through.

3. Taking logarithms, $\ln\langle\sigma v\rangle = \text{constant} - \frac{2}{3}\ln T - \tau_G$. Since $\tau_G \propto T^{-1/3}$, $d\tau_G/d\ln T = -\tau_G/3$. Therefore $\zeta = -\frac{2}{3} + \tau_G/3 = (\tau_G - 2)/3$.
4. Carbon-12 is destroyed by proton capture at the rate $n_H n_{12} \langle\sigma v\rangle_{12}$ per unit volume, and nitrogen-14 at the rate $n_H n_{14} \langle\sigma v\rangle_{14}$, where n_H is the number of protons per unit volume. In a steady state every step of the cycle runs at the same rate, so these are equal, and $n_{12}/n_{14} = \langle\sigma v\rangle_{14}/\langle\sigma v\rangle_{12}$. The capture by nitrogen-14 is the slowest step, so $\langle\sigma v\rangle_{14}$ is the smallest of the capture rates, and nitrogen-14 is the most abundant of the catalysts.
5. From the condition for the minimum, $\sqrt{E_G/E_0} = E_G^{1/2} E_0^{-1/2} = 2E_0/k_B T$, so the tunnelling factor is $e^{-2E_0/k_B T}$ while the Maxwell factor is $e^{-E_0/k_B T}$; their exponents add to $\tau_G = 3E_0/k_B T$, of which the tunnelling supplies two thirds. The mean of E over $f(E)$ is $(2/\sqrt{\pi})(k_B T)^{-3/2} \int_0^\infty E^{3/2} e^{-E/k_B T} dE$; integrating by parts, with $E^{3/2}$ differentiated, reduces it to $\frac{3}{2}k_B T$ times the normalisation integral of f , which is one. The peak energy E_0 is several times this mean, so the reactions come from the high-energy tail of the distribution.

§4.5 The Nucleosynthesis of the Elements

1. Taking logarithms, $\ln \epsilon_{3\alpha} = \text{const} - 3\ln T - Q_{3\alpha}/k_B T$, and $d(1/T)/d\ln T = -1/T$, so the derivative is $Q_{3\alpha}/k_B T - 3$. A fractional rise δ changes $\ln T$ by approximately δ , and so changes $\ln \epsilon_{3\alpha}$ by $(Q_{3\alpha}/k_B T - 3)\delta$, which for small changes is the fractional change in the rate.
2. The energy released per unit mass of hydrogen is $q_H c^2$ with $q_H = (4m_H - m_{\text{He}})/4m_H$, and per unit mass of helium it is $q_{\text{He}} c^2$ with $q_{\text{He}} = (3m_\alpha - m_{\text{C}})/3m_\alpha$. The factor c^2 cancels in the ratio, which is q_{He}/q_H .
3. The temperature must be proportional to $E_G \propto \mu_r Z_1^2 Z_2^2$. For two carbon nuclei $Z_1 Z_2 = 36$ and $\mu_r = 6m_u$, and for two oxygen nuclei $Z_1 Z_2 = 64$ and $\mu_r = 8m_u$. The ratio is $(64/36)^2 (8/6) = (4/3)^4 (4/3) = (4/3)^5$.
4. Setting $d\mathcal{N}_A/dt = 0$ gives $\sigma_A \mathcal{N}_A = \sigma_{A-1} \mathcal{N}_{A-1}$ for each A , so

the product is the same along the path. If σ_A is smaller by a factor s than the neighbouring cross-sections, $\mathcal{N}_A$ must be larger by the factor s to keep the product equal.

5. Capture and decay are equally likely when $t_{\text{cap}} = t_\beta$, that is when $n_n = 1/(\langle\sigma v\rangle t_\beta)$. Decay and capture are independent processes with rates $1/t_\beta$ and $1/t_{\text{cap}}$, so the probability that decay comes first is the ratio of its rate to the total rate, $(1/t_\beta)/(1/t_\beta + 1/t_{\text{cap}}) = 1/(1 + t_\beta/t_{\text{cap}})$, which is close to one in the s-process and close to zero in the r-process.

§4.6 The Sun

1. Each conversion turns two protons into neutrons and releases two electron neutrinos, so neutrinos are produced at twice the rate of conversions. If the neutrinos carry away a fraction f_{loss} of Q , only $Q(1 - f_{\text{loss}})$ of each conversion appears as luminosity, so the rate of conversions is $L_\odot/Q(1 - f_{\text{loss}})$, and dividing twice this by $4\pi d^2$ gives the stated flux.
2. Hydrostatic equilibrium gives $dP/dr = -\rho GM_\odot/r^2$, and $\rho = P\mu m_H/k_B T$, so $d \ln P/dr = -(GM_\odot\mu m_H/k_B T)/r^2$. Integrating from r_0 to r gives the stated result. As $r \rightarrow \infty$ the pressure tends to the finite value $P_0 \exp(-GM_\odot\mu m_H/k_B T r_0)$. At coronal temperatures $k_B T$ is not much smaller than the gravitational energy $GM_\odot\mu m_H/r_0$ of a particle at the base, so the exponential is not very small, and this limit is far larger than the pressure of the interstellar gas, which therefore cannot hold the corona in place. The corona must flow outwards.
3. Pressure balance gives $P_{\text{in}} + B^2/2\mu_0 = P_{\text{out}}$. At the same temperature and mean molecular weight the density is proportional to the gas pressure, so $\rho_{\text{in}}/\rho_{\text{out}} = P_{\text{in}}/P_{\text{out}} = 1 - B^2/2\mu_0 P_{\text{out}}$. The gas pressure inside cannot be negative, so the largest field is $B = (2\mu_0 P_{\text{out}})^{1/2}$.
4. The difference of angular speeds is $\Omega_{\text{eq}}\delta \cos^2 \theta$, and the equator gains one full turn in the time $2\pi/(\Omega_{\text{eq}}\delta \cos^2 \theta)$. Near the pole $\cos^2 \theta \rightarrow 1$, which gives $2\pi/\delta\Omega_{\text{eq}}$.
5. The number of parents falls as $N_p = N_0 e^{-t \ln 2/t_{1/2}}$, where N_0 is the initial number, and each decay produces one daughter, so

$N_d = N_0 - N_p$. Then $N_d/N_p = e^{t \ln 2 / t_{1/2}} - 1$, and taking the logarithm of $1 + N_d/N_p$ gives the stated age.

§4.7 Stellar Evolution on and beyond the Main Sequence

1. The lifetime is proportional to $M/L \propto M/M^{3.5} = M^{-2.5}$, so $t_{\text{MS}} = t_{\text{MS},\odot}(M/M_\odot)^{-2.5}$. Setting $t_{\text{MS}} = t$ gives $M = M_\odot(t/t_{\text{MS},\odot})^{-0.4}$.
2. Burning a mass dM_c of gas in the shell turns a mass $X dM_c$ of hydrogen into helium and releases $X q_{\text{H}} c^2 dM_c$, which is radiated as the luminosity, so $L dt = X q_{\text{H}} c^2 dM_c$. At constant luminosity the time to grow from M_1 to M_2 is $X q_{\text{H}} c^2 (M_2 - M_1)/L$.
3. The total energy is $-E_{\text{th}}$ by the virial theorem, and it changes at the rate $L_{\text{nuc}} - L$, so $dE_{\text{th}}/dt = L - L_{\text{nuc}}$. If a small rise in temperature makes L_{nuc} exceed L , the thermal energy falls, the core cools, and the reaction rate falls back, while a small fall in temperature has the opposite effect. The burning therefore returns to the state in which $L_{\text{nuc}} = L$.
4. From $L = 4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4$ at constant L , $R \propto T_{\text{eff}}^{-2}$. A rise of the effective temperature by a factor k reduces the radius by a factor k^2 .
5. The volume of the shell is $4\pi(v_s t)^2 \Delta r$, and the number of particles is $M_s/\mu m_{\text{H}}$, so $n = M_s/(4\pi\mu m_{\text{H}} v_s^2 t^2 \Delta r)$, which goes as t^{-2} .

§4.8 Star Clusters and Their Ages

1. From $L_{\text{to}} = L_\odot (M_{\text{to}}/M_\odot)^{3.5}$ and $t = t_{\text{MS},\odot} (M_{\text{to}}/M_\odot)^{-2.5}$, eliminating $M_{\text{to}}/M_\odot = (L_{\text{to}}/L_\odot)^{1/3.5}$ gives $t = t_{\text{MS},\odot} (L_{\text{to}}/L_\odot)^{-2.5/3.5}$, and $2.5/3.5 = 5/7$. From $L = 4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4$, $T_{\text{eff}} \propto (L/R^2)^{1/4} \propto M^{(3.5-1.6)/4} = M^{0.475}$, and $M_{\text{to}} \propto t^{-0.4}$, so the effective temperature of the turn-off goes as $t^{-0.19}$.
2. The luminosity is found as $L_{\text{to}} \propto 10^{-0.4\mathcal{M}}$ with $\mathcal{M} = m - (m - \mathcal{M})$, so an error δ in the distance modulus multiplies the luminosity by $10^{0.4\delta} = e^{0.4 \ln 10 \delta} \approx 1 + 0.4 \ln 10 \delta$. Since $t \propto L_{\text{to}}^{-5/7}$, the fractional error in the age is $-\frac{5}{7}$ times that in the luminosity, which is $-(2 \ln 10 / 7) \delta$.

3. A star now on the giant branches left the main sequence at some time between $t - t_g$ and t , so its main-sequence lifetime lies between $t - t_g$ and t . With $M \propto t_{\text{MS}}^{-0.4}$ its mass lies between M_{to} and $M_{\text{to}}(1 - t_g/t)^{-0.4} \approx M_{\text{to}}(1 + 0.4 t_g/t)$, using the binomial expansion.
4. The distance is $d = 10^{(m_V - \mathcal{M}_V)/5} \times 10$ pc. The true distance modulus is $m_V - (\mathcal{M}_V + \Delta)$, smaller by Δ than the one used, so the true distance is $10^{-\Delta/5}$ times the distance found. The distance found is therefore too large by the factor $10^{\Delta/5}$.

Chapter 5. Stellar Death and Compact Objects

§5.1 Degenerate Matter and White Dwarfs

1. Allowing two spin states, $N_e = 2V \cdot \frac{4}{3}\pi p_F^3/h^3$. With $h = 2\pi\hbar$ this gives $n_e = p_F^3/3\pi^2\hbar^3$, so $p_F = \hbar(3\pi^2 n_e)^{1/3}$, as found from the octant of Figure 68.
2. Since $h = 2\pi\hbar$, $n_Q^{2/3} = m_e k_B T / 2\pi\hbar^2$, so $\hbar^2/2m_e k_B T = 1/4\pi n_Q^{2/3}$. Then $E_F/k_B T = (3\pi^2 n_e)^{2/3}/4\pi n_Q^{2/3} = \frac{1}{4} 3^{2/3} \pi^{1/3} (n_e/n_Q)^{2/3}$. The numerical factor is close to one, so $E_F \gg k_B T$ when $n_e \gg n_Q$.
3. At the minimum $R = 2C_1/C_2$, so $E_K = C_1/R^2 = C_2/2R$ and $E_{\text{grav}} = -C_2/R = -2E_K$. The total energy is $E_K - 2E_K = -E_K = -C_2^2/4C_1$. Since $C_2 \propto M^2$ and $C_1 \propto M^{5/3}$, the energy goes as $-M^{7/3}$.
4. Setting $\hbar(3\pi^2 n_e)^{1/3} = m_e c$ with $n_e = \rho/\mu_e m_H$ gives $\rho_{\text{rel}} = \mu_e m_H (m_e c/\hbar)^3/3\pi^2$. Write the mass-radius relation as $R^3 = K_R^3/M$ with $K_R^3 = (9\pi/4)^2 \hbar^6/[G^3 m_e^3 (\mu_e m_H)^5]$. The mean density $3M/4\pi R^3 = 3M^2/4\pi K_R^3$ equals ρ_{rel} when $M^2 = (9\pi/4)(\hbar c/G)^3/(\mu_e m_H)^4$, in which the electron mass has cancelled. Then $M = (9\pi/4)^{1/2}(\hbar c/G)^{3/2}/(\mu_e m_H)^2$, and dividing by $M_{\text{Ch}} = (5/4)^{3/2}(9\pi/4)^{1/2}(\hbar c/G)^{3/2}/(\mu_e m_H)^2$ gives $(4/5)^{3/2}$, about seven tenths.
5. Dividing the equation of hydrostatic equilibrium by the equation of radiative diffusion gives $dP/dT = 64\pi\sigma_{\text{SB}}GMT^3/3\kappa L$. With $\rho = \mu m_H P/k_B T$, the opacity is $\kappa = \kappa_0 \mu m_H P T^{-9/2}/k_B$, so

$P dP = C_{\text{env}}(M/L)T^{15/2} dT$ with $C_{\text{env}} = 64\pi\sigma_{\text{SB}}Gk_B/3\kappa_0\mu m_{\text{H}}$. Integrating from the surface gives $P^2 = \frac{4}{17}C_{\text{env}}(M/L)T^{17/2}$, so $P \propto (M/L)^{1/2}T^{17/4}$ and $\rho \propto P/T \propto (M/L)^{1/2}T^{13/4}$. At the base of the layer this density equals a constant times $T_c^{3/2}$, so $(M/L)^{1/2} \propto T_c^{3/2-13/4} = T_c^{-7/4}$, and therefore $L \propto MT_c^{7/2}$.

§5.2 Core Collapse and Supernovae

1. The binding energy is the energy needed to break a nucleus into its nucleons. Breaking iron-56 completely requires $E_{\text{b}}(^{56}\text{Fe})$, of which $13E_{\text{b}}(^4\text{He})$ would be released again if the nucleons were reassembled into thirteen helium nuclei, so the first reaction absorbs $E_{\text{b}}(^{56}\text{Fe}) - 13E_{\text{b}}(^4\text{He})$. The second reaction breaks a helium nucleus into nucleons and absorbs $E_{\text{b}}(^4\text{He})$. Breaking iron completely absorbs $E_{\text{b}}(^{56}\text{Fe})$ per nucleus, or $E_{\text{b}}(^{56}\text{Fe})/56m_{\text{u}}$ per unit mass. The ratio of this to the thermal energy per unit mass is $2E_{\text{b}}(^{56}\text{Fe})/3k_B T$, which is much larger than one at any temperature at which $k_B T$ is small compared with the binding energy per nucleus, so photodisintegration can remove a large part of the thermal energy of the core.
2. Setting $N_{\text{det}} = 1$ gives $d^2 = f_{\text{a}}E_{\text{grav}}\sigma N_{\text{p}}/4\pi\bar{E}$, so the greatest distance is $d = (f_{\text{a}}E_{\text{grav}}\sigma N_{\text{p}}/4\pi\bar{E})^{1/2}$. It goes as $N_{\text{p}}^{1/2}$, and doubling the mass of the detector doubles N_{p} and multiplies the distance by $\sqrt{2}$.
3. The number of nucleons burned is $M_{\text{b}}/m_{\text{u}}$, so the energy released is $\Delta E_{\text{b}}M_{\text{b}}/m_{\text{u}}$. The binding energy of a uniform sphere is $\frac{3}{5}GM^2/R$. The white dwarf is unbound if $\Delta E_{\text{b}}M_{\text{b}}/m_{\text{u}} > \frac{3}{5}GM^2/R$, that is if $M_{\text{b}}/M > 3GMm_{\text{u}}/5R\Delta E_{\text{b}}$, in which GMm_{u}/R is the magnitude of the gravitational potential energy of a nucleon at the surface.
4. Setting $dN_{\text{Co}}/dt = 0$ gives $e^{-t/t_{\text{Co}}}/t_{\text{Co}} = e^{-t/t_{\text{Ni}}}/t_{\text{Ni}}$, or $e^{t(1/t_{\text{Ni}}-1/t_{\text{Co}})} = t_{\text{Co}}/t_{\text{Ni}}$. Taking the logarithm and using $1/t_{\text{Ni}} - 1/t_{\text{Co}} = (t_{\text{Co}} - t_{\text{Ni}})/t_{\text{Ni}}t_{\text{Co}}$ gives the time stated.
5. The flux is $L/4\pi d^2$, and on the late part of the curve $\ln L = \text{constant} - t/t_{\text{Co}}$. Since $\log_{10} F = \log_{10} e \ln F$, the magnitude is $m = \text{constant} + 2.5 \log_{10} e t/t_{\text{Co}}$, whose derivative is $2.5 \log_{10} e/t_{\text{Co}}$. The distance and the number of nickel nuclei enter L and F

only as constant factors, which become additive constants in m and do not affect its rate of change.

§5.3 Neutron Stars and Pulsars

1. The white dwarf has $R \propto \hbar^2/[Gm_e(\mu_e m_H)^{5/3}]$ and the neutron star $R \propto \hbar^2/Gm_n^{8/3}$, with the same constant and the same factor $M^{-1/3}$. The ratio of the neutron star to the white dwarf is $m_e(\mu_e m_H)^{5/3}/m_n^{8/3} = (m_e/m_n)(\mu_e m_H/m_n)^{5/3}$. With $\mu_e = 2$ and $m_n \approx m_H$ this is $2^{5/3}/1840 \approx 1/580$.
2. The angular momentum $\frac{2}{5}MR^2\Omega$ is constant, so $\Omega \propto R^{-2}$, and the rotational kinetic energy $\frac{1}{5}MR^2\Omega^2$ goes as R^{-2} ; it grows by the factor $(R_i/R)^2$. The field goes as R^{-2} , so B^2R^3 goes as R^{-1} , and the magnetic energy grows by the factor R_i/R . The rotational and the magnetic energy both come from the gravitational potential energy released in the collapse, since the matter has to be pulled inwards against the centrifugal effect of the rotation and against the pressure of the compressed field.
3. From $\dot{\Omega} = -K_\Omega\Omega^3$, differentiating with respect to time gives $\ddot{\Omega} = -3K_\Omega\Omega^2\dot{\Omega}$, so $\Omega\ddot{\Omega}/\dot{\Omega}^2 = -3K_\Omega\Omega^3/\dot{\Omega} = 3$. With $\Omega = 2\pi/P_s$, the relation $\dot{\Omega} \propto -\Omega^3$ becomes $P_s\dot{P}_s = \text{constant}$, and differentiating with the product rule gives $\ddot{P}_s = -\dot{P}_s^2/P_s$.
4. From $\Omega^{-2} - \Omega_0^{-2} = 2K_\Omega t$ with $\Omega = 2\pi/P_s$, we have $P_s^2 - P_0^2 = 8\pi^2 K_\Omega t$. Since $K_\Omega = P_s\dot{P}_s/4\pi^2$, this gives $t = (P_s^2 - P_0^2)/2P_s\dot{P}_s = (P_s/2\dot{P}_s)[1 - (P_0/P_s)^2]$. The bracket is less than one, so the true age is less than t_{char} .
5. Since $\sin i \leq 1$, $f_m \leq M_2^3/(M_1 + M_2)^2$, and since $M_1 \geq 0$, $(M_1 + M_2)^2 \geq M_2^2$, so $f_m \leq M_2$. For $\sin i = 1$ and $M_1 = 0$ the mass function is M_2 , so $M_2 = f_m$. For $M_1 = f_m$, let $M_2 = xf_m$; then $x^3 = (1+x)^2$, whose root is $x \approx 2.15$, which is where the accent curve of Figure 80 crosses $M_1 = f_m$.

Chapter 6. The Interstellar Medium

§6.1 Gas, Dust, and the Phases of the Interstellar Medium

1. With $A_\lambda = C/\lambda$ for a constant C , $A_V = C/\lambda_V$ and $E_{B-V} = A_B - A_V = C(1/\lambda_B - 1/\lambda_V)$. The ratio is $\mathcal{R}_V = (1/\lambda_V)/(1/\lambda_B - 1/\lambda_V)$, and multiplying the numerator and denominator by $\lambda_B\lambda_V$ gives $\lambda_B/(\lambda_V - \lambda_B)$.
2. The flux at the grain is $L/4\pi r^2$, so the grain absorbs $\pi a_g^2 L/4\pi r^2$ per unit time. It radiates $4\pi a_g^2 \sigma_{\text{SB}} T_g^4$ from its whole surface. Equating the power absorbed to the power radiated gives $T_g^4 = L/16\pi\sigma_{\text{SB}}r^2$, in which a_g has cancelled. A grain that radiates poorly at the long wavelengths where a cold body emits most of its energy must reach a higher temperature to radiate what it absorbs, so real grains are hotter than T_g .
3. The flux per unit frequency leaving the surface is πB_ν , and dividing by $h\nu$ gives the number of photons, so $\dot{N}_{\text{ion}} = 4\pi R^2 \int_{\nu_0}^{\infty} (\pi B_\nu/h\nu) d\nu$ with $\nu_0 = \chi_{\text{H}}/h$. With $B_\nu \approx (2h\nu^3/c^2)e^{-h\nu/k_B T_{\text{eff}}}$ and $x = h\nu/k_B T_{\text{eff}}$ this becomes $(8\pi^2 R^2/c^2)(k_B T_{\text{eff}}/h)^3 \int_{x_0}^{\infty} x^2 e^{-x} dx$. Integrating by parts twice, each time differentiating the power of x and integrating the exponential, gives $\int_{x_0}^{\infty} x^2 e^{-x} dx = (x_0^2 + 2x_0 + 2)e^{-x_0}$. For the cooler star x_0 is doubled and T_{eff}^3 is divided by eight, so the ratio is $8(x_0^2 + 2x_0 + 2)e^{x_0}/(4x_0^2 + 4x_0 + 2)$, which grows exponentially with x_0 and exceeds a hundred when x_0 is about four, as it is for an O star.
4. Since $r_S^3 \propto n_{\text{H}}^{-2}$, doubling the density reduces r_S by the factor $2^{2/3}$. The mass of ionised gas is proportional to $r_S^3 n_{\text{H}} \propto n_{\text{H}}^{-1}$, so it halves. The thickness of the front is a few times $1/n_{\text{H}}\sigma_{\text{ion}}$, so its ratio to r_S goes as $1/n_{\text{H}}r_S \propto n_{\text{H}}^{-1/3}$, and it is reduced by the factor $2^{1/3}$.
5. Equal pressures give $n_{\text{cold}}k_B T_{\text{cold}} = n_{\text{warm}}k_B T_{\text{warm}}$, so $n_{\text{cold}}/n_{\text{warm}} = T_{\text{warm}}/T_{\text{cold}}$. Equal masses mean equal numbers of atoms, $n_{\text{cold}}V_{\text{cold}} = n_{\text{warm}}V_{\text{warm}}$, so $V_{\text{warm}}/V_{\text{cold}} = n_{\text{cold}}/n_{\text{warm}} = T_{\text{warm}}/T_{\text{cold}}$, and the warm gas fills the larger volume by the ratio of the temperatures.

Chapter 7. The Milky Way

§7.1 The Milky Way

1. For a flat curve $(dv_c/ds)_0 = 0$, so $\mathcal{A} = v_0/2s_0$ and $\mathcal{B} = -v_0/2s_0$. For the Keplerian curve $(dv_c/ds)_0 = -v_0/2s_0$, so $\mathcal{A} = 3v_0/4s_0$ and $\mathcal{B} = -v_0/4s_0$. For rigid rotation $(dv_c/ds)_0 = v_0/s_0$, so $\mathcal{A} = 0$ and $\mathcal{B} = -v_0/s_0$. The radial velocity $\mathcal{A}d \sin 2l$ vanishes for rigid rotation, since then ω is the same at every radius and the distances between stars do not change.
2. The mass inside radius s is $\int_0^s 4\pi s'^2 \rho ds' = v_c^2 s/G$, so the halo has mass $v_c^2 s_h/G$. Beyond s_h the halo acts as a point mass, by the shell theorem, and the circular speed is $\sqrt{GM/s} = v_c \sqrt{s_h/s}$, which falls as $s^{-1/2}$.
3. For $90^\circ < l < 180^\circ$, $\cos l < 0$, so $s^2 = s_0^2 - 2s_0 d \cos l + d^2 > s_0^2$ at every point, and every point lies outside the solar circle. For a flat curve $\omega = v_0/s$ is then smaller than ω_0 , and $\sin l > 0$, so $v_r = (\omega - \omega_0)s_0 \sin l < 0$. In Figure 87 the line of sight in this range points away from the centre, so s increases steadily along it and never has a minimum.
4. The star is furthest from the black hole on the sky when it crosses the plane of the sky, so $a = \theta_m d$. The whole of its orbital speed $2\pi a/P_{\text{orb}}$ lies along the line of sight when it passes in front of or behind the black hole, so $K = 2\pi a/P_{\text{orb}}$. Therefore $a = KP_{\text{orb}}/2\pi$ and $d = KP_{\text{orb}}/2\pi\theta_m$. Kepler's third law then gives $M = 4\pi^2 a^3/GP_{\text{orb}}^2 = K^3 P_{\text{orb}}/2\pi G$.
5. The mass inside s is $M(s) = \frac{4}{3}\pi s^3 \bar{\rho}$, and $M(s) = sv_c^2/G$, so $v_c^2 = \frac{4}{3}\pi G s^2 \bar{\rho}$. The period is $2\pi s/v_c = 2\pi/\sqrt{4\pi G \bar{\rho}/3} = \sqrt{3\pi/G\bar{\rho}}$, which depends on the rotation curve only through $\bar{\rho}$.

Chapter 8. Galaxies

§8.1 Galaxies and Their Scaling Relations

1. Integrating by parts as in the text,

$$\begin{aligned} L(s) &= 2\pi\mathcal{J}_0 \int_0^s s' e^{-s'/s_d} ds' \\ &= 2\pi\mathcal{J}_0 s_d^2 [1 - (1 + s/s_d)e^{-s/s_d}] \end{aligned} \quad (\text{S.2})$$

. Setting $L(s_{1/2})$ equal to half of $L = 2\pi\mathcal{J}_0 s_d^2$ gives $1 - (1 + x)e^{-x} = \frac{1}{2}$, which is the stated equation.

2. The disc is tilted about its major axis, so lengths along that axis appear unchanged, while a length R along the perpendicular direction in the disc makes the angle i with the plane of the sky and appears shortened to $R \cos i$. The image is therefore an ellipse with semi-axes R and $R \cos i$, and $q = \cos i$. Then $\sin i = \sqrt{1 - q^2}$, and $v_c = W/2 \sin i = W/2\sqrt{1 - q^2}$.
3. The luminosity is $L = 2\pi\mathcal{J}_e e^{k_V} \int_0^\infty e^{-k_V(s/s_e)^{1/4}} s ds$. With $s = s_e(u/k_V)^4$, $ds = 4s_e u^3 du/k_V^4$, so $s ds = 4s_e^2 u^7 du/k_V^8$, and the integral is $4s_e^2 k_V^{-8} \times 7!$. Therefore $L = 8 \times 7! \pi e^{k_V} k_V^{-8} \mathcal{J}_e s_e^2 = 8! \pi e^{k_V} k_V^{-8} \mathcal{J}_e s_e^2$.
4. Equal circular speeds give equal luminosities, and hence equal absolute magnitudes. The distance moduli then differ by $\Delta m = 5 \log_{10}(d_2/d_1)$, so $d_2/d_1 = 10^{\Delta m/5}$.
5. The mass inside s is sv_c^2/G , and by Problem 1 the luminosity inside s is $2\pi\mathcal{J}_0 s_d^2 [1 - (1 + s/s_d)e^{-s/s_d}]$. The ratio is $\Upsilon(s) = sv_c^2 / \{2\pi G \mathcal{J}_0 s_d^2 [1 - (1 + s/s_d)e^{-s/s_d}]\}$. For $s \gg s_d$ the bracket tends to one, and $\Upsilon(s) \approx sv_c^2 / 2\pi G \mathcal{J}_0 s_d^2$, which is proportional to s .

Chapter 9. Cosmological Models, Parameters, and the Dark Sector

§9.1 The Expanding Universe

1. For galaxies of luminosity L_q alone, the calculation in the text gives $N_q(> F) = (4\pi n f_q/3)(L_q/4\pi F)^{3/2}$, and summing over q

gives $N(> F) = (4\pi n/3)(4\pi F)^{-3/2} \sum_q f_q L_q^{3/2}$, which is proportional to $F^{-3/2}$. Since $F \propto 10^{-0.4m}$, we have $F^{-3/2} \propto 10^{0.6m}$. The number of galaxies with fluxes between F and $F + dF$ is $|dN/dF| dF \propto F^{-5/2} dF$, and they deliver a flux proportional to $F^{-3/2} dF$; integrating from F upwards gives a total proportional to $F^{-1/2}$, which grows without limit as $F \rightarrow 0$. The faint, distant galaxies together contribute an unlimited flux, which is Olbers' paradox in another form.

2. The separation $d \propto (t - t_b)^\zeta$ gives $\dot{d}/d = \zeta/(t - t_b)$, and since $v_r = \dot{d}$ this is the Hubble constant at time t , $H = \zeta/(t - t_b)$. At t_0 this gives $t_0 - t_b = \zeta/H_0 = \zeta t_H$. The speed $\dot{d} \propto (t - t_b)^{\zeta-1}$ decreases with time when $\zeta < 1$, so these values describe an expansion that slows down, and they give an age less than t_H .
3. The components of $\omega \times \mathbf{r}$ are $v_1 = \omega_2 r_3 - \omega_3 r_2$ and the two obtained by cyclic permutation, so $H_{12} = -\omega_3$, $H_{21} = \omega_3$, and so on, with zeros on the diagonal, and $H_{kj} = -H_{jk}$. An antisymmetric matrix has zeros on its diagonal and is fixed by the three entries H_{32} , H_{13} , and H_{21} , which can be taken as ω_1 , ω_2 , and ω_3 . The field is linear, so by the argument of the text the observer at B measures $\mathbf{v}(\mathbf{r}_B + \mathbf{r}') - \mathbf{v}(\mathbf{r}_B) = \omega \times \mathbf{r}'$, a circulation about an axis through B parallel to ω . The radial velocity of a galaxy at relative position $\mathbf{r}'$ is $\mathbf{r}' \cdot (\omega \times \mathbf{r}')/r' = 0$, since a scalar triple product with two equal vectors vanishes.
4. The energy radiated is Lt_H , which requires a mass $Lt_H/f_{\text{He}}c^2$ of hydrogen to be turned into helium. Dividing by M gives $\Delta Y = Lt_H/f_{\text{He}}Mc^2 = t_H/f_{\text{He}}Yc^2$. With the measured values of the quantities and a mass-to-light ratio of a few in solar units, ΔY is a few per cent, far below the observed quarter.
5. The luminosity per unit volume is $nL = n4\pi R^2\sigma_{\text{SB}}T_{\text{eff}}^4$, and with no absorption the energy density grows at this rate, so it reaches u_s after a time $u_s/nL = 4\sigma_{\text{SB}}T_{\text{eff}}^4/(cn4\pi R^2\sigma_{\text{SB}}T_{\text{eff}}^4) = 1/(n\pi R^2c) = \ell/c$.

§9.2 The Evidence for Dark Matter

1. With $P = \rho\sigma_v^2$, hydrostatic equilibrium gives $\sigma_v^2 d\rho/dr = -GM\rho/r^2$. For $\rho = \sigma_v^2/2\pi Gr^2$ we have $d\rho/dr = -2\rho/r$, so

the left-hand side is $-2\sigma_v^2\rho/r$, and this equals the right-hand side if $M = 2\sigma_v^2 r/G$. The same mass follows from integrating $dM/dr = 4\pi r^2\rho = 2\sigma_v^2/G$, so the density is a solution. The circular speed is $v_c^2 = GM/r = 2\sigma_v^2$.

2. From Hubble's law $d = v_r/H_0 \propto 1/H_0$. The radius is the angular radius multiplied by d , so $R \propto 1/H_0$, and the virial mass $M \propto \sigma_v^2 R$ is proportional to $1/H_0$, since σ_v is measured directly. The luminosity is $L = 4\pi d^2 F \propto 1/H_0^2$. Dividing gives $M/L \propto H_0$.
3. Let the particle move along the x -axis, passing the body at $x = 0$. The perpendicular component of the acceleration is $GMb/(b^2 + x^2)^{3/2}$, and with $dt = dx/c$ the change of perpendicular velocity is $(GM/c) \int b(b^2 + x^2)^{-3/2} dx$. The substitution $x = b \tan \phi$ turns the integral into $b^{-1} \int_{-\pi/2}^{\pi/2} \cos \phi d\phi = 2/b$, so the change of velocity is $2GM/bc$, and dividing by c gives the angle $2GM/c^2 b$.
4. The roots of $\theta^2 - \beta\theta - \theta_E^2 = 0$ have product $-\theta_E^2$ and difference $\sqrt{\beta^2 + 4\theta_E^2} \geq 2\theta_E$. For $\beta \gg \theta_E$ the binomial expansion gives $\sqrt{\beta^2 + 4\theta_E^2} \approx \beta + 2\theta_E^2/\beta$, so $\theta_+ \approx \beta + \theta_E^2/\beta$, close to the source, and $\theta_- \approx -\theta_E^2/\beta$, close to the lens.
5. For $r_c \ll R$ the gas gives $M(R) \approx 3\beta_g k_B T R / G \mu m_H$, and the virial theorem gives $M = 5\sigma_v^2 R / G$. Equating the two and cancelling R/G gives $k_B T = 5\mu m_H \sigma_v^2 / 3\beta_g$.

§9.3 Dark Matter Candidates and Their Detection

1. For $\eta \ll 1$, $A \approx 2/(\eta \cdot 2) = 1/\eta$. At $\eta = 1$, $A = 3/(1 \cdot \sqrt{5}) = 3/\sqrt{5}$. For $\eta \gg 1$, write $A = (\eta^2 + 2)/[\eta^2(\eta^2 + 4)]^{1/2} = (1 + 2/\eta^2)(1 + 4/\eta^2)^{-1/2}$; the binomial expansion to second order in $1/\eta^2$ gives $(1 + 2/\eta^2)(1 - 2/\eta^2 + 6/\eta^4) = 1 + 2/\eta^4$ to that order.
2. With constant ρ_{DM} , $\int_0^{D_S} D_L(D_S - D_L) dD_L/D_S = D_S^2/2 - D_S^2/3 = D_S^2/6$, so $\tau_{ml} = (4\pi G \rho_{DM}/c^2)(D_S^2/6) = 2\pi G \rho_{DM} D_S^2/3c^2$. Substituting the density and $D_S = s_0$ gives $v_c^2/6c^2$. The square of the Einstein radius in the plane of the lens is $(4GM/c^2)D_L(D_S - D_L)/D_S$, and the product $D_L(D_S - D_L)$ has its maximum at $D_L = D_S/2$, where its derivative $D_S - 2D_L$ vanishes.
3. A recoil E_R requires $2\mu_I^2 v^2/m_{nuc} \geq E_R$, which gives $v \geq$

$(m_{\text{nuc}} E_{\text{R}}/2\mu_{\text{r}}^2)^{1/2}$. For $m_{\text{nuc}} \gg m_{\text{DM}}$ the reduced mass is $\mu_{\text{r}} \approx m_{\text{DM}}$, so the smallest speed is $(m_{\text{nuc}} E_{\text{R}}/2m_{\text{DM}}^2)^{1/2}$, which grows as the square root of the mass of the nucleus.

4. Substituting ρ and $v_{\text{max}} = 2\sigma_v$ into the bound gives $m_\nu^4 \geq 3\sigma_v^2 h^3 / (2\pi G r^2 \cdot 4\pi g_\nu \cdot 8\sigma_v^3) = 3h^3 / 64\pi^2 g_\nu G r^2 \sigma_v$. Small galaxies have small r and small σ_v , so the right-hand side is largest for them; a small halo must pack its neutrinos densely into a small range of momenta, which the exclusion principle forbids unless each neutrino is heavy.
5. With $\Gamma \propto v_{\text{h}}$ and $v_{\text{h}} \approx v_\odot + v_\oplus \cos \psi \cos[2\pi(t - t_{\text{J}})/P_{\text{orb}}]$, dividing by the mean gives $\Gamma/\Gamma_0 = 1 + (v_\oplus \cos \psi / v_\odot) \cos[2\pi(t - t_{\text{J}})/P_{\text{orb}}]$. The perpendicular component $w_\perp$ enters v_{h}^2 only through $w_\perp^2$, so it changes v_{h} by about $w_\perp^2/2v_\odot$, and the rate by a fraction $w_\perp^2/2v_\odot^2$, which is of order $v_\oplus^2/v_\odot^2$.